

The Collatz Conjecture and Its Local Mathematical Corpus

A Literature-First Reconstruction

Working critical reconstruction

Live edition – not a completion claim

Abstract

This live edition reconstructs Collatz and adjacent mathematics from primary literature and an authorized local corpus. It separates sourced theorems, independent deductions, finite calculations, definitions, choices, analogies, conjectures, missing arrows, contradictions, retractions, and nonclaims. The document remains incomplete until its external coverage, claim, dependency, morphism, proof, formalization, and recovery ledgers all pass their release gates.

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1 Scope, authority, and exactness

1.1 Subject gate

The subject is the Collatz problem and mathematics joined to it by an explicit mathematical edge. An edge must be exhibited by a common typed object, an explicit map, a theorem whose hypotheses include an admitted Collatz object, or a proved reduction or obstruction. Shared vocabulary, analogous diagrams, and co-occurrence in a local folder do not suffice.

The zeta project supplied a workflow pattern and a frozen routing index. It supplies no mathematical zeta–Collatz claim. Any such claim, if encountered, must pass the same typed-map and exact-proof gate as every other proposed connection.

1.2 Source order

Published literature is read first and has no invented admission threshold. Primary source T_EX controls when it is available; a PDF is read when it is the sole or controlling witness. Modern local Chatnotes are reconstructed only after the relevant literature has fixed definitions and known results. Residual Gemini-science material is read last as unverified intake. Its correct bounded

results may survive independent proof, but its attribution or rhetorical conclusion supplies no evidence.

1.3 Typed records

Each mathematical record receives a stable identifier and one of the following roles: sourced theorem, independent theorem, calculation, definition, choice, analogy, conjecture, missing arrow, impossibility result, contradiction, retraction, or explicit nonclaim. These distinctions are epistemic and do not assert that the underlying mathematics is disconnected.

For every map the edition records its domain, codomain, coordinate formula, well-definedness, preserved operations or relations, kernel or fibres where applicable, injectivity and surjectivity status, inverse formula and identities where applicable, exceptional locus, information loss, hypotheses, and proof dependencies. A claimed nonexistence is admitted only with an explicit obstruction.

1.4 Propagation rule

When a source correction, counterexample, repaired proof, or new bounded result changes a dependent statement, the change is propagated through the live text, claim and dependency records, morphism ledger, programme reconstruction, formal certificates, exact gap records, and action receipts before any later consequence is used.

1.5 Current status

This is a live artifact. At the present checkpoint it fixes method and scope only; it does not yet assert content-level literature or local-corpus coverage, and it makes no claim about the Collatz conjecture’s endpoint.

2 Foundational literature spine

The same phrase “the Collatz map” is used for maps with different state spaces and different clocks. This section does not normalize those presentations. It first records the source symbols and then introduces project-local subscripts only where two source symbols would otherwise collide.

2.1 Three clocks and their exact first-return relation

Let $\mathbb{N}_{>0} = \{1, 2, \dots\}$ and let $\mathbb{N}_{>0}^{\text{odd}}$ be its odd elements. The current Tao source writes $\text{Col} : \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$ for the unshortened map [32, Sec. 1.1]; here it is displayed as

$$U(n) = \begin{cases} 3n + 1, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases} \quad (1)$$

Böhm–Sontacchi write $u : \mathbb{Z} \rightarrow \mathbb{Z}$, and Bernstein–Lagarias write T , for the shortened map

$$T(n) = \begin{cases} (3n + 1)/2, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases} \quad (2)$$

The same branch formula extends continuously to $T : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ [14, Eqs. (1.1)–(1.3)]. Crandall instead writes $C : D^+ \rightarrow D^+$, where $D^+ = \mathbb{N}_{>0}^{\text{odd}}$, for the odd-to-odd map

$$e(x) = \nu_2(3x + 1), \quad C_{\text{Cr}}(x) = \frac{3x + 1}{2^{e(x)}}. \quad (3)$$

The subscript in C_{Cr} is editorial disambiguation; the source symbol is C [1, p. 1281, Eq. (1.1)]. Tao calls this same odd-to-odd map Syr [32, Sec. 1.2, Eq. (S-def)].

Proposition 2.1 (the two exact first-return clocks). *For $x \in \mathbb{N}_{>0}^{\text{odd}}$, put $a = e(x)$. Then $a \geq 1$ and*

$$T^j(x) = \frac{3x + 1}{2^j} \quad (1 \leq j \leq a), \quad (4)$$

$$U^{j+1}(x) = \frac{3x + 1}{2^j} \quad (0 \leq j \leq a). \quad (5)$$

The values in (4) are even for $1 \leq j < a$ and odd for $j = a$. Consequently

$$C_{\text{Cr}}(x) = T^{e(x)}(x) = U^{e(x)+1}(x), \quad (6)$$

and these are respectively the first returns of the shortened and unshortened orbits to $\mathbb{N}_{>0}^{\text{odd}}$.

Proof. Because x is odd, $3x + 1$ is a positive even integer, so $a = \nu_2(3x + 1) \geq 1$. The odd branch of T gives the case $j = 1$ of (4). As long as $j < a$, the quotient has 2-adic valuation $a - j > 0$, so the next branch of T divides it by 2. Induction proves the first identity and shows that no earlier value is odd. For U , its first branch sends x to $3x + 1$ and each of the next a branches divides by 2, proving the second identity and its first-return claim. The terminal quotient is exactly (3). $\square$

Explicit nonclaim 2.2. Equation (6) is a variable-time first-return identity. It does not identify one iterate of C_{Cr} with one iterate of T or U , and it does not permit stopping times, periods, or density statements to be transferred without translating the clock.

2.2 Krasikov–Lagarias predecessor counts and the whole-orbit clock

Krasikov–Lagarias use exactly the shortened map T in (2). For a fixed positive target $a \not\equiv 0 \pmod{3}$, they define

$$\pi_a(x) = \#\{n \in \mathbb{N}_{>0} : n \leq x, \exists j \in \mathbb{N}_0, T^j(n) = a\}, \quad (7)$$

$$\pi_a^*(x) = \#\{n \in \mathbb{N}_{>0} : n \leq x, \exists j \in \mathbb{N}_0, T^j(n) = a, T^i(n) \leq x \ (0 \leq i \leq j)\}. \quad (8)$$

Thus $\pi_a^*(x) \leq \pi_a(x)$. For $k \geq 2$, a residue $m \bmod 3^k$ not divisible by 3, and $y \geq 0$, the typed auxiliary function is

$$\phi_k^m(y) = \inf_{\substack{a \equiv m \pmod{3^k} \\ a \text{ not contained in a finite } T\text{-cycle}}} \pi_a^*(2^y a). \quad (9)$$

The printed exponent 3^j in the congruence is an unbound-index source defect; Source Issue 2.6 records the manifestation and the forced repair.

Proposition 2.3 (shortened-to-unshortened clock embedding). *Fix $N \in \mathbb{N}_{>0}$, put $n_j = T^j(N)$, and define $\iota_N : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ by*

$$s_0 = 0, \quad s_{j+1} = s_j + \begin{cases} 1, & n_j \text{ even}, \\ 2, & n_j \text{ odd}, \end{cases} \quad \iota_N(j) = s_j. \quad (10)$$

Then

$$U^{\iota_N(j)}(N) = T^j(N) \quad (j \in \mathbb{N}_0). \quad (11)$$

The map ι_N is strictly increasing. Its fibres are singletons on its image and its equality kernel is the diagonal. The times outside its image are exactly $s_j + 1$ for those j with n_j odd, and the omitted state is

$$U^{s_j+1}(N) = 3n_j + 1 \geq 4. \quad (12)$$

Consequently

$$\exists j \in \mathbb{N}_0, T^j(N) = 1 \iff \exists r \in \mathbb{N}_0, U^r(N) = 1. \quad (13)$$

Proof. For even n , $T(n) = U(n)$. For odd n ,

$$U(n) = 3n + 1, \quad U^2(n) = \frac{3n + 1}{2} = T(n).$$

Induction using the two increments in (10) proves (11). Every increment is positive, which proves strict increase, singleton fibres, and the diagonal equality kernel. An increment of two contributes precisely the intermediate time $s_j + 1$; an increment of one contributes none. This proves the asserted complement of the image and (12). Every U -time is therefore either an image time or one of those intermediate times. The latter cannot carry the value 1, while image times carry exactly the T -orbit values, proving (13). $\square$

Since every U -orbit value is a positive integer, $\text{Col}_{\min}(N) = 1$ if and only if the U -orbit reaches 1. Proposition 2.3 therefore gives the exact set identity

$$\{N \leq x : \text{Col}_{\min}(N) = 1\} = \{N \leq x : \exists j \in \mathbb{N}_0, T^j(N) = 1\}, \quad (14)$$

where both sets are subsets of $\mathbb{N}_{>0}$. The cardinality on the right is $\pi_1(x)$. The clock map loses the intermediate state after an odd input, but (12) proves that this information loss does not affect target-1 reachability.

Theorem 2.4 (Krasikov–Lagarias, source theorem). *For every fixed $a \in \mathbb{N}_{>0}$ with $a \not\equiv 0 \pmod{3}$, there exists $x_0(a)$ such that*

$$\pi_a(x) \geq x^{0.84} \quad (x \geq x_0(a)). \quad (15)$$

This is Theorem 6.1 of [23, p. 254]. Its exponent is exactly $0.84 = 21/25$. The paper reports a computer-found positive feasible solution of $L_{11}^{NT}(\lambda)$ at $\lambda = 1.7922310$. The exact comparison

$$1792231^{25} - 2^{21}10^{150} > 0 \quad (16)$$

proves $\log_2(1.7922310) > 21/25$ without floating-point arithmetic. The paper does not print the $k = 11$ coordinate vector or an independently checkable feasibility certificate. Thus (16) is independently certified, while the feasibility premise and Theorem 2.4 retain their status as a published computer-assisted result.

Proposition 2.5 (typed final-target handoff). *Let $k \geq 2$, let $1 < \lambda \leq 2$, and suppose the explicitly defined linear program $L_k^{NT}(\lambda)$ has a feasible solution with principal coordinates c_k^m . Put $\gamma = \log_2 \lambda$. For every $0 < \beta < \gamma$ and every fixed $a \in \mathbb{N}_{>0}$ with $a \not\equiv 0 \pmod{3}$, there exists $x_0(a)$ such that*

$$\pi_a(x) \geq x^\beta \quad (x \geq x_0(a)).$$

Proof. Theorem 2.2 of the source gives, for $m \in [3^k]$,

$$\phi_k^m(y) \geq \Delta_1 c_k^m \lambda^y, \quad \Delta_1 = \frac{1}{4 \max_r c_k^r}, \quad y \geq 0.$$

First suppose that a is not in a finite T -cycle and $a \equiv 2 \pmod{3}$. With $m = a \bmod 3^k$ and $y = \log_2(x/a)$, equations (8) and (9) give, for $x \geq a$,

$$\pi_a(x) \geq \Delta_1 c_k^m a^{-\gamma} x^\gamma.$$

If instead $a \equiv 1 \pmod{3}$, source equation (2.1), $\phi_k^m(y) = \phi_k^{2m}(y-1)$ for $y \geq 1$, moves to the residue $2m \equiv 2 \pmod{3}$ and contributes the further fixed factor λ^{-1} . In either case the positive fixed prefactor dominates $x^{\beta-\gamma}$ once x exceeds an a -dependent threshold.

Now suppose a lies in a finite T -cycle, and let M be the maximum on that cycle. Choose $r \geq 1$ with $b = 2^r a > M$. Then $T^r(b) = a$. If b lay in a finite cycle, its orbit would reach a , so it would lie in the same cycle and satisfy $b \leq M$, a contradiction. The noncyclic case applies to b , and every orbit reaching b reaches a . Therefore $\pi_a(x) \geq \pi_b(x) \geq x^\beta$ beyond $x_0(b)$. $\square$

Source issue 2.6 (Krasikov–Lagarias manifestation and proof-text defects). The published paper prints 3^j , with j unbound, in (9); prints the real variable y , rather than the residue m , in a congruence modulo 3; prints $\delta = \beta_2 - \beta_1 > 0$ immediately after $\beta_1 > \beta_2 > \dots$, although the halting contradiction requires $\delta < 0$; closes the Theorem 5.1 induction with $k \in [3^m]$ rather than $m \in [3^k]$; reverses the indices in $\Delta c_m^k \lambda^y$; and omits a parenthesis from $T^{(j)}$ in Theorem 6.1. The arXiv source additionally prints the two three-lift exponents in (L4) as 3^k , while the journal and the source’s own subsequent minimum use 3^{k-1} , and substitutes a tree-vertex macro for ν in the initial interval of Theorem 5.1. The exact manifestations, typed repairs, and dependency reasons are recorded in `qa/KRASIKOV-LAGARIAS-2003-F029-difference-inequality-audit.md`.

Explicit nonclaim 2.7. Theorem 2.4 does not prove the Collatz conjecture, positive density, density one, a target-uniform threshold, or the claim for targets divisible by 3. Since $x^{0.84}/x \rightarrow 0$, it is not an *almost all* theorem. The paper does not prove the stronger rounded exponent 0.8417560, strict increase or convergence to 2 of its optimal λ_k , or optimality of the difference-inequality method.

2.3 Terras–Allouche–Korec: exact finite coordinates and density-one power descent

The four sources in this subsection use the shortened map

$$T(y) = \begin{cases} (3y+1)/2, & y \equiv 1 \pmod{2}, \\ y/2, & y \equiv 0 \pmod{2}, \end{cases}$$

which is exactly (2). It is not one iterate of the unshortened map U on an odd input. The transfer to the U -orbit is made only after the source results have been stated and audited in their own clock.

2.3.1 Terras's parity coordinate

Terras works on $\mathbb{N}_0$. Put

$$X(y) = \begin{cases} 1, & y \text{ odd}, \\ 0, & y \text{ even}, \end{cases} \quad X_i(y) = X(T^i(y)), \quad (17)$$

$$S_k(y) = \sum_{i=0}^{k-1} X_i(y), \quad \lambda_k(y) = 2^{-k} 3^{S_k(y)}, \quad (18)$$

$$E_k(y) = (X_0(y), \dots, X_{k-1}(y)) \in \{0, 1\}^k. \quad (19)$$

Terras's Theorem 1.1 gives the exact affine expression

$$T^k(y) = \lambda_k(y)y + \rho_k(y), \quad \rho_k(y) = \frac{\lambda_k(y)}{2} \sum_{i=0}^{k-1} X_i(y) \lambda_{i+1}(y)^{-1}. \quad (20)$$

In particular $\rho_k(y) \geq 0$; it is not an unspecified error term [24, pp. 241–242, Thm. 1.1].

Proposition 2.8 (Terras's exact parity coordinate). *For every $k \in \mathbb{N}_0$ and $x, y \in \mathbb{N}_0$,*

$$E_k(x) = E_k(y) \iff x \equiv y \pmod{2^k}. \quad (21)$$

Consequently E_k induces a bijection

$$\mathbb{Z}/2^k\mathbb{Z} \xrightarrow{\sim} \{0, 1\}^k.$$

Thus, on every complete interval of 2^k consecutive integers, each chronological parity word occurs exactly once.

Proof. The assertion is trivial for $k = 0$. Suppose it is known for k , and write $\epsilon = X(x) = X(y)$. On the common parity branch,

$$T(x) - T(y) = \frac{3^\epsilon(x - y)}{2}. \quad (22)$$

If $x \equiv y \pmod{2^{k+1}}$, then x, y have the same parity and (22) gives $T(x) \equiv T(y) \pmod{2^k}$. The inductive hypothesis applied to their next k iterates proves $E_{k+1}(x) = E_{k+1}(y)$.

Conversely, if $E_{k+1}(x) = E_{k+1}(y)$, then x, y have the same ϵ , and the final k coordinates give $E_k(T(x)) = E_k(T(y))$. Hence $T(x) \equiv T(y) \pmod{2^k}$. Equation (22) and the oddness of 3^ϵ imply $2^{k+1} \mid x - y$. This proves (21) by induction. Both finite sets in the induced map have 2^k elements, so the injection is a bijection. Translating any complete interval to a complete residue system proves the last assertion. $\square$

This proposition is the arithmetic content behind the independent Bernoulli coordinates in Terras's Corollaries 1.3–1.4. No random surrogate has replaced the orbit: probabilities below are normalized, exact counts on $\mathbb{Z}/2^k\mathbb{Z}$.

2.3.2 Coefficient stopping and actual stopping

Terras defines the two positive-time stopping functions

$$\chi(y) = \min\{j \in \mathbb{N}_{>0} : T^j(y) < y\}, \quad (23)$$

$$\tau(y) = \min\{j \in \mathbb{N}_{>0} : \lambda_j(y) < 1\}, \quad (24)$$

with value ∞ if the relevant set is empty. Time zero is excluded. Since $\rho_j(y) \geq 0$, (20) gives $\tau(y) \leq \chi(y)$. If $\tau(y) = j$, Terras's Theorem 1.9 proves that all sufficiently large $z \equiv y \pmod{2^j}$ satisfy $\chi(z) = j$. It follows that, for each fixed j , the two sets $\{\tau = j\}$ and $\{\chi = j\}$ differ by only finitely many integers [24, pp. 243–244].

For $k \geq 1$, define the finite-coordinate tail

$$F(k) = \mathbb{P}[\tau \geq k] = 2^{-k} \#\{1 \leq y \leq 2^k : \tau(y) \geq k\}. \quad (25)$$

The final equality is independent of the chosen complete residue block by Proposition 2.8.

Theorem 2.9 (Terras 1976 coefficient-tail theorem, with the printed support bound repaired). *The sequence $F(k)$ is nonincreasing and*

$$\lim_{k \rightarrow \infty} F(k) = 0.$$

More exactly, if $\gamma = \log 2 / \log 3$, then every length- k parity word counted by $F(k)$ has at most

$$q_k = \lfloor (k-1)(1-\gamma) \rfloor + 1 \quad (26)$$

zero coordinates. Hence

$$F(k) \leq 2^{-k} \sum_{a=0}^{q_k} \binom{k}{a} \xrightarrow{k \rightarrow \infty} 0. \quad (27)$$

Proof. The tail events $\{\tau \geq k+1\} \subseteq \{\tau \geq k\}$ prove monotonicity. Consider a word counted by $F(k)$, and let b' be the number of ones among its first $k-1$ coordinates. Since no coefficient drop has occurred before time k ,

$$\lambda_{k-1} = \frac{3^{b'}}{2^{k-1}} \geq 1.$$

For $k > 1$ equality is impossible because a positive power of 2 is not a power of 3; thus $b' > (k-1)\gamma$. The first $k-1$ positions therefore contain fewer than $(k-1)(1-\gamma)$ zeros, and the last coordinate contributes at most one more. Integrality gives (26).

Proposition 2.8 turns the word count into (27). Finally

$$3^2 > 2^3 \implies \gamma > \frac{1}{2} \implies \frac{q_k}{k} \longrightarrow 1 - \gamma < \frac{1}{2}.$$

The lower binomial tail in (27) therefore tends to zero, by the standard de Moivre–Laplace estimate used in the source (or by any equivalent fixed-distance binomial tail bound). $\square$

Source issue 2.10 (Terras 1976 proof-text defects and forced local repairs). Five source distinctions are required.

1. Proposition 1.7 prints, without a sign hypothesis,

$$T^k(y) < y \iff \frac{\rho_k(y)}{1 - \lambda_k(y)} < y.$$

It is false as printed: for $y = 1, k = 1$,

$$T(1) = 2 \not< 1, \quad \lambda_1(1) = \frac{3}{2}, \quad \rho_1(1) = \frac{1}{2},$$

but $\rho_1(1)/(1 - \lambda_1(1)) = -1 < 1$. Division is valid only under $\lambda_k(y) < 1$. Theorem 1.9 invokes it under $\tau(y) = k$, which supplies precisely that missing hypothesis.

2. The proof of Theorem 1.9 cites Proposition 1.9 for $\tau \leq \chi$. The relevant result is Proposition 1.8.
3. Definition 1.12 prints the proper-prefix range $1 \leq i \leq k - 2$. Formula (9), the stopping event, and the subsequent recurrence require every proper nonempty prefix, $1 \leq i \leq k - 1$. The literal range already loses the decisive prefix when $k = 2$.
4. In Theorem 1.17, a terminal word of length k , with a zeros and $b = k - a$ ones, has last digit zero and an active length- $k - 1$ prefix. Therefore

$$\frac{3^b}{2^{k-1}} > 1, \quad a < 1 + (k - 1)(1 - \gamma).$$

The source drops $+1$, and then uses $a \leq \lfloor k(1 - \gamma) \rfloor$. The terminal word $(1, 0)$ at $k = 2$ has $a = 1$, while the printed cutoff is 0. The repaired cutoff is exactly (26); its ratio to k has the same strict limit below $1/2$, so the theorem's limiting conclusion survives.

5. Definition 1.10 introduces point masses and lower-tail sums before Theorem 1.11 uses $\mathbb{P}[\tau \geq k]$; formula (6) supplies the direct complete-block definition only afterwards. Equation (25) states the tail on its finite block before use.

These are local repairs. None is treated as text literally printed by Terras [24, pp. 243–247].

Theorem 2.11 (Terras 1979 finite-complement density identity). *For every $k \in \mathbb{N}_{>0}$, the natural density exists and satisfies*

$$\delta\{y \in \mathbb{N}_{>0} : \chi(y) \geq k\} = \mathbb{P}[\tau \geq k] = F(k). \quad (28)$$

Both source inequalities are non-strict $\geq$.

Proof. For each fixed j , the 1976 coset theorem recalled above says that $\{\chi = j\}$ and $\{\tau = j\}$ differ by a finite set. The second set is a union of residue classes modulo 2^j ; hence

$$\delta\{\chi = j\} = \mathbb{P}[\tau = j].$$

Only the finite set of indices $1 \leq j < k$ is now summed:

$$\delta\{\chi < k\} = \sum_{j=1}^{k-1} \mathbb{P}[\tau = j] = \mathbb{P}[\tau < k].$$

Taking the complements in $\mathbb{N}_{>0}$ and in the finite probability space $\mathbb{Z}/2^k\mathbb{Z}$ gives (28). This is the explicit finite-complement closure supplied by the 1979 note; it does not interchange natural density with an infinite sum [25, pp. 101–102]. $\square$

Corollary 2.12 (Terras density-one finite stopping). *The set $\{y \in \mathbb{N}_{>0} : \chi(y) < \infty\}$ has natural density one.*

Proof. For every k ,

$$\{\chi = \infty\} \subseteq \{\chi \geq k\}.$$

Theorem 2.11 and Theorem 2.9 therefore give

$$\bar{\delta}\{\chi = \infty\} \leq F(k) \xrightarrow[k \rightarrow \infty]{} 0.$$

The exceptional set has upper density zero, so its complement has natural density one. $\square$

Explicit nonclaim 2.13. Corollary 2.12 leaves open a zero-density exceptional set and does not prove the Collatz conjecture. It is a theorem about first descent below the starting value, not eventual arrival at 1. The 1979 argument proves existence of the displayed natural densities by a finite complement; it supplies no license to sum arbitrary density limits.

2.3.3 Allouche's unnormalized residue system

Allouche begins with positive integers n, d , with $d \geq 2$ and $\gcd(n, d) = 1$, and a chosen complete representative system

$$A_d = \{0, r_1, \dots, r_{d-1}\} \subset \mathbb{Z} \quad (29)$$

modulo d . Let $\varphi : \mathbb{Z} \rightarrow A_d$ send an integer to its chosen representative. The map is

$$g(\ell) = \begin{cases} \ell/d, & \ell \equiv 0 \pmod{d}, \\ (n\ell - \varphi(n\ell))/d, & \ell \not\equiv 0 \pmod{d}. \end{cases} \quad (30)$$

After the introduction, the source assumes $n > d \geq 2$. The data (n, d, A_d, φ) are retained: in the classical specialization

$$n = 3, \quad d = 2, \quad A_2 = \{0, -1\},$$

(30) is the shortened Collatz map on $\mathbb{Z}$. Replacing the source representative -1 by 1 would alter the affine remainder and is not done.

For $k \geq 1$, set

$$\alpha(k, j) = \#\{0 \leq v < k : g^v(j) \not\equiv 0 \pmod{d}\}, \quad (31)$$

$$P_k(j) = g^k(j) - \frac{n^{\alpha(k, j)}}{d^k} j. \quad (32)$$

Also put

$$R = \max_{a \in A_d} |a|, \quad M = \frac{R}{n - d}. \quad (33)$$

Theorem 2.14 (Allouche's affine coordinate and exact branch counts). *For $k \geq 1$, $j, \lambda \in \mathbb{Z}$, and $0 \leq i \leq k$,*

$$g^k(d^k \lambda + j) = n^{\alpha(k, j)} \lambda + g^k(j), \quad (34)$$

$$\alpha(k, d^k \lambda + j) = \alpha(k, j), \quad (35)$$

$$|P_k(j)| \leq M \frac{n^k}{d^k}, \quad (36)$$

$$\#\{0 \leq j < d^k : \alpha(k, j) = i\} = \binom{k}{i} (d - 1)^i. \quad (37)$$

Thus $j \bmod d^k$ is the residue-fibre coordinate, λ is its integral quotient coordinate, and the affine multiplier is constant on the fibre.

More quantitatively, let $A < B$ be rational numbers, neither equal to any n^μ/d^ν with $\mu \in \mathbb{N}_0$, $\nu \in \mathbb{N}_{>0}$. Let C be the maximum of the absolute denominators of A, B , and $\vartheta = \mathbf{1}_{(A,B)}$. Then the finite-coordinate comparison used in Allouche's Theorem 1 is

$$\left| \sum_{1 \leq \ell \leq x} \vartheta \left(\frac{g^k(\ell)}{\ell} \right) - \frac{x}{d^k} \sum_{i=0}^k \binom{k}{i} (d-1)^i \vartheta \left(\frac{n^i}{d^k} \right) \right| \leq \frac{5}{2} d^k + MCn^k. \quad (38)$$

For every $\varepsilon > 0$, there are k_0 and $\eta \in (0, 1)$, depending only on the fixed source data and ε , such that

$$\frac{1}{d^k} \sum_{|i - \frac{d-1}{d}k| > \varepsilon k} \binom{k}{i} (d-1)^i \leq \eta^k \quad (k \geq k_0). \quad (39)$$

Proof. For $a \in \mathbb{Z}$, $t \in \mathbb{Z}$, and $\epsilon = 1_{\{a \not\equiv 0 \pmod{d}\}}$, the representative is unchanged after adding dt , and direct substitution in (30) gives

$$g(a + dt) = g(a) + n^\epsilon t. \quad (40)$$

Apply this first with $t = d^{k-1}\lambda$, and then induct on k . The first branch symbol is unchanged, the next $k-1$ symbols are those of $g(j)$, and their multipliers multiply. This proves (34) and (35).

At a nondivisible step the additive correction in (30) has absolute value at most R/d ; at a divisible step it is zero. Expanding k iterates and bounding every later branch multiplier by n/d gives

$$|P_k(j)| \leq \frac{R}{d} \sum_{r=0}^{k-1} \left(\frac{n}{d} \right)^r = \frac{R}{n-d} \left(\frac{n^k}{d^k} - 1 \right) \leq M \frac{n^k}{d^k},$$

which proves (36).

It remains to count branches without replacing them by a stochastic model. The map

$$j \bmod d^k \mapsto (j \bmod d, g(j) \bmod d^{k-1})$$

is a bijection. Indeed, for first residue $a = 0$, write $j = dq$, so $g(j) = q$, which determines $q \bmod d^{k-1}$. For $a \neq 0$, write $j = a + dq$; then

$$g(j) = \frac{na - \varphi(na)}{d} + nq,$$

and $\gcd(n, d) = 1$ makes $q \bmod d^{k-1}$ unique. There is one divisible first residue and $d-1$ nondivisible ones. Induction therefore gives

$$\binom{k-1}{i} (d-1)^i + (d-1) \binom{k-1}{i-1} (d-1)^{i-1} = \binom{k}{i} (d-1)^i,$$

proving (37). Finally, (38) is the exact proposition on pp. 9-08–9-10 of the source, obtained by partitioning the summation into residue fibres, using (34)–(36), and bounding the two rational interval endpoints. Equation (39) is source Lemma 4 [26, pp. 9-04–9-11]. $\square$

For completeness, the hypotheses actually consumed by Allouche's Theorem 1 are recorded rather than compressed into the phrase *a large-deviation argument*. Given $a > 1$, put

$$k = \left\lfloor \frac{\log x}{a \log n} \right\rfloor, \quad \mathcal{F}(x) = \sum_{1 \leq \ell \leq x} \vartheta \left(\frac{g^k(\ell)}{\ell} \right).$$

The theorem has three separate cases:

1. if $A < B < 0$, then $\mathcal{F}(x) = O(Cx^{1/a})$;

2. if $A < B$, $B > 0$, and

$$\frac{d-1}{d} - \frac{\log d}{\log n} - \frac{\log B}{k \log n} \geq \varepsilon,$$

then, for $k \geq k_0(\varepsilon)$,

$$\mathcal{F}(x) = O \left(Cx^{1/a} + x^{1 - \frac{|\log n|}{a \log n}} \right);$$

3. if $0 < A < B$ and

$$\frac{\log d}{\log n} + \frac{\log A}{k \log n} - \frac{d-1}{d} \geq \varepsilon,$$

the same estimate holds.

The implied constants depend on n, d, A_d . These are distinct hypothesis packages; no one of them is silently substituted for another.

Theorem 2.15 (Allouche's branch-slope obstruction, with the displayed domain completed). *Let $d < n$ be coprime positive integers. After the local completion below, there are functions $F, G : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying*

$$\begin{aligned} F(\ell) &= \ell/d + O(1), & G(\ell) &= \ell/d + O(1) & (\ell \equiv 0 \pmod{d}), \\ F(\ell) &= n\ell/d + O(1), & G(\ell) &= n\ell/d + O(1) & (\ell \not\equiv 0 \pmod{d}), \end{aligned}$$

such that every F -orbit is eventually periodic and only finitely many periods occur, whereas

$$|G^k(\ell)| \rightarrow \infty$$

for every ℓ outside a finite set.

Exact source audit and independent domain repair. Allouche chooses u with $d < n < d^u$. The displayed divisible branch of F is specified only for

$$\ell \equiv d^i \pmod{d^u}, \quad 1 \leq i \leq u.$$

These classes do not exhaust the multiples of d : for $d = 2, u = 3$, the class $\ell \equiv 6 \pmod{8}$ is missing. Thus the printed cases do not literally define the announced function $F : \mathbb{Z} \rightarrow \mathbb{Z}$.

For every $\ell \equiv 0 \pmod{d}$, let ρ_ℓ be the unique integer with

$$0 \leq \rho_\ell < d^{u-1}, \quad \frac{\ell}{d} + \rho_\ell \equiv 0 \pmod{d^{u-1}}, \quad (41)$$

and define the completed divisible branch by

$$F(\ell) = \frac{\ell}{d} + \rho_\ell. \quad (42)$$

Existence and uniqueness are the unique representative of $-\ell/d \bmod d^{u-1}$ in $[0, d^{u-1})$. On a source class $\ell \equiv d^i \pmod{d^u}$, it gives

$$\rho_\ell = \begin{cases} d^{u-1} - d^{i-1}, & i < u, \\ 0, & i = u, \end{cases}$$

so (42) agrees exactly with every printed divisible case. It also proves, on every formerly missing divisible residue,

$$F(\ell) \equiv 0 \pmod{d^{u-1}}, \quad \left| F(\ell) - \frac{\ell}{d} \right| < d^{u-1}. \quad (43)$$

Retain the printed nondivisible branches unchanged. The remainder of the source proof uses the divisible branch only through (43): pp. 9-12–9-14 obtain a uniform bound for the F -iterates, set $G = F + 1$, obtain an expanding lower bound for the G -iterates, and use finite predecessor fibres to isolate the finite exceptional set. Therefore that proof applies to the completed map [26, pp. 9-04, 9-12–9-14]. Formula (42) is an edition reconstruction, not a formula silently attributed to the printed source. $\square$

Proposition 2.16 (Allouche’s exact classical threshold extracted from the affine and counting lemmas). *Put*

$$c_A = \frac{3}{2} - \log_3 2. \quad (44)$$

For every real $c > c_A$, the set

$$\mathcal{M}_c^T = \{y \in \mathbb{N}_{>0} : \exists r \in \mathbb{N}_0, T^r(y) < y^c\} \quad (45)$$

has natural density one.

Proof. In the classical specialization of Theorem 2.14, write

$$A_m(y) = \alpha(m, y) = \sum_{i=0}^{m-1} X_i(y).$$

Equations (32) and (36), with $M = 1$, give

$$T^m(y) = \frac{3^{A_m(y)}}{2^m} y + P_m(y), \quad |P_m(y)| \leq \left(\frac{3}{2}\right)^m. \quad (46)$$

Choose $\varepsilon > 0$ with $c_A + \varepsilon < c$. On the ternary-height block $3^m \leq y < 3^{m+1}$, call y good when

$$A_m(y) \leq \left(\frac{1}{2} + \varepsilon\right) m.$$

For a good y ,

$$\frac{3^{A_m(y)}}{2^m} y \leq y \left(\frac{3^{1/2+\varepsilon}}{2}\right)^m.$$

If $q = 3^{1/2+\varepsilon}/2$, the ratio $y/3^m \in [1, 3)$ gives the uniform comparison

$$q^m \leq 3^{|\log_3 q|} y^{\log_3 q}.$$

Since $1 + \log_3 q = c_A + \varepsilon$, the main term is at most

$$C_\varepsilon y^{c_A + \varepsilon}, \quad C_\varepsilon = 3^{|\log_3 q|}.$$

The remainder has strictly smaller exponent:

$$\left(\frac{3}{2}\right)^m \leq y^{\log_3(3/2)}, \quad \log_3(3/2) = c_A - \frac{1}{2} < c_A.$$

Hence $T^m(y) < y^c$ for all sufficiently large good y .

By (37), every complete residue block of length 2^m contains exactly

$$\sum_{i > (1/2 + \varepsilon)m} \binom{m}{i}$$

bad residues. Its normalized value tends to zero by (39). The interval $[3^m, 3^{m+1})$ is a union of complete 2^m -blocks plus at most two end fragments, of total length below 2^{m+1} . Their relative contribution is

$$O\left(\frac{2^m}{3^m}\right) = O((2/3)^m).$$

Thus the proportion outside $\mathcal{M}_c^T$ in each ternary-height block tends to zero. Summing these geometrically increasing blocks proves that the exceptional proportion up to an arbitrary x tends to zero: after a fixed initial segment, each block error is at most $\epsilon_0 3^m$, and $\sum_{m \leq M} 3^m < \frac{3}{2} 3^M$, while $x \geq 3^M$ on the last block. Since ϵ_0 is arbitrary, $\delta(\mathcal{M}_c^T) = 1$. $\square$

Explicit nonclaim 2.17. Proposition 2.16 is an edition reconstruction from Allouche's printed Lemmas 1–4 and quantitative proposition; the decimal exponent is not printed there as a standalone theorem. Korec explicitly records $3/2 - \log_3 2 = 0.86907\dots$ as the Allouche threshold [27, p. 86]. The endpoint $c = c_A$ is not included. Theorem 2.15 shows, after its explicit domain repair, that the two leading branch slopes and bounded additive errors do not determine global orbit behaviour; it is not a result about the endpoint of the classical Collatz map.

2.3.4 Korec's improved power threshold

For real c , Korec defines

$$M_c = \{y \in \mathbb{N}_0 : \exists r \in \mathbb{N}_0, T^r(y) < y^c\}. \quad (47)$$

With X_i, E_m, S_m as in (17)–(19), he also defines, with a non-strict boundary,

$$U(m, d) = \#\{0 \leq y < 2^m : S_m(y) \leq md\}. \quad (48)$$

Proposition 2.8 gives the exact complete-block identity

$$U(m, d) = \sum_{0 \leq i \leq md} \binom{m}{i},$$

where an empty or nonintegral endpoint is interpreted by the displayed integer condition. Korec's Lemma 2 consequently gives

$$d > \frac{1}{2} \implies \frac{U(m, d)}{2^m} \longrightarrow 1. \quad (49)$$

Theorem 2.18 (Korec 1994 density estimate, with three local proof-text repairs). *If*

$$c > \log_4 3,$$

then M_c has natural density one. The inequality is strict.

Proof. It suffices to prove the assertion for $\log_4 3 < c < 1$: if $c \geq 1$, choose $\log_4 3 < c' < 1$, and $M_{c'} \subseteq M_c$ away from the immaterial inputs 0, 1.

For a large cutoff a , choose the least positive integer m such that

$$a \leq m^2 2^m. \quad (50)$$

Define the repaired parameter

$$d_* = \frac{1}{2} \left(\frac{c}{\log_2 3} + \frac{1}{2} \right). \quad (51)$$

Since

$$c > \log_4 3 = \frac{1}{2} \log_2 3,$$

the two endpoints whose arithmetic mean occurs in (51) satisfy

$$\frac{1}{2} < \frac{c}{\log_2 3};$$

hence

$$\frac{1}{2} < d_* < \frac{c}{\log_2 3}. \quad (52)$$

Consider $m2^m \leq y < a$, and put $k = S_m(y)$. For $0 \leq p < m$, every branch of T divides its input by at most 2, so

$$T^p(y) \geq \frac{y}{2^p} > m.$$

At each of the k odd steps the exact ratio is $(3T^p(y) + 1)/(2T^p(y))$, while at each even step it is $1/2$. Therefore

$$\begin{aligned} T^m(y) &< y \left(\frac{3m+1}{2m} \right)^k \left(\frac{1}{2} \right)^{m-k} \\ &= y \frac{3^k}{2^m} \left(1 + \frac{1}{3m} \right)^k \\ &\leq y \frac{3^k}{2^m} \left(1 + \frac{1}{3m} \right)^m < y \frac{3^k}{2^{m-1}}. \end{aligned} \quad (53)$$

The final inequality uses $(1 + 1/(3m))^m < e^{1/3} < 2$.

Because $c < 1$, equations (50) and (53) show that $T^m(y) < y^c$ whenever

$$(m^2 2^m)^{1-c} 3^k \leq 2^{m-1}. \quad (54)$$

Taking base-two logarithms and preserving every denominator proves the exact equivalence

$$\frac{k}{m} \leq \frac{c}{\log_2 3} - \frac{1 + 2(1-c) \log_2 m}{m \log_2 3}. \quad (55)$$

The right side tends to $c/\log_2 3 > d_*$. Hence, for all sufficiently large m ,

$$S_m(y) \leq md_* \implies T^m(y) < y^c.$$

The non-strict boundary is the one counted in (48).

By (49) and (52), the proportion of such y in every complete interval of 2^m consecutive integers tends to one. The minimality in (50) gives

$$a > (m-1)^2 2^{m-1}.$$

Consequently the discarded initial interval below $m2^m$ and the single terminal fragment shorter than 2^m have relative sizes bounded by

$$\frac{m2^m}{a} < \frac{2m}{(m-1)^2} \longrightarrow 0, \quad \frac{2^m}{a} < \frac{2}{(m-1)^2} \longrightarrow 0.$$

All remaining points lie in complete 2^m -blocks, where the exceptional proportion is

$$1 - \frac{U(m, d_*)}{2^m} \longrightarrow 0.$$

Thus $\#(M_c \cap [0, a))/a \rightarrow 1$, proving the theorem. $\square$

Source issue 2.19 (Korec 1994: the three printed proof defects). The theorem above is the published conclusion, but its printed proof requires three visible repairs [27, pp. 86–88].

1. The source sets

$$d = \frac{1}{2} \left(\frac{c}{\log_4 3} + \frac{1}{2} \right)$$

and immediately asserts $d < c/\log_2 3$. Since

$$\frac{1}{2} \left(\frac{c}{\log_4 3} + \frac{1}{2} \right) = \frac{c}{\log_2 3} + \frac{1}{4},$$

the assertion is false. Equation (51) replaces the denominator $\log_4 3$ by the $\log_2 3$ required by the ensuing comparison and proves the claimed open interval.

2. The source calls

$$\frac{k}{m} \leq \frac{c}{\log_2 3} - \frac{1 + 2(1-c)\log_2 m}{m}$$

equivalent to (54). Exact rearrangement gives (55), with an additional $\log_2 3$ in the final denominator. Because the numerator is positive and $\log_2 3 > 1$, the printed condition subtracts more; it remains a stronger sufficient condition, but it is not equivalent.

3. The proof text uses $S_m(y) < md$, whereas the definition (48) and the final block count use $S_m(y) \leq md$. The strict margin in (52) proves the implication with $S_m(y) \leq md_*$ for all sufficiently large m , so the defined block count applies including its boundary.

The witness time is $r = m$, and $m \leq \log_2 y$ on the retained interval. It is controlled by the starting value, not by a constant independent of that value.

Source issue 2.20 (Tao line 165: sign-inconsistent Allouche threshold). Tao's v5 and v7 source line 165, and the 2022 journal p. 2, print

$$\theta > \frac{3}{2} - \frac{\log 3}{\log 2} \approx 0.869. \quad (56)$$

The symbolic expression and decimal cannot both be correct. Indeed,

$$3^2 > 2^3 \implies \frac{\log 3}{\log 2} > \frac{3}{2},$$

so the printed symbolic right side is negative. The exact correction, proved independently in Proposition 2.16 and explicitly reported by Korec, is

$$\boxed{\theta > \frac{3}{2} - \log_3 2 = \frac{3}{2} - \frac{\log 2}{\log 3}} \approx 0.8690702464. \quad (57)$$

Tao's subsequent Korec threshold

$$\theta > \log_4 3 \approx 0.7924812504$$

is printed correctly [32, Sec. 1.1, source line 165]. This repair is not inferred from the decimal alone: the sign obstruction, Allouche-coordinate derivation, and Korec's exact retrospective formula agree.

Proposition 2.21 (exact Tao–Korec threshold and clock crosswalk). *Let*

$$c_K = \log_4 3, \quad c_A = \frac{3}{2} - \log_3 2.$$

Then:

1. $c_K < c_A$, with exact difference

$$c_A - c_K = \frac{(\log_2 3 - 1)(2 - \log_2 3)}{2 \log_2 3} > 0. \quad (58)$$

2. For every $y \in \mathbb{N}_{>0}$ and real c ,

$$\exists j \in \mathbb{N}_0, T^j(y) < y^c \iff \exists r \in \mathbb{N}_0, U^r(y) < y^c. \quad (59)$$

Consequently the density statements of Proposition 2.16 and Theorem 2.18 transfer to Tao's whole-orbit minimum without any change in their sets of starting values.

3. For $N \in \mathbb{N}_{>0}^{\text{odd}}$, define

$$a_j = \nu_2(3 \text{Syr}^j(N) + 1), \quad A_r = \sum_{j=0}^{r-1} a_j.$$

Then the exact accelerated affine coordinate is

$$\text{Syr}^r(N) = \frac{3^r}{2^{A_r}} N + \sum_{j=0}^{r-1} \frac{3^{r-1-j}}{2^{A_r-A_j}}, \quad A_0 = 0. \quad (60)$$

For $N > 1, r \geq 1$, the exponent of its multiplicative term is exactly

$$\log_N \left(\frac{3^r}{2^{A_r}} N \right) = 1 + \frac{r}{\log N} \left(\log 3 - \frac{A_r}{r} \log 2 \right). \quad (61)$$

At the parameter coordinate

$$\frac{A_r}{r} = 2, \quad \frac{r}{\log N} = \frac{1}{\log 4},$$

the right side of (61) is exactly $\log_4 3$.

Proof. Put $L = \log_2 3$. Since $2 < 3 < 4$, one has $1 < L < 2$, and

$$c_A - c_K = \frac{3}{2} - \frac{1}{L} - \frac{L}{2} = \frac{(L-1)(2-L)}{2L} > 0,$$

proving (58).

Proposition 2.3 proves that every shortened state $T^j(y)$ is the unshortened state $U^{\iota_y(j)}(y)$, giving the forward implication in (59). Conversely, every unshortened time outside the image of ι_y is the intermediate state

$$U^{s_j+1}(y) = 3T^j(y) + 1$$

following an odd shortened state. Its next unshortened state is

$$U^{s_j+2}(y) = \frac{3T^j(y) + 1}{2} = T^{j+1}(y).$$

If the intermediate state is below y^c , then its half is also below y^c . Thus every unshortened witness produces a shortened witness, which proves the reverse implication and the exact equality of starting sets.

Finally, the first-return identity gives

$$\text{Syr}^{j+1}(N) = \frac{3 \text{Syr}^j(N) + 1}{2^{a_j}}.$$

Induction expands this recurrence to (60); every summand in its remainder is positive and its denominator records the exact remaining valuation sum. Taking logarithms of the multiplicative term gives (61). Substitution of the two displayed parameter coordinates gives

$$1 + \frac{\log 3 - 2 \log 2}{\log 4} = \frac{\log 3}{\log 4} = \log_4 3.$$

□

Explicit nonclaim 2.22. The exact coordinate calculation in (61) explains Tao's statement that optimizing the descent parameters recovers Korec [32, source line 695]; it is not a substitute for Korec's natural-density proof. Tao's locally displayed loose choices $n_0 = 0.1 \log_2 x$ and $A_{n_0} > 1.9n_0$ do not themselves yield the optimal exponent. The separate $1.9n$ versus $1.9n_0$ source defect is audited elsewhere and is not conflated with Source Issue 2.20.

None of the Terras, Allouche, or Korec density-one theorems includes its strict endpoint or removes a possible zero-density exceptional set. None proves the Collatz conjecture. Equation (59) identifies the existence of a descent witness, not one iterate of T with one iterate of U ; their clocks remain related by the variable-time map ι_y .

2.4 Run lengths and chronological parity words

Put

$$\mathcal{A} = (\mathbb{Z}_{\geq 1})^{\mathbb{N}_0}, \quad \mathcal{B} = \{(b_k)_{k \geq 0} \in \{0, 1\}^{\mathbb{N}_0} : b_0 = 1 \text{ and infinitely many } b_k \text{ equal } 1\}.$$

Let $\sigma_{\mathcal{A}}$ and $\sigma_{\mathcal{B}}$ denote the left shifts. For $\mathbf{a} = (a_0, a_1, \dots) \in \mathcal{A}$, set

$$s_0 = 0, \quad s_j = \sum_{i=0}^{j-1} a_i \quad (j \geq 1),$$

and define $B : \mathcal{A} \rightarrow \mathcal{B}$ by

$$B(\mathbf{a})_k = 1 \iff k = s_j \text{ for some } j \geq 0. \quad (62)$$

Proposition 2.23 (run-length/parity-word isomorphism). *The map $B : \mathcal{A} \rightarrow \mathcal{B}$ is a bijection. If the one positions of $\mathbf{b} \in \mathcal{B}$ are $0 = s_0 < s_1 < s_2 < \dots$, then*

$$B^{-1}(\mathbf{b})_j = s_{j+1} - s_j. \quad (63)$$

Moreover, for every $\mathbf{a} \in \mathcal{A}$,

$$B(\sigma_{\mathcal{A}} \mathbf{a}) = \sigma_{\mathcal{B}}^{a_0} B(\mathbf{a}). \quad (64)$$

Proof. Every a_i is positive, so the partial sums s_j are strictly increasing, start at zero, and are unbounded. Thus (62) defines an element of $\mathcal{B}$. Conversely, the one positions of an element of $\mathcal{B}$ form a strictly increasing unbounded sequence beginning at zero; their successive positive gaps define (63). Taking partial sums of those gaps recovers the original one positions, which proves both inverse identities. Removing the first gap translates every later one position by $-a_0$, exactly as the right side of (64) does. $\square$

For $x \in \mathbb{N}_{>0}^{\text{odd}}$, define the accelerated exponent code and shortened parity code by

$$E(x) = (e(C_{\text{Cr}}^j(x)))_{j \geq 0} \in \mathcal{A}, \quad (65)$$

$$P(x) = (T^k(x) \bmod 2)_{k \geq 0} \in \mathcal{B}. \quad (66)$$

Both maps are total on $\mathbb{N}_{>0}^{\text{odd}}$: every accelerated value remains a positive odd integer and each return time is finite.

Theorem 2.24 (typed first-return coding morphism). *For every $x \in \mathbb{N}_{>0}^{\text{odd}}$,*

$$P(x) = B(E(x)). \quad (67)$$

The dynamics satisfy

$$E(C_{\text{Cr}} x) = \sigma_{\mathcal{A}} E(x), \quad P(C_{\text{Cr}} x) = \sigma_{\mathcal{B}}^{e(x)} P(x). \quad (68)$$

The fibres of E have cardinality at most one. No surjectivity of $E : \mathbb{N}_{>0}^{\text{odd}} \rightarrow \mathcal{A}$ is asserted.

Proof. Proposition 2.1 shows that the shortened parity block beginning at $C_{\text{Cr}}^j(x)$ is

$$10^{e(C_{\text{Cr}}^j(x))-1}.$$

Concatenating these blocks gives exactly (67). The first equality in (68) follows directly from the definition of E ; the second follows either from the first equality, Proposition 2.23, and (67), or directly from $C_{\text{Cr}}(x) = T^{e(x)}(x)$.

For injectivity, let

$$Q_3(z) = \sum_{k \geq 0} (T^k(z) \bmod 2) 2^k \quad (z \in \mathbb{Z}_2)$$

be the Bernstein–Lagarias chronological parity encoder. Their theorem, reproved below as Proposition 2.90, makes Q_3 injective. If $E(x) = E(y)$, then (67) gives $P(x) = P(y)$, hence $Q_3(x) = Q_3(y)$ and $x = y$. The proof identifies the image only as $E(\mathbb{N}_{>0}^{\text{odd}})$; it supplies no realization of an arbitrary positive-integer run sequence by a positive orbit. $\square$

2.5 The accelerated odd 2-adic survivor and its countable shift

Urata writes

$$\mathbb{U}_2 = 1 + 2\mathbb{Z}_2 = \mathbb{Z}_2^\times$$

for the closure of the odd-type rationals. His continuous extension is the map

$$\varphi : \mathbb{Z}_2 \setminus \{0, -1/3\} \longrightarrow \mathbb{U}_2$$

given, without merging its two branches, by

$$\varphi(x) = \begin{cases} (3x+1)/2^{\nu_2(3x+1)}, & x \in \mathbb{U}_2 \setminus \{-1/3\}, \\ x/2^{\nu_2(x)}, & x \in \mathbb{Z}_2 \setminus (\mathbb{U}_2 \cup \{0\}). \end{cases} \quad (69)$$

The restriction to odd units is the accelerated map used below; on positive odd integers it is exactly C_{Cr} . The full even-input branch in (69) is retained because it belongs to the source definition, but it is not smuggled into the odd first-return system [20, printed pp. 5–8].

Source issue 2.25 (typing Urata’s valuation-continuity sentence). After defining the exponent of zero to be infinity, the text calls the exponent continuous on $\mathbb{Q}_2$ without specifying a topology on $\mathbb{Z} \cup \{\infty\}$. Continuity at zero is therefore not yet a typed statement. What is used here is the exact local constancy of ν_2 on $\mathbb{Q}_2^\times$; compactness of $\mathbb{Z}_2$ is invoked independently.

For $p \geq 1$, define

$$F_p : \mathbb{U}_2 \longrightarrow \mathbb{U}_2, \quad F_p(y) = \frac{2^p y - 1}{3}. \quad (70)$$

The literal identity

$$3F_p(y) + 1 = 2^p y$$

shows that $\nu_2(3F_p(y) + 1) = p$ and $\varphi(F_p(y)) = y$. Conversely every odd-unit preimage has this form with its unique valuation. Thus

$$\varphi^{-1}(y) \cap \mathbb{U}_2 = \{F_p(y) : p \geq 1\}, \quad (71)$$

an explicitly labelled countably infinite fibre [20, printed p. 8, Example 2].

For an actual odd-unit orbit $x_k = \varphi^k(x_0)$, put

$$p_k = \nu_2(3x_{k-1} + 1), \quad S_0 = 0, \quad S_n = \sum_{k=1}^n p_k,$$

and retain the coordinate polynomial

$$C_n(p_1, \dots, p_n) = \sum_{j=1}^n 3^{n-j} 2^{S_{j-1}}. \quad (72)$$

This is a reindexing of Urata's displayed $C(p_1, \dots, p_{n-1})$, not a replacement source formula. Successive substitution gives

$$2^{S_n} x_n = C_n + 3^n x_0. \quad (73)$$

Urata Proposition 1 records this relation and the resulting complete n -step inverse set. Propositions 2–3 show that common first n exponents give $|x - y|_2 \leq 2^{-S_n}$ and that a complete exponent sequence determines at most one survivor. In fact subtraction of (73), together with the even difference of two odd n th iterates, gives the sharper bound

$$|x - y|_2 \leq 2^{-(S_n+1)}. \quad (74)$$

Source issue 2.26 (the Cauchy bound in Urata Proposition 4). Proposition 4 constructs approximants x_n whose first n exponents are prescribed. For $m \leq n$, the pair x_m, x_n shares only the first m prescribed exponents, but the printed comparison uses S_n rather than S_m . The stronger printed bound is false. For $p_1 = 1, p_2 = 3$, the first two approximants are

$$x_1 = \frac{1}{3}, \quad x_2 = \frac{11}{9}, \quad |x_2 - x_1|_2 = \left| \frac{8}{9} \right|_2 = 2^{-3} > 2^{-4} = 2^{-S_2}.$$

This is a proof defect in the displayed Cauchy estimate, not a counterexample to the proposition. The official formula (72) itself is consistent; no correction from the later unauthenticated retypesetting is attributed to Urata.

Define the survivor domain

$$X = \mathbb{U}_2 \setminus \bigcup_{j \geq 0} \varphi^{-j} \{-1/3\}. \quad (75)$$

Theorem 2.27 (repaired countable-alphabet conjugacy). *The exponent encoder*

$$E_U : X \longrightarrow (\mathbb{Z}_{\geq 1})^{\mathbb{N}_0}, \quad E_U(x)_j = \nu_2(3\varphi^j(x) + 1), \quad (76)$$

is a homeomorphism. If $\mathbf{p} = (p_1, p_2, \dots)$ and $S_j = p_1 + \dots + p_j$, its inverse is the unnormalized coordinate series

$$E_U^{-1}(\mathbf{p}) = - \sum_{j=1}^{\infty} \frac{2^{S_{j-1}}}{3^j}. \quad (77)$$

Moreover

$$E_U \circ \varphi = \sigma_{\mathcal{A}} \circ E_U. \quad (78)$$

Urata Propositions 3–4 supply the uniqueness and existence statements; the explicit inverse, repaired proof, and conjugacy formulation here are separate constructive reconstructions with no novelty claim.

Proof. The series in (77) converges because $S_{j-1} \geq j-1$. More generally define, for $k \geq 0$,

$$x_k = - \sum_{r=1}^{\infty} \frac{2^{p_{k+1} + \dots + p_{k+r-1}}}{3^r},$$

where the empty exponent sum is zero. Its first term is $-1/3$ and every later term is even, so x_k is an odd unit. Direct cancellation gives

$$3x_k + 1 = 2^{p_{k+1}} x_{k+1}. \quad (79)$$

Therefore the valuation is exactly p_{k+1} , $\varphi(x_k) = x_{k+1}$, and no x_k is $-1/3$. This proves existence in X without the defective Cauchy estimate. Urata Proposition 3, or (74) with $n \rightarrow \infty$, proves uniqueness.

Every finite exponent prefix is locally constant on X . Conversely, (74) shows that common prefixes tending to infinite length force the inverse points together. This proves continuity in both directions. Dropping the first valuation from the definition proves (78). $\square$

Corollary 2.28 (exact exponent cylinders and Haar law). *For $w = (p_1, \dots, p_n)$, put $F_w = F_{p_1} \circ \dots \circ F_{p_n}$. Then*

$$F_w(y) = \frac{2^{S_n} y - C_n}{3^n}, \quad (80)$$

$$F_w(\mathbb{U}_2) = F_w(1) + 2^{S_n+1} \mathbb{Z}_2. \quad (81)$$

The exponent cylinder in X is $F_w(X)$; its ambient closure is the ball (81). For normalized Haar measure μ_U on $\mathbb{U}_2$,

$$\mu_U(F_w(X)) = 2^{-S_n} = \prod_{j=1}^n 2^{-p_j}. \quad (82)$$

Consequently E_U carries Haar measure to the Bernoulli product law $\Pr(p_j = k) = 2^{-k}$ for $k \geq 1$, and the accelerated survivor system is Bernoulli and strongly mixing modulo its countable null exceptional set.

Proof. Composition of (70) gives (80). Since $\mathbb{U}_2 = 1 + 2\mathbb{Z}_2$ and 3^n is a unit, its difference set is exactly $2^{S_n+1} \mathbb{Z}_2$, proving (81). This ball has ambient $\mathbb{Z}_2$ -Haar measure $2^{-(S_n+1)}$ and hence normalized $\mathbb{U}_2$ -measure 2^{-S_n} . The removed preimage tree of $-1/3$ is countable, so it has measure zero. Cylinder products prove the product law; the Bernoulli shift consequences follow through (78). $\square$

Proposition 2.29 (exact induced-system bridge). *For $x \in X$ and $S_j = p_1 + \dots + p_j$ from $E_U(x)$,*

$$\varphi^j(x) = T^{S_j}(x), \quad (T^k(x) \bmod 2)_{k \geq 0} = B(E_U(x)). \quad (83)$$

Thus X is the first-return system of the shortened binary shift on odd inputs whose parity sequence contains infinitely many ones. The excluded word $1000\dots$ is the parity word of $-1/3$, and words with only finitely many ones are its accelerated preimages.

Proof. Equation (79) says that one odd branch followed by $p_j - 1$ even divisions reaches the next odd unit, so the elapsed shortened time is exactly p_j . Summing the retained times proves the first identity; the intervening parity block is 10^{p_j-1} , which proves the second through Proposition 2.23. The final exceptional-set description follows from the total Bernstein–Lagarias parity conjugacy. $\square$

Proposition 2.30 (periodic exponent words). *For a genuine survivor orbit with first n exponents,*

$$\varphi^n(x) = x \iff (2^{S_n} - 3^n)x = C_n \iff x = \frac{C_n}{2^{S_n} - 3^n}. \quad (84)$$

The orbit returns after n accelerated steps exactly when its full exponent sequence is n -periodic. Its least accelerated period equals the least shift period; neither need equal a displayed nonleast n .

Proof. The first equivalence is (73) with $x_n = x_0$; $2^{S_n} \neq 3^n$ gives the second. Repetition of a returning orbit repeats the valuations. Conversely a periodic exponent sequence determines by Theorem 2.27 the same point after one word. These are Urata Propositions 5–6 with the least-period qualification made explicit [20, printed pp. 10–11]. $\square$

Explicit nonclaim 2.31. Urata’s arbitrary exponent realization holds on the enlarged domain $X \subset \mathbb{Z}_2^\times$, not on $\mathbb{N}_{>0}^{\text{odd}}$. Thus the earlier $E : \mathbb{N}_{>0}^{\text{odd}} \rightarrow \mathcal{A}$ remains only injective. Rationality of (84) does not imply integrality or a new positive cycle; printed p. 11 asks when that fraction is integral and leaves the question open. The rational eventual-periodicity paragraph on printed p. 6 is conjectural. The variable cylinder modulus is 2^{S_n+1} , not 2^n , and the Haar statements do not transfer to the countable Haar-null ordinary integers.

2.6 Chronological affine words and fixed returns

On $\mathbb{Q}$ put

$$h_0(z) = \frac{z}{2}, \quad h_1(z) = \frac{3z+1}{2}.$$

Let $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{L-1})$ be a chronological binary word of length $L \geq 1$. Suppose it has m ones, at positions

$$0 \leq v_0 < \dots < v_{m-1} < L.$$

Proposition 2.32 (formal chronological composition). *For every $z \in \mathbb{Q}$,*

$$h_{\varepsilon_{L-1}} \circ \dots \circ h_{\varepsilon_0}(z) = \frac{3^m z + \sum_{r=0}^{m-1} 3^{m-r-1} 2^{v_r}}{2^L}. \quad (85)$$

The sum is empty when $m = 0$.

Proof. After the first j chronological branches, let

$$m_j = \sum_{i=0}^{j-1} \varepsilon_i, \quad A_j = \sum_{\substack{0 \leq r < m \\ v_r < j}} 3^{m_j-r-1} 2^{v_r}.$$

The induction invariant is

$$h_{\varepsilon_{j-1}} \circ \dots \circ h_{\varepsilon_0}(z) = \frac{3^{m_j} z + A_j}{2^j}.$$

It is the identity at $j = 0$. A zero branch only adds one power of 2 to the denominator. A one branch multiplies the old numerator by 3 and adds 2^j ; this raises every old exponent of 3 by one and inserts the new term $3^0 2^j$. The invariant at $j = L$ is (85). $\square$

The formal word equals $T^L(z)$ exactly when every prefix value has the parity prescribed by the next ε_j . Algebra alone does not supply that itinerary condition.

Source issue 2.33 (Böhm–Sontacchi Proposition 4). Böhm–Sontacchi Proposition 1 supplies the formal affine-composition algebra. Their printed Proposition 4 displays its specialization as $u^n(x)$ while quantifying over every $x \in \mathbb{Z}$ and every run tuple [13, printed p. 263, Props. 1 and 4]. That unrestricted iterate statement is false. With $x = 0$, $m = 1$, and $n_0 = n_1 = 1$, the formal chronological word is $(0, 1, 0)$ and

$$h_0 \circ h_1 \circ h_0(0) = \frac{1}{4}, \quad T^3(0) = 0.$$

Thus the right side survives as a formal composition; equality with an iterate requires a separate branch-correctness proof.

Source issue 2.34 (Böhm–Sontacchi Proposition 5 position range). The printed statement gives $v_{i-1} < v_i$ only for $1 \leq i < m$ and then $v_m = n$; it does not visibly print the necessary final inequality $v_{m-1} < v_m$ [13, printed p. 263, Prop. 5]. If the display is read literally, $m = n = 1$ and $(v_0, v_1) = (1, 1)$ give the candidate $x = -2$, while $T(-2) = -1$. The formal run construction does impose $v_m - v_{m-1} = 1 + n_m > 0$. This edition therefore states positions directly as $0 \leq v_0 < \dots < v_{m-1} < L$; that final inequality is an explicit reconstruction, not silently attributed printed text.

Lemma 2.35 (wrong branches cannot close in $\mathbb{Z}_2$). *For $z \in \mathbb{Z}_2$, exactly one of $h_0(z), h_1(z)$ lies in $\mathbb{Z}_2$, namely $h_{z \bmod 2}(z)$. If $v_2(y) < 0$, then*

$$v_2(h_0(y)) = v_2(h_1(y)) = v_2(y) - 1.$$

Proof. If z is even, $z/2$ is integral and $(3z+1)/2$ is not; if z is odd, the roles reverse. When $v_2(y) < 0$, the valuations of $3y$ and 1 are unequal, so the ultrametric equality gives $v_2(3y + 1) = v_2(y)$; division by 2 proves the second assertion for both branches. $\square$

Theorem 2.36 (recovered fixed-return theorem). *Define*

$$R(\varepsilon) = \frac{\sum_{r=0}^{m-1} 3^{m-r-1} 2^{v_r}}{2^L - 3^m} \in \mathbb{Q}. \quad (86)$$

If $R(\varepsilon) \in \mathbb{Z}_2$, every branch of the formal word is correct and

$$T^L(R(\varepsilon)) = R(\varepsilon).$$

Conversely, if $z \in \mathbb{Z}_2$, $T^L(z) = z$, and ε is its chronological parity word, then $z = R(\varepsilon)$. If $z \in \mathbb{Z}$, this is an integer return. Its exact period divides L and need not equal L .

Proof. The denominator in (86) is nonzero: a positive power of 2 cannot equal a nonnegative power of the odd integer 3 . Solving the fixed-point equation for (85) gives exactly (86). Start its right-to-left evaluation at $R \in \mathbb{Z}_2$. If a wrong branch occurs, Lemma 2.35 sends the current value outside $\mathbb{Z}_2$ with negative valuation, and every later formal branch decreases that valuation. The final value cannot then equal the initial integral value. Hence all branches are correct and the formal fixed point is a genuine T -return. The converse substitutes the actual itinerary into Proposition 2.32 and solves the same affine equation. A return after L iterates says only that the least positive period divides L . This last scope agrees with the unnumbered paragraph following Böhm–Sontacchi Proposition 5 [13, printed p. 264]. $\square$

Corollary 2.37 (Crandall’s accelerated cycle identity). *Let $x \in \mathbb{N}_{>0}^{\text{odd}}$, let $a_j = e(C_{\text{Cr}}^j(x))$, and put*

$$A_0 = 0, \quad A_j = \sum_{r=0}^{j-1} a_r \quad (1 \leq j \leq k).$$

If $C_{\text{Cr}}^k(x) = x$, then

$$x(2^{A_k} - 3^k) = \sum_{j=0}^{k-1} 2^{A_j} 3^{k-1-j}. \quad (87)$$

Conversely, if $0 = A_0 < A_1 < \dots < A_k$ and an odd positive integer x satisfies (87), then the corresponding formal shortened word fixes x ; by Theorem 2.36 it is a genuine T -return, and its odd returns give $C_{\text{Cr}}^k(x) = x$.

Proof. Theorem 2.24 places the k ones of the shortened itinerary at $A_0, \dots, A_{k-1}$ and gives total shortened length A_k . Equation (86), with $L = A_k$ and $m = k$, is equivalent to (87). Conversely, the same identity makes x the formal fixed point of that word. Its membership in $\mathbb{Z} \subset \mathbb{Z}_2$ supplies branch correctness, and the one positions split the genuine itinerary into the prescribed k first-return blocks. This is Crandall’s Eq. (7.3) and Corollary 7.1, including its return-time rather than exact-period direction [1, pp. 1287–1288]. $\square$

2.7 Crandall’s complete accelerated programme

This subsection reconstructs all twelve printed pages of Crandall’s 1978 article from the published witness [1, pp. 1281–1292]. The source uses only the accelerated odd map (3). Its trajectory is the indexed sequence

$$\mathcal{T}_m = (C_{\text{Cr}}^j(m))_{j \geq 1},$$

stopped after the least index j for which $C_{\text{Cr}}^j(m) = 1$, if such an index exists, and left infinite otherwise. Its height is that least index, or ∞ when there is none. Thus the braces used in the source’s display denote an ordered sequence, not its underlying set; repeated values in a nonterminating periodic trajectory do not make its height finite. In particular, the source table has $\mathcal{T}_1 = (1)$ and $h(1) = 1$; this is not the zero-step hitting-time convention.

For a segment $x_i = C_{\text{Cr}}^i(x_0)$, put

$$e_i = \nu_2(3x_i + 1), \quad A_0 = 0, \quad A_j = \sum_{i=0}^{j-1} e_i. \quad (88)$$

Repeated use of Proposition 2.1 gives

$$C_{\text{Cr}}^j(x_0) = T^{A_j}(x_0) = U^{A_j+j}(x_0). \quad (89)$$

Thus every height, trajectory position, and period in Crandall’s paper is accelerated time j . Neither A_j nor $A_j + j$ may be substituted for it.

2.7.1 Large finite excursions and the random-walk nonclaim

Proposition 2.38 (Crandall, Theorem 2.2). *For $k \geq 2$, let $m_k = 2^k - 1$. Then*

$$C_{\text{Cr}}^j(m_k) = 3^j 2^{k-j} - 1 \quad (0 \leq j < k), \quad (90)$$

and hence

$$\frac{\sup \mathcal{T}_{m_k}}{m_k} > \left(\frac{3}{2}\right)^{k-1}. \quad (91)$$

Consequently the family of ratios

$$\left\{ \frac{\sup \mathcal{T}_m}{m} : m \in \mathbb{N}_{>0}^{\text{odd}} \right\}$$

is unbounded. For the ratios with denominator m^α , this construction proves unboundedness only for $\alpha < \log_2 3$.

Proof. Equation (90) is immediate at $j = 0$. If it holds with $j < k - 1$, then

$$3(3^j 2^{k-j} - 1) + 1 = 2(3^{j+1} 2^{k-j-1} - 1),$$

whose second factor is odd; the next accelerated exponent is therefore one. At $j = k - 1$, the trajectory contains $2 \cdot 3^{k-1} - 1$. Direct cross-multiplication gives

$$\frac{2 \cdot 3^{k-1} - 1}{2^k - 1} > \frac{3^{k-1}}{2^{k-1}}$$

because $3^{k-1} > 2^{k-1}$. This proves the stated bound. Dividing the same member by m_k^α yields an exponential factor $3^{k-1}/2^{\alpha k}$, which this construction forces upward exactly when $\alpha < \log_2 3$ [1, pp. 1282–1283, Thm. 2.2 and following paragraph]. $\square$

Section 3 assigns the heuristic probabilities $\Pr(e(m) = a) = 2^{-a}$ for $a \geq 1$. Its model logarithmic drift is

$$\sum_{a \geq 1} 2^{-a} \log(3/2^a) = -\log(4/3). \quad (92)$$

It then proposes the individual estimate $h(m) \sim \log m / \log(4/3)$. For the source average

$$H(x) = \frac{2}{x} \sum_{\substack{m \leq x \\ m \in \mathbb{N}_{>0}^{\text{odd}}}} h(m), \quad (93)$$

Conjecture 3.1 is

$$H(x) \sim \frac{2 \log x}{\log(16/9)}. \quad (94)$$

Explicit nonclaim 2.39. Equations (92)–(94) are explicitly heuristic or conjectural. They prove neither termination nor unboundedness. The source also reports Everett’s almost-everywhere descent theorem, but does not prove it. Everett’s primary proof and the exact variable-clock transfer are now reconstructed in Theorem 2.79 and Proposition 2.81; neither follows from the heuristic drift displayed here. The sentence after Conjecture 3.1 calls it “(3.1)”, duplicating the number of the individual-height heuristic [1, pp. 1282–1284].

2.7.2 Backward coordinates and the exact finite-height address

Crandall defines, for $a \in \mathbb{Z}_{>0}$ and $n \in \mathbb{Q}$,

$$B_a(n) = \frac{2^a n - 1}{3}, \quad B_{a_j \dots a_1} = B_{a_j} \circ \dots \circ B_{a_1}. \quad (95)$$

The subscript order is backward: when all displayed values are integral, C_{Cr} removes the leftmost applied branch B_{a_j} .

For a natural-order address $\mathbf{a} = (a_1, \dots, a_j)$, define

$$n_0 = 1, \quad n_i = B_{a_i}(n_{i-1}) \quad (1 \leq i \leq j). \quad (96)$$

Let $G^\sharp$ be the set of such finite addresses satisfying

$$a_1 > 2, \quad (97)$$

$$2^{a_i} n_{i-1} \equiv 1 \pmod{3} \quad (1 \leq i \leq j), \quad (98)$$

$$2^{a_i} n_{i-1} \not\equiv 1 \pmod{9} \quad (1 \leq i < j). \quad (99)$$

Theorem 2.40 (Crandall's finite-height address theorem, repaired). *Let*

$$\mathcal{M} = \{m \in \mathbb{N}_{>0}^{\text{odd}} : m > 1, h(m) < \infty\}.$$

The map

$$\text{ev} : G^\sharp \longrightarrow \mathcal{M}, \quad (a_1, \dots, a_j) \longmapsto B_{a_j \dots a_1}(1) \quad (100)$$

is a bijection. Its inverse sends $m \in \mathcal{M}$ to the length- $h(m)$ address

$$a_i = e(C_{Cr}^{h(m)-i}(m)) \quad (1 \leq i \leq h(m)). \quad (101)$$

The exact trajectory of $n_j = \text{ev}(\mathbf{a})$ is

$$n_j \longmapsto n_{j-1} \longmapsto \dots \longmapsto n_1 \longmapsto 1,$$

so its height is exactly j .

Proof. Suppose first that (98) holds through index i . Then $n_i = (2^{a_i} n_{i-1} - 1)/3$ is integral. Since the numerator is odd and the denominator is odd, n_i is odd. Moreover

$$n_i \equiv 0 \pmod{3} \iff 2^{a_i} n_{i-1} \equiv 1 \pmod{9}.$$

Thus (99) says exactly that every nonterminal n_i is prime to 3, which is necessary for another inverse branch to be integral. From $3n_i + 1 = 2^{a_i} n_{i-1}$ and oddness of n_{i-1} , one has $\nu_2(3n_i + 1) = a_i$ and hence $C_{Cr}(n_i) = n_{i-1}$. Finally, $a_1 > 2$ ensures $n_1 \neq 1$, so the first occurrence of 1 is terminal and the height is j .

Conversely, a finite trajectory from $m > 1$ supplies the valuations in (101). Reversing its branch equations gives (96). Every nonterminal orbit value must be prime to 3: if $3 \mid n_i$, then no equation $3u + 1 = 2^a n_i$ is possible modulo 3. Hence all three gates hold. The determinism of C_{Cr} recovers every a_i uniquely, so evaluation and (101) are literal two-sided inverses. This is the corrected content of Lemmas 4.1–4.3 and Theorem 4.1 [1, pp. 1284–1285]. $\square$

Source issue 2.41 (two finite-address endpoint defects). The printed definition of G imposes a non-1 (mod 9) condition on a_1 and a terminal mod-3 condition on a_j . At length $j = 1$, both apply to the same letter. It thereby excludes the valid height-one example

$$B_6(1) = 21, \quad C_{Cr}(21) = 1, \quad 2^6 \equiv 1 \pmod{9}.$$

The mod-9 gate belongs only to nonterminal indices, as in (99). Independently, the printed Lemma 4.3 includes “an integer of height j ” while $a_1 > 2$ excludes $B_2(1) = 1$; the source table nevertheless has $h(1) = 1$. Restricting the lemma to the $m > 1$ set used by Theorem 4.1 repairs this endpoint. Likewise, inside $\mathcal{M}$ the height-one family $(4^s - 1)/3$ has $s \geq 2$, not merely $s \geq 1$ [1, pp. 1282, 1284–1285].

The address theorem is genuinely a bijection, not a merely suggestive parameterization: its domain, codomain, evaluation, inverse, and exact clock are all displayed. It is also a finite-orbit theorem. It provides no infinite address space and no conclusion about nonterminating positive orbits.

2.7.3 Fixed-height and total termination counts

Let

$$\pi_h(x) = \#\{m \in \mathcal{M} : h(m) = h, m \leq x\}.$$

The elementary size estimate is

$$B_{a_j \dots a_1}(1) < \frac{2^{a_1 + \dots + a_j}}{3^j}. \quad (102)$$

It follows by induction from (95), retaining the discarded negative translation rather than replacing the branch by an equality.

Source issue 2.42 (domain of the address-counting lemma). Printed Lemma 5.2 says that for every real $z > 0$, the number of length- j addresses in G with coordinate sum at most z is at least

$$\left(2 \left\lfloor \frac{z-2}{6j} \right\rfloor\right)^j. \quad (103)$$

At $z = 1, j = 2$, the right side is 4, while no two positive coordinates have sum at most 1. The proof’s blocks of six require a nonnegative number of blocks. The statement is correct at the scope used later after requiring $z \geq 2$, or globally after replacing the floor by its nonnegative part.

Theorem 2.43 (Crandall, fixed accelerated height). *There are real constants $r, x_0 > 0$, independent of h , such that for every $h \in \mathbb{Z}_{>0}$,*

$$x > \max\{x_0, 2^{h/r}\} \quad (104)$$

implies

$$\pi_h(x) > \frac{(\log_2(x^r))^h}{h!}. \quad (105)$$

In particular, infinitely many positive odd integers have each prescribed positive accelerated height h .

Proof. Theorem 2.40, (102), and the repaired form of (103) give the source’s explicit preliminary bound

$$\pi_h(x) \geq \left(\frac{\log_2(3^h x/4)}{3h} - 2 \right)^h$$

once the displayed base is positive. Choose x_0 so that $x/4 > x^{1/2}$ for $x > x_0$. Choose $0 < r < 1$ with

$$r \log_2(3/64) + \frac{1}{2} > 3er > 0. \quad (106)$$

If also $x > 2^{h/r}$, then $h < r \log_2 x$. Substituting this strict inequality in the preliminary bound and using $h!e^h > h^h$ yields (105). This follows the source proof while making its strict threshold and the repaired Lemma 5.2 domain explicit [1, p. 1286, Thm. 5.1]. $\square$

Let

$$\pi(x) = \#\{m \in \mathcal{M} : m \leq x\} = \sum_{h \geq 1} \pi_h(x). \quad (107)$$

The exact handbook route. Crandall’s remark after Lemma 6.1 says only that “various formulas” pertaining to the lemma occur in reference 4; it gives no handbook section, page, or formula number [1, p. 1287]. The relevant formulas can nevertheless be identified exactly. Abramowitz–Stegun use $P(z)$ for the standard normal lower tail, but also use $P(\chi^2 \mid \nu)$ and $Q(\chi^2 \mid \nu) = 1 - P(\chi^2 \mid \nu)$ for the lower and upper chi-square probabilities. Their formulas 26.2.2–26.2.6 give $P(0) = Q(0) = 1/2$; formula 26.2.29 relates this P to erf; formula 26.4.11 gives the large- ν normal coordinate

$$\frac{\chi^2 - \nu}{\sqrt{2\nu}};$$

and formula 26.4.21 gives the exact finite Poisson sum below [3, printed pp. 931, 934, 940–941]. These are identified relevant formulas, not locators supplied by Crandall.

Proposition 2.44 (exact Poisson–gamma–chi-square coordinate). *For $n \in \mathbb{N}_0$ and $t \geq 0$, set*

$$F_n(t) = e^{-t} \sum_{j=0}^n \frac{t^j}{j!}. \quad (108)$$

Then, with $\Gamma(a, t)$ denoting the upper incomplete gamma function,

$$F_n(t) = \frac{\Gamma(n+1, t)}{\Gamma(n+1)} = Q(2t \mid 2n+2). \quad (109)$$

For fixed n ,

$$F'_n(t) = -e^{-t} \frac{t^n}{n!} \quad (t > 0), \quad (110)$$

so F_n is a continuous strictly decreasing bijection $[0, \infty) \rightarrow (0, 1]$. Its inverse is the unique upper-tail quantile on this fixed- n fibre; no elementary inverse formula is asserted.

Proof. Integration by parts gives

$$\Gamma(n+1, t) = n!e^{-t} \sum_{j=0}^n \frac{t^j}{j!}$$

by induction from $\Gamma(1, t) = e^{-t}$, proving the first equality in (109). In Abramowitz–Stegun 26.4.21 take $c = n + 1$, $\nu = 2c$, $m = t$, and $\chi^2 = 2t$ to obtain the second equality literally [3, printed p. 941, Eq. 26.4.21]. Differentiating the finite sum makes every term cancel except the displayed last term in (110). Finally $F_n(0) = 1$, while $F_n(t) \rightarrow 0$ because an exponential dominates the fixed polynomial; strict monotonicity supplies the stated fibres and inverse. $\square$

Lemma 2.45 (completion of Crandall’s Poisson lemma). *Let $k(t)$ be integer-valued and suppose*

$$\frac{k(t) - t}{\sqrt{t}} \rightarrow 0. \quad (111)$$

Then

$$\lim_{t \rightarrow \infty} e^{-t} \sum_{u=0}^{k(t)} \frac{t^u}{u!} = \frac{1}{2}. \quad (112)$$

Consequently the same limit holds after deleting $u = 0$ for each of the following three exact upper cutoffs:

$$k(t) = \lfloor t \rfloor - 1, \quad k(t) = \lfloor t \rfloor, \quad k(t) = \lceil t \rceil - 1. \quad (113)$$

The first is Crandall’s printed strict cutoff $u < \lfloor t \rfloor$; the third is the maximal integer cutoff uniformly authorized by the strict height hypothesis in Theorem 5.1.

Proof. The handbook route to the source case is now literal. Put $n = \lfloor t \rfloor$, $\nu = 2n$, and $\chi^2 = 2t$ in Proposition 2.44. Formula 26.4.21 identifies the sum $u = 0, \dots, n - 1$ with $Q(2t \mid 2n)$, while the normal coordinate in 26.4.11 is

$$\frac{2t - 2n}{\sqrt{4n}} = \frac{t - n}{\sqrt{n}} \rightarrow 0.$$

Together with $P(0) = Q(0) = 1/2$, this is the exact classical formula path to Crandall’s limit. Formula 26.4.11 prints neither an error term nor a proof on that page, however, so it is not used as an unspoken uniformity assertion. The following argument controls all three real- t endpoints directly.

Let X_t have the Poisson distribution with mean t . The characteristic function of $Y_t = (X_t - t)/\sqrt{t}$ is

$$\mathbf{E}(e^{isY_t}) = \exp\left(t\left(e^{is/\sqrt{t}} - 1 - is/\sqrt{t}\right)\right).$$

The Taylor remainder is $O_s(t^{-1/2})$, so this converges to $e^{-s^2/2}$. The Lévy continuity theorem gives $Y_t \Rightarrow N(0, 1)$. Write $a_t = (k(t) - t)/\sqrt{t}$. For every fixed $\delta > 0$, condition (111) gives

$$\Pr(Y_t \leq -\delta) \leq \Pr(Y_t \leq a_t) \leq \Pr(Y_t \leq \delta)$$

for all sufficiently large t . Weak convergence at the two fixed continuity points gives lower and upper limits between $\Phi(-\delta)$ and $\Phi(\delta)$; letting $\delta \downarrow 0$ proves (112). Every cutoff in (113) differs from t by at most a fixed constant. Removing the $u = 0$ term changes the probability by exactly $e^{-t} \rightarrow 0$.

The high-resolution source page prints $u < \lfloor t \rfloor$, not $u \leq \lfloor t \rfloor$. Crandall points to error-function asymptotics but prints no proof [1, p. 1287, Lem. 6.1]. The argument above supplies the all-real analytic completion while explicitly retaining its invocation of Lévy’s continuity theorem. $\square$

Theorem 2.46 (Crandall’s power-law theorem, with an explicit witness). *There are $c > 0$ and $X_0 > 0$ such that*

$$\pi(x) > x^c \quad (x > X_0). \quad (114)$$

The source leaves c unspecified. Its proof, with the endpoint repaired, permits the concrete edition choice $c = 1/100$.

Proof. For $x > x_0$, Theorem 2.43 may be summed only over $h < r \log_2 x$. The printed proof sums through $\lfloor r \log_2 x \rfloor$, which includes one unauthorized equality term when $r \log_2 x$ is integral. Summing instead through $\lceil r \log_2 x \rceil - 1$ and applying Lemma 2.45 gives, for every fixed $d \in (0, 1/2)$ and all sufficiently large x ,

$$\pi(x) > (1/2 - d)e^{r \log_2 x} = (1/2 - d)x^{r/\log 2}. \quad (115)$$

This proves the source theorem for every $c < r/\log 2$ after increasing the threshold.

For an exact explicit witness, take $r = 1/100$. Since $3/64 > 2^{-5}$ and $e < 3$, the left side of (106) is greater than $45/100$, while its middle term $3e/100$ is less than $9/100$. Thus this r is admissible. Since $\log 2 < 1$, one has $r/\log 2 > 1/100$; the fixed positive factor in (115) is eventually absorbed by the excess power. Hence $c = 1/100$ works. This is an edition-supplied sharpening of the source’s existential conclusion, not a number printed by Crandall. $\square$

Explicit nonclaim 2.47. Theorem 2.46 is a lower bound of order x^c for some $c < 1$. It is compatible with natural density zero and is not a positive-density theorem. It proves neither $\pi(x) \gg x$ nor convergence for almost every odd starting value. The frozen route text that calls this positive density is routing metadata, not mathematical evidence.

2.7.4 Membership coordinates and the return obstruction

For a backward word $m = B_{b_k \dots b_1}(n)$, Crandall uses the reverse cumulative sums

$$A_0 = 0, \quad A_i = \sum_{j=k-i+1}^k b_j \quad (1 \leq i \leq k). \quad (116)$$

Expanding the affine composition gives

$$2^{A_k} n - 3^k m = \sum_{j=0}^{k-1} 2^{A_j} 3^{k-1-j}. \quad (117)$$

Theorem 2.48 (Crandall, exact trajectory-membership coordinates). *For $m, n \in \mathbb{N}_{>0}^{\text{odd}}$, one has $n \in \mathcal{T}_m$ if and only if there are an integer $k \geq 1$ and integers*

$$0 = A_0 < A_1 < \dots < A_k$$

satisfying (117). If such data exist, then $n = 1$ or n is the k -th displayed member of $\mathcal{T}_m$.

Proof. If $n = C_{\text{Cr}}^k(m)$, reverse its k branch equations and take $b_i = e(C_{\text{Cr}}^{k-i}(m))$; expansion gives (117). Conversely, define

$$d_i = A_{k-i+1} - A_{k-i} \quad (1 \leq i \leq k). \quad (118)$$

The strict increase makes every d_i positive, and solving (117) gives $m = B_{d_k \dots d_1}(n)$. Crandall’s Lemma 4.1 then proves that every formal inverse branch is integral, odd, and removed by the actual accelerated map. Hence $C_{\text{Cr}}^k(m) = n$, with $n = 1$ treated as the terminating case [1, pp. 1287–1288, Eq. (7.2), Thm. 7.1]. $\square$

Source issue 2.49 (Theorem 7.1 difference index). The printed converse assigns $d_i = A_{k-i+1} - A_{k-1}$. For $i = 2$ this can give zero and it does not recover successive coordinate gaps. Equation (118) is the forced correction. It is exactly the difference needed to recover A_i from the d -word and makes the printed conclusion $m = B_{d_k \dots d_1}(n)$ valid.

Taking $n = m$ in (117) gives precisely (87). The forward clause of Crandall's printed Corollary 7.1 assumes that the trajectory has period k ; its converse says that the displayed equation produces a return at the k -th element. Combining that converse with Theorem 2.48 yields the edition's slightly more general equivalence: the equation with strict A -coordinates characterizes a return after k accelerated steps. It does not assert that k is the least period. Positivity of the right side forces

$$2^{A_k} > 3^k. \quad (119)$$

Proposition 2.50 (Crandall's explicit Diophantine obstruction). *Suppose integers a, b satisfy*

$$b > 1, \quad 0 < 2^a - 3^b \mid 2^{a-b} - 1. \quad (120)$$

Put

$$d = \frac{2^{a-b} - 1}{2^a - 3^b}, \quad m = 2^b d - 1. \quad (121)$$

Then $m \in \mathbb{N}_{>0}^{\text{odd}}$, $m > 1$, and $C_{\text{Cr}}^b(m) = m$. Therefore the Collatz conjecture would be false. The least period is allowed to divide b .

Proof. The positive divisor relation forces $a > b$, so d is a positive integer and m is positive odd. If $m = 1$, then $d = 2^{1-b}$ is not an integer because $b > 1$; hence $m > 1$. Multiplying the definition of d by 2^b gives

$$m(2^a - 3^b) = 3^b - 2^b = \sum_{j=0}^{b-1} 2^j 3^{b-1-j}.$$

This is the return equation with $k = b$, $A_j = j$ for $0 \leq j < b$, and $A_b = a$. The strictness $A_{b-1} < A_b$ follows from $a > b$. Corollary 2.37 supplies the genuine return [1, p. 1288]. $\square$

2.7.5 Pillai's fixed-difference citation and the exact Pólya route

Crandall reports two external Diophantine results. The first says that, for each fixed integer z , the equation

$$2^x - 3^y = z \quad (122)$$

has only finitely many integer solutions (x, y) [1, ref. 9 and p. 1288]. Pillai's cited primary paper has now been read in full. Its Corollary 1 states the corresponding positive- exponent, positive-difference conclusion [4, p. 4]. The paper also identifies the qualitative result as a special case of Pólya's earlier fixed-prime gap theorem [4, p. 2]. Using Pólya's sourced gap theorem yields exactly the finiteness required by Crandall and bypasses Pillai's defective quantitative derivation. The proof-dependency audit of Pólya's explicit Thue citations is recorded separately below.

Source issue 2.51 (Pillai's literal quantitative proof). Pillai quotes Siegel's approximation exponent correctly but calls its constant a positive *integer*; Siegel asserts only a positive real constant [10, p. 174]. More seriously, Pillai puts $\alpha = (A/B)^{1/r}$ and asserts that α has degree r whenever A and B are not both perfect r th powers [4, p. 3]. This is false: $(4/1)^{1/4} = \sqrt{2}$ has degree two, and $(2/2)^{1/2} = 1$ has degree one although neither displayed coefficient is a square. His next

comparison bounds Y/X only from above, although the step which replaces Y^{r-1} by a constant multiple of X^{r-1} needs a lower bound. Finally, the proof of Theorem I applies the lemma to every residue pair without checking its “not both powers” hypothesis; the residue pair $h = \ell = 0$ with outer coefficients one violates it. Thus the printed derivation is not admitted literally.

The defect is repairable on the exact source theorem rather than by changing Pillai’s conclusion. For $q \geq 1$ write

$$\mu(q) = \min_{1 \leq \lambda \leq q} \left(\frac{q}{\lambda + 1} + \lambda \right). \quad (123)$$

Theorem 2.52 (Siegel’s rational-approximation theorem). *Let ξ be algebraic of degree $q \geq 2$ and let $\epsilon > 0$. There is a positive real constant $c = c(\xi, \epsilon)$ such that, for all rational integers u, v with $v > 0$,*

$$\left| \xi - \frac{u}{v} \right| > \frac{c}{v^{\mu(q) + \epsilon}}.$$

The constant is not asserted to be an integer or numerically effective.

Proof. This is the denominator form printed on Siegel’s p. 174 [10, p. 174]. His Hauptsatz is stated in height form on printed p. 178 and proved through p. 191; Zusätze 1–3 give a uniform positive constant and extend the conclusion to nonintegral algebraic ξ [10, pp. 178, 191]. Although the Hauptsatz minimizes over $1 \leq s < q$, this is the same minimum as in (123): the excluded endpoint has value $q + q/(q + 1) > q$, whereas the candidate $s = 1$ has value $q/2 + 1 \leq q$. If $u/v = p/s$ is reduced, then $s \leq v$. In the error- < 1 case, $H(p/s) = \max\{|p|, s\} \leq \max\{1, |\xi| + 1\}s$; the complementary error case is absorbed by decreasing the positive constant. This is the exact height-to-denominator map, with no assertion of an effective constant. $\square$

Lemma 2.53 (Monotonicity of Siegel’s exponent). *The sequence $\mu(q)$ is nondecreasing on $\mathbb{N}_{>0}$.*

Proof. For an old candidate $1 \leq \lambda \leq q$, replacing q by $q + 1$ increases $q/(\lambda + 1) + \lambda$ by $1/(\lambda + 1)$. The only new candidate is $\lambda = q + 1$, whose value $(q + 1)/(q + 2) + q + 1$ is greater than $q + 1$. But $\mu(q) \leq q/2 + 1 \leq q + 1$, using the old candidate $\lambda = 1$. Every candidate for $\mu(q + 1)$ is therefore at least $\mu(q)$. $\square$

Proposition 2.54 (Actual-degree repair of Pillai’s lemma). *Fix $A, B, r \in \mathbb{N}_{>0}$ with $r \geq 2$ and fix $\eta > 0$. There is a constant $C = C(A, B, r, \eta) > 0$ such that, for all $X, Y \in \mathbb{N}_{>0}$ with $AX^r > BY^r$,*

$$AX^r - BY^r > CX^{r - \mu(r) - \eta}. \quad (124)$$

Consequently, for every $\delta > 0$ the coefficient-free inequality $AX^r - BY^r > X^{r - \mu(r) - \delta}$ holds for all sufficiently large X . No hypothesis that A and B are not both r th powers is required.

Proof. Put $\alpha = (A/B)^{1/r}$ and $d = [\mathbb{Q}(\alpha) : \mathbb{Q}] \leq r$. The exact factorization is

$$AX^r - BY^r = B(\alpha X - Y) \sum_{j=0}^{r-1} (\alpha X)^{r-1-j} Y^j. \quad (125)$$

If $d \geq 2$, Siegel’s theorem at the *actual* degree gives a constant $c > 0$ for which

$$\left| \alpha - \frac{Y}{X} \right| > cX^{-\mu(d) - \eta}.$$

The difference is positive, and Lemma 2.53 gives $\mu(d) \leq \mu(r)$. Hence $\alpha X - Y > cX^{1-\mu(r)-\eta}$. Retaining the positive $j = 0$ term in the sum in (125) gives

$$AX^r - BY^r > Bc\alpha^{r-1}X^{r-\mu(r)-\eta}.$$

If $d = 1$, write $\alpha = P/Q$ in lowest positive terms. Strict positivity excludes $PX = QY$, so $PX - QY \geq 1$ and $\alpha X - Y \geq 1/Q$. Since $r \geq 2$ and $Y > 0$, the factor sum in (125) is strictly larger than its $j = 0$ term. Hence

$$AX^r - BY^r > \frac{B\alpha^{r-1}}{Q}X^{r-1}.$$

Moreover $\mu(r) > 1$, because every candidate in (123) has $\lambda \geq 1$ and a strictly positive first summand. The displayed bound therefore implies (124) for $X \geq 1$. Finally, with $\eta = \delta/2$, the ratio of the established lower bound to $X^{r-\mu(r)-\delta}$ is $CX^{\delta/2}$, which exceeds one for $X > C^{-2/\delta}$. $\square$

Theorem 2.55 (Repaired quantitative Pillai theorem). *Fix $A, B, M, N \in \mathbb{N}_{>0}$ with $M, N > 1$ and $\delta > 0$, and assume, as Pillai does, that $\log M / \log N \notin \mathbb{Q}$. Then there is an integer $x_0 = x_0(A, B, M, N, \delta) \geq 0$ such that*

$$AM^x - BN^y > M^{x(1-\delta)} \tag{126}$$

whenever $x, y \in \mathbb{N}_{>0}$, $x > x_0$, and $AM^x > BN^y$.

Proof. Replace δ by $\delta_0 = \min(\delta, 1/2)$. The required large r exists constructively: for $k = \lceil \sqrt{r} \rceil$ and $\lambda = k - 1 \geq 1$,

$$\mu(r) \leq \frac{r}{k} + k - 1 < 2\sqrt{r},$$

so $\mu(r)/r < 2/\sqrt{r}$. Choose, for example, an integer $r > (4/\delta_0)^2$, increasing it to at least two if necessary. Then $\mu(r)/r < \delta_0/2$. Choose $\eta > 0$ with $\eta/r < \delta_0/4$. Write uniquely

$$x = rs + h, \quad y = rt + \ell, \quad 0 \leq h, \ell < r.$$

For the fixed residue pair (h, ℓ) , put $A_h = AM^h$, $B_\ell = BN^\ell$, $X = M^s$, and $Y = N^t$. Proposition 2.54 gives

$$AM^x - BN^y > C_{h,\ell}M^{(x-h)(1-(\mu(r)+\eta)/r)}.$$

Put

$$C_* = \min_{0 \leq h, \ell < r} C_{h,\ell} > 0, \quad \gamma = 1 - \frac{\mu(r) + \eta}{r}, \quad \varepsilon = \delta_0 - \frac{\mu(r) + \eta}{r} > \frac{\delta_0}{4}.$$

Here $\gamma > 1 - 3\delta_0/4 > 0$. For every residue pair,

$$\frac{C_{h,\ell}M^{(x-h)\gamma}}{M^{x(1-\delta_0)}} = C_{h,\ell}M^{x\varepsilon-h\gamma} \geq C_*M^{x\varepsilon-(r-1)\gamma}.$$

The last quantity exceeds one whenever

$$x > \frac{(r-1)\gamma - \log_M C_*}{\varepsilon},$$

a threshold independent of y, h, ℓ . This proves $AM^x - BN^y > M^{x(1-\delta_0)} \geq M^{x(1-\delta)}$. The preceding proof establishes the same quantified conclusion for all $M, N > 1$, without the irrational-log hypothesis. This stronger statement is an edition result; Pillai's sourced theorem remains recorded with his hypothesis. $\square$

We now type the independent qualitative route used for Crandall. For a finite ordered tuple $S = (p_1, \dots, p_r)$ of pairwise distinct primes define

$$\mathcal{A}_S = \{p_1^{e_1} \cdots p_r^{e_r} : e_i \in \mathbb{N}_0\}.$$

Proposition 2.56 (Prime-exponent coordinate morphism). *The map*

$$E_S : (\mathbb{N}_0^r, +) \longrightarrow (\mathcal{A}_S, \cdot), \quad E_S(e_1, \dots, e_r) = \prod_{i=1}^r p_i^{e_i},$$

is a commutative-monoid isomorphism. Its inverse is $a \mapsto (\nu_{p_1}(a), \dots, \nu_{p_r}(a))$; every fibre is a singleton. Thus the coordinate loses no exponent information while the prime set S is retained.

Proof. Multiplication in $\mathcal{A}_S$ adds every valuation, so E_S preserves the monoid law. Existence and uniqueness of prime factorization prove both surjectivity and the displayed two-sided inverse. $\square$

Theorem 2.57 (Thue's fixed-value theorem at Pólya's exact scopes). *Let $c \in \mathbb{Z} \setminus \{0\}$.*

1. *If $U \in \mathbb{Z}[X, Y]$ is homogeneous and irreducible of degree greater than two, then $U(x, y) = c$ has only finitely many solutions in $\mathbb{Z}^2$.*
2. *More generally, if a homogeneous integral binary form F has infinitely many integer solutions of $F(x, y) = c$, then, up to multiplication by a nonzero scalar, F is a power of a linear form or an indefinite quadratic form.*

Proof. Thue's Theorem IV is printed for positive x, y [9, pp. 303–304]. On the off-axis domain, the map

$$(x, y) \longmapsto (\operatorname{sgn} x, \operatorname{sgn} y, |x|, |y|)$$

is a bijection

$$(\mathbb{Z} \setminus \{0\})^2 \longrightarrow \{\pm 1\}^2 \times \mathbb{N}_{>0}^2,$$

with inverse $(\epsilon, \delta, X, Y) \mapsto (\epsilon X, \delta Y)$. For each of the four sign pairs, the transformed form $U_{\epsilon, \delta}(X, Y) = U(\epsilon X, \delta Y)$ remains integral, homogeneous, irreducible, and of the same degree, so Thue's positive-quadrant theorem gives a finite set. If $y = 0$, irreducibility in degree greater than one forces the coefficient u_d of X^d to be nonzero, and the remaining equation is $u_d x^d = c$; the other axis is identical. This proves the exact all-integer extension in part 1.

Part 2 is the exceptional-form formulation quoted by Pólya [6, pp. 144–145]. His locator is Thue II, printed p. 3. Both available primary scans omit printed pp. 2–3, so that page is not represented as visually verified. Thue II printed p. 1 gives the equivalent dehomogenized theorem and its degree-one/degree-two power exceptions [8, p. 1]. The manifestation gap, Pólya's exact quotation, and the relation between the two formulations are recorded in `qa/THUE-1908-1909-F017-binary-form-audit.md`. $\square$

Proposition 2.58 (Pólya's residue-cell cubic morphism). *Let*

$$L_1(n) = an + b, \quad L_2(n) = cn + d,$$

where $a, b, c, d \in \mathbb{Z}$, $a, c \neq 0$, and $\Delta = ad - bc \neq 0$. Fix pairwise distinct primes $p_1, \dots, p_t$ and residue vectors $\mathbf{r}, \mathbf{s} \in \{0, 1, 2\}^t$. Let $D_{\mathbf{r}, \mathbf{s}}$ be the set of integers n for which $L_1(n), L_2(n) > 0$, both values have prime support in $\{p_1, \dots, p_t\}$, and

$$\nu_{p_i}(L_1(n)) \equiv r_i \pmod{3}, \quad \nu_{p_i}(L_2(n)) \equiv s_i \pmod{3} \quad (1 \leq i \leq t).$$

Put

$$R = \prod_{i=1}^t p_i^{r_i}, \quad S_0 = \prod_{i=1}^t p_i^{s_i}, \quad G_{\mathbf{r},\mathbf{s}}(X, Y) = aS_0Y^3 - cRX^3.$$

Then

$$\Phi_{\mathbf{r},\mathbf{s}} : D_{\mathbf{r},\mathbf{s}} \longrightarrow \mathcal{Z}_{\mathbf{r},\mathbf{s}} = \{(X, Y) \in \mathbb{N}_{>0}^2 : G_{\mathbf{r},\mathbf{s}}(X, Y) = \Delta\},$$

defined by

$$X = \prod_{i=1}^t p_i^{(\nu_{p_i}(L_1(n)) - r_i)/3}, \quad Y = \prod_{i=1}^t p_i^{(\nu_{p_i}(L_2(n)) - s_i)/3},$$

is well defined and injective. Its inverse on its image is

$$n = \frac{RX^3 - b}{a} = \frac{S_0Y^3 - d}{c}.$$

The target $\mathcal{Z}_{\mathbf{r},\mathbf{s}}$ is finite.

Proof. The residue hypotheses make the displayed exponents nonnegative integers and give the exact identities

$$an + b = RX^3, \quad cn + d = S_0Y^3.$$

Multiplying the second identity by a , the first by c , and subtracting gives

$$aS_0Y^3 - cRX^3 = ad - bc = \Delta.$$

The two displayed recovery formulas agree by this equation and recover the original n , proving injectivity and singleton fibres on the image.

The cubic has

$$\text{Disc}(G_{\mathbf{r},\mathbf{s}}) = -27(aS_0)^2(cR)^2 \neq 0.$$

It need not be irreducible over $\mathbb{Q}$, so part 1 of Theorem 2.57 cannot be invoked alone. Suppose instead that, over a characteristic-zero field,

$$G_{\mathbf{r},\mathbf{s}}(X, Y) = \lambda(uX + vY)^3.$$

Its nonzero X^3 and Y^3 coefficients force $\lambda, u, v \neq 0$, whereas its zero X^2Y coefficient would give $3\lambda u^2v = 0$, a contradiction. It is therefore not a scalar linear cube. A quadratic power cannot have degree three. Part 2 of Theorem 2.57 now proves finiteness. If one uses Pólya's positive-right-side phrasing and $\Delta < 0$, the exact involution

$$(G_{\mathbf{r},\mathbf{s}}, \Delta, X, Y) \longmapsto (-G_{\mathbf{r},\mathbf{s}}, -\Delta, X, Y)$$

preserves the solution set and exceptional-form status. $\square$

Theorem 2.59 (Pólya's fixed-prime gap theorem). *Let $r \geq 2$ and enumerate $\mathcal{A}_S$ increasingly as $a_0 < a_1 < a_2 < \dots$. Then*

$$\lim_{j \rightarrow \infty} (a_{j+1} - a_j) = \infty. \quad (127)$$

Proof. This is the exact statement in Pólya's Section 4, equations (12)–(14) [6, pp. 147–148]. Pólya proves it equivalent to his Satz I for products of two essentially distinct rational linear factors. His proof of Satz I reduces finitely supported prime-exponent vectors modulo three, fixes a residue vector, and invokes Thue's finiteness theorem for the resulting binary cubic [6, pp. 144–145]. Proposition 2.58 verifies the exact dependency, including the reducible-cubic cells not covered by Thue's irreducible-form theorem alone. Thue I, Satz 12, records the same modulo-three proof template [7, pp. 30–31]; Pólya supplies the residue-cell reduction used here. No effective gap threshold is imported. $\square$

Corollary 2.60 (Finite fixed-difference fibres). *For every nonempty finite prime set S and every fixed $z \in \mathbb{Z} \setminus \{0\}$, the fibre*

$$\Delta_S^{-1}(z) = \{(u, v) \in \mathcal{A}_S^2 : u - v = z\}, \quad \Delta_S(u, v) = u - v,$$

is finite.

Proof. If $|S| \geq 2$, choose an index after which every adjacent gap in (127) exceeds $|z|$. The difference between any two distinct later terms is a sum of one or more adjacent positive gaps, hence also exceeds $|z|$. Only pairs with at least one earlier coordinate remain. If $a_0, \dots, a_{J-1}$ are the earlier terms, then choosing one of these values and which coordinate of (u, v) it occupies leaves the other coordinate uniquely determined by $u - v = z$; hence at most $2J$ pairs remain. If $S = \{p\}$, use directly $p^{j+1} - p^j = (p - 1)p^j \rightarrow \infty$. $\square$

Proposition 2.61 (Exact fixed- z consequence of Pólya's sourced theorem). *For every $z \in \mathbb{Z}$, equation (122) has only finitely many solutions $(x, y) \in \mathbb{Z}^2$.*

Proof. First prove the exponent-domain reduction. If $x = -a < 0$ and $y \geq 0$, then

$$2^x - 3^y = \frac{1 - 2^a 3^y}{2^a}$$

has odd numerator and is not an integer. If $x \geq 0$ and $y = -b < 0$, the numerator of $(3^b 2^x - 1)/3^b$ is -1 modulo 3. If both exponents are negative, then

$$2^{-a} - 3^{-b} = \frac{3^b - 2^a}{2^a 3^b}$$

has absolute value strictly between zero and one: its numerator is nonzero by unique factorization and has absolute value smaller than the denominator. Thus an integer-valued difference forces $x, y \in \mathbb{N}_0$.

Define

$$I : \mathbb{N}_0^2 \longrightarrow \mathcal{A}_{\{2,3\}}^2, \quad I(x, y) = (2^x, 3^y).$$

The valuation map

$$V(u, v) = (\nu_2(u), \nu_3(v))$$

satisfies $V \circ I = \text{id}_{\mathbb{N}_0^2}$, so I is injective. For $z \neq 0$, the solution set is exactly

$$I^{-1}\left(\Delta_{\{2,3\}}^{-1}(z)\right).$$

Corollary 2.60 makes the inner fibre finite, and injectivity makes its preimage finite. For $z = 0$, unique factorization gives only $2^x = 3^y = 1$, hence $(x, y) = (0, 0)$. This proves Crandall's literal integer-exponent statement; it is not merely the positive- z statement printed in Pillai's Corollary 1. $\square$

2.7.6 Herschfeld's eventual-uniqueness theorem

Crandall's reference 10 spells the author's surname "Herschfeld." The primary paper's title and byline give *Aaron Herschfeld* [5, pp. 231–234]. Crandall reports at most one integer solution for sufficiently large z [1, p. 1288]. The primary theorem is stronger in the sign parameter but narrower in the exponent domain: it treats sufficiently large $|d|$ and positive exponents. The exact domain bridge is proved below rather than hidden in the citation.

Source issue 2.62 (Herschfeld source boundaries). On printed p. 232, a congruence argument for the specialized equation (2) ends by referring to equation (1); the objects and displayed congruence make the intended label unambiguous, but the defect is retained. Printed p. 233 reports a further $|d| \leq 100$ calculation from an unpublished paper; it is not used here. The final paragraph's general at-most-nine result for $a^x - b^y = d$ has a separate Siegel-1929 dependency and is not a premise of the special (2, 3) theorem. That general result remains outside the admitted scope until its exact cited theorem and coefficient hypotheses are read.

Lemma 2.63 (The two order coordinates). *For $r \geq 1$ and $s \geq 3$,*

$$\text{ord}_{3^r}(2) = 2 \cdot 3^{r-1}, \quad \text{ord}_{2^s}(3) = 2^{s-2}. \quad (128)$$

Proof. If $2^n \equiv 1 \pmod{3^r}$, reduction modulo three shows that $n = 2m$ is even. The lifting-the-exponent identity gives

$$v_3(2^{2^m} - 1) = v_3(4^m - 1) = 1 + v_3(m).$$

Thus $3^r \mid 2^n - 1$ exactly when $2 \cdot 3^{r-1} \mid n$. If $3^n \equiv 1 \pmod{2^s}$ with $s \geq 3$, reduction modulo eight shows that n is even, and

$$v_2(3^n - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(n) - 1 = 2 + v_2(n).$$

Therefore $2^s \mid 3^n - 1$ exactly when $2^{s-2} \mid n$. This also makes precise Herschfeld's older phrase that 3 "belongs to" 2^{s-2} modulo 2^s [5, pp. 232–234]. $\square$

Theorem 2.64 (Herschfeld's positive-exponent theorem, with dependency repaired). *There is a real number $D > 0$ such that, for every $d \in \mathbb{Z}$ with $|d| > D$,*

$$\#\{(x, y) \in \mathbb{N}_{>0}^2 : 2^x - 3^y = d\} \leq 1. \quad (129)$$

The proof supplies no effective numerical value of D .

Proof. Fix $0 < \delta < 1/2$. Apply Theorem 2.55 first to bases (2, 3) and then to (3, 2). Let $x_1, y_1 \in \mathbb{N}_0$ be thresholds such that

$$0 < 2^u - 3^v, \quad u > x_1 \implies 2^u - 3^v > 2^{u(1-\delta)}, \quad (130)$$

$$0 < 3^v - 2^u, \quad v > y_1 \implies 3^v - 2^u > 3^{v(1-\delta)}. \quad (131)$$

Set

$$D = \max\{2^{x_1+5}, 3^{y_1+2}\}. \quad (132)$$

Suppose two distinct solutions have the same d . They cannot agree in one coordinate without agreeing in both. Relabel them so that $X > x$; then

$$2^X - 2^x = 3^Y - 3^y > 0,$$

so $Y > y$. Factoring and using coprimality gives

$$2^x(2^{X-x} - 1) = 3^y(3^{Y-y} - 1), \quad (133)$$

and hence

$$2^{X-x} \equiv 1 \pmod{3^y}, \quad 3^{Y-y} \equiv 1 \pmod{2^x}. \quad (134)$$

Lemma 2.63 yields

$$X - x \geq 2 \cdot 3^{y-1}, \quad Y - y \geq 2^{x-2} \quad (x > 2). \quad (135)$$

If $d > 0$, then $d > 2^{x_1+5}$ and $d = 2^x - 3^y < 2^x$, so $x > x_1 + 5 > 5$. Therefore

$$2^X = 3^Y + d > 3^Y \geq 3^{2^{x-2}} > 2^{2^{x-2}}, \quad X > 2^{x-2} > 2x.$$

Applying (130) at the larger solution gives

$$d = 2^X - 3^Y > 2^{X(1-\delta)} > 2^{X/2} > 2^x > d,$$

a contradiction.

If $d < 0$, then $|d| > 3^{y_1+2}$ and $|d| = 3^y - 2^x < 3^y$, so $y > y_1 + 2 > 2$. Now

$$3^Y = 2^X - d > 2^X \geq 2^{2 \cdot 3^{y-1}} > 3^{3^{y-1}}, \quad Y > 3^{y-1} > 2y.$$

Applying (131) at the larger solution gives

$$|d| = 3^Y - 2^X > 3^{Y(1-\delta)} > 3^{Y/2} > 3^y > |d|,$$

again a contradiction. The elementary inequalities used above are strict for exactly the displayed ranges: $2^{x-2} > 2x$ for $x > 5$ and $3^{y-1} > 2y$ for $y > 2$. This reconstructs Herschfeld's printed proof while replacing only its reliance on Pillai's defective literal derivation by the separately attributed repair already proved in this edition [5, pp. 233–234]. $\square$

The next statement is an edition consequence, not part of Herschfeld's printed exponent domain.

Proposition 2.65 (Exact all-integer extension). *For the same D as in (132) and every $d \in \mathbb{Z}$ with $|d| > D$,*

$$\#\{(x, y) \in \mathbb{Z}^2 : 2^x - 3^y = d\} \leq 1. \quad (136)$$

Proof. The exponent-domain argument in Proposition 2.61 proves that any integer-valued difference has $x, y \in \mathbb{N}_0$. It remains to include a possible zero coordinate without importing it into Herschfeld's statement.

Suppose two nonnegative solutions are ordered as in the preceding proof. If $d > 0$, then $d < 2^x$ still forces $x > x_1 + 5 > 5$, even if $y = 0$. The second congruence in (134) and the second bound in (135) remain valid, and the larger pair (X, Y) has positive coordinates. The positive- d contradiction is unchanged. If $d < 0$, then $|d| < 3^y$ still forces $y > y_1 + 2 > 2$, even if $x = 0$. The first congruence and first order bound remain valid, the larger pair is positive, and the negative- d contradiction is unchanged. Thus the nonnegative fibre has size at most one. Composing this conclusion with the proved integer-to-nonnegative exponent reduction gives (136). This proves the exact domain used by Crandall and strengthens his one-sided large- z wording to Herschfeld's two-sided large- $|d|$ parameter without changing either source statement. $\square$

Explicit nonclaim 2.66. Herschfeld's eventual uniqueness is not inferred from Pólya's fixed- d finiteness, and fixed- d finiteness is not inferred from eventual uniqueness. The threshold in (132) is ineffective; the theorem supplies neither an effective search bound nor a Collatz endpoint or cycle exclusion.

Proposition 2.67 (Exact primitive-root solution-to-return construction). *Put $t = \log_2 3$. Let p be an odd prime such that $\text{ord}_p(2) = p - 1$, and suppose*

$$(x, y) \in \mathbb{Z}^2, \quad 2^x - 3^y = p, \quad y \geq \frac{p}{t-1}. \quad (137)$$

Then one can construct integers

$$0 = A_0 < A_1 < \cdots < A_y = x$$

and a positive odd integer $m > 1$ such that

$$m(2^{A_y} - 3^y) = \sum_{j=0}^{y-1} 2^{A_j} 3^{y-1-j}. \quad (138)$$

Consequently m occurs in its own Crandall trajectory, at accelerated return time y ; its least period is only asserted to divide y .

Proof. The integer-exponent denominator argument in the proof of Proposition 2.61 gives $x, y \in \mathbb{N}_0$. Under the last hypothesis in (137), $y > 0$. The equality then excludes $p = 3$ after reduction modulo 3, so $p \geq 5$. Since $0 < t - 1 < 1$, the last hypothesis also gives $y > p \geq 5$; in particular all index ranges below are nonempty.

Because $2^x > 3^y$, one has $x > ty$. Hence the integer

$$L = x - y > (t - 1)y \geq p \quad \implies \quad L \geq p + 1. \quad (139)$$

Equivalently, the assumed lower bound on y is the exact integer inequality $3^y \geq 2^{p+y}$; no decimal approximation to t is being used.

First put $A_j = j$ for $0 \leq j \leq y - 2$, and set

$$R_0 = \sum_{j=0}^{y-2} 2^j 3^{y-1-j}.$$

If $R_0 \not\equiv 0 \pmod{p}$, choose A_{y-1} to be the least integer in $[y - 1, x - 1]$ satisfying

$$2^{A_{y-1}} \equiv -R_0 \pmod{p}. \quad (140)$$

Such an exponent exists: the interval contains at least $p - 1$ consecutive integers by (139), and $\text{ord}_p(2) = p - 1$ says exactly that the powers of 2 over any complete exponent-residue system modulo $p - 1$ biject onto $(\mathbb{Z}/p\mathbb{Z})^\times$.

If instead $R_0 \equiv 0 \pmod{p}$, keep $A_j = j$ for $0 \leq j \leq y - 3$, replace A_{y-2} by $y - 1$, and write

$$R_1 = \sum_{j=0}^{y-3} 2^j 3^{y-1-j} + 2^{y-1} 3.$$

The replacement changes the partial sum by

$$R_1 - R_0 = 3 \cdot 2^{y-2} \not\equiv 0 \pmod{p},$$

so $R_1 \not\equiv 0 \pmod{p}$. Choose A_{y-1} to be the least integer in $[y, x-1]$ satisfying

$$2^{A_{y-1}} \equiv -R_1 \pmod{p}. \quad (141)$$

This interval again contains at least $p-1$ consecutive integers, since its cardinality is $L \geq p+1$. Thus the selector is defined in both cases, and the resulting sequence is strictly increasing with $A_0 = 0$ and $A_y = x$.

Let

$$S = \sum_{j=0}^{y-1} 2^{A_j} 3^{y-1-j}.$$

The selected congruence gives $p \mid S$. The $j=0$ summand is odd and every other summand is even, so S and $m = S/p$ are odd. Moreover $0 < t-1 < 1$ gives $y > p$, whence

$$S \geq 3^{y-1} \geq 3^p > p;$$

The last strict inequality follows by induction from $3^n > n$ for $n \geq 1$; therefore m is a positive odd integer greater than 1. Finally, $2^{A_y} - 3^y = 2^x - 3^y = p$, so $mp = S$ is exactly (138). Corollary 7.1, in its exact form proved above, now gives the asserted self-return. $\square$

Corollary 2.68 (Crandall's conditional primitive-root bound). *Assume Conjecture 2.1. If p is a prime for which 2 is a primitive root, then every integer solution of $2^x - 3^y = p$ satisfies*

$$y < \frac{p}{\log_2 3 - 1}.$$

Proof. A solution violating the strict bound satisfies the hypotheses of Proposition 2.67 and therefore produces a positive odd $m > 1$ of infinite height, contrary to Conjecture 2.1. $\square$

Explicit nonclaim 2.69. The source prints the conditional bound but omits this construction. The proposition does not assert that any solution violating the bound exists, and it is not the adjacent sufficient obstruction with $A_j = j$. The selected return time y need not be the least period, and the corollary remains conditional on the Collatz conjecture.

2.7.7 Continued fractions and the repaired cycle bound

Put $t = \log_2 3$, and let p_n/q_n be its continued-fraction convergents, with

$$p_{-1} = 1, \quad q_{-1} = 0, \quad p_0 = a_0, \quad q_0 = 1, \quad p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2}. \quad (142)$$

Lemma 2.70 (Exact continued-fraction transfer for the Crandall variables). *The number $t = \log_2 3$ is irrational. Let p_n/q_n be its simple continued-fraction convergents, with the recurrence (142). Then, for $n \geq 2$,*

$$\frac{1}{q_n + q_{n+1}} < |p_n - q_n t| < \frac{1}{q_{n+1}}. \quad (143)$$

If $x \in \mathbb{Z}$ and $0 < y < q_n$, then

$$|p_n - q_n t| < |x - yt|. \quad (144)$$

For $n \geq 4$, if $y = q_n$ and $x \in \mathbb{Z}$ with $x \neq p_n$, then the same strict inequality holds with $y = q_n$.

Proof. Since $t > 1$, a rational representation $t = a/b$ with $b > 0$ would have $a > 0$ and $2^a = 3^b$, contradicting unique factorization. The continued fraction is therefore infinite. Steuding's Theorems 3.1–3.2 [12, printed p. 38] give the displayed recurrence and

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1},$$

so each p_n/q_n is reduced. Steuding's Theorem 3.6 [12, printed p. 42], applied to the complete tail α_{n+1} of t , gives

$$t - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n(\alpha_{n+1}q_n + q_{n-1})}.$$

Because $a_{n+1} < \alpha_{n+1} < a_{n+1} + 1$ and $q_{n+1} = a_{n+1}q_n + q_{n-1}$, multiplying by q_n yields (143). For $x > 0$, Steuding's Theorem 3.8 [12, printed pp. 44–45], with $(p, q) = (x, y)$, gives (144); the equality case $x/y = p_n/q_n$ cannot occur because p_n/q_n is reduced and $0 < y < q_n$. If $x \leq 0$, then $|x - yt| \geq yt > 1$, since $t = \log_2 3 > 1$, while the upper bound in (143) is < 1 .

For the final assertion, (143) gives $|p_n - q_n t| < 1/q_{n+1} < 1/2$ for $n \geq 4$. If $x \neq p_n$, integrality gives $|x - p_n| \geq 1$, and the triangle inequality gives $|x - q_n t| \geq 1 - |p_n - q_n t| > |p_n - q_n t|$. $\square$

Source issue 2.71 (four repairs in the continued-fraction chain). The proof chain on printed pp. 1289–1290 needs four distinct repairs [1, pp. 1289–1290].

- (i) Lemma 7.1 omits $0 < y$. As printed, $(x, y) = (0, 0)$ contradicts its strict best-approximation inequality. The used form has $0 < y < q_n$ and is proved from Lemma 2.70.
- (ii) Lemma 7.3 introduces p_n/q_n but then writes p_m, q_m , leaves x unquantified, and uses $|e^u - 1| > |u|$. This analytic inequality holds for $u > 0$, not for $u < 0$. The exact form needed for cycles is

$$\left. \begin{array}{l} x, y \in \mathbb{Z}, \quad 0 < y < q_n, \\ x - y \log_2 3 > 0 \end{array} \right\} \implies |2^x - 3^y| > 3^y \log 2 |p_n - q_n t|. \quad (145)$$

Indeed, for $u = (x - yt) \log 2 > 0$ the positive-term power series gives $e^u - 1 - u = \sum_{j \geq 2} u^j/j! > 0$, and best approximation supplies $|x - yt| > |p_n - q_n t|$. In the cycle application, the missing sign is exactly (119).

- (iii) Theorem 7.2 explicitly reduces to $k \leq q_n$, but then invokes Lemma 7.3, whose repaired denominator hypothesis is strict: $y < q_n$. The final assertion of Lemma 2.70 handles $k = q_n$ when $A_k \neq p_n$. If $A_k = p_n$, the positive exponential inequality is itself strict. Thus the same denominator estimate holds at the equality boundary that the printed proof does not justify.
- (iv) Corollary 7.2 claims that its minimum becomes unbounded as n increases. For a fixed cycle its second entry instead tends to zero. The correct proof first fixes n and only then chooses a sufficiently large cycle minimum; this quantifier order is proved below.

The later references to Eq. (7.5) when computing q_n are correct only as references to the printed continued-fraction list; the recurrence itself is Eq. (7.6).

Theorem 2.72 (Crandall's cycle bound, with proof gaps repaired). *Let $m > 1$ be the least member of a C_{Cr} -cycle of least accelerated period k . For every integer $n \geq 4$,*

$$k > \min \left\{ q_n, \frac{2m}{q_n + q_{n+1}} \right\}. \quad (146)$$

Consequently, only finitely many cycles have any prescribed accelerated period [1, p. 1290].

Proof. Minimality of m and the branch equation give, for the coordinates (88),

$$2^{A_j} \leq (3 + 1/m)^j. \quad (147)$$

The return equation and this bound imply

$$m < \frac{k(3 + 1/m)^{k-1}}{2^{A_k} - 3^k}. \quad (148)$$

If $k > q_n$, (146) is immediate. Otherwise the repaired best-approximation argument in Source issue 2.71, including its equality case, gives

$$2^{A_k} - 3^k > 3^k \log 2 |p_n - q_n t|.$$

Combining this with (148), and writing $v = 1/(3m) > 0$, first yields

$$k > m(3 + 1/m)(\log 2)|p_n - q_n t|(1 + v)^{-k}.$$

The function $f(v) = (1 + v)^{-k}$ has $f''(v) = k(k + 1)(1 + v)^{-k-2} > 0$; strict convexity above its tangent at zero therefore gives $f(v) > f(0) + f'(0)v = 1 - kv$. Hence

$$k > m(\log 2)(3 + 1/m)|p_n - q_n t| \left(1 - \frac{k}{3m}\right). \quad (149)$$

For $n \geq 4$, the recurrence gives $q_n + q_{n+1} \geq q_4 + q_5 = 12 + 41 = 53 > 20$. If $k \geq m/10$, the second member of the minimum in (146) is strictly below $m/10 \leq k$. Otherwise $k < m/10$, so $1 - k/(3m) > 29/30$, and the continued-fraction lower bound $|p_n - q_n t| > (q_n + q_{n+1})^{-1}$ turns (149) into

$$k > \frac{(3m + 1) \log 2}{q_n + q_{n+1}} \frac{29}{30} > \frac{2m}{q_n + q_{n+1}}.$$

The final strict inequality is exact: the positive-term expansion $\log 2 = 2(1/3 + 1/(3^3 \cdot 3) + \dots)$ gives $\log 2 > 56/81 > 20/29$, while $(3m + 1)/m > 3$. This proves (146).

For finiteness, fix a proposed period L and choose an integer $n \geq 4$ with $q_n > L$; the convergent denominators are unbounded. An infinite family of distinct period- L cycles would have unbounded distinct least members, so choose one with $m > L(q_n + q_{n+1})/2$. Both entries of the minimum would then exceed L , contradicting (146) with $k = L$. $\square$

Printed p. 1282 says that Conjecture (2.1) had been verified for $m < 10^9$, citing references [1] and [5] [1, p. 1282]. Those references are not two independent witnesses. Reference [1] is Conway's *Unpredictable iterations*; its correct proceedings pages are 49–52, not Crandall's 45–52. Reference [5], printed as “*Zentralblatt für Mathematik*, Band 233, p. 10041,” is a defective rendering of *Zbl* 0337.10041, the *Zentralblatt* record for that same Conway paper: 10041 is part of the accession identifier, not a printed page. Thus the two routes collapse to one work, rather than corroborating one another [1, p. 1292].

Conway defines the ordinary one-step map g by $g(n) = n/2$ for even n and $g(n) = 3n + 1$ for odd n , literally the map U in (1). On the first page of his article he reports the inclusive range $n \leq 10^9$ and attributes its verification to D. H. Lehmer, Emma Lehmer, and J. L. Selfridge [2, p. 219]. He supplies no citation for that sentence and no program, machine, range partition, stopping criterion, independent check, checksum, log, or preserved output. His proved undecidability theorem is expressly said to say nothing about the particular Collatz game; the verification sentence is historical background, not a theorem of the article. The published contemporaneous report and its attribution are therefore content-verified, while the underlying computation remains unauthenticated and nonreproducible at this checkpoint.

Proposition 2.73 (One-step verification to accelerated exclusion). *Let $N \in \mathbb{N}_{>0}$. Suppose that for every $n \in \mathbb{N}_{>0}$ with $n \leq N$ there exists $r \in \mathbb{N}_0$ such that $U^r(n) = 1$. Then for every $m \in \mathbb{N}_{>0}^{\text{odd}}$ with $m \leq N$ there exists $j \in \mathbb{N}_0$ such that $C_{\text{Cr}}^j(m) = 1$. Consequently no $m \in \mathbb{N}_{>0}^{\text{odd}}$ with $1 < m \leq N$ can satisfy $C_{\text{Cr}}^k(m) = m$ for any $k \in \mathbb{N}_{>0}$.*

Proof. Starting from an odd m , the successive odd states of the U -orbit are exactly the C_{Cr} -orbit, with the j th odd return occurring at the unshortened time $A_j + j$ by (89). Since 1 is odd, any occurrence $U^r(m) = 1$ is one of those successive odd returns; hence $C_{\text{Cr}}^j(m) = 1$ for a unique first such index j . Finally $C_{\text{Cr}}(1) = 1$. If $m > 1$ also satisfied $C_{\text{Cr}}^k(m) = m$, its accelerated orbit would be periodic from m and could not enter the fixed point 1, a contradiction. $\square$

Printed p. 1290 uses the “known result” that no $1 < m < 10^9$ appears in its own trajectory, without a new inline citation [1, p. 1290]. Proposition 2.73 proves the exact clock and domain transfer from Conway’s stronger inclusive ordinary-map report to that accelerated statement; it does not authenticate the computation.

For a hypothetical nontrivial cycle, its least member m_0 lies in $\mathbb{N}_{>0}^{\text{odd}}$. Conway’s inclusive report, after the proved transfer, directly excludes $m_0 \leq 10^9$. Independently, Crandall’s weaker strict-range form first gives $m_0 \geq 10^9$, and oddness excludes equality with the even endpoint. Both routes therefore give, exactly as Crandall writes,

$$m_0 > 10^9, \quad \text{equivalently} \quad m_0 \geq 1,000,000,001.$$

With

$$q_{10} = 31867, \quad q_{11} = 79335, \quad (150)$$

Theorem 2.72 gives the least accelerated period ℓ the bound

$$\ell > \min \left\{ 31867, \frac{2 \cdot 10^9}{111202} \right\} > 17985.$$

Every positive return exponent k is a multiple of ℓ , so this recovers Crandall’s Theorem 7.3: $C_{\text{Cr}}^k(m) = m$ with $m > 1$ implies $k > 17985$, conditional on the published 1972 report reused by Crandall in 1978. The source variable is an accelerated return time, not a current or reproducible verification record. For the same return, (89) gives $A_k \geq k > 17985$ shortened steps and $A_k + k > k > 17985$ unshortened steps; these are transferred inequalities, not an identification of the three clocks.

The opening of Section 7 also says every member of a bounded trajectory is strictly below its supremum. The correct relation is $\leq$. The intended finite pigeonhole argument survives: boundedness plus absence of nontrivial self-returns would force the trajectory to contain 1.

2.7.8 The generalized $qx + r$ programme

For positive odd q, r , $q > 1$, Crandall defines

$$C_{q,r}(m) = \frac{qm + r}{2^{e_{q,r}(m)}}, \quad e_{q,r}(m) = \nu_2(qm + r), \quad m \in \mathbb{N}_{>0}^{\text{odd}}. \quad (151)$$

Conjecture 8.1 predicts that, except for $(q, r) = (3, 1)$, some positive odd starting value fails to reach 1. The exceptional pair is not an assertion that the ordinary conjecture is true.

Proposition 2.74 (Crandall’s proved generalized failures). *The following statements hold.*

(a) If $r > 1$, every $m \in \mathbb{N}_{>0}^{\text{odd}}$ divisible by r fails to reach 1.

(b) For $(q, r) = (5, 1)$,

$$13 \mapsto 33 \mapsto 83 \mapsto 13, \quad (e_0, e_1, e_2) = (1, 1, 5).$$

(c) For $(q, r) = (181, 1)$,

$$27 \mapsto 611 \mapsto 27, \quad (e_0, e_1) = (3, 12).$$

Proof. If $r \mid m$, then $qm + r \equiv 0 \pmod{r}$; division by a power of 2, a unit modulo odd r , preserves divisibility by r . Induction excludes the value 1. The two cycles follow by exact substitution:

$$5(13) + 1 = 2(33), \quad 5(33) + 1 = 2(83), \quad 5(83) + 1 = 2^5(13),$$

and

$$181(27) + 1 = 2^3(611), \quad 181(611) + 1 = 2^{12}(27).$$

The associated differences are $2^7 - 5^3 = 3$ and $2^{15} - 181^2 = 7$. The printed starting value in the second cycle is 27; 21 is an OCR error [1, p. 1291]. $\square$

Proposition 2.75 (the omitted finite certificate for $q = 1093$). *For $C_{1093,1}$, a positive odd integer has height one exactly when*

$$m = \frac{2^{364p} - 1}{1093} \quad (p \in \mathbb{Z}_{>0}). \quad (152)$$

No positive odd integer has height two. Every positive odd integer not in (152) therefore has infinite height, and this infinite-height set has natural density one in $\mathbb{N}_{>0}^{\text{odd}}$.

Proof. Exact modular exponentiation gives

$$\text{ord}_{1093}(2) = 364, \quad 2^{364} \equiv 1 \pmod{1093^2}. \quad (153)$$

For the first equality it suffices to check the proper divisors of $364 = 2^2 \cdot 7 \cdot 13$; convenient witnesses include

$$2^{182} \equiv 1092, \quad 2^{52} \equiv 27, \quad 2^{28} \equiv 121 \pmod{1093}.$$

A height-one value satisfies $1093m + 1 = 2^a$, so the order calculation is equivalent to $364 \mid a$ and proves (152). The congruence modulo 1093^2 shows that every such m is itself divisible by 1093. If it had a predecessor u , then $1093u + 1 = 2^b m$ would be simultaneously congruent to 1 and 0 modulo 1093, impossible. Thus height two does not occur. Any finite trajectory of height greater than one would contain a height-two tail, so no other finite height occurs. Finally, the number of parameters p producing $m \leq x$ is $O(\log x)$; division by the number of odd integers at most x gives density zero for the height-one family.

The source states the classification and calls the height-two impossibility easy, but omits the modular certificate [1, p. 1291]. The calculation is reproduced by `certificates/crandall_1978_checks.py`. $\square$

Explicit nonclaim 2.76. Infinite height need not mean an unbounded orbit; it may arise from a nontrivial cycle. Crandall explicitly states that no unbounded trajectory was then known for any $qx + 1$ problem. The $7x + 1$ trajectory from 3 exceeding 10^{2000} is a finite empirical excursion only. The heuristic drift $\log(q/4) > 0$ for $q > 3$ is not an unboundedness theorem [1, pp. 1291–1292].

2.8 Everett's finite parity coordinates and density-one descent

Everett works on $\mathbb{N}_0$ with the map

$$f_{\text{Ev}}(m) = \begin{cases} m/2, & m \text{ even}, \\ (3m+1)/2, & m \text{ odd}. \end{cases} \quad (154)$$

The subscript is editorial: the source writes f . Thus $f_{\text{Ev}} = T|_{\mathbb{N}_0}$, exactly; it is neither the unshortened map U nor Crandall's odd first-return map. Put

$$m_n = f_{\text{Ev}}^n(m), \quad x_n = m_n \bmod 2 \in \{0, 1\}. \quad (155)$$

The source word is chronological. When encoded as an integer below, x_0 is the least significant digit [11, pp. 42–43].

Theorem 2.77 (Everett, Theorem 1: finite parity fibres). *Let $N \geq 1$, let $w = (x_0, \dots, x_{N-1}) \in \{0, 1\}^N$, and put*

$$X = X(w) = \sum_{n=0}^{N-1} x_n.$$

There are unique integers $a(w), b(w)$ satisfying

$$0 \leq a(w) < 2^N, \quad 0 \leq b(w) < 3^X,$$

such that the complete nonnegative fibre of w and its N -th iterate are

$$m = a(w) + 2^N Q, \quad m_N = b(w) + 3^X Q, \quad Q \in \mathbb{N}_0. \quad (156)$$

Consequently every length- N binary word occurs at one and only one representative $0 \leq m < 2^N$.

Proof. This is the source induction with its implicit quotient domain made explicit. For $N = 1$, $x_0 = 0$ gives

$$m = 2Q, \quad m_1 = Q, \quad (a, b, X) = (0, 0, 0),$$

whereas $x_0 = 1$ gives

$$m = 1 + 2Q, \quad m_1 = 2 + 3Q, \quad (a, b, X) = (1, 2, 1).$$

Assume (156) at length N , and abbreviate its coordinates to a, b, X, Q_N . Because 3^X is odd, the parity of $m_N = b + 3^X Q_N$ fixes the parity of Q_N once b and the next required bit x_N are fixed. There are exactly four cases.

If b is even and $x_N = 0$, write $Q_N = 2Q_{N+1}$. Then

$$a_N = a, \quad b_N = b/2, \quad X_N = X.$$

If b is even and $x_N = 1$, write $Q_N = 1 + 2Q_{N+1}$. Then

$$a_N = a + 2^N, \quad b_N = \frac{3b + 3^{X+1} + 1}{2}, \quad X_N = X + 1.$$

Since $b \leq 3^X - 1$,

$$b_N \leq \frac{3(3^X - 1) + 3^{X+1} + 1}{2} = 3^{X+1} - 1 < 3^{X+1}.$$

If b is odd and $x_N = 0$, write $Q_N = 1 + 2Q_{N+1}$. Then

$$a_N = a + 2^N, \quad b_N = \frac{b + 3^X}{2}, \quad X_N = X,$$

and $b < 3^X$ gives $b_N < 3^X$. Finally, if b is odd and $x_N = 1$, write $Q_N = 2Q_{N+1}$. Then

$$a_N = a, \quad b_N = \frac{3b + 1}{2}, \quad X_N = X + 1,$$

and

$$b_N < \frac{3^{X+1} + 1}{2} < 3^{X+1}.$$

In each case $0 \leq a_N < 2^{N+1}$, and the forced parity decomposition of Q_N is unique. This proves existence, uniqueness, both coefficient bounds, and the retained quotient coordinate at stage $N + 1$ [11, pp. 43–44, Eq. (4) and proof]. $\square$

Proposition 2.78 (the exact finite parity morphism and inverse residue). *For $N \geq 1$, define*

$$P_N : \mathbb{Z}/2^N\mathbb{Z} \longrightarrow \{0, 1\}^N, \quad P_N([m]) = (T^n(m) \bmod 2)_{0 \leq n < N}. \quad (157)$$

Then P_N is a bijection. If

$$c(w) = \sum_{j=0}^{N-1} x_j 2^j 3^{\sum_{i=j+1}^{N-1} x_i},$$

its inverse is the unique residue

$$P_N^{-1}(w) \equiv -3^{-X(w)}c(w) \pmod{2^N}. \quad (158)$$

Reduction $\mathbb{Z}/2^{N+1}\mathbb{Z} \rightarrow \mathbb{Z}/2^N\mathbb{Z}$ corresponds to deletion of the last word digit. On $\mathbb{N}_0$, the fibre over w is exactly $a(w) + 2^N\mathbb{N}_0$, and (156) records the quotient that the word forgets.

Proof. Theorem 2.77 proves well-definedness on residue classes, singleton fibres at residue level, surjectivity, and the stated nonnegative fibres. Iterating the two branch equations gives

$$2^N T^N(m) = 3^{X(w)}m + c(w) \quad (159)$$

whenever the first N parities are w . This follows by induction: the next zero branch multiplies the existing numerator by no new factor of 3, while the next one branch multiplies it by 3 and adds 2^N . Divisibility of the right side by 2^N gives (158); $3^{X(w)}$ is odd and therefore a unit modulo 2^N . The same prefix read at levels $N + 1$ and N proves the reduction identity. $\square$

The source corollary uses the 2^N positive representatives $1, \dots, 2^N$, replacing the zero residue by 2^N . This implicit representative change is valid, but it must be stated. Numerically encoding the target word identifies P_N exactly with the finite chronological parity map $\overline{Q}_{3,N}$ in (189). Everett proves the bijection and its integer affine fibres. Later sources add permutation-order, lifting, inverse-limit, and graph-relation structure. Everett's quotient Q_N in the proof of Theorem 1 is not the later parity encoder Q_3 .

If two nonnegative integers have the same infinite parity sequence, the finite theorem makes their difference divisible by 2^N for every N , so they are equal. This is the source's infinite injectivity conclusion. It does not assert that every infinite binary sequence is realized by an ordinary nonnegative integer.

Theorem 2.79 (Everett, Theorem 2: natural-density-one descent). *Let*

$$A(M) = \#\{1 \leq m \leq M : \exists k \geq 1, f_{\text{Ev}}^k(m) < m\}. \quad (160)$$

Then

$$\lim_{M \rightarrow \infty} \frac{A(M)}{M} = 1. \quad (161)$$

Thus the source phrase “almost every” means ordinary natural density one in the intervals $[1, M]$.

Proof. First take $M = 2^N$. For a length- N word put

$$X = \sum_{n=0}^{N-1} x_n, \quad L = \frac{\log 2}{\log(10/3)}, \quad \varepsilon = L - \frac{1}{2}. \quad (162)$$

Here $\varepsilon > 0$, since $L > 1/2$ is equivalent to $4 > 10/3$. Let H_N be the set of words satisfying

$$\frac{1}{2} - \varepsilon < \frac{X}{N} < \frac{1}{2} + \varepsilon = L. \quad (163)$$

Under uniform counting on $\{0, 1\}^N$, each coordinate is Bernoulli with mean $1/2$, and every prescribed pair of coordinate values has 2^{N-2} extensions. Hence

$$\mathbb{E}X = N/2, \quad \text{Var}(X) = N/4.$$

Chebyshev’s inequality gives the exact estimate cited, but not proved, in the source:

$$\frac{\#H_N}{2^N} \geq 1 - \frac{1}{4\varepsilon^2 N}. \quad (164)$$

This is a finite uniform count transported by Proposition 2.78; it does not assume that one ordinary orbit has independent random steps.

Let D_N contain the words with $X/N < L$. Then $H_N \subseteq D_N$. If $x_n = 0$, then $m_{n+1}/m_n = 1/2$. If $x_n = 1$ and $m_n > 1$, write $m_n = 2Q + 1$, where $Q \geq 1$; then

$$\frac{m_{n+1}}{m_n} = \frac{3Q + 2}{2Q + 1} \leq \frac{5}{3}. \quad (165)$$

Moreover,

$$\frac{X}{N} < L \iff (1/2)^{N-X} (5/3)^X < 1. \quad (166)$$

Use the source corollary’s representatives $1 \leq m \leq 2^N$. For a word in D_N , either some $m_n = 1 < m$ with $n < N$, or no such value occurs. In the latter case every odd factor satisfies (165), so

$$m_N = m_0 \prod_{n=0}^{N-1} \frac{m_{n+1}}{m_n} \leq (1/2)^{N-X} (5/3)^X m_0 < m_0.$$

The one excluded starting value is $m = 1$. Therefore

$$A(2^N) \geq \#D_N - 1 \geq \#H_N - 1, \quad (167)$$

and (164) gives

$$\frac{A(2^N)}{2^N} \geq 1 - \frac{1}{4\varepsilon^2 N} - 2^{-N} \longrightarrow 1. \quad (168)$$

It remains to retain the source's interpolation, rather than claim that a dyadic subsequence proves the full limit. Put $A_N = A(2^N)$ and

$$n_N = 2^N + ((2^{N+1} - 2^N) - (A_{N+1} - A_N)) = A_N + (2^{N+1} - A_{N+1}), \quad 2^N \leq n_N \leq 2^{N+1}. \quad (169)$$

The difference $n_N - 2^N$ is exactly the number of failures in the dyadic shell $(2^N, 2^{N+1}]$. Its bounds follow because that shell contains 2^N integers. For $2^N < M \leq n_N$, monotonicity gives

$$\frac{A(M)}{M} \geq \frac{A_N}{n_N}.$$

For $M = n_N + k$, the dyadic interval has exactly $n_N - 2^N$ failures in total, so its first $n_N - 2^N + k$ members contain at least k successes. Consequently

$$\frac{A(M)}{M} \geq \frac{A_N + k}{n_N + k} \geq \frac{A_N}{n_N},$$

where the last inequality is equivalent to $k(n_N - A_N) \geq 0$. Finally,

$$\frac{A_N}{n_N} = \frac{A_N/2^N}{A_N/2^N + 2(1 - A_{N+1}/2^{N+1})} \longrightarrow 1 \quad (170)$$

by (168). The two interval estimates and (170) prove (161) [11, pp. 44–45, Eqs. (5)–(12) and Thm. 2]. $\square$

Calculation 2.80 (an exact rational concentration witness). The strict inequality $L > 23/40$ is equivalent to

$$2^{40}3^{23} > 10^{23},$$

and direct integer subtraction gives

$$2^{40}3^{23} - 10^{23} = 3511519796081828298752 > 0.$$

Thus $\varepsilon > 3/40$. Applying the same Chebyshev proof with the smaller rational window $3/40$ gives the completely explicit edition bound

$$\frac{A(2^N)}{2^N} \geq 1 - \frac{400}{9N} - 2^{-N}. \quad (171)$$

This is a proof-checking witness for the source limit, not an attribution of the displayed constant to Everett.

Proposition 2.81 (Everett–Crandall first-descent crosswalk). *Define*

$$D_T = \{m \in \mathbb{N}_{>0}^{\text{odd}} : \exists k \geq 1, T^k(m) < m\},$$

$$D_C = \{m \in \mathbb{N}_{>0}^{\text{odd}} : \exists j \geq 1, C_{\text{Cr}}^j(m) < m\}.$$

Then $D_T = D_C$ pointwise. Moreover,

$$\lim_{M \rightarrow \infty} \frac{\#(D_C \cap [1, M])}{\#(\mathbb{N}_{>0}^{\text{odd}} \cap [1, M])} = 1. \quad (172)$$

This is the exact primary verification of the theorem reported by Crandall.

Proof. If $C_{Cr}^j(m) < m$, then $C_{Cr}^j(m) = T^{A_j}(m)$ by (89), so $m \in D_T$. Conversely, suppose $T^k(m) < m$. If $T^k(m)$ is odd, it is one of the successive odd-return states. If it is even, put $r = \nu_2(T^k(m)) \geq 1$. The next r shortened steps are divisions by two, and

$$T^{k+r}(m) = \frac{T^k(m)}{2^r} < m$$

is odd. It is therefore a later C_{Cr} -iterate. This proves the pointwise equality without identifying shortened time k with accelerated time j .

Every positive even m belongs to the full shortened descent set because $T(m) = m/2 < m$. Put

$$B(M) = \#(D_C \cap [1, M]), \quad O(M) = \#(\mathbb{N}_{>0}^{\text{odd}} \cap [1, M]) = \lceil M/2 \rceil.$$

The exact finite-cutoff identity is therefore

$$A(M) = \lfloor M/2 \rfloor + B(M). \quad (173)$$

Consequently

$$1 - \frac{B(M)}{O(M)} = \frac{M}{O(M)} \left(1 - \frac{A(M)}{M} \right).$$

The multiplier tends to 2. Hence Everett's ordinary-density statement and Crandall's relative-density statement are equivalent, and (172) follows. $\square$

Explicit nonclaim 2.82. Everett proves first descent below the starting value, not convergence to 1. The exceptional set may be infinite of density zero. The proof gives no uniform descent-time or total-stopping-time bound and no accelerated-time bound. Its finite parity surjectivity is not ordinary-integer surjectivity onto infinite binary sequences. It proves no statement about U , generalized $qx + r$ maps, nontrivial cycles, or unbounded trajectories. The introductory verification “up to the millions” is historical context without a supplied computational certificate. *Source issue 2.83* (manifestation and presentation boundaries). The four raster pages are clear, but their OCR layer corrupts formulas and is not used as mathematical evidence. The publisher identity is fixed by DOI and PII; the inspected content witness is a matching VOR facsimile, while a publisher-hosted PDF hash remains unavailable. In the article itself, the abstract's phrase “every integer” must be read as every positive integer: the fixed point 0 has no iterate equal to 1. The body correctly uses $m \geq 1$ for that conjecture. The positive-representative swap and $Q_N \in \mathbb{N}_0$ are implicit, and the probability estimate is cited rather than proved. The preceding reconstruction makes each step explicit. No theorem-level mathematical error was found.

2.9 Steuding's orbit continued fraction: an exact endpoint recoding

Steuding's starred Exercise 5.22 uses the unshortened positive-integer recursion (1), asks for an equivalence-class reformulation of the Collatz conjecture, and asks for a Markoff constant [12, physical PDF p. 100; printed p. 87, Ex. 5.22]. It is an exercise prompt, not a theorem accompanied by a proof. The propositions and calculation below supply an edition solution without a novelty claim.

The target printed in the exercise is the purely periodic irrational

$$\beta = \overline{[4, 2, 1]},$$

with a visible overbar. Ordinary PDF text extraction loses that bar. The unbarred finite fraction is $[4, 2, 1] = 13/3$ and cannot be substituted: Steuding's rational and irrational equivalence classes are disjoint [12, printed p. 79].

Proposition 2.84 (whole orbit to continued-fraction morphism). *For $m \in \mathbb{N}_{>0}$, define*

$$x_j(m) = U^j(m), \quad \Theta(m) = [x_0(m), x_1(m), x_2(m), \dots]. \quad (174)$$

Every $\Theta(m)$ is an irrational real number. If

$$\mathcal{G}(y) = \frac{1}{y - \lfloor y \rfloor}$$

is the Gauss tail map on nonintegral reals, then

$$\lfloor \Theta(m) \rfloor = m, \quad \mathcal{G}(\Theta(m)) = \Theta(U(m)). \quad (175)$$

Consequently Θ is a bijection from $\mathbb{N}_{>0}$ onto its image, its inverse there is the floor map, and it conjugates U to $\mathcal{G}$ restricted to that image.

Proof. Every $x_j(m)$ is a positive integer. The finite convergents of (174) have denominator continuants tending to infinity, and consecutive convergents differ by the reciprocal of the product of their denominators. Their alternating nested intervals therefore have a unique common limit. That limit cannot be rational: if it were the reduced fraction p/q , choose n so large that $q_n > q$ and $q_{n+1} > q$. Since each convergent p_n/q_n is reduced, it is then distinct from p/q . The source bound

$$\left| \frac{p}{q} - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

would make the left-hand side strictly smaller than $1/(qq_n)$. But two distinct reduced rationals with denominators q and q_n are separated by at least $1/(qq_n)$. This is the constructive content of Steuding's infinite-fraction discussion [12, Sec. 3.4, printed pp. 42–43].

The tail $[x_1(m), x_2(m), \dots]$ is greater than one, so

$$\Theta(m) = m + \frac{1}{[x_1(m), x_2(m), \dots]}$$

has floor m . Applying $\mathcal{G}$ deletes the first digit, while the recursion gives $x_{j+1}(m) = x_j(U(m))$. This proves both identities in (175). The floor identity proves injectivity and supplies the displayed inverse; surjectivity is only onto the stated image. $\square$

Steuding defines $\alpha \sim \gamma$ by

$$\alpha = \frac{a\gamma + b}{c\gamma + d}, \quad a, b, c, d \in \mathbb{Z}, \quad ad - bc = \pm 1. \quad (176)$$

Serret's Theorem 5.7 says that two irrational simple continued fractions are equivalent exactly when finite prefixes can be removed from both to leave the same infinite tail [12, printed pp. 79–80].

Source issue 2.85 (the matrix step in the printed Serret proof). Printed p. 78 calls all integer matrices of determinant ± 1 the special linear group $\mathrm{SL}_2(\mathbb{Z})$; the group is $\mathrm{GL}_2(\mathbb{Z})$, since $\mathrm{SL}_2(\mathbb{Z})$ has determinant $+1$. Lemma 5.6 also suppresses the choice of simultaneous matrix sign and leaves its constructive sufficiency direction to an induction.

The converse proof on printed p. 80 then (i) invokes $c > d > 0$ for the original matrix before that reduction has been proved, (ii) duplicates the numerator entry R by printing $S = ap_{m-1} + bq_{m-1}$ instead of $S = cp_{m-1} + dq_{m-1}$, and (iii) writes α where the convergents and error terms belong to γ (called β in the source). The factors $c\alpha + d$ on the same page must likewise be $c\gamma + d$.

Here is the exact repair. Start with an arbitrary matrix M in (176) and replace it by $-M$, if necessary, so that $D = c\gamma + d > 0$. For convergents p_m/q_m to γ , put $\varepsilon_m = p_m - \gamma q_m$. The lower row of the composite matrix is

$$\begin{aligned} Q_m &= cp_m + dq_m = Dq_m + c\varepsilon_m, \\ S_m &= cp_{m-1} + dq_{m-1} = Dq_{m-1} + c\varepsilon_{m-1}. \end{aligned}$$

Since $\varepsilon_m \rightarrow 0$ and

$$q_m - q_{m-1} = (a_m - 1)q_{m-1} + q_{m-2} \longrightarrow \infty,$$

one has $Q_m > S_m > 0$ for every sufficiently large m . The Euclidean algorithm on the coprime pair (Q_m, S_m) constructs its positive continued-fraction denominator word. The two finite presentations obtained by splitting the last partial quotient choose the required determinant sign. After that choice, any two numerator rows with the same lower row and the same determinant differ by an integral multiple of (Q_m, S_m) ; changing the first partial quotient absorbs precisely that multiple. Thus the composite matrix is a finite continued-fraction prefix, and the two numbers have a common tail. This proves the needed Serret direction without the printed circular step. The complete locator audit is in `qa/STEUDING-2005-F011-exercise-522-audit.md`.

Theorem 2.86 (edition solution of Steuding's Exercise 5.22). *Let $b_{3r} = 4$, $b_{3r+1} = 2$, and $b_{3r+2} = 1$. The following statements are equivalent:*

- for every $m \in \mathbb{N}_{>0}$, the U -orbit of m eventually enters the cycle $4, 2, 1$;*
- for every $m \in \mathbb{N}_{>0}$, $\Theta(m) \sim [4, 2, 1]$.*

More precisely, for each fixed m ,

$$\Theta(m) \sim [4, 2, 1] \iff \exists r, s \in \mathbb{N}_0 \forall j \in \mathbb{N}_0 : U^{r+j}(m) = b_{s+j}. \quad (177)$$

Proof. If the orbit eventually enters the cycle, its digit sequence in (174) is eventually a cyclic shift of $(4, 2, 1, 4, 2, 1, \dots)$. Serret's theorem therefore gives the forward implication of (177). Conversely, Serret's theorem turns the equivalence of the two irrational continued fractions into the displayed equality of tails for some r, s . Hence $U^r(m) \in \{4, 2, 1\}$. The literal identities

$$U(4) = 2, \quad U(2) = 1, \quad U(1) = 4$$

then force every subsequent value. Universal quantification over m gives the two global statements. The phase variable s is retained; no preferred entry point into the cycle is assumed. $\square$

Calculation 2.87 (the requested Markoff constant). Let $d = \sqrt{229}$. The exact period matrix is

$$\begin{pmatrix} 4 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 13 & 9 \\ 3 & 2 \end{pmatrix}.$$

Therefore

$$3\beta^2 - 11\beta - 9 = 0, \quad \beta = \frac{11 + d}{6}. \quad (178)$$

Index its partial quotients by $a_{3r} = 4$, $a_{3r+1} = 2$, $a_{3r+2} = 1$. Formula (5.11) of the source is

$$\lambda(\beta) = \liminf_{n \rightarrow \infty} ([a_{n+1}, a_{n+2}, \dots] + [0, a_n, a_{n-1}, \dots, a_1])^{-1}. \quad (179)$$

The three phases, with no cyclic normalization, are

$n \bmod 3$	$[a_{n+1}, a_{n+2}, \dots]$	$\lim[0, a_n, \dots, a_1]$	sum	reciprocal
0	$(13 + d)/10$	$(d - 13)/10$	$d/5$	$5/d$
1	$(7 + d)/18$	$(d - 7)/18$	$d/9$	$9/d$
2	$(11 + d)/6$	$(d - 11)/6$	$d/3$	$3/d$

The reversed finite fractions converge on each residue subsequence to the displayed reversed periodic tail; equivalently, Steuding's Galois formula identifies each with the negative conjugate of the corresponding cyclic complete quotient [12, Thm. 5.3 and Eq. (5.11)]. Taking the liminf in (179) gives

$$\lambda([4, 2, 1]) = \min \left\{ \frac{5}{\sqrt{229}}, \frac{9}{\sqrt{229}}, \frac{3}{\sqrt{229}} \right\} = \frac{3}{\sqrt{229}}. \quad (180)$$

The exact quadratic-field identities are reproduced by `certificates/steuding_522_checks.py`.

Proposition 2.88 (exact Crandall clock crosswalk). *For an odd start y_0 , put $y_i = C_{\text{Cr}}^i(y_0)$, $e_i = \nu_2(3y_i + 1)$, and $A_j = \sum_{i < j} e_i$. Then the Steuding digit sequence and the accelerated orbit satisfy*

$$y_j = U^{A_j+j}(y_0), \quad (181)$$

and, for $1 \leq r \leq e_j + 1$,

$$U^{A_j+j+r}(y_0) = \frac{3y_j + 1}{2^{r-1}}. \quad (182)$$

For an arbitrary $m = 2^s y_0$ with y_0 odd, the first s Steuding digits are the exact halving prefix before this odd-return expansion begins.

Proof. Equation (181) is (89). Starting from the odd state y_j , the first U step forms $3y_j + 1$ and the next e_j steps divide successively by 2, proving (182); at $r = e_j + 1$ the value is y_{j+1} . The even-start statement is the literal iteration $U^r(2^s y_0) = 2^{s-r} y_0$ for $0 \leq r \leq s$. $\square$

Explicit nonclaim 2.89. Theorem 2.86 is a lossless endpoint recoding, not a proof of either side: the full orbit is stored as the full digit sequence and the initial integer is recovered by the floor map. Steuding's Theorem 5.8 gives only

$$\Theta(m) \sim \beta \implies \lambda(\Theta(m)) = \frac{3}{\sqrt{229}}.$$

Equality of Markoff constants is not a converse and cannot certify Collatz termination. The orbit-digit continued fraction $\Theta(m)$ is also not Crandall's Section 7 continued fraction for the fixed number $\log_2 3$; no map between those two constructions is evidenced here. Finally, (181) is a variable-time expansion, not a one-step identification of U and C_{Cr} .

2.10 The 2-adic parity conjugacy and notation orientation

Let $S : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ be the digit shift

$$S(z) = \frac{z - (z \bmod 2)}{2}.$$

Proposition 2.90 (Bernstein–Lagarias, with maps oriented). *Define*

$$Q_3(x) = \sum_{i=0}^{\infty} (T^i(x) \bmod 2) 2^i \quad (x \in \mathbb{Z}_2). \quad (183)$$

If $z \neq 0$ has one positions $0 \leq d_1 < d_2 < \dots$ (a finite or infinite list), put

$$\Phi_3(z) = - \sum_r 3^{-r} 2^{d_r}, \quad \Phi_3(0) = 0. \quad (184)$$

Then Q_3 and Φ_3 are mutually inverse 2-adic isometries and

$$Q_3 \circ T = S \circ Q_3, \quad T = \Phi_3 \circ S \circ \Phi_3^{-1}. \quad (185)$$

Proof. If $x \equiv y \pmod{2^N}$, induction through their common parity branches shows that the first N digits of $Q_3(x)$ and $Q_3(y)$ agree. Conversely, if those digits agree and contain r ones, the common affine word gives

$$T^N(x) - T^N(y) = \frac{3^r}{2^N}(x - y).$$

The left side is in $\mathbb{Z}_2$ and 3^r is a 2-adic unit, so $x \equiv y \pmod{2^N}$. Hence Q_3 preserves every 2-adic congruence distance and is injective.

The series (184) converges because its term valuations tend to infinity. Splitting off the first digit gives

$$\Phi_3(2w) = 2\Phi_3(w), \quad \Phi_3(1 + 2w) = \frac{2\Phi_3(w) - 1}{3}.$$

The first value is even and the second odd, and applying the corresponding branch of T gives $T(\Phi_3(z)) = \Phi_3(Sz)$. Thus the chronological parities of $\Phi_3(z)$ are the digits of z , so $Q_3\Phi_3 = \text{id}_{\mathbb{Z}_2}$. The injective Q_3 is therefore bijective with inverse Φ_3 . Removing the first parity digit proves the first identity in (185); conjugating by the inverse proves the second. This reproduces Bernstein–Lagarias Eqs. (1.3), (1.5), and (1.6) with the inverse orientation explicit [14, author-copy pp. 1–2]. $\square$

Corollary 2.91 (periodic parity word to fixed-return fraction). *With the word data of Proposition 2.32, put*

$$q_\varepsilon = \sum_{i=0}^{L-1} \varepsilon_i 2^i, \quad z_\varepsilon = \frac{q_\varepsilon}{1 - 2^L} \in \mathbb{Z}_2.$$

Then

$$\Phi_3(z_\varepsilon) = R(\varepsilon) = \frac{\sum_{r=0}^{m-1} 3^{m-r-1} 2^{v_r}}{2^L - 3^m}. \quad (186)$$

Thus the Böhm–Sontacchi return candidate is the inverse parity image of the purely periodic chronological word. Its exact T -period is the least shift period of that infinite word. In particular, a length- L return has exact period L precisely when the repeated block has least shift period L .

Proof. The one positions of z_ε are $v_r + sL$ for $0 \leq r < m$ and $s \geq 0$. Formula (184) gives

$$\begin{aligned}\Phi_3(z_\varepsilon) &= - \sum_{s \geq 0} \sum_{r=0}^{m-1} 3^{-(sm+r+1)} 2^{v_r+sL} \\ &= - \left(\sum_{r=0}^{m-1} 3^{-(r+1)} 2^{v_r} \right) \sum_{s \geq 0} \left(\frac{2^L}{3^m} \right)^s \\ &= \sum_{r=0}^{m-1} 3^{m-r-1} 2^{v_r} \\ &= \frac{2^L - 3^m}{2^L - 3^m},\end{aligned}$$

where the geometric series converges in $\mathbb{Z}_2$ because $|2^L/3^m|_2 = 2^{-L} < 1$. The conjugacy $Q_3 T = S Q_3$ and bijectivity of Q_3 identify the least positive return time of the point with the least positive shift period of its parity sequence. $\square$

Laarhoven–de Weger use the symbol Φ for the chronological series (183), first modulo 2^k and then on $\mathbb{Z}_2$. Their identities are $T_k = \Phi_k^{-1} \sigma_{2,k} \Phi_k$ and $T = \Phi^{-1} S \Phi$ [19, Secs. 2–3]. Consequently their Φ is Q_3 in the notation of Proposition 2.90, not Bernstein–Lagarias’s inverse Φ_3 . This is an exact notation crosswalk, not a mathematical disagreement.

2.11 Finite De Bruijn relations and the projective parity map

Laarhoven–de Weger’s finite objects must first be typed as relations. For $k \geq 1$ put

$$V_k = \mathbb{Z}/2^k \mathbb{Z}, \quad \mathcal{P}(V_k) = \{A : A \subseteq V_k\}.$$

Their modular Collatz graph has an edge $[a]_k \rightarrow [b]_k$ precisely when integer lifts a_1, b_1 exist with $T(a_1) = b_1$. Its forward relation is therefore

$$\mathcal{T}_k([a]_k) = \{[T(a)]_k, [T(a) + 2^{k-1}]_k\} \quad : V_k \longrightarrow \mathcal{P}(V_k). \quad (187)$$

Here a may be any integer representative. All representatives have the same parity, and changing a by 2^k changes its branch value by an odd multiple of 2^{k-1} . Thus the two displayed residues are exactly all possible outputs and are distinct.

Write a binary word least-significant digit first and identify $b_0 \cdots b_{k-1}$ with $\sum_{i < k} b_i 2^i \in V_k$. The binary De Bruijn forward relation is

$$\sigma_{2,k}(b_0 \cdots b_{k-1}) = \{b_1 \cdots b_{k-1} c : c \in \{0, 1\}\}. \quad (188)$$

Numerically, this is the right shift together with that value plus 2^{k-1} . Neither (187) nor (188) is a deterministic self-map.

The source calls its finite encoder Φ_k . To prevent its dimension subscript from colliding with Bernstein–Lagarias’s multiplier subscript in $\Phi_3 = Q_3^{-1}$, this edition writes the same map as

$$\overline{Q}_{3,k} : V_k \longrightarrow V_k, \quad \overline{Q}_{3,k}([a]_k) = \sum_{i=0}^{k-1} (T^i(a) \bmod 2) 2^i. \quad (189)$$

Thus $\overline{Q}_{3,k}$ is source Φ_k without a normalization or a change of orientation. The symbol 3 in Q_3 and Φ_3 denotes the $3x + 1$ multiplier; it is not the graph dimension.

Everett’s earlier Theorem 1 is exactly the bijectivity of this same finite chronological map on the representatives $0 \leq a < 2^k$, together with the stronger integer-fibre coordinates in Theorem 2.77. Laarhoven–de Weger’s later theorem adds the two-successor graph relation; that relation is not retroactively attributed to Everett.

For a point map $f : V_k \rightarrow V_k$ and $A \subseteq V_k$, write $f[A] = \{f(a) : a \in A\}$. This elementwise extension is required in the next statement.

Theorem 2.92 (Laarhoven–de Weger finite graph theorem, reconstructed). *For every $k \geq 1$, $\overline{Q}_{3,k}$ is a bijection and*

$$\overline{Q}_{3,k}[\mathcal{T}_k(a)] = \sigma_{2,k}(\overline{Q}_{3,k}(a)) \quad (a \in V_k). \quad (190)$$

Consequently it is a directed-graph isomorphism from the binary modular Collatz graph to $B(2, k)$. Its inverse is the reduction modulo 2^k of Bernstein–Lagarias’s $\Phi_3 = Q_3^{-1}$.

Proof. Proposition 2.90 gives the exact congruence equivalence

$$x \equiv y \pmod{2^k} \iff Q_3(x) \equiv Q_3(y) \pmod{2^k}.$$

It proves that (189) is well-defined and injective; finiteness makes it bijective. It also identifies it pointwise as the reduction of Q_3 , so reduction of Φ_3 supplies the inverse.

Let a_0 and $a_0 + 2^k$ represent the two lift classes modulo 2^{k+1} . Their first k parity digits agree, while their next parity digits differ, again by the exact isometry. Applying T deletes the common first digit. The two resulting length- k itinerary words therefore have common first $k - 1$ digits and both possible terminal digits. They are precisely $\sigma_{2,k}(\overline{Q}_{3,k}([a_0]_k))$. On the state side their residues are exactly the two values in (187). This proves (190) as equality of two-element subsets. $\square$

The active source theorem has no proof environment. Its preceding prose depends on an earlier Lagarias congruence, and source line 131 names the next digit as x_{k+1} although the definitions force x_k . A proof at source lines 160–166 is commented out; its first sentence reverses the image edge, although its following equivalence has the orientation used above. These are source defects, not alternative graph conventions [19, Sec. 2].

Proposition 2.93 (exact path multiplicities). *Let A_k be the adjacency matrix of either graph in Theorem 2.92, and let J be the 2^k -by- 2^k all-ones matrix. Then*

$$A_k^k = J, \quad A_k^\ell = 2^{\ell-k} J \quad (\ell \geq k). \quad (191)$$

Proof. After k De Bruijn steps all k initial digits have been discarded. For a specified terminal word, exactly one ordered choice of the k appended digits reaches it. After $\ell \geq k$ steps the first $\ell - k$ appended digits are discarded and arbitrary, while the last k are forced by the terminal word. This gives respectively 1 and $2^{\ell-k}$ directed paths. The graph isomorphism transports the count to the modular Collatz graph. $\square$

Explicit nonclaim 2.94. Equation (191) yields a uniform endpoint modulo 2^k only after a probability model is specified—for example, independent uniform choices of the two outgoing finite-graph edges, equivalently uniform unresolved higher input bits. It does not make the deterministic integer $T^k(n)$ random, and it does not prove that the known low bits are statistically independent of an integer iterate. The source’s phrase that they tell us “absolutely nothing” is not used without this qualification.

2.12 Edge projections, the inverse limit, and literal integer edges

Let $\rho_{k+1,k} : V_{k+1} \rightarrow V_k$ be reduction. Directly from itinerary prefixes,

$$\rho_{k+1,k} \circ \overline{Q}_{3,k+1} = \overline{Q}_{3,k} \circ \rho_{k+1,k}. \quad (192)$$

Let $E_k \subseteq V_k \times V_k$ be the edge relation of $\mathcal{T}_k$. Division by two loses exactly one congruence digit, so

$$([a]_k, [b]_k) \in E_k \iff b \equiv T(a) \pmod{2^{k-1}}. \quad (193)$$

The right side is independent of the representative a because its parity and branch value modulo 2^{k-1} are fixed by $[a]_k$.

Proposition 2.95 (exact graph inverse limit and natural-edge recovery). *Reduction of both endpoints maps E_{k+1} onto E_k . The inverse limit of the vertex systems is $\mathbb{Z}_2$, and its compatible edge relation is exactly the graph of the deterministic map $T : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$. Under (192), the inverse-limit graph isomorphism is Q_3 , and the target relation is the graph of the deterministic digit shift S .*

Moreover, for fixed literal $a, b \in \mathbb{N}_0$, once both lie in $\{0, \dots, 2^k - 1\}$,

$$\begin{aligned} b = T(a) &\iff ([a]_k, [b]_k) \in E_k \text{ for all sufficiently large } k, \\ &\iff ([a]_k, [b]_k) \in E_k \text{ for infinitely many } k. \end{aligned} \quad (194)$$

Proof. Equation (193) shows that every higher edge reduces to a lower edge. Conversely, either lift of the initial vertex fixes one more input bit and hence selects one of the two lower successors; choosing that lift and either next higher bit supplies a higher edge. Thus edge projection is onto. Notice that for one fixed higher vertex its two successors reduce to a singleton, not to the entire lower successor set.

Suppose compatible $x, y \in \mathbb{Z}_2$ form an edge at every level. Then (193) gives $y \equiv T(x) \pmod{2^{k-1}}$ for every k , so $y = T(x)$. The converse follows by reduction of an actual edge. The same argument for suffixes gives the graph of S , while (192) and Proposition 2.90 identify the limiting map as Q_3 .

If $b = T(a)$, every sufficiently large literal pair is an edge. Conversely, an occurrence in E_k gives $b \equiv T(a) \pmod{2^{k-1}}$. Infinitely many occurrences include one with $2^{k-1} > |b - T(a)|$, when the congruence forces equality in $\mathbb{Z}$. This proves (194). $\square$

This is the typed repair of the source phrase “edges that appear in infinitely many” changing finite graphs. It distinguishes the common ambient pair $(a, b) \in \mathbb{N}_0^2$ from compatible residue vertices and does not call the set-valued $\mathcal{T}_k$ a deterministic quotient map.

2.13 Periodic components, predecessor fibres, and source corrections

Proposition 2.96 (exact binary cycle count). *Let M_k be the number of components of $C(\mathbb{Z}_2)$ containing a cycle of least length k . Then*

$$2^k = \sum_{d|k} dM_d, \quad M_k = \frac{1}{k} \sum_{d|k} \mu(d) 2^{k/d} = \frac{2^k}{k} + O(2^{k/2}). \quad (195)$$

In particular,

$$(M_k)_{k \geq 1} = (2, 1, 2, 3, 6, 9, \dots).$$

Proof. The Q_3 conjugacy identifies T -cycles with shift orbits of purely periodic binary sequences. The infinite periodic repetition of every length- k block has a unique least shift period $d \mid k$. A shift orbit of least period d contains exactly d distinct phases, so counting all length- k blocks gives the first identity. Möbius inversion gives the second. In the error sum every proper divisor contributes at most $2^{k/2}$, and the number of divisors is at most k , which gives the displayed asymptotic. Direct substitution gives the listed values. $\square$

The source prints the correct divisor identity and Möbius formula, but its asymptotic changes the bound variable from k to an unbound n . Its printed sequence $(1, 2, 1, 2, 3, 6, \dots)$ contradicts both the formula and its own two fixed-point components; Proposition 2.96 records the forced correction. Likewise, a cyclic De Bruijn sequence gives a Hamiltonian cycle. The source calls it a path while its parenthesized wrap-around digits supply the closing edge [19, Secs. 2 and 4].

Proposition 2.97 (exact predecessor fibres). *For every $y \in \mathbb{Z}_2$,*

$$T^{-1}(y) = \left\{ 2y, \frac{2y-1}{3} \right\}, \quad S^{-1}(y) = \{2y, 2y+1\}. \quad (196)$$

Both fibres have two distinct elements. In particular, the forward Collatz map is not a function automorphism on any weak component.

Proof. The point $2y$ is even and maps to y . Since 3 is a unit in $\mathbb{Z}_2$, $(2y-1)/3$ is a 2-adic integer; it is odd and its odd branch also maps to y . The two branch equations give no other preimages. Equality of the two would imply $4y = -1$, impossible in $\mathbb{Z}_2$. The shift formula is immediate from its deleted first digit. Both predecessors of a point lie in its weak component, proving the final assertion. $\square$

Source line 241 says the forward mapping on an aperiodic component is an automorphism. Equation (196) contradicts that sentence under the ordinary meaning of function automorphism. The separate source assertion that all aperiodic components are pairwise isomorphic is not used downstream.

2.14 Generalized maps: exact unit gate and scaling quotient

For odd a, b , Laarhoven–de Weger define

$$T^{(a,b)}(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (an+b)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

At retained modulus 2, this is precisely the Matthews presentation

$$(m_0, r_0) = (1, 0), \quad (m_1, r_1) = (a, -b).$$

Oddness of a, b supplies the branch congruence and makes both multipliers units. Hence the finite and infinite parity encoders in source Section 5.1 are the $d = 2$ specialization of Theorem 2.110; their orientation is $\text{It}_{T^{(a,b)}}$, not its inverse. Only at $(a, b) = (3, 1)$ does this encoder specialize to Q_3 .

For a retained base $d \geq 2$, consider instead

$$F(n) = \frac{a_i n + b_i}{d} \quad (n \equiv i \pmod{d}). \quad (197)$$

The exact hypotheses for the full digit-shift conclusion are

$$a_i i + b_i \equiv 0 \pmod{d}, \quad \gcd(a_i, d) = 1 \quad (0 \leq i < d). \quad (198)$$

The first is branch integrality. For $n = i + dt$, the next digit is

$$F(i + dt) \equiv \frac{a_i i + b_i}{d} + a_i t \pmod{d}. \quad (199)$$

The second condition makes this an affine permutation of all d next digits with singleton fibres. With $m_i = a_i$ and $r_i = -b_i$, conditions (198) are exactly the retained Matthews congruence and unit gate. Theorem 2.110 therefore proves the full finite and d -adic De Bruijn conclusions. If a multiplier is not a unit, Proposition 2.112 gives the missing digits and non-singleton fibres explicitly.

The source's Section 5.2 says only "appropriately chosen" coefficients and does not print (198). If that phrase means merely integer-valued branches, choose $a_i = d, b_i = 0$ for every i . Then $F = \text{id}_{\mathbb{Z}}$, whose finite graph is not a De Bruijn graph. The source also leaves $x_i^{(f)}$ undefined and does not say that p is prime despite writing $\mathbb{Z}_p$. The edition admits the claim only at the exact scope above; for prime $d = p$ it is the usual p -adic statement. The source's ternary example has multipliers 2, 4, 4, all units modulo 3, and lies inside the repaired scope.

Proposition 2.98 (rational cycles and primitive $3n + b$ data). *Let $(x_0, \dots, x_{L-1})$ be a rational T -cycle in $\mathbb{Z}_2$, with cyclic indices, and let $b > 0$ be a common odd denominator. Put $n_j = bx_j$. Then (n_j) is an integer cycle of $T^{(3,b)}$, and*

$$T(x_j) = \frac{1}{b} T^{(3,b)}(n_j). \quad (200)$$

Conversely every such integer cycle scales to a rational T -cycle. The resulting bijection is between rational cycles and the equivalence classes generated by the scalings

$$(b, (n_j)) \sim (cb, (cn_j)) \quad (c \in \mathbb{N}_{>0} \text{ odd}), \quad (201)$$

or, equivalently, their unique representatives with $b > 0$ and $\gcd(b, n_0, \dots, n_{L-1}) = 1$. For a rational T -cycle this primitive b is automatically coprime to 3.

Proof. Multiplication by odd b preserves parity in $\mathbb{Z}_2$. On an even phase, $bT(x_j) = n_j/2$. On an odd phase, $bT(x_j) = (3n_j + b)/2$. This proves (200) and both cycle directions. Scaling numerator and denominator by a common odd positive integer leaves every x_j fixed. Dividing by the common gcd terminates at a primitive representative, and any two primitive positive-denominator representatives of the same rational tuple are equal. If the cycle has return length L and contains m odd steps, the fixed-return equation shows that the reduced denominator of every phase divides $2^L - 3^m$. Hence their primitive common denominator also divides $2^L - 3^m$. If $m \geq 1$, this integer is coprime to 3. If $m = 0$, every phase uses $x \mapsto x/2$, so periodicity forces the zero cycle and the primitive common denominator is 1. Thus it is coprime to 3 in both cases. $\square$

The source's first, unqualified "one-to-one" sentence requires (201): for every odd b , the integer cycle $\{b, 2b\}$ of $T^{(3,b)}$ scales to the same rational cycle $\{1, 2\}$. Its subsequent phrase "having denominator b " can instead be read as the exact reduced common denominator; on that reading it already selects the primitive representative and is correct. Lagarias's primary treatment explicitly retains the raw gcd strata and then defines this primitive layer [17, pp. 39–40]. The proposition makes both scopes explicit without weakening the scaling identity or suppressing the raw scaling fibres.

Source issue 2.99 (remaining prose and hypothesis boundaries). The source introduction describes absence of divergent paths by $\lim_k T^k(n) < \infty$, although the $1 \leftrightarrow 2$ orbit has no limit; the boundedness condition is $\sup_k T^k(n) < \infty$. Source line 79 says “for all n ” where its fixed-input argument requires “for all k ”. The rational-periodicity statement in Sections 4 is explicitly a conjecture. The average-multiplier discussion and the predicted $5n + 1$ components are heuristics; no theorem is attributed to either discussion. Finally, the source’s positive-integer reformulation through $\frac{1}{3}\mathbb{Z} \setminus \mathbb{Z}$ depends on earlier cited literature and is not enlarged here before that exact dependency is checked.

Explicit nonclaim 2.100. The finite graph theorem, exact path counts, inverse limit, cycle enumeration, and generalized unit-gated conjugacy do not characterize $Q_3(\mathbb{N}_{>0}) \subset \mathbb{Z}_2$. They therefore do not prove that a positive integer reaches 1. The arithmetic difficulty lies in the embedded subset and its labels, not in the existence of the ambient De Bruijn isomorphism.

2.15 Primary parity and rational-cycle dependencies

The De Bruijn source invokes three earlier results whose exact scopes now have controlling primary witnesses. Everett already proves the finite parity bijection and its nonnegative affine fibres [11]; Lagarias adds the finite permutation structure and proves the infinite parity map [16]; Bernstein proves the inverse-series inclusion [15], and Lagarias’s rational-cycle coordinates [17]. This section keeps all three map orientations and both the raw and reduced denominator layers visible.

Put

$$\mathcal{R} = \mathbb{Q} \cap \mathbb{Z}_2 = \mathbb{Z}_{(2)}.$$

Lagarias 1985 denotes this ring by $\mathbf{Q}_2$; it is the set of rationals with odd reduced denominator, not the full 2-adic field. Its elements are exactly the finite or eventually periodic binary expansions in $\mathbb{Z}_2$.

Proposition 2.101 (Everett’s finite bijection; Lagarias’s permutation structure). *For $k \geq 1$, define*

$$\overline{Q}_k([n]_k) = \left[\sum_{i=0}^{k-1} (T^i(n) \bmod 2) 2^i \right]_k.$$

Then $\overline{Q}_k$ is a permutation of $\mathbb{Z}/2^k\mathbb{Z}$. Its order is a power of two, in fact it divides 2^k . Every permutation cycle of length r at level k lifts at level $k + 1$ either to one cycle of length $2r$ or to two cycles of length r .

Proof. Everett’s Theorem 1 proves the bijection by the complete four-case affine induction in Theorem 2.77. Lagarias’s Theorem B states the permutation and power-of-two order [16, p. 7, Thm. B]; its proof is explicitly a sketch and omits the induction details. Bernstein independently states the lift alternative [15, p. 407]. For the later structure at the scope used here, Proposition 2.90 gives

$$a \equiv b \pmod{2^k} \iff Q_3(a) \equiv Q_3(b) \pmod{2^k}.$$

Thus $\overline{Q}_k$ is the reduction of the same isometric bijection Q_3 and is a permutation. Each residue has two lifts. A lifted permutation cycle either returns each chosen lift after r steps, yielding two r -cycles, or exchanges the two lifts after r steps, yielding one $2r$ -cycle. Induction from level one proves that every cycle length and the permutation order divide 2^k . $\square$

This is the finite labeling called Φ_k by Laarhoven–de Weger. Its cycles are cycles of a permutation of labels. They are not cycles of the two-successor modular Collatz relation $\mathcal{T}_k$, and Proposition 2.101 alone is not the graph isomorphism of Theorem 2.92.

Source issue 2.102 (Lagarias 1985 finite table and examples). The sentence immediately before Table 2 on printed p. 7 says that one-cycles are omitted; the table itself is on printed p. 8. Its $k = 6$ row also omits the nontrivial transposition $(14, 46)$. Direct evaluation gives

$$\overline{Q}_6(14) = 46, \quad \overline{Q}_6(46) = 14.$$

The displayed permutation order 4 remains correct. The calculation certificate `certificates/1 agarias_dependency_checks.py` enumerates all residues for $k \leq 6$ and verifies that this is the only nontrivial cycle missing from the printed $k = 6$ row.

On printed p. 18 the series for $Q_\infty(1)$ has summation index i but exponent $2n$; the intended exponent is $2i$. More substantially, the printed value $Q_\infty(3) = -20/3$ is inconsistent with the paper’s own map: the trajectory $3, 5, 8, 4, 2, 1, 2, \dots$ has parity word $11000(10)^\infty$, and hence

$$Q_\infty(3) = 1 + 2 + \frac{2^5}{1 - 2^2} = -\frac{23}{3}.$$

The printed value is not used.

Lagarias’s Eq. (2.33) defines the infinite map in the forward orientation

$$Q_\infty(x) = \sum_{i \geq 0} (T^i(x) \bmod 2) 2^i.$$

Proposition 2.103 (Lagarias infinite parity theorem, proof completed). *The map Q_∞ is a continuous, bijective, measure-preserving self-map of $\mathbb{Z}_2$. It is Q_3 and Laarhoven–de Weger Φ ; it is not the inverse map Φ_3 .*

Proof. Theorem L states the result [16, pp. 17–18]. Its printed argument proves prefix preservation, injectivity, surjectivity through finite inverse residues, and inverse continuity. Compatibility of those inverse residues follows because every $\overline{Q}_k$ is a reduction of the same prefix map. Every residue cylinder modulo 2^k is therefore permuted onto one cylinder of the same Haar measure, which supplies the measure-preservation step not printed at the end of the source proof. $\square$

Proposition 2.104 (one proved half of Periodicity). *Let $\Phi_3 = Q_3^{-1}$. Then*

$$\Phi_3(\mathcal{R}) \subseteq \mathcal{R}, \tag{202}$$

or equivalently

$$\mathcal{R} \subseteq Q_3(\mathcal{R}). \tag{203}$$

The Periodicity Conjecture is the equality

$$Q_3(\mathcal{R}) = \mathcal{R}. \tag{204}$$

Proof. Bernstein writes a rational parity address as a finite or eventually periodic sequence of one-positions and substitutes it into his explicit series for Φ_3 . The resulting geometric relation proves that the image is rational [15, p. 407, Cor. 1]. Applying the bijection Q_3 gives the equivalent forward inclusion. Lagarias prints (204) as an unproved conjecture on printed p. 18 [16]; no reverse inclusion is added. $\square$

The positive-integer “third form” in Lagarias 1985 is likewise a conjecture:

$$Q_3(\mathbb{N}_{>0}) \subseteq \frac{1}{3}\mathbb{Z} \setminus \mathbb{Z}. \quad (205)$$

Bernstein instead proves that the inverse-oriented assertion $\mathbb{Z}^+ \subseteq \Phi_3((1/3)\mathbb{Z})$ is equivalent to the ordinary conjecture [15, pp. 406–407, Thms. 2–3]. An equivalence to an endpoint conjecture is not a proof of either formulation.

Lagarias’s last exploratory value on printed p. 18 is left as $Q_3(-82/7) = ?/15$. Exact iteration gives a prefix of length 13 followed by the parity block 1100, hence the independent finite calculation

$$Q_3(-82/7) = \frac{18098}{5} = \frac{54294}{15}. \quad (206)$$

The same certificate reproduces this value. It is not attributed to the source and does not prove Periodicity.

Theorem 2.105 (Lagarias rational fixed-return coordinates). *Let $n \geq 1$ and $v = (v_0, \dots, v_{n-1}) \in \{0, 1\}^n$, and put*

$$m(v) = \sum_{i=0}^{n-1} v_i, \quad D(v) = 2^n - 3^{m(v)} \neq 0,$$

$$N(v) = \sum_{j=0}^{n-1} v_j 2^j 3^{v_{j+1} + \dots + v_{n-1}}.$$

There is exactly one $x(v) \in \mathcal{R}$ with $T^n(x(v)) = x(v)$ and initial parity word v , namely

$$x(v) = \frac{N(v)}{D(v)}. \quad (207)$$

Its least T -period is n exactly when v is not a proper repeated block.

Proof. The branch composition determined by v is

$$U_{v_{n-1}} \circ \dots \circ U_{v_0}(x) = \frac{3^{m(v)}x + N(v)}{2^n},$$

and $D(v) \neq 0$ because 2^n is even while $3^{m(v)}$ is odd. Thus its unique formal fixed point is (207). Lagarias proves that this formal point has the prescribed actual branches: for each $y \in \mathcal{R}$, exactly one of the two forward branch expressions $U_0(y) = y/2$ and $U_1(y) = (3y+1)/2$ lies in $\mathcal{R}$, with the choice determined by the parity of y ; if $y \notin \mathcal{R}$, both expressions remain outside $\mathcal{R}$ [17, pp. 36–37, Thm. 2.1]. The denominator $D(v)$ is odd, so the formal fixed point lies in $\mathcal{R}$. If its prescribed composition ever selected the other forward expression, it would leave $\mathcal{R}$ and could never return, contradicting that the composition ends again at $x(v) \in \mathcal{R}$. Thus every prescribed branch is actual. For every $d \geq 1$, the conjugacy in Proposition 2.90 gives $Q_3(T^d(x(v))) = S^d(Q_3(x(v)))$. Hence a T -return after d steps implies a shift return after d steps. Conversely, a shift return and the injectivity of Q_3 from Proposition 2.103 imply $T^d(x(v)) = x(v)$. The two sets of positive return times are equal, so their least elements agree; this least period is n exactly when v is not a proper repeated block. $\square$

Write the positive reduced common denominator as

$$\delta(v) = \frac{|D(v)|}{\gcd(|N(v)|, |D(v)|)}. \quad (208)$$

It is always odd and prime to 3. The proof cannot omit the all-even word. If $m(v) \geq 1$, then $3 \nmid D(v)$. If $m(v) = 0$, the return equation is $x = x/2^n$, so $x = 0$ and $\delta(v) = 1$. Thus

$$\delta(v) \equiv \pm 1 \pmod{6}. \quad (209)$$

Proposition 2.106 (raw scaling fibres and the primitive layer). *For odd b , let*

$$T^{(3,b)}(n) = \begin{cases} (3n+b)/2, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases}$$

Multiplication by a common positive odd denominator gives

$$bT(x_j) = T^{(3,b)}(bx_j).$$

Consequently rational T -cycles correspond to common-odd-scaling classes of integer $T^{(3,b)}$ -cycles. Without identifying the raw representatives, the unique member with b equal to the positive reduced common denominator is characterized by

$$\gcd(b, n_0, \dots, n_{L-1}) = 1.$$

It is exactly a primitive integer $T^{(3,b)}$ -cycle, and division by b is the two-sided inverse. Its parameter satisfies $b \equiv \pm 1 \pmod{6}$.

Proof. The branch identity is the calculation in Proposition 2.98. Lagarias explicitly defines the primitive condition and the inverse correspondence on printed pp. 39–40 [17]. For the raw strata, if $d \mid b$ and $n = dm$, $b = dc$, then

$$T^{(3,b)}(dm) = dT^{(3,c)}(m).$$

This proves every scaling fibre without suppressing it. Choosing the reduced common denominator gives the displayed gcd condition and a literal inverse. Equation (209) supplies the parameter restriction. $\square$

Source issue 2.107 (Lagarias 1990 local coordinate defects). Six printed statements require exact local repair.

First, the itinerary sentence at the top of printed p. 34 defines $v = (v_0, \dots, v_{n-1})$ but quantifies $T^i(x(v)) \equiv v_i \pmod{2}$ for $1 \leq i \leq n$, which skips v_0 and invokes the undefined v_n . The correct range is $0 \leq i \leq n-1$. Second, printed p. 39 says that T never changes the denominator of an arbitrary element of $\mathcal{R}$, but $T(1/3) = 1$. The correct preserved scope is a reduced odd denominator prime to 3; every periodic rational has that property by the case split above. Third, the printed formula for $\delta(v)$ needs $|D(v)|$ for a positive denominator unless a signed-gcd convention is supplied. Fourth, printed p. 40 writes $T^{(3,b)}(n) = T^{(3,-b)}(-n)$, whereas direct branch evaluation gives

$$T^{(3,b)}(n) = -T^{(3,-b)}(-n).$$

Fifth, printed p. 34 ends one correspondence sentence with $b \equiv 1 \pmod{6}$; the preceding sentence, printed p. 40, and the examples require $b \equiv \pm 1 \pmod{6}$. Sixth, the invariant gcd stratum on p. 40 uses the standing $\gcd(b, 3) = 1$ hypothesis and is not a statement for arbitrary odd b .

Explicit nonclaim 2.108. The Periodicity Conjecture does not follow from the one-sided Bernstein inclusion. Lagarias's primitive-cycle existence and finiteness statements are conjectures. A periodic parity address gives a rational periodic state, but this does not classify the positive integers inside $\mathbb{Z}_2$, exclude nontrivial integer cycles, or prove convergence to 1.

2.16 Relatively-prime generalized maps and the full d -adic shift

Matthews's author form first permits a retained presentation integer $d \geq 2$, nonzero integers $m_0, \dots, m_{d-1}$, and integers $r_i \equiv im_i \pmod{d}$, and writes

$$\mathcal{T}(x) = \frac{m_i x - r_i}{d} \quad (x \equiv i \pmod{d}). \quad (210)$$

The source symbol is T ; the calligraphic letter here prevents collision with the already fixed shortened $3x+1$ map. The source separately reserves $d(\mathcal{T})$ for the least modulus on whose residue classes the map is affine. On an arbitrary retained presentation impose the unit condition

$$\gcd(m_0 m_1 \cdots m_{d-1}, d) = 1, \quad (211)$$

equivalently when every m_i is a unit modulo d . This condition forces the retained d to equal $d(\mathcal{T})$: if affine data at a smaller modulus q described the same map, equality of rational slopes on each nonempty intersection of a q -class and a d -class would give $qm_i = dm'_j$; coprimality gives $d \mid q$, a contradiction. At the least modulus this is exactly Matthews's *relatively-prime type* [21, author-form pp. 2–3].

The 1984 Matthews–Watts paper uses a nonidentical presentation: positive unit multipliers $m_1, \dots, m_d$, a complete residue set $R_d = \{x_1, \dots, x_d\}$, and branch data indexed by those representatives. The exact crosswalk sends the later digit i to the unique $x_j \equiv i \pmod{d}$ and relabels the corresponding m_j, r_j . This reindexing does not erase the earlier positivity hypothesis.

Proposition 2.109 (source extension and cylinder theorems). *Every map (210) at the retained d has a unique continuous extension $\hat{\mathcal{T}} : \mathbb{Z}_d \rightarrow \mathbb{Z}_d$ with the same branch formula. Relative primeness is not needed for this assertion.*

Assume (211). For a word $\mathbf{i} = (i_0, \dots, i_{\alpha-1}) \in \{0, \dots, d-1\}^\alpha$, put

$$b_\alpha(\mathbf{i}) = \sum_{j=0}^{\alpha-1} \frac{r_{i_j} d^j}{m_{i_0} m_{i_1} \cdots m_{i_j}} \in \mathbb{Z}_d.$$

Then its itinerary cylinder is exactly

$$\left\{ x \in \mathbb{Z}_d : \hat{\mathcal{T}}^j(x) \equiv i_j \pmod{d} \text{ for } 0 \leq j < \alpha \right\} = b_\alpha(\mathbf{i}) + d^\alpha \mathbb{Z}_d. \quad (212)$$

Thus the d^α length- α cylinders are exactly the d^α residue classes modulo d^α .

Proof. On the clopen cylinder $x \equiv i \pmod{d}$, the congruence $r_i \equiv im_i \pmod{d}$ puts $m_i x - r_i$ in $d\mathbb{Z}_d$; division by d therefore gives a continuous branch into $\mathbb{Z}_d$. The branches assemble continuously, and uniqueness follows from density of $\mathbb{Z}$ in $\mathbb{Z}_d$. When $d = d(\mathcal{T})$ this is Matthews's Theorem 6.1 [21, author-form p. 12]; the displayed clopen-branch proof also establishes the stated retained-presentation version.

For the second assertion, every denominator in b_α is a unit. Put

$$X_k = \sum_{j=k}^{\alpha-1} \frac{r_{i_j} d^{j-k}}{m_{i_k} \cdots m_{i_j}} \quad (0 \leq k < \alpha).$$

Then $X_k \equiv r_{i_k}/m_{i_k} \equiv i_k \pmod{d}$ and $(m_{i_k}X_k - r_{i_k})/d = X_{k+1}$ for $k < \alpha - 1$. Adding any element of $d^\alpha \mathbb{Z}_d$ to $X_0 = b_\alpha$ preserves these first α digits, because the difference after j common branches remains divisible by $d^{\alpha-j}$. Conversely, repeated rearrangement of the branch equation for an x with the displayed itinerary gives

$$x = b_\alpha(\mathbf{i}) + \frac{d^\alpha \widehat{\mathcal{T}}^\alpha(x)}{m_{i_0} \cdots m_{i_{\alpha-1}}},$$

whose second term lies in $d^\alpha \mathbb{Z}_d$. This proves (212). It reconstructs author-form Theorem 2.2(ii) and is the later indexing of Matthews–Watts Eq. (2.4) and Section 3, Lemma 4 [21, author-form p. 4] [22, pp. 169, 172–173]. $\square$

Let $S_d : \mathbb{Z}_d \rightarrow \mathbb{Z}_d$ be the digit shift

$$S_d \left(\sum_{j \geq 0} a_j d^j \right) = \sum_{j \geq 0} a_{j+1} d^j.$$

For $x \in \mathbb{Z}_d$, let $i_j(x) \in \{0, \dots, d-1\}$ be defined by $\widehat{\mathcal{T}}^j(x) \equiv i_j(x) \pmod{d}$.

Theorem 2.110 (constructive generalized itinerary conjugacy). *Under (211), the map*

$$\text{It}_{\mathcal{T}} : \mathbb{Z}_d \longrightarrow \mathbb{Z}_d, \quad \text{It}_{\mathcal{T}}(x) = \sum_{j \geq 0} i_j(x) d^j, \quad (213)$$

is an isometric homeomorphism for the digit-prefix metric. Its inverse is the coordinate series

$$\text{It}_{\mathcal{T}}^{-1} \left(\sum_{j \geq 0} i_j d^j \right) = \sum_{j \geq 0} \frac{r_{i_j} d^j}{m_{i_0} m_{i_1} \cdots m_{i_j}}. \quad (214)$$

It conjugates the extended map to the full shift:

$$\text{It}_{\mathcal{T}} \circ \widehat{\mathcal{T}} = S_d \circ \text{It}_{\mathcal{T}}, \quad \widehat{\mathcal{T}} = \text{It}_{\mathcal{T}}^{-1} \circ S_d \circ \text{It}_{\mathcal{T}}. \quad (215)$$

Every fibre is a singleton. No additive or ring-homomorphism property is asserted, so no group-theoretic kernel is attached to this map.

Proof. For a prescribed digit sequence $(i_j)_{j \geq 0}$, define the convergent suffix series

$$X_k = \sum_{j=k}^{\infty} \frac{r_{i_j} d^{j-k}}{m_{i_k} m_{i_{k+1}} \cdots m_{i_j}}.$$

The unit condition gives

$$X_k \equiv \frac{r_{i_k}}{m_{i_k}} \equiv i_k \pmod{d}, \quad \frac{m_{i_k} X_k - r_{i_k}}{d} = X_{k+1}.$$

Hence X_0 has exactly the prescribed itinerary, proving that (214) is a right inverse.

For an arbitrary x , repeated rearrangement of its actual branch equations gives, for every $\alpha \geq 1$,

$$x = \sum_{j=0}^{\alpha-1} \frac{r_{i_j(x)} d^j}{m_{i_0(x)} \cdots m_{i_j(x)}} + \frac{d^\alpha \widehat{\mathcal{T}}^\alpha(x)}{m_{i_0(x)} \cdots m_{i_{\alpha-1}(x)}}.$$

The final term tends to zero in the digit-prefix metric. Thus (214) is also a left inverse.

If $x \equiv y \pmod{d^\alpha}$, common-branch induction gives $\widehat{\mathcal{T}}^j(x) \equiv \widehat{\mathcal{T}}^j(y) \pmod{d^{\alpha-j}}$ for $0 \leq j \leq \alpha$; hence their first α itinerary digits agree. The converse follows from Proposition 2.109. Therefore

$$x \equiv y \pmod{d^\alpha} \iff \text{It}_{\mathcal{T}}(x) \equiv \text{It}_{\mathcal{T}}(y) \pmod{d^\alpha},$$

which proves the isometry and continuity in both directions. Finally, $i_j(\widehat{\mathcal{T}}x) = i_{j+1}(x)$, so deleting the first itinerary digit proves (215). $\square$

This is a constructive proof, with no novelty claim, of the general- d statement that the author form presents only as something that “should hold” [21, author-form p. 14]. At $d = 2$ for the shortened $3x + 1$ map, (213) is pointwise the Bernstein–Lagarias Q_3 and Laarhoven–de Weger’s Φ orientation. It is an explicit choice having the orientation of Matthews’s C_T . Matthews’s Theorem 6.3 alone does not state a coordinate normalization that would identify its named C_T pointwise merely from the conjugacy equation.

Corollary 2.111 (Haar cylinder independence and strong mixing). *The map $\text{It}_{\mathcal{T}}$ preserves normalized Haar measure μ_d . If $A = a + d^\alpha \mathbb{Z}_d$ and $B = b + d^\beta \mathbb{Z}_d$, then for $K \geq \beta$,*

$$\mu_d(\widehat{\mathcal{T}}^{-K}(A) \cap B) = \mu_d(A)\mu_d(B) = d^{-\alpha-\beta}. \quad (216)$$

Consequently $\widehat{\mathcal{T}}$ is strongly mixing and ergodic on $(\mathbb{Z}_d, \mu_d)$.

Proof. The isometry permutes the d^α residue balls at every level, so it preserves their measures and hence Haar measure on the Borel sigma-algebra. Under the conjugacy, B fixes the first β digits and $S_d^{-K}(A)$ fixes digits K through $K + \alpha - 1$. For $K \geq \beta$ these blocks are disjoint, giving exactly $d^{-\alpha-\beta}$. Finite unions of residue balls approximate every Borel set in Haar measure. Replacing two Borel sets by such unions makes the error in the intersection at most the sum of the two symmetric-difference measures, uniformly in K because the maps preserve measure. Letting that error tend to zero proves strong mixing; ergodicity follows by applying the mixing identity to an invariant set. This reproduces Matthews–Watts Lemma 5 and Corollary 1 and Matthews’s later Theorem 6.2 [22, pp. 173–174] [21, author-form pp. 12–13]. $\square$

Proposition 2.112 (exact nonunit obstruction). *Suppose instead that $g_i = \gcd(m_i, d) > 1$ for some branch i . Among the d lifts $i + dt$ modulo d^2 , the next itinerary digit assumes exactly d/g_i values, each with g_i lifts. Hence some two-digit words beginning with i are absent and the canonical itinerary encoder is not a bijection onto the full d -shift.*

Proof. The branch equation gives

$$\mathcal{T}(i + dt) = \mathcal{T}(i) + m_i t.$$

Modulo d , the image of multiplication by m_i has cardinality d/g_i and every image element has g_i preimages. These are exactly the possible second digits and their lift multiplicities. $\square$

For the unshortened Collatz map in Matthews’s presentation, $d = 2, m_1 = 6, r_1 = -2$, so every odd input maps to an even value and the word 11 is absent. Proposition 2.112 obstructs the canonical itinerary map from being a full-shift conjugacy; it does not assert categorical nonexistence of every differently defined conjugacy.

Remark 2.113 (formal certificate boundary). The pinned Lean certificate checks the literal lift identity, the explicit unit-affine inverse, the equality between the residue gcd and the ordinary gcd of the retained integer multiplier, the d/g_i attainable-digit count, and the g_i -element fibre over every attainable digit. In particular, the range and fibre assertions in Proposition 2.112 are kernel-checked. The artifact deliberately does not certify the inverse limit, infinite series, homeomorphism, or full shift conjugacy. Those assertions therefore remain outside the formal certificate; extending the formalization is project work, not an open mathematical problem.

Explicit nonclaim 2.114. The Haar-almost-every conclusions live in $\mathbb{Z}_d$. The embedded ordinary integers are countable and Haar-null. Neither the conjugacy nor strong mixing proves an endpoint statement for positive integer Collatz orbits; it transports the unresolved embedding of that special subset into shift coordinates.

2.17 Integer rcwa presentations: exact crosswalk and a signed-slope defect

Kohl calls a map $f : \mathbb{Z} \rightarrow \mathbb{Z}$ *residue-class-wise affine* (rcwa) when some positive integer q admits coefficients $a_i, b_i, c_i \in \mathbb{Z}$ for which

$$f(n) = \frac{a_i n + b_i}{c_i} \quad (n \equiv i \pmod{q}). \quad (217)$$

The least such q is $\text{Mod}(f)$. At the least modulus the source stores each coefficient triple in primitive form and takes $c_i > 0$ [28, Defs. 1.1 and 2.1]. Algorithm 2.2 separately permits a nonminimal modulus and unreduced input before reduction. The following bridge therefore retains its presentation modulus; it does not normalize either source.

Proposition 2.115 (fixed-modulus Matthews–rcwa crosswalk). *Fix $d \geq 2$. The presentation-level assignment*

$$(m_i, r_i)_{0 \leq i < d} \longmapsto (a_i, b_i, c_i)_{0 \leq i < d} = (m_i, -r_i, d)_{0 \leq i < d} \quad (218)$$

sends the Matthews presentation (210) to an rcwa presentation of the same function $\mathbb{Z} \rightarrow \mathbb{Z}$. Its image consists of the fixed-modulus, common-denominator presentations with nonzero branch slopes, and its inverse on that presentation class is $m_i = a_i, r_i = -b_i$.

Conversely, let (217) be a reduced rcwa coefficient list at the retained modulus d , and assume $a_i \neq 0$ for every i . Then

$$m_i = \frac{a_i d}{c_i}, \quad r_i = -\frac{b_i d}{c_i} \quad (219)$$

are integers, satisfy $r_i \equiv im_i \pmod{d}$, and reproduce the same branch map in Matthews form. Returning through (218) scales the i th reduced triple by d/c_i . Reducing that scaled triple recovers the starting reduced triple exactly; the literal common-denominator list and the reduced list are distinct but reversibly related while d is retained.

Proof. The Matthews congruence puts $m_i n - r_i$ in $d\mathbb{Z}$ throughout $i + d\mathbb{Z}$, which proves the first assertion and its displayed inverse. For the converse, Kohl’s reduced-list divisibility gives $c_i \mid d$ and $c_i \mid a_i i + b_i$. Hence the integers in (219) obey

$$im_i - r_i = \frac{d(a_i i + b_i)}{c_i} \equiv 0 \pmod{d},$$

and direct substitution gives $(m_i n - r_i)/d = (a_i n + b_i)/c_i$.

For Matthews data, put $g_i = \gcd(m_i, d)$. The congruence $r_i \equiv im_i \pmod{d}$ gives

$$\gcd(m_i, r_i, d) = g_i.$$

Therefore branchwise reduction sends $(m_i, -r_i, d)$ exactly to $(m_i/g_i, -r_i/g_i, d/g_i)$. Applying (219) with the retained d recovers (m_i, r_i) . Conversely, primitivity and $c_i \mid a_i i + b_i$ imply $\gcd(a_i, c_i) = 1$; hence for the reconstructed data $\gcd(a_i d/c_i, d) = d/c_i$, so reduction returns precisely (a_i, b_i, c_i) . This proves exact reversibility at fixed d . $\square$

This bridge separates three conditions which are not interchangeable. In the common-denominator presentation, the unit condition is $\gcd(m_i, d) = 1$ for every branch. As proved above it forces d to be the least modulus, makes every triple $(m_i, -r_i, d)$ primitive, and leaves every reduced denominator equal to d . Hence Kohl's divisor is exactly d . Kohl calls a reduced rcwa map *integral* when this divisor is one, so for $d \geq 2$ Matthews relatively-prime type excludes Kohl integrality. Neither condition says that the integer map is bijective. For example, the shortened $3x + 1$ map has reduced triples $(1, 0, 2)$ and $(3, 1, 2)$: it is Matthews-relatively-prime, has Kohl divisor two, and is surjective but not injective on $\mathbb{Z}$.

Source issue 2.116 (signed slopes in Kohl Remark 1.2). For a reduced coefficient list at modulus q , the printed remark uses

$$\frac{1}{q} \sum_{i=0}^{q-1} \frac{c_i}{a_i} \tag{220}$$

and asserts that it is at most one for an injection, at least one for a surjection, and equal to one for a bijection [28, Rem. 1.2]. As printed, the expression is undefined on a zero-slope branch and false for negative slopes. The same article lists $f(n) = -n$ as an rcwa permutation; at modulus one its reduced triple is $(-1, 0, 1)$, so (220) equals -1 , not 1 [28, Def. 2.4 and following example].

Proposition 2.117 (corrected rcwa image-density statistic). *For a reduced rcwa list at modulus q , put*

$$D(f) = \frac{1}{q} \sum_{\substack{0 \leq i < q \\ a_i \neq 0}} \frac{c_i}{|a_i|}. \tag{221}$$

If f is injective, then $D(f) \leq 1$; if f is surjective, then $D(f) \geq 1$; and if f is bijective, then $D(f) = 1$.

Proof. On $i + q\mathbb{Z}$ with $a_i \neq 0$, the image of the branch is one residue class of modulus $|a_i|q/c_i$ and therefore has natural density $c_i/(|a_i|q)$. A zero-slope branch has a singleton image and density zero. If f is injective, the branch images are disjoint, so their density sum is at most one. If f is surjective, their finite union, together with any singleton images, is all of $\mathbb{Z}$, so the sum of their densities counted with overlap is at least one. A bijection satisfies both bounds. $\square$

The absolute value and omission of zero slopes are forced, while the positive- slope Collatz calculation in the source remains unchanged. This repaired integer-image statistic neither proves the d -adic itinerary conjugacy nor transfers a Haar-almost-every statement to ordinary integer orbits.

2.18 Siegel's Hydra and numen formalism: typed source boundary

The controlling witness in this subsection is Siegel's version-pinned source for arXiv:2601.17030v3 [31]. Its Hydra and numen sections are unchanged from v2 to v3; the v3 change is confined to the later Fourier calculation audited at the end of this subsection. The source permits both number fields and finite extensions of $\mathbb{F}_q(T)$. Accordingly, $\mathcal{O}_K$ is a finite free module over the integer ring $\mathcal{O}_{\mathbb{F}}$ of the base field; it has a finite $\mathbb{Z}$ -basis only in the number-field case.

Let $K/\mathbb{F}$ be such a global field, let $\Lambda \subsetneq \mathcal{O}_K$ be a nonzero ideal, and retain an enumeration of the $p = |\mathcal{O}_K/\Lambda|$ cosets by $\{0, \dots, p-1\}$, with 0 denoting Λ . If ℓ is a place, an $\mathbb{F}$ -linear map used on K_{ℓ} will always be assumed bounded at ℓ . This assumption is necessary: an arbitrary $\mathbb{F}$ -linear endomorphism of K need not extend continuously to a fixed completion. The operator norm is

$$\|r\|_{\ell} = \sup_{z \in K \setminus \{0\}} \frac{|r(z)|_{\ell}}{|z|_{\ell}},$$

when this supremum is finite. The source display includes $z = 0$ in a quotient by $|z|_{\ell}$ and its following remark asserts extendibility without this boundedness condition.

Definition 2.118 (Siegel's printed branch data). A pre-Hydra has, for every $j \in \mathcal{O}_K/\Lambda$, an invertible affine $\mathbb{F}$ -linear branch

$$H_j : K \longrightarrow K, \quad H_j(z) = r_j(z) + c_j, \quad r_j \in \text{End}_{\mathbb{F}}^{\times}(K), \quad c_j \in K.$$

The selected piecewise evaluator is

$$H(z) = H_{[z]_{\Lambda}}(z) \quad (z \in \mathcal{O}_K).$$

The source's literal Definition 5.3 additionally requires $H_j(z) \in \mathcal{O}_K$ for every $z \in \mathcal{O}_K$ and every j .

The enumeration in the source should read $i = \Lambda_{\beta(i)}$ for a coset $i \in \mathcal{O}_K/\Lambda$, not $i \in \Lambda_{\beta(i)}$. After the cosets have been replaced by digit labels, DigSum_p uses those labels directly; applying β to them a second time is ill-typed. We retain the enumeration rather than normalize it away.

Source issue 2.119 (the two printed integrality predicates). Definition 5.3 requires

$$H_j(\mathcal{O}_K) \subseteq \mathcal{O}_K \quad \text{for every } j, \tag{G}$$

whereas Definition 5.4 calls the same Hydra *integral* when

$$H_j(z) \in \mathcal{O}_K \iff z \in j + \Lambda \quad (z \in \mathcal{O}_K). \tag{E}$$

These conditions cannot coexist. Since Λ is proper, choose a coset different from $j + \Lambda$ and an integer z in it. Condition (G) says $H_j(z)$ is integral and condition (E) says it is not. The source's own shortened Collatz example displays the intended distinction: $(3z+1)/2$ is integral exactly on odd z , so it satisfies (E) but not (G).

We therefore use three explicitly different predicates:

$$\begin{aligned} \textit{selected-branch admissible} & & H_j(j + \Lambda) &\subseteq \mathcal{O}_K, \\ \textit{exact-integral on global integers} & & H_j^{-1}(\mathcal{O}_K) \cap \mathcal{O}_K &= j + \Lambda, \\ \textit{inverse-integral at a completion } v & & H_j^{-1}(\mathcal{O}_{K_v}) &= j + \Lambda \mathcal{O}_{K_v}. \end{aligned}$$

The first makes the piecewise evaluator a map $\mathcal{O}_K \rightarrow \mathcal{O}_K$. The second detects the assigned coset among global integers. The third is the stronger local inverse-image statement needed by a backward branch-correctness argument. It is not implied merely by writing the second predicate.

Proposition 2.120 (scalar Hydra–Matthews bridge at retained modulus). *Let $K = \mathbb{Q}$, $\mathcal{O}_K = \mathbb{Z}$, and $\Lambda = D\mathbb{Z}$, with the residue labels $0, \dots, D-1$ retained. Scalar selected-branch-admissible data*

$$H_i(n) = a_i n + b_i$$

are in exact two-sided correspondence with Matthews data

$$H_i(n) = \frac{m_i n - s_i}{D}, \quad m_i, s_i \in \mathbb{Z}, \quad s_i \equiv i m_i \pmod{D},$$

by

$$m_i = D a_i, \quad s_i = -D b_i.$$

The complete integrality locus of the i th extension is

$$i + \frac{D}{\gcd(m_i, D)} \mathbb{Z}.$$

Consequently, exact integrality on global integers is equivalent to $\gcd(m_i, D) = 1$ for every i .

Proof. Admissibility gives $H_i(i), H_i(i+D) \in \mathbb{Z}$. Subtracting gives $D a_i \in \mathbb{Z}$, and then

$$D b_i = D H_i(i) - (D a_i) i \in \mathbb{Z}.$$

Thus m_i and s_i are integral, while

$$m_i i - s_i = D(a_i i + b_i) = D H_i(i)$$

proves the congruence. Conversely, that congruence makes $(m_i n - s_i)/D$ integral on $i + D\mathbb{Z}$, and the two displayed assignments are literal inverses.

For arbitrary $n \in \mathbb{Z}$, branch integrality is equivalent to $m_i(n - i) \equiv 0 \pmod{D}$. Writing $g_i = \gcd(m_i, D)$, this is equivalent to $n - i \in (D/g_i)\mathbb{Z}$, which proves the locus and the final assertion. $\square$

This proposition applies only to scalar branches. A genuinely non-scalar $\mathbb{F}$ -linear Hydra is not silently placed in the Matthews or rcwa classes. For the scalar subclass, the further reduced rcwa crosswalk is Proposition 2.115; neither step minimizes the retained modulus.

Source issue 2.121 (uniform separation is not bounded finiteness). Source Proposition 4.2 claims that if Y is uniformly separated in an arbitrary valued field, every orbit of every map $T : Y \rightarrow Y$ is preperiodic or norm-divergent. Its proof uses the stronger assertion that every bounded intersection with Y is finite.

For a nontrivially valued counterexample, take $K = \mathbb{Q}((t))$ with the t -adic absolute value, $Y = \mathbb{Z} \subset K$, and $T(n) = n + 1$. Distinct integers have distance one, but the orbit of zero is injective and has norm one after the first term. It is neither preperiodic nor divergent.

The proof becomes valid under the exact replacement

$$\#\{y \in Y : |y| \leq R\} < \infty \quad \text{for every } R > 0. \quad (\text{BF})$$

For a number field, $\mathcal{O}_K$ is boundedly finite in its full Minkowski embedding. It need not be discrete at one archimedean embedding: $\mathbb{Z}[\sqrt{2}]$ has nonzero elements arbitrarily close to zero there. Nor does escape in the full Minkowski norm imply escape at every individual archimedean absolute value.

For the numen, let Σ_p^* be the finite words in $\{0, \dots, p-1\}$, including the empty word. The source fixes least-significant digit first:

$$\text{DigSum}_p(j_1, \dots, j_N) = \sum_{a=1}^N j_a p^{a-1}.$$

Proposition 2.122 (finite affine word and numen cocycle). *For $\mathbf{j} = (j_1, \dots, j_N)$, define*

$$H_{\mathbf{j}} = H_{j_1} \circ \dots \circ H_{j_N}, \quad H_{\mathbf{j}}(z) = M_{\mathbf{j}}(z) + X_{\mathbf{j}}.$$

Then

$$M_{\mathbf{j}} = r_{j_1} \circ \dots \circ r_{j_N}, \quad X_{\mathbf{j}} = \sum_{m=0}^{N-1} (r_{j_1} \circ \dots \circ r_{j_m})(c_{j_{m+1}}), \quad (\text{W})$$

with the empty operator product equal to I . For concatenation,

$$M_{\mathbf{i} \wedge \mathbf{j}} = M_{\mathbf{i}} \circ M_{\mathbf{j}}, \quad X_{\mathbf{i} \wedge \mathbf{j}} = M_{\mathbf{i}}(X_{\mathbf{j}}) + X_{\mathbf{i}}. \quad (\text{C})$$

Proof. The rightmost branch in $H_{j_1} \circ \dots \circ H_{j_N}$ acts first. Composing two affine maps gives

$$(M_i, X_i) \circ (M_j, X_j) = (M_i \circ M_j, M_i(X_j) + X_i).$$

Induction proves both (W) and (C); the empty word gives $(I, 0)$. $\square$

Here and below a word is an address. It is not automatically an actual piecewise-Hydra itinerary.

Proposition 2.123 (centered numen descent and multiplier non-descent). *Let $\mathbf{i} \sim \mathbf{j}$ when their DigSum values agree. If $c_0 = H_0(0) = 0$, then $X_{\mathbf{i}} = X_{\mathbf{j}}$ whenever $\mathbf{i} \sim \mathbf{j}$, so the finite numen factors as*

$$\Sigma_p^*/\sim \xrightarrow[\cong]{\text{DigSum}_p} \mathbb{N}_0 \xrightarrow{X_H} K.$$

The multiplier does not generally factor through this quotient:

$$M_{\mathbf{j} \wedge (0)} = M_{\mathbf{j}} \circ r_0$$

need not equal $M_{\mathbf{j}}$.

Proof. Every finite DigSum fibre consists of the canonical base- p word and its finitely many terminal-zero extensions. Since the appended zero is the rightmost map in the composition,

$$X_{\mathbf{j} \wedge (0)} = H_{\mathbf{j}}(H_0(0)) = H_{\mathbf{j}}(0) = X_{\mathbf{j}}.$$

Iteration proves descent. The displayed multiplier identity follows directly from Proposition 2.122 and supplies the obstruction. $\square$

Thus every occurrence of $M_H(n)$ requires a canonical digit word and its retained length. The source proves descent only for X_H .

Proposition 2.124 (finite digit functional equations). *Functions $f : \mathbb{N}_0 \rightarrow K$ satisfying*

$$f(pn + j) = r_j(f(n)) + c_j \quad (n \in \mathbb{N}_0, 0 \leq j < p) \quad (222)$$

are in bijection with the affine fibre

$$\{a \in K : (I - r_0)a = c_0\},$$

by $a = f(0)$. If $I - r_0$ is invertible, there is one solution, with $a = (I - r_0)^{-1}c_0$. If $n = \sum_{m=0}^{N-1} d_m p^m$ is its canonical digit expansion and

$$P_0 = I, \quad P_m = r_{d_0} \circ \cdots \circ r_{d_{m-1}},$$

then

$$f(n) = P_N(a) + \sum_{m=0}^{N-1} P_m(c_{d_m}). \quad (223)$$

Proof. Putting $n = j = 0$ in (222) gives $(I - r_0)f(0) = c_0$. Conversely, once an admissible a is chosen, repeated Euclidean division by p reaches zero and recursively defines one value at every nonnegative integer. Applying the recursion along the canonical digits proves (223). $\square$

The finite theorem is sound. It also isolates the term omitted in the source's proof of its continuation lemma.

Source issue 2.125 (Lemma 6.1 as printed). Lemma 6.1 assumes only that $I - r_0$ is invertible, but its finite formula uses the centered word sum and omits $P_N(a)$ from (223). Its equation (6.29) prints equality between the norm of a sum and the sum of termwise norm bounds; only $\leq$ follows. Its pointwise density definition uses a limsup frequency when $\log\|r_j\|_\ell \leq 0$ and a liminf when that logarithm is positive. These are the reverse of the choices needed for an upper Lyapunov bound. Almost-everywhere convergence leaves a null nonconvergence set on which no value is assigned. The proof establishes uniqueness on $\mathbb{N}_0$, not uniqueness of arbitrary functions on $\mathbb{Z}_p$ satisfying the same recursion. Finally, in the nonarchimedean case the displayed bound omits $|a|_\ell$.

The last omission has a concrete instance. Over $\mathbb{Q}_2$, take $r_0 = 3$ and $c_0 = 1$. Then $a = (1 - 3)^{-1} = -1/2$, so $|a|_2 = 2 > |c_0|_2 = 1$.

Theorem 2.126 (corrected numen continuation). *Fix a completion K_ℓ at which all r_j are bounded, assume $I - r_0$ is invertible, and put $a = (I - r_0)^{-1}c_0$. For*

$$\mathfrak{z} = \sum_{m \geq 0} d_m p^m \in \mathbb{Z}_p,$$

define

$$f_N(\mathfrak{z}) = P_N(a) + \sum_{m=0}^{N-1} P_m(c_{d_m}), \quad P_m = r_{d_0} \circ \cdots \circ r_{d_{m-1}}. \quad (224)$$

For $q_{j,m} = m^{-1} \#\{0 \leq k < m : d_k = j\}$ and $\alpha_j = \log\|r_j\|_\ell$, set

$$L^+(\mathfrak{z}) = \sum_{\alpha_j < 0} \alpha_j \liminf_{m \rightarrow \infty} q_{j,m} + \sum_{\alpha_j > 0} \alpha_j \limsup_{m \rightarrow \infty} q_{j,m}. \quad (225)$$

Then:

- (i) if $L^+(\mathfrak{z}) < 0$, the sequence (224) converges;
- (ii) if $\prod_{j=0}^{p-1} \|r_j\|_\ell < 1$, it converges for Haar-almost every $\mathfrak{z}$ and defines a measurable function on its conull convergence domain;
- (iii) if $\rho = \max_j \|r_j\|_\ell < 1$, it converges uniformly on $\mathbb{Z}_p$ to a continuous function;
- (iv) whenever $\mathfrak{z}$ lies in the convergence domain, $p\mathfrak{z} + j$ also lies there and

$$f(p\mathfrak{z} + j) = r_j(f(\mathfrak{z})) + c_j;$$

- (v) if ℓ is nonarchimedean and $\max_j \|r_j\|_\ell \leq 1$, every finite truncation satisfies

$$|f_N(\mathfrak{z})|_\ell \leq \max\{|a|_\ell, \max_j |c_j|_\ell\}.$$

In case (iii), the continuous extension is the unique continuous function with the digit recursion and initial value a .

Proof. Submultiplicativity gives

$$\|P_m\|_\ell \leq \prod_{j=0}^{p-1} \|r_j\|_\ell^{mq_{j,m}}.$$

For a negative α_j , the limsup of $\alpha_j q_{j,m}$ is $\alpha_j \liminf q_{j,m}$; for a positive α_j , it is $\alpha_j \limsup q_{j,m}$. Since there are finitely many digits,

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \log \|P_m\|_\ell \leq L^+(\mathfrak{z}).$$

Under (i), there are $\eta < 1$ and m_0 such that $\|P_m\|_\ell \leq \eta^m$ for $m \geq m_0$. The translations form a finite set, so the series in (224) is absolutely convergent and $P_N(a) \rightarrow 0$.

Haar-almost every base- p digit sequence has each digit frequency $1/p$. Its Lyapunov exponent is bounded above by

$$\frac{1}{p} \sum_j \log \|r_j\|_\ell = \frac{1}{p} \log \prod_j \|r_j\|_\ell,$$

which proves (ii); a pointwise limit of measurable truncations is measurable on the convergence domain. Under (iii), $\|P_m\|_\ell \leq \rho^m$ for every digit sequence, and the geometric tail estimate is uniform. The truncations are locally constant, hence their uniform limit is continuous.

The finite identity

$$f_N(p\mathfrak{z} + j) = r_j(f_{N-1}(\mathfrak{z})) + c_j$$

proves (iv) on passage to the limit. In the nonarchimedean case, the ultrametric inequality and $\|P_m\|_\ell \leq 1$ give (v). Finally, $\mathbb{N}_0$ is dense in $\mathbb{Z}_p$, and Proposition 2.124 fixes all its values; two continuous extensions therefore agree everywhere. $\square$

Functional equations alone do not give literal uniqueness among arbitrary functions on $\mathbb{Z}_p$. For example, with $p = 2$, $r_0 = r_1 = 1/2$, and $c_0 = c_1 = 0$, choose a non-eventually-periodic digit sequence. Its tail-equivalence class is countable and contains no shift cycle. A nonzero seed value propagates consistently over that class by the invertible branch factors, while the function may be set to zero off the class. The result is Borel and satisfies the recursion, but is not the zero function.

Proposition 2.127 (operator-typed periodic word). *For a nonempty finite word $\mathbf{w}$, put $M = M_{\mathbf{w}}$ and $X = X_{\mathbf{w}}$. Then*

$$X_{\mathbf{w}^{\wedge m}} = \sum_{k=0}^{m-1} M^k(X). \quad (226)$$

If $\|M\|_{\ell} < 1$, then $I - M$ is invertible on K_{ℓ} and

$$z_{\mathbf{w}} = (I - M)^{-1}X$$

is the unique fixed point of the affine extension $H_{\mathbf{w}}$. It is a point of an actual N -step piecewise-Hydra cycle only if, for $\mathbf{w} = (w_1, \dots, w_N)$,

$$\begin{aligned} z_{\mathbf{w}} &\in w_N + \Lambda, \\ H_{w_N}(z_{\mathbf{w}}) &\in w_{N-1} + \Lambda, \\ &\vdots \\ H_{w_2} \circ \dots \circ H_{w_N}(z_{\mathbf{w}}) &\in w_1 + \Lambda. \end{aligned} \quad (227)$$

Proof. The cocycle gives $X_{\mathbf{w}^{\wedge m}} = M(X_{\mathbf{w}^{\wedge(m-1)}}) + X$; induction proves (226). The Neumann series $\sum_{k \geq 0} M^k$ converges in operator norm and is the inverse of $I - M$. Thus $M(z_{\mathbf{w}}) + X = z_{\mathbf{w}}$, and subtracting two fixed-point equations proves uniqueness. Since the rightmost branch acts first, conditions (227) are exactly the successive branch-selection requirements. Without them, the calculation concerns only the affine extension. $\square$

The source's fractions $(1 - M^m)/(1 - M)$ and $X/(1 - M)$ must be replaced by operator composition with $(I - M)^{-1}$. Its equality $\|M^m\| = \|M\|^m$ is false for a general operator norm; the sufficient relation is $\|M^m\| \leq \|M\|^m$.

Source issue 2.128 (Siegel Theorem 6.1 as stated). The source assumes a proper, centered, integral Λ -Hydra and the two conditions

1. for every $n \geq 1$, some place ℓ_n satisfies $\|M_H(n)\|_{\ell_n} < 1$;
2. some nontrivial nonarchimedean place v satisfies $\|r_j\|_v > 1$ for every digit j and $v(\Lambda) \geq 1$.

It then states:

1. an element $z \in \mathcal{O}_K$ is periodic for H if and only if $z = X_H(\mathfrak{z})$ for some $\mathfrak{z} \in (\mathbb{Z}_p \setminus \mathbb{N}_0) \cap \mathbb{Q}$;
2. if K is a number field and $X_H(\mathfrak{z}) \in \mathcal{O}_K$ for an irrational $\mathfrak{z} \in \mathbb{Z}_p \setminus \mathbb{Q}$, then the source calls $X_H(\mathfrak{z})$ a trajectory divergent with respect to every nontrivial archimedean absolute value.

This is a source-statement record, not an admitted theorem of this edition.

The proof sketch does not certify either conclusion. The exact blocking points are as follows.

1. The literal Hydra and integral hypotheses have no simultaneous models, by Source Issue 2.119.
2. M_H does not descend through finite DigSum equivalence. Thus $M_H(n)$ is not defined until a canonical word and its retained length are specified. The displayed $\lambda_p(n)$ is likewise not defined.

3. Two formulas use an undefined M_q . Scalar fractions such as $(1 - M^m)/(1 - M)$ are not typed for general endomorphisms, and the equality $\|M^m\| = \|M\|^m$ is false. The operator-typed, submultiplicative replacement is Proposition 2.127.
4. Passing from an arbitrary finite word to the canonical digit word of its DigSum may remove terminal zeroes. Repeating those two words need not give the same multiplier or affine map. The centered all-zero word and the fixed point zero also lie outside the displayed $n \geq 1$ construction.
5. The notation $v(\Lambda)$, the *correctness* predicate, and the claimed extension of the branches to $\mathcal{O}_{K_v}$ are not defined or proved in this source. A formal affine fixed point is not an actual piecewise orbit until every condition in (227) is established.
6. The converse uses integrality backwards: from an integral later value it infers that a possibly nonintegral predecessor is integral. The printed global predicate does not imply the completion-level inverse-image identity needed for this step, nor is a local-to-global arrow supplied.
7. The forward direction gives, at most, an existential rational address for a periodic point. It does not prove that the same numen fibre contains no irrational address. Consequently it cannot support the later inference that an irrational address has a nonpreperiodic value. The zero fibre is also unclassified.
8. Part II begins by invoking a nonexistent hypothesis (iii), then imports a zero-fibre assertion and a correctness result without rebuilding their hypotheses. It also depends on the false uniformly-discrete orbit classification repaired in Source Issue 2.121.
9. A numen value is a point, not a trajectory. Even after replacing it by its forward orbit, escape in the full Minkowski embedding would not by itself prove divergence at every individual archimedean place.

Accordingly, the periodic-word fixed-point proposition is retained at its proved scope, while the source's rational-fibre correspondence and irrational-address divergence conclusions are not proved and are not used. The legacy audit identifier **PO-COL-000006** records that gap; it is not a mathematical problem or a commitment attributed to the source or reader.

Source issue 2.129 (the v3 nonarchimedean Fourier inversion). Let $L = K_\ell$ be a finite extension of $\mathbb{Q}_q$, let $\mathcal{O}_L$ be its integer ring, let π be a uniformizer, and let $\mathfrak{D} = \mathfrak{D}_{L/\mathbb{Q}_q}$ be its different. The source uses the trace character

$$e_\ell(x) = \exp\left(2\pi i \{\mathrm{Tr}_{L/\mathbb{Q}_q}(x)\}_q\right).$$

The annihilator of $\mathcal{O}_L$ under the pairing $(t, x) \mapsto e_\ell(tx)$ is

$$\mathfrak{D}^{-1} = \{t \in L : \mathrm{Tr}_{L/\mathbb{Q}_q}(t\mathcal{O}_L) \subseteq \mathbb{Z}_q\},$$

not $\mathcal{O}_L$ in general. Thus the source's asserted dual $L/\mathcal{O}_L$ is correct for this character only after a conductor normalization, for example when the relevant different is trivial. With the printed character the dual is $L/\mathfrak{D}^{-1}$.

For comparison, if an L -valued random variable X has first been proved to lie in $\mathcal{O}_L$ almost surely, define

$$\hat{\mu}(t) = \mathbb{E}[e_\ell(-tX)] \quad (t \in L/\mathfrak{D}^{-1}).$$

Writing e and $f = [L : \mathbb{Q}_q]/e$ for the ramification index and residue degree, finite character orthogonality gives the correctly typed identity

$$\Pr(X \in w + \pi^n \mathcal{O}_L) = q^{-nf} \sum_{t \in \pi^{-n} \mathfrak{D}^{-1}/\mathfrak{D}^{-1}} \hat{\mu}(t) e_\ell(tw) \quad (w \in \mathcal{O}_L, n \geq 0). \quad (228)$$

Indeed, the displayed frequency quotient is the complete character group of $\mathcal{O}_L/\pi^n \mathcal{O}_L$, which has q^{nf} elements. Averaging those characters is one on the zero residue class and zero on every other class, proving (228).

This correction does not certify source Proposition 7.6. Its surrounding Fourier chain has further independent defects:

1. for a nonscalar bounded local branch r_j , the frequency map is the trace adjoint $r_j^\dagger$, defined by

$$\mathrm{Tr}_{L/\mathbb{Q}_q}(t r_j x) = \mathrm{Tr}_{L/\mathbb{Q}_q}(r_j^\dagger t x),$$

not $t \mapsto r_j t$ unless the required self-adjoint or scalar identity is proved;

2. the characteristic function is naturally defined first on L ; its descent to a quotient requires an almost-sure support statement matched to the annihilator;
3. source Proposition 7.5 multiplies values in an object assumed only to be an abelian group, shadows its summation variable, and leaves another variable unbound, so it cannot justify the later rearrangement;
4. $B_{H,\ell} = e \max_j \log_q |c_j|_\ell$ is undefined when every translation is zero and may be negative, although Proposition 7.5 is invoked only for a nonnegative integer parameter;
5. the correct size relation is $q^{B/e} = |\pi^{-B}|_\ell$, not $|\pi^B|_\ell$; a real absolute value cannot be compared to the field element π^B ; and division by π^B cannot be replaced by division by the real number $q^{B/e}$.

The visible v3 changes repair neither the dual lattice nor these typing and scaling defects. No proposition depending on this Fourier chain is used downstream. The legacy audit identifier PO-COL-000007 records the resulting dependency gap; it does not convert the defective chain into an open mathematical problem.

Explicit nonclaim 2.130. The finite numen is an address-to-state construction. It is not identified with a state-to-itinerary conjugacy, and its fibres are not classified. Neither the corrected continuation theorem nor the formal periodic-word fixed point proves an ordinary-integer Collatz endpoint. The source's Correspondence Principle and Fourier consequences are recorded as open audits, not imported conclusions.

2.19 Current almost-bounded-orbit theorem: registered boundary

For $A \subset \mathbb{N}_{>0}$, Tao's logarithmic density is the limit, when it exists, of

$$\frac{\sum_{n \in A \cap [1,x]} 1/n}{\sum_{n \in \mathbb{N}_{>0} \cap [1,x]} 1/n}.$$

The current source is the version-pinned arXiv source 1909.03562v7, not the older routed local witness.

Theorem 2.131 (Tao, source theorem). *Let $f : \mathbb{N}_{>0} \rightarrow \mathbb{R}$ satisfy $f(N) \rightarrow +\infty$. With U as in (1) and*

$$U_{\min}(N) = \inf_{j \in \mathbb{N}_0} U^j(N),$$

one has $U_{\min}(N) < f(N)$ on a subset of $\mathbb{N}_{>0}$ of logarithmic density one.

This is Theorem 1.3 of the source [32, Sec. 1.1, Thm. 1.3]. Its quantifier over every divergent f and its logarithmic-density conclusion are retained exactly; it is not a natural-density theorem and not a bounded-orbit theorem with a universal constant. At the checkpoint at which this theorem was first registered, its proof-dependency audit was still open. The exact repaired source chain through Proposition 1.11, Theorem 3.1, Theorem 1.6, and this theorem is now closed in Subsection 2.23; no local claim may use more than the displayed conclusion.

Proposition 2.132 (exact Siegel–Tao Syracuse coordinate). *Let*

$$\Omega = (\mathbb{N}_{>0})^{\mathbb{N}_{>0}}, \quad \mathbb{P}_{\text{geom}}(a_k = r) = 2^{-r},$$

with independent coordinates, and put $A_k = a_1 + \dots + a_k$. The map

$$\mathcal{R} : \Omega \longrightarrow \mathbb{Z}_2, \quad \mathcal{R}((a_k)_{k \geq 1}) = \sum_{k \geq 1} 2^{A_k - 1}, \quad (229)$$

is a bijection onto the set $\mathbb{Z}_2^{\infty 1}$ of 2-adic digit strings having infinitely many one digits. Its inverse sends the increasing one positions $0 \leq n_1 < n_2 < \dots$ to

$$a_1 = n_1 + 1, \quad a_k = n_k - n_{k-1} \quad (k \geq 2). \quad (230)$$

It pushes $\mathbb{P}_{\text{geom}}$ to normalized Haar measure on $\mathbb{Z}_2$; the complement of $\mathbb{Z}_2^{\infty 1}$ is Haar-null.

For Siegel’s numen $\chi_3 : \mathbb{Z}_2 \rightarrow \mathbb{Z}_3$, one has the pointwise identity

$$\chi_3(\mathcal{R}((a_k))) = \sum_{k \geq 1} 3^{k-1} 2^{-A_k} \quad \text{in } \mathbb{Z}_3. \quad (231)$$

For $n \geq 1$, define the pointwise realization

$$\text{Syrac}_n(a_1, \dots, a_n) := F_n(a_n, \dots, a_1) \pmod{3^n}.$$

Consequently,

$$\chi_3(\mathcal{R}((a_k))) \pmod{3^n} = \text{Syrac}_n(a_1, \dots, a_n) = \sum_{k=1}^n 3^{k-1} 2^{-A_k} \pmod{3^n}, \quad (232)$$

and $\text{Syrac}_n(a_1, \dots, a_n)$, under $\mathbb{P}_{\text{geom}}$, has exactly Tao’s law $\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$. Thus the equality is pointwise for this explicitly declared coupling, rather than an unnamed equality in distribution.

Proof. The partial sums in (229) converge in $\mathbb{Z}_2$, because $A_k \rightarrow \infty$. Their one positions are exactly $A_k - 1$, which proves (230), both inverse identities, and singleton fibres.

Fix $(a_1, \dots, a_k)$. Its run-length cylinder has product mass

$$\prod_{j=1}^k 2^{-a_j} = 2^{-A_k}.$$

Its image fixes every binary digit from position zero through position $A_k - 1$, with ones exactly at $A_j - 1$; that Haar cylinder also has mass 2^{-A_k} . These cylinders generate the product sigma-algebra. Hence $\mathcal{R}_*\mathbb{P}_{\text{geom}}$ is Haar measure. A 2-adic string with finitely many ones is a nonnegative integer, so the omitted set is countable and Haar-null.

Siegel’s historical v1 one-position formula is

$$\chi_p \left(\sum_{k \geq 1} 2^{n_k} \right) = \sum_{k \geq 1} \frac{p^{k-1}}{2^{n_k+1}}.$$

The current v3 source obtains the same formula from its finite word cocycle and q -adic rising limit [29, historical v1 source lines 405–568] [30, current v3 source lines 1851–2000]. Substitution of $p = 3$ and $n_k = A_k - 1$ proves (231). Tao’s source formula is the same series with $A_k = a_{[1,k]}$ [32, source lines 384–393, 436–440]. Modulo 3^n , every term with $k > n$ vanishes. Expanding the source definition of F_n after the displayed reversal $(a_1, \dots, a_n) \mapsto (a_n, \dots, a_1)$ gives the middle expression in (232). Reversal preserves the iid product law, which proves the final law statement without identifying two deterministic functions that the source defines only in distribution. $\square$

This proves, with sample space, measure, coordinate order, inverse, fibres, and projections exposed, the equality of laws stated in Siegel v3 source lines 3180–3184. It does not identify the shortened and accelerated maps or their clocks. The same indexed Siegel source contains no first-passage interval, Proposition 5.2 event chain, or Section 5 function c_n ; it is a coordinate source for the later Fourier layer, not a repair of that argument. The exact version and source audit is `qa/SIEGEL-2007-V1-V3-F023-syracuse-crosswalk-audit.md`; the finite exact certificate is `certificates/siegel_v1_tao_syracuse_crosswalk_checks.py`.

Source issue 2.133 (current Siegel v3 Fourier presentation). The targeted v3 passage has three forced local defects that remain quarantined from later use. Source line 3146 writes $\hat{\mathbb{Z}}_2$ where the denominators and the declared characteristic-function domain require $\hat{\mathbb{Z}}_3$. With the source convention $\varphi_3(t) = \mathbb{E}e^{-2\pi i\{tX\}_3}$, line 3144’s inversion exponential requires the opposite sign; its printed sign reconstructs the mass at $-m$. Already modulo three it swaps the masses $1/3$ at 1 and $2/3$ at 2. Finally, line 3164’s last characteristic function omits its subscript 3. None affects Proposition 2.132, which was proved directly from the coordinate series.

Source issue 2.134 (current v7 proof-text audit). A line-level audit of the pinned v7 source records several forced local corrections: a function/value omission and the printed $N - 1$ in place of the induction index $n - 1$ in Lemma 2.1; a scalar/vector mismatch and an $n^{-1/2}$ factor where a d -dimensional Gaussian integral gives $n^{-d/2}$ in Lemma 2.2; undefined or inconsistent indices in the proofs of Proposition 1.9 and Proposition 5.2; a first-passage location written where a first-passage time is required; an unsigned $\exp(O(\log^{0.6} x))$ used as a lower bound; the malformed expression $q^{-1} \log O(M_1/M_0)$ where the preceding calculation gives $q^{-1}(1 + \log(M_1/M_0))$; and malformed formulas in the black-set and renewal arguments. The v7 text does correct an older time/location interchange at the end of Proposition 5.2 and an older complemented-event mismatch in Section 6. The latter is forced: the factorized Fourier coefficient contains $1_{E_k \cap B_k \cap C_{k,l}}$, so v5’s replacement by $1_{\bar{E}_k \cap B_k \cap C_{k,l}}$ has no equality or domination behind it.

The complete v5–v7 comparison has 28 non-whitespace hunks. The journal agrees with v7 on 15 distinguishing hunks, with v5 on 12, and is hybrid on the remaining Lemma 7.9 propagation hunk; it is not a uniform intermediate version. In particular, its equations (6.7)–(6.8) retain the v5 reserve, while equation (6.11) contains the v7 event correction.

The reserve change is substantive. With $a = \log(4/3)$ and $b = \log 2$, the optimized fluctuation cost is

$$\frac{b^2}{4a} C_A^2 \log n = 0.4175208154 \dots C_A^2 \log n.$$

The v5/journal half-window retains only $\frac{1}{2} b C_A^2 \log n = 0.3465735903 \dots C_A^2 \log n$, and its printed conversion also drops the factor b ; an explicit admissible family makes one claimed sub- 3^n summand exceed 3^n . V7's $0.99b C_A^2 \log n$ reserve closes with exact remaining coefficient $0.2686948933 \dots$. The stopping-time derivation and Archimedean separation are proved in Source Issue 2.138 and Proposition 2.142. This identifies a failure of the printed v5 argument, not a proof that its stated corollary is false by every method.

Two issues are not merely typographical. Lines 758–763 try to infer $T_x(\mathbf{N}_y) \in I_y$ from an additive endpoint restriction $\mathbf{N}_y \in [y + 2 \log^{0.8} x, y^\alpha - 2 \log^{0.8} x]$. Since the preceding time estimate depends on $\log(\mathbf{N}_y/x)$, this additive displacement does not supply a margin of order $\log^{0.8} x$. The following proposition proves a replacement and fixes its exact downstream boundary. In Lemma 5.3, lines 909–914 replace summable terms by $2^{-(a_1 + \dots + a_{m_0})/2}$ and then call the resulting product uniformly bounded. Its exact sum is $(1 + \sqrt{2})^{m_0}$, which grows with x ; a second proposition below retains the stronger pre-weakening estimates and proves the uniform bound. Separately, the Lemma 7.9 Markov event must be intersected with $\{R \leq \mathbf{r}\}$; the next line works under exactly that restriction. The complete v7 renewal dependency is reconstructed and certified at its declared scope in Corollary 2.148, while the stronger v5/journal form remains outside that proof. These findings neither enlarge the theorem nor constitute a claim that its conclusion is false.

At the final proof step, v7 line 1851 and journal p. 54 name $E_{p, 4^A(1+p^3)}$, whereas E_* was defined using $E_{p, 4^A(1+p)^3}$ and the ensuing inequality again uses $4^A(1+p)^3$. The edition retains the definitionally supported latter event and does not assert an identity between the two parameters. The terminal boundary, all 28 source hunks, the journal matrix, and the active citation census are recorded in `qa/TA0-V5-V7-JOURNAL-F028-whole-version-citation-audit.md`.

Proposition 2.135 (Multiplicative endpoint repair for v7 Proposition 5.2). *Retain the source constants*

$$\alpha = 1.001, \quad \lambda = \log(4/3), \quad L = \log x,$$

let $y \in \{x^\alpha, x^{\alpha^2}\}$, and let $\mathbf{N}_y$ have the source's logarithmic distribution on the odd integers in $[y, y^\alpha]$. Let

$$\mathcal{G}_x = \left\{ \vec{a}^{(n_0)}(\mathbf{N}_y) \in \mathcal{A}^{(n_0)} \right\}$$

be the good valuation event in source lines 733–740, and retain the source interval

$$I_y = \left[\frac{\log(y/x)}{\lambda} + L^{0.8}, \frac{\log(y^\alpha/x)}{\lambda} - L^{0.8} \right].$$

For all sufficiently large x , the interval

$$R_y = \left[y \exp(L^{0.8}), y^\alpha \exp(-L^{0.8}) \right]$$

is nonempty and

$$\mathcal{G}_x \cap \{\mathbf{N}_y \in R_y\} \subseteq \{T_x(\mathbf{N}_y) \in I_y\}.$$

Uniformly for the two displayed values of y ,

$$\mathbb{P}(\mathbf{N}_y \notin R_y) = O(L^{-1/5}) \quad \text{and hence} \quad \mathbb{P}(T_x(\mathbf{N}_y) \in I_y) = 1 - O(L^{-1/5}).$$

Proof. On $\mathcal{G}_x$, source lines 741–752 give, with one absolute constant K_0 uniform in the two choices of y ,

$$\left| \log \frac{\text{Syr}^j(\mathbf{N}_y)}{\mathbf{N}_y} + j\lambda \right| \leq K_0 L^{0.6} \quad (0 \leq j \leq n_0). \quad (233)$$

This estimate reaches the threshold before n_0 . Indeed, $\mathbf{N}_y \leq x^{\alpha^3}$ and

$$\log \frac{\text{Syr}^{n_0}(\mathbf{N}_y)}{x} \leq (\alpha^3 - 1)L - \lambda n_0 + K_0 L^{0.6}.$$

Here $\alpha^3 - 1 < 1/300$, whereas

$$\frac{\lambda}{10 \log 2} > \frac{1}{40} : \quad \log(4/3) > \frac{1/3}{1 + 1/3} = \frac{1}{4}, \quad \log 2 < 1.$$

Since $n_0 = \lfloor L/(10 \log 2) \rfloor$, the preceding upper bound is negative for all sufficiently large x . Thus $1 \leq T_x(\mathbf{N}_y) \leq n_0$ on $\mathcal{G}_x$. Apply (233) at $j = T_x(\mathbf{N}_y)$ and at $j = T_x(\mathbf{N}_y) - 1$. The defining inequalities

$$\text{Syr}^{T_x(\mathbf{N}_y)}(\mathbf{N}_y) \leq x < \text{Syr}^{T_x(\mathbf{N}_y)-1}(\mathbf{N}_y)$$

then give an absolute $K \geq K_0$ such that

$$\left| T_x(\mathbf{N}_y) - \frac{\log(\mathbf{N}_y/x)}{\lambda} \right| \leq K L^{0.6}. \quad (234)$$

Since $\lambda < 1$, the constant $\delta = 1/\lambda - 1$ is positive. If $\mathbf{N}_y \in R_y$, then

$$\begin{aligned} \frac{\log(\mathbf{N}_y/x)}{\lambda} - \left(\frac{\log(y/x)}{\lambda} + L^{0.8} \right) &\geq \delta L^{0.8}, \\ \left(\frac{\log(y^\alpha/x)}{\lambda} - L^{0.8} \right) - \frac{\log(\mathbf{N}_y/x)}{\lambda} &\geq \delta L^{0.8}. \end{aligned}$$

Because $\delta L^{0.8}/L^{0.6} = \delta L^{0.2} \rightarrow \infty$, both margins dominate the error in (234); this proves the event inclusion. The logarithmic width of R_y is $(\alpha - 1) \log y - 2L^{0.8} > 0$ for large x .

For $1 \leq a < b$, subtracting the even terms from the ordinary harmonic sum gives the uniform integral estimate

$$\sum_{\substack{a \leq m \leq b \\ m \equiv 1 \pmod{2}}} \frac{1}{m} = \frac{1}{2} \log \frac{b}{a} + O(1/a).$$

The denominator of the law of $\mathbf{N}_y$ is consequently $\frac{1}{2}(\alpha - 1) \log y + O(1/y)$, while the two discarded intervals have combined mass $L^{0.8} + O(1/y)$. Their probability is $O(L^{-0.2}) = O(L^{-1/5})$. Source line 737 gives $\mathbb{P}(\mathcal{G}_x^c) = O(L^{-10})$, and the last assertion follows. $\square$

The multiplicative restriction is used only to replace source lines 759–762. Every subsequent line through the end of the section at line 925 uses the unchanged interval I_y and the resulting probability estimate, not either endpoint restriction. In particular, lines 816–820 have an $L^{0.8}$ interval margin against an $O(L^{0.7})$ reconstruction error, and $\#(I_y \cap \mathbb{Z})$ in lines 841–845 is unchanged. One other local inequality in this path must be written with its sign exposed: on $\mathcal{G}_x$, for $0 \leq j \leq n - m_0$, there is an absolute K_1 such that

$$\text{Syr}^j(\mathbf{N}_y) \geq e^{-2K_1 L^{0.6}} (4/3)^{n-m_0-j} \text{Syr}^{n-m_0}(\mathbf{N}_y).$$

If the last factor lies in E' , its source lower bound is $e^{-L^{0.7}}(4/3)^{m_0}x$; since $m_0 = \lfloor (\alpha - 1)L/100 \rfloor$, the resulting lower bound is strictly larger than x for sufficiently large x . This is the typed replacement for the unsigned $\exp(O(L^{0.6}))$ comparison at lines 790–793. Likewise, the calculation immediately preceding source line 889 gives, when $q \leq M_0$,

$$\sum_{\substack{M_0 \leq M \leq M_1 \\ M \equiv a \pmod{q}}} \frac{1}{M} \leq \frac{1}{q} \left(1 + \log \frac{M_1}{M_0} \right),$$

which is the exact expression used in the following estimate.

Proposition 2.136 (completion of the v7 Section 5 internal argument). *Retain the notation of the pinned source and of Proposition 2.135. Put*

$$r = n - m_0, \quad G_x = \mathcal{G}_x = \left\{ \vec{a}^{(n_0)}(\mathbf{N}_y) \in \mathcal{A}^{(n_0)} \right\}.$$

For the chosen $E \subseteq 2\mathbb{N} + 1$, write

$$E' = \{M \in 2\mathbb{N} + 1 : T_x(M) = m_0, \text{ Pass}_x(M) \in E, \\ e^{-L^{0.7}}(4/3)^{m_0}x \leq M \leq e^{L^{0.7}}(4/3)^{m_0}x\}.$$

For $n \in I_y \cap \mathbb{Z}$, define

$$B_{n,y} = \{T_x(\text{Syr}^r(\mathbf{N}_y)) = m_0\} \\ \cap \{\text{Pass}_x(\text{Syr}^r(\mathbf{N}_y)) \in E\} \cap G_x.$$

Then the source event $B_{n,y}$ satisfies the exact equality

$$B_{n,y} = \{\text{Syr}^r(\mathbf{N}_y) \in E'\} \cap G_x. \quad (235)$$

For a fixed positive tuple $\vec{a} \in \mathcal{A}^{(r)}$ and fixed $M \in E'$, the final equivalence needed in the approximate formula is

$$\{\text{Syr}^r(\mathbf{N}_y) = M, \vec{a}^{(r)}(\mathbf{N}_y) = \vec{a}\} = \{\text{Aff}_{\vec{a}}(\mathbf{N}_y) = M\}. \quad (236)$$

In particular, the displayed approximate formula of source Proposition 5.2 follows with $n - m_0$ in place of the undefined $n - m$.

For $X \in \mathbb{Z}/3^r\mathbb{Z}$, define exactly as in source line 854

$$c_n(X) = 3^r \sum_{\substack{M \in E' \\ M \equiv X \pmod{3^r}}} \frac{1}{M}. \quad (237)$$

There is an absolute constant C such that

$$0 \leq c_n(X) \leq C \quad (238)$$

for every sufficiently large x , both choices of y , every $n \in I_y \cap \mathbb{Z}$, and every $X \in \mathbb{Z}/3^r\mathbb{Z}$.

To type the remaining dependency exactly, put

$$p_r(Y) = \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^r\mathbb{Z}) = Y).$$

For $n \in I_y$, the source parameter inequalities give $r = n - m_0 \geq m_0 \geq 1$ for all sufficiently large x , so the two levels below lie in the declared domain of the mixing proposition. The fine-scale-mixing input used by the source is, for every fixed $B > 0$,

$$\sum_{Y \bmod 3^r} \left| p_r(Y) - 3^{m_0-r} \sum_{\substack{Y' \bmod 3^r \\ Y' \equiv Y \pmod{3^{m_0}}} p_r(Y') \right| \ll_B m_0^{-B}. \quad (239)$$

After the indicator of $\mathcal{A}^{(r)}$ is removed by a correctly typed Chernoff estimate, the exact implication from (239) changes the expectation by $O_B(m_0^{-B})$. Its resulting main term collapses, without a further estimate, to the source quantity

$$Z = \sum_{M \in E'} \frac{3^{m_0}}{M} \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^{m_0}\mathbb{Z}) \equiv M \pmod{3^{m_0}}). \quad (240)$$

Proof. Assume first that $B_{n,y}$ holds. Thus $T_x(\text{Syr}^r(\mathbf{N}_y)) = m_0$, $\text{Pass}_x(\text{Syr}^r(\mathbf{N}_y)) \in E$, and G_x holds. Time $n = r + m_0$ is a crossing:

$$\text{Syr}^n(\mathbf{N}_y) \leq x < \text{Syr}^{n-1}(\mathbf{N}_y).$$

The good-orbit estimate used in (234) places both this crossing time and the actual first-passage time within $O(L^{0.6})$ of $\log(\mathbf{N}_y/x)/\log(4/3)$. Hence their difference is $O(L^{0.6})$. Since $m_0 = \lfloor L/100000 \rfloor$, this gives $T_x(\mathbf{N}_y) \geq n - m_0 = r$ for all sufficiently large x . The semigroup identity for first passage may therefore be used at time r , and gives

$$T_x(\mathbf{N}_y) = r + T_x(\text{Syr}^r(\mathbf{N}_y)) = n.$$

Writing $M = \text{Syr}^r(\mathbf{N}_y)$, the same signed good-orbit comparison between times r and n gives a fixed absolute K_2 . Moreover, the crossing at time n and the two adjacent good-prefix bounds imply $e^{-K_2 L^{0.6}} x \leq \text{Syr}^n(\mathbf{N}_y) \leq x$. Consequently,

$$e^{-K_2 L^{0.6}} (4/3)^{m_0} x \leq M \leq e^{K_2 L^{0.6}} (4/3)^{m_0} x.$$

Since $K_2 L^{0.6} \leq L^{0.7}$ for all sufficiently large x , this lies in the defining window for E' . The two other defining conditions for E' , namely $T_x(M) = m_0$ and $\text{Pass}_x(M) \in E$, are already part of $B_{n,y}$. Hence the right side of (235) holds.

Conversely, suppose that $M = \text{Syr}^r(\mathbf{N}_y) \in E'$ and G_x holds. The signed lower comparison displayed immediately before this proposition gives, for every $0 \leq j \leq r$,

$$\text{Syr}^j(\mathbf{N}_y) \geq e^{-2K_1 L^{0.6}} (4/3)^{r-j} M \geq e^{-2K_1 L^{0.6} - L^{0.7}} (4/3)^{n-j} x.$$

The exponent at $j = r$ is

$$m_0 \log(4/3) - 2K_1 L^{0.6} - L^{0.7},$$

which tends to $+\infty$ because $m_0 = \lfloor L/100000 \rfloor$; for $j < r$ the exponent is larger. Thus the displayed quantity is $> x$, and no passage has occurred by time r . The defining properties $T_x(M) = m_0$ and $\text{Pass}_x(M) \in E$ now give passage time n and passage location in E , proving the reverse inclusion in (235). Replacing the full good event by its prefix of length r changes a fixed- n probability by at most $O(L^{-10})$; summing over the $O(L)$ integers in I_y remains $O(L^{-9})$. Finally, the prefix-valuation uniqueness lemma says that $\text{Aff}_{\bar{a}}(\mathbf{N}_y) = M$, with M an odd positive

integer, is equivalent to the conjunction on the left of (236). This proves the repaired approximate formula.

It remains to prove (238). Write

$$m = m_0, \quad A = a_1 + \cdots + a_m, \quad b = a_m,$$

where every $a_i \geq 1$; the source's $\mathbb{N}^{m_0}$ at lines 862 and 865 must therefore be $(\mathbb{N} + 1)^{m_0}$. For $\vec{a} = (a_1, \dots, a_m)$ define

$$c_{n,\vec{a}}(X) = 3^r \sum_{\substack{M \in E' \\ M \equiv X \pmod{3^r} \\ \vec{a}^{(m)}(M) = \vec{a}}} \frac{1}{M}.$$

Uniqueness of the valuation prefix gives the exact decomposition

$$c_n(X) = \sum_{\vec{a} \in (\mathbb{N}+1)^m} c_{n,\vec{a}}(X).$$

For an M in the summand, first passage at time m gives

$$3^m 2^{-A} M + F_m(\vec{a}) \leq x < 3^{m-1} 2^{-(A-b)} M + F_{m-1}(a_1, \dots, a_{m-1}).$$

The coefficient in the second term is 3^{m-1} , not the printed 3^m . Since $0 \leq F_{m-1} \leq 3^{m-1} = o(x)$, the second inequality gives $M > 3^{-m} 2^{A-b} x$ for all sufficiently large x ; the first gives the upper bound below. Thus every such M lies in the definite interval

$$M_0 \leq M \leq M_1, \quad M_0 = 3^{-m} 2^{A-b} x, \quad M_1 = 3^{-m} 2^A x.$$

Multiplying the m -step affine identity by 2^A shows that M lies in one residue class modulo 2^{A+1} . Together with $M \equiv X \pmod{3^r}$, the Chinese remainder theorem gives one residue class modulo

$$q = 2^{A+1} 3^r.$$

If $2^A \leq x^{1/2}$, then $b \leq A$, $n \leq n_0$, and $3^n \leq x^{4/25}$ give

$$\frac{q}{M_0} = \frac{2^{b+1} 3^n}{x} \leq 2x^{-17/50} = o(1), \quad \frac{M_1}{M_0} = 2^b.$$

Thus $q \leq M_0$ for all sufficiently large x . The exact progression estimate already displayed above gives

$$c_{n,\vec{a}}(X) \ll b 2^{-A}. \tag{241}$$

If $2^A > x^{1/2}$, the exact last-step valuation identity is

$$b = \nu_2 \left(3 \left(3^{m-1} 2^{-(A-b)} M + F_{m-1}(a_1, \dots, a_{m-1}) \right) + 1 \right).$$

The inner coefficient 3^{m-1} corrects the second occurrence of the printed 3^m defect at source line 899. Put $H = 3^{m-1} 2^{-(A-b)} M$. The preceding lower bound gives $H > x - F_{m-1}$, hence $F_{m-1} = o(H)$. Because 2^b divides the positive integer $3(H + F_{m-1}) + 1$, one has

$$2^b \leq 3(H + F_{m-1}) + 1 \leq 5H$$

for all sufficiently large x . Therefore $M \geq (1/5)3^{1-m}2^A \gg 3^{-m}2^A$. Applying the unsimplified progression estimate with the larger of the two lower bounds yields

$$c_{n,\bar{a}}(X) \ll 3^n 2^{-A} + b 2^{-A}. \quad (242)$$

The b -terms in both branches have the exact total mass

$$\sum_{a_1, \dots, a_m \geq 1} b 2^{-A} = \left(\sum_{a \geq 1} 2^{-a} \right)^{m-1} \sum_{b \geq 1} b 2^{-b} = 2.$$

On the high- A branch,

$$\sum_{2^A > x^{1/2}} 2^{-A} \leq x^{-1/4} \sum_{a_1, \dots, a_m \geq 1} 2^{-A/2} = x^{-1/4} (1 + \sqrt{2})^m.$$

Since $3^5 < 2^8$, $1 + \sqrt{2} < e$, $n \leq n_0$, and $m \leq L/100000$,

$$3^n x^{-1/4} (1 + \sqrt{2})^m \leq x^{4/25 - 1/4 + 1/100000} = x^{-8999/100000}.$$

Equations (241) and (242) now prove the absolute uniform bound. This is the step missing from the printed summation: its weakened majorant has total mass $(1 + \sqrt{2})^m$, which is not uniformly bounded.

For the iid geometric tuple in source line 852, the same union–Chernoff calculation used earlier gives

$$\mathbb{P}((\mathbf{a}_1, \dots, \mathbf{a}_r) \notin \mathcal{A}^{(r)}) = O(L^{-10}).$$

This is the typed replacement for citing source line 737, which concerns the actual valuation tuple. Boundedness of c_n permits removal of the indicator with the same error. Applying the displayed sourced ℓ^1 input (239) and multiplying by c_n changes the expectation by $O_B(m_0^{-B})$. Projective consistency of the source variables gives, for each residue $X \bmod 3^{m_0}$,

$$\sum_{\substack{Y \equiv X \pmod{3^{m_0}} \\ Y \bmod 3^r}} p_r(Y) = \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^{m_0}\mathbb{Z}) = X);$$

this is source line 355, obtained by reducing the defining finite series. Therefore the main term is

$$\sum_{X \bmod 3^r} c_n(X) 3^{m_0-r} \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^{m_0}\mathbb{Z}) \equiv X \pmod{3^{m_0}}).$$

Substitution of (237) and interchange of the finite sums multiplies 3^r by 3^{m_0-r} , leaves one $X = M \bmod 3^r$ for every $M \in E'$, and gives (240) exactly. Taking, for example, $B = 12$ makes the uniform mixing-output error more than sufficient after the $O(L)$ passage-time sum. This last paragraph certifies the exact dependency use of the stated mixing estimate; it does not supply a proof of that estimate. $\square$

Thus Proposition 5.2 and the uniform c_n lemma are certified at their stated scope, and the Section 5 deduction is checked through source line 925. At this dependency boundary the fine-scale-mixing, Fourier, and renewal branch was still open; the next subsection reconstructs and discharges that branch without treating the durable whole-paper release gate as complete. In the source’s nonperiodicity remark, line 699 additionally requires an exponent representing $-\mathbf{m}$ modulo the least positive period, or a period multiple at least $\mathbf{m}$; the counting conclusion then remains unchanged. The earlier descent estimate also requires $1.9n_0$, not the undefined $1.9n$, at source line 689.

2.20 The fine-scale Fourier and renewal branch

This subsection reconstructs the branch left open after Proposition 2.136. The controlling text is the version-pinned arXiv v7 source, lines 926–1860 [32]. ArXiv v5 and the published paper are comparison manifestations, not silent replacements for v7. Exact source hashes, line-level defects, and the proof calculations abbreviated here are recorded in the F024 Fourier–renewal audit and the F028 whole-version audit in the external QA register.

Source issue 2.137 (version boundary in the many-triangles estimate). ArXiv v5, source lines 1654–1659, and the journal’s Lemma 7.9 state the unconditional estimate

$$\mathbb{E} \exp \left(- \sum_{p=1}^{t_{\min(r,R)}} 1_W(X_p) + \varepsilon \min(r, R) \right) \leq e^\varepsilon. \quad (243)$$

ArXiv v7, source lines 1660–1665, states only

$$\mathbb{E} 1_{\{r \geq R\}} \exp \left(- \sum_{p=1}^{t_R} 1_W(X_p) + \varepsilon R \right) \leq e^\varepsilon. \quad (244)$$

On $r \geq R$ the integrands agree, but the v7 formula deletes every path with $r < R$; the two statements are not identical. Moreover, the displayed v5/journal induction crosses the first triangle’s top after t_1 . On the event $r = 1 < R$, the left side of (243) stops at t_1 , so the additional factors used by that induction are absent. We therefore record the stronger statement as printed but do not use it. The weaker v7 statement is proved below.

Source issue 2.138 (Section 6 reserve is a substantive version repair). Write $a = \log(4/3)$, $b = \log 2$, and

$$q = \frac{b^2}{4a} = 0.4175208154 \dots$$

The optimized square-root fluctuation cost in Corollary 6.3 is $qC_A^2 \log N$. ArXiv v5 and journal equations (6.7)–(6.8) retain only $\frac{1}{2}bC_A^2 \log N = 0.3465735903 \dots C_A^2 \log N$; the v5 calculation also drops the factor b when converting 2^l to an exponential. Its printed size estimate therefore has the wrong leading sign. An explicit family satisfying the broad v5 concentration conditions makes one of its natural-number summands exceed 3^N ; the construction and its exact scope are recorded in F028.

V7 uses $0.99bC_A^2 \log N = 0.6862157088 \dots C_A^2 \log N$, and the stopping event supplies precisely that tightened window after enlarging C_A . More generally, a retained coefficient ρ closes this calculation exactly when

$$\rho > \frac{\log 2}{4 \log(4/3)} = 0.6023552099 \dots$$

The v5 failure is a defect in the printed proof. It neither produces two colliding coordinate tuples nor proves the separation corollary false by a different argument.

Proposition 2.139 (exact Section 6 reserve boundary). *Put*

$$a = \log(4/3), \quad b = \log 2, \quad \rho_* = \frac{b}{4a}.$$

Let $n \geq 2$, $L = \log n$, $C_A > 0$, and let $a_1, \dots, a_K$ be positive integers satisfying

$$a_{[1,j]} \geq 2j - C_A(\sqrt{jL} + L) \quad (1 \leq j \leq K). \quad (245)$$

If

$$l = a_{[1,K]} \leq \frac{n \log 3}{b} - \rho C_A^2 L$$

for a fixed $\rho > \rho_*$, then, after first choosing C_A sufficiently large in terms of $\rho - \rho_*$ and then n sufficiently large in terms of C_A, ρ ,

$$\sum_{j=1}^K 3^{j-1} 2^{l-a_{[1,j]}} < 3^n. \quad (246)$$

Here

$$\rho_* = 0.6023552099 \dots, \quad 0.99 > \rho_*.$$

For the v5 value $\rho = \frac{1}{2}$, the universal conclusion (246) is false even on sequences satisfying the source's all-interval concentration inequalities. By contrast, on the v7 stopping event $E_k \cap B_k$, $C_A \geq 600$ gives

$$a_{[1,k+1]} \leq \frac{n \log 3}{\log 2} - 0.99 C_A^2 \log n.$$

Proof. From (245),

$$S := \sum_{j=1}^K 3^{j-1} 2^{l-a_{[1,j]}} \ll 3^n e^{-\rho b C_A^2 L + b C_A L} \sum_{j \geq 1} e^{-aj + b C_A \sqrt{j} L}.$$

The exact completion of the square is

$$-aj + b C_A \sqrt{j} L = -a \left(\sqrt{j} - \frac{b C_A \sqrt{L}}{2a} \right)^2 + \frac{b^2}{4a} C_A^2 L.$$

Splitting the sum into unit intervals around its maximum, or comparing it with the corresponding Gaussian integral after $u = \sqrt{j}$, gives

$$\sum_{j \geq 1} e^{-aj + b C_A \sqrt{j} L} \ll (1 + C_A^2 L) e^{b^2 C_A^2 L / (4a)}.$$

The coefficient of $C_A^2 L$ in the resulting exponent is

$$-\rho b + \frac{b^2}{4a} = -b(\rho - \rho_*).$$

It is strictly negative under the stated hypothesis. It absorbs first the linear $b C_A L$ term after C_A is chosen, and then the polynomial factor after n is chosen. This proves $S < 3^n$.

For the v5 counterexample, set

$$\lambda = \frac{2a}{b}, \quad \gamma = \lambda^{-2}, \quad J = \lfloor \gamma C_A^2 L \rfloor,$$

and, for $1 \leq i \leq J$, put

$$d_i = \lfloor \lambda i \rfloor - \lfloor \lambda(i-1) \rfloor, \quad a_i = 2 - d_i;$$

put $d_i = 0$ and $a_i = 2$ afterwards. Thus $a_i \in \{1, 2\}$, and the deficit on every interval of length r is at most $\lambda r + 1$. For $r \leq J$, $\lambda r \leq C_A \sqrt{r} L$; for $r > J$, the entire deficit is at most $C_A \sqrt{r} L + O(1)$. Hence the source's additional $C_A L$ tolerance makes every printed interval inequality valid.

Let $D = \lfloor \lambda J \rfloor$. Choose l , with $l + D$ even, within two of

$$\frac{n \log 3}{b} - \frac{1}{2} C_A^2 L,$$

and put $K = (l + D)/2$. For sufficiently large n , $K \geq J$ and $a_{[1,K]} = 2K - D = l$. The $j = J$ summand satisfies

$$\log \frac{3^{J-1} 2^{l-a_{[1,J]}}}{3^n} = \left(\frac{b^2}{4a} - \frac{1}{2} b \right) C_A^2 L + O(1).$$

The coefficient is

$$0.4175208154 \dots - 0.3465735903 \dots = 0.0709472252 \dots > 0,$$

so that single summand eventually exceeds 3^n .

Finally put $T = n \log 3/b - C_A^2 L$. On B_k , $a_{[1,k]} \leq T < a_{[1,k+1]}$. The $i = k, j = k + 1$ inequality on E_k and $a_k \geq 1$ yield

$$a_{[1,k+1]} \leq T + 1 + C_A(\sqrt{L} + L).$$

For $n \geq 2$, the last overshoot is at most $6C_A L$, and for $C_A \geq 600$ this is at most $0.01C_A^2 L$. This gives the v7 0.99-window. $\square$

Explicit nonclaim 2.140. Proposition 2.139 refutes the universal Archimedean size assertion in the printed v5 proof, not the separation corollary by every possible proof. The construction supplies no pair of colliding tuples. The coefficient 0.99 is convenient rather than canonical, and the threshold ρ_* belongs to this exact optimization-and-absorption argument. No Fourier-decay theorem, main theorem, or Collatz endpoint follows from this finite margin calculation.

Definition 2.141 (uniform lifting and fibre averaging). For $0 \leq r \leq N$, let

$$L_{r \rightarrow N} : \mathbb{R}^{\mathbb{Z}/3^r \mathbb{Z}} \longrightarrow \mathbb{R}^{\mathbb{Z}/3^N \mathbb{Z}}, \quad (L_{r \rightarrow N} \mu)(y) = 3^{r-N} \mu(y \bmod 3^r).$$

For the reduction map $\pi_{N,r} : \mathbb{Z}/3^N \mathbb{Z} \rightarrow \mathbb{Z}/3^r \mathbb{Z}$, its fibre above x is $x + 3^r \mathbb{Z}/3^N \mathbb{Z}$. On signed measures on the larger group define the fibre-constant projection

$$P_{r,N} : \mathbb{R}^{\mathbb{Z}/3^N \mathbb{Z}} \longrightarrow \mathbb{R}^{\mathbb{Z}/3^N \mathbb{Z}}, \quad (P_{r,N} \nu)(y) = 3^{r-N} \sum_{\substack{z \bmod 3^N \\ z \equiv y \pmod{3^r}}} \nu(z).$$

Both maps preserve total mass, $P_{r,N}$ is an ℓ^1 -contraction, and $L_{r \rightarrow N}$ is an ℓ^1 -isometry. If μ_N is the law of Tao's N -Syracuse random variable, projective consistency says $(\pi_{N,r})_* \mu_N = \mu_r$. Substitution in the two displayed maps gives the typed identity

$$P_{r,N} \mu_N = L_{r \rightarrow N} \mu_r. \tag{247}$$

Proposition 2.142 (exact reduction from characteristic decay to fine-scale mixing). *For every $q \in \mathbb{N}_{>0}$, let μ_q be the law on $\mathbb{Z}/3^q \mathbb{Z}$ just defined. Assume the v7 characteristic-function estimate, uniformly for $\xi \in \mathbb{Z}/3^q \mathbb{Z}$ with $3 \nmid \xi$,*

$$\left| \mathbb{E} e^{-2\pi i \xi \text{Syrac}(\mathbb{Z}/3^q \mathbb{Z})/3^q} \right| \ll_B q^{-B} \tag{248}$$

for every fixed $B > 0$, with an implied constant depending only on B . After correcting source line 1051 from 3^q to 3^{q-1} , and line 1093 from m to $k + 1$, the source argument proves

$$\|\mu_N - L_{r \rightarrow N} \mu_r\|_1 \ll_A r^{-A} \quad (N, r \in \mathbb{N}_{>0}, 1 \leq r \leq N).$$

This is exactly Proposition 1.14, including its domain and uniformity.

Proof. First suppose $0.9N \leq r \leq N$. Let E be the source's all-interval concentration event. For a fixed k , let $E_k \supseteq E$ retain only the concentration inequalities involving $a_1, \dots, a_{k+1}$. The source's concentration estimate gives $\mathbb{P}(E_k \setminus E) = O_{A, C_A}(N^{-A-1})$, so the triangle inequality permits E to be replaced by E_k at the stated error scale. On E_k , stop at the unique k satisfying

$$a_{[1,k]} \leq \frac{N \log 3}{\log 2} - C_A^2 \log N < a_{[1,k+1]},$$

and condition also on $a_{[1,k+1]} = l$. Write B_k for this stopping event and $C_{k,l} = \{a_{[1,k+1]} = l\}$. Put

$$T_N = \frac{N \log 3}{\log 2} - C_A^2 \log N.$$

The $i = k, j = k + 1$ inequality in E_k , together with $a_k \geq 1$, gives

$$a_{k+1} \leq 1 + C_A(\sqrt{\log N} + \log N),$$

and hence

$$l = a_{[1,k+1]} \leq T_N + 1 + C_A(\sqrt{\log N} + \log N) \leq \frac{N \log 3}{\log 2} - 0.99C_A^2 \log N \quad (249)$$

after choosing $C_A \geq 600$ and then $N \geq 2$. The last displayed sufficient choice follows from $\sqrt{\log N} \leq 2 \log N$, $2 \leq 3 \log N$, and $6C_A \leq 0.01C_A^2$. The prefix and suffix offset maps then have the exact typed decomposition

$$X_N = F_{k+1}(a_{k+1}, \dots, a_1) + 3^{k+1}2^{-l}F_{N-k-1}(a_N, \dots, a_{k+2}) \pmod{3^N}.$$

The events $E_k, B_k, C_{k,l}$ and the prefix offset depend only on $a_1, \dots, a_{k+1}$, and hence are independent of the suffix variables. Let $\eta \in \mathbb{Z}/3^N\mathbb{Z}$ be a frequency surviving fibre averaging, so $\eta \notin 3^{N-r}\mathbb{Z}/3^N\mathbb{Z}$. There are unique

$$0 \leq v < N - r, \quad \bar{\eta}' \in (\mathbb{Z}/3^{N-v}\mathbb{Z})^\times,$$

where $v = \nu_3(\eta)$ and

$$\eta/3^v = 2^l \bar{\eta}' \pmod{3^{N-v}}.$$

Thus the quotient is asserted unique only in the correct quotient ring, not as a residue modulo 3^N . Put

$$q = N - k - v - 1.$$

The concentration range for k , together with $v < N - r \leq 0.1N$, gives

$$q \geq 0.9N - \frac{N \log 3}{2 \log 2} - O(C_A \sqrt{N \log N}) - 1 \gg N$$

for sufficiently large N . Let $\eta'_q \in (\mathbb{Z}/3^q\mathbb{Z})^\times$ be the reduction of $\bar{\eta}'$. Cancelling the exact powers of three and two in the suffix character gives the typed frequency morphism

$$\begin{aligned} & \mathbb{E} \exp \left(-\frac{2\pi i \eta 3^{k+1} 2^{-l} F_{N-k-1}(a_N, \dots, a_{k+2})}{3^N} \right) \\ &= \mathbb{E} \exp \left(-\frac{2\pi i \eta'_q F_{N-k-1}(a_N, \dots, a_{k+2})}{3^q} \right) \\ &= \mathbb{E} \exp \left(-\frac{2\pi i \eta'_q \text{Syrac}(\mathbb{Z}/3^q\mathbb{Z})}{3^q} \right). \end{aligned}$$

The final equality is projective consistency of the offset law from $N - k - 1$ to q , and (248) bounds this factor by $O_B(N^{-B})$, uniformly in every surviving frequency and admissible (k, l) .

Write $\mathcal{E}_{r,N;k,l}^2$ for the squared oscillation at this fixed pair (k, l) . Fourier inversion, removal of the frequencies divisible by 3^{N-r} , the preceding suffix bound, and Plancherel give

$$\mathcal{E}_{r,N;k,l}^2 \ll_B N^{-2B} 3^N \sum_y p_y^2, \quad (250)$$

where, with the source events $E_k, B_k, C_{k,l}$,

$$p_y = \mathbb{P}(F_{k+1}(a_{k+1}, \dots, a_1) \equiv y \pmod{3^N}, E_k \cap B_k \cap C_{k,l}).$$

Thus $(p_y)_y$ is the unnormalised restricted joint mass, not a conditional probability distribution.

For positive coordinates,

$$F_q(a_1, \dots, a_q) = \sum_{u=1}^q 3^{q-u} 2^{-a_{[u,q]}}.$$

The unique term of least 2-adic valuation is $3^{q-1} 2^{-a_{[1,q]}}$. Hence equality of two rational offsets forces equality of their total valuation sums. Subtraction gives equality of the two $(q-1)$ -coordinate tails, and induction proves that F_q has singleton fibres on $\mathbb{N}_{>0}^q$.

Under the fixed-total and concentration conditions, multiplying a congruence of two prefix offsets by 2^l gives a congruence modulo 3^N between two positive integers. Their prefix inequalities and (249) meet the hypotheses of Proposition 2.139 with $\rho = 0.99$. Hence both positive integers are strictly below 3^N . They are therefore equal as integers. Dividing by 2^l and applying rational injectivity proves residue-level separation. Thus

$$0 \leq p_y \leq 2^{-l}, \quad \sum_y p_y \leq 1,$$

and consequently

$$3^N \sum_y p_y^2 \leq 3^N (\max_y p_y) \sum_y p_y \leq 3^N 2^{-l} = N^{O(C_A^2)}.$$

Taking B sufficiently large in (248) absorbs this precise polynomial and closes (250) with an arbitrary prescribed N^{-2A-4} reserve. There are $O_A(N)$ admissible stopping indices k , and, from the displayed fixed-total window, $O_A(\log N)$ admissible totals l . Cauchy–Schwarz converts the squared bound back to the fixed-pair ℓ^1 -oscillation, and the union bound over these (k, l) uses less than the two reserved powers of N . The complement of the all-interval concentration event is $O_A(N^{-A-1})$ after enlarging C_A . Adding these disjoint fixed-pair contributions proves the restricted ℓ^1 estimate, rather than silently identifying a conditioned estimate with the unconditional one.

For general $10 \leq r \leq N$, construct $N = n_0 > n_1 > \dots > n_t = r$ by setting $n_{i+1} = \max(r, \lceil 0.9n_i \rceil)$. Because every $n_i \geq 10$ before termination, this sequence decreases strictly, and every adjacent pair is in the proved range. By (247), functoriality of the lifts, and preservation of the ℓ^1 -norm,

$$\begin{aligned} \|\mu_N - L_{r \rightarrow N} \mu_r\|_1 &\leq \sum_{i=0}^{t-1} \|L_{n_i \rightarrow N} \mu_{n_i} - L_{n_{i+1} \rightarrow N} \mu_{n_{i+1}}\|_1 \\ &= \sum_{i=0}^{t-1} \|\mu_{n_i} - L_{n_{i+1} \rightarrow n_i} \mu_{n_{i+1}}\|_1. \end{aligned}$$

Applying the restricted estimate with exponent $A + 1$, and summing the geometrically separated levels, gives $O_A(r^{-A})$. This supplies the lifting sequence implicit in source lines 932–936. When $1 \leq r < 10$, the triangle inequality gives a bound of two, which is $O_A(r^{-A})$ after enlarging the constant by at most 29^A . The finitely many N excluded while proving the restricted large- N estimate are absorbed in the same A -dependent constant. These two finite-scale clauses are separate from the strictly decreasing chain. $\square$

For $d \in \mathbb{N}_{>0}$, $t \in \mathbb{R}_{\geq 0}$, and $y \in \mathbb{R}^d$, we use Tao's Gaussian–exponential kernel

$$G_t(y) = \begin{cases} \exp(-\|y\|^2/t) + \exp(-\|y\|), & t > 0, \\ \exp(-\|y\|), & t = 0. \end{cases} \quad (251)$$

Lemma 2.143 (the corrected two-dimensional local factor). *In the proof of Tao's Lemma 2.2, the source-line-516 factor is $n^{-d/2}$, not $n^{-1/2}$. In particular, let $\mathbf{H}_1, \mathbf{H}_2, \dots$ be iid copies of the holding vector $\mathbf{H} \in \mathbb{Z}^2$ of Proposition 2.144, put $\mathbf{H}_{[1,q]} = \sum_{i=1}^q \mathbf{H}_i$ with the empty sum at $q = 0$, and recall $\mathbb{E}\mathbf{H} = (4, 16)$. There exist constants $c, C > 0$, depending only on the law of $\mathbf{H}$, such that for every $q \in \mathbb{N}_0$ and $(x, t) \in \mathbb{Z}^2$,*

$$\mathbb{P}(\mathbf{H}_{[1,q]} = (x, t)) \leq C(q+1)^{-1} G_q(c((x, t) - q(4, 16))).$$

Proof. After the source's contour shift, the Fourier integral contains $e^{-cn|u|^2}$ on a bounded subset of $\mathbb{R}^d$. The change of variables $v = \sqrt{n}u$ gives

$$\int e^{-cn|u|^2} du \leq n^{-d/2} \int_{\mathbb{R}^d} e^{-c|v|^2} dv.$$

All remaining source steps preserve this dimension. Setting $d = 2$ gives the displayed factor. $\square$

Proposition 2.144 (pairing, phase, and exact holding vector). *Let $n \in \mathbb{N}_{>0}$, let $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ satisfy $3 \nmid \xi$, and define*

$$\chi : \mathbb{Z}[1/2] \longrightarrow \mathbb{C}, \quad \chi(x) = \exp\left(-\frac{2\pi i \xi(x \bmod 3^n)}{3^n}\right).$$

Fix $0 < \varepsilon \leq e^{-10}$. For $(r, l) \in \mathbb{N}_{>0} \times \mathbb{Z}$, define the signed fractional phase

$$\theta(r, l) = \left\{ \frac{\xi 3^{2r-2} (2^{-l+1} \bmod 3^n)}{3^n} \right\} \in (-1/2, 1/2],$$

and define the typed partition of the strip

$$B = \{(r, l) : 1 \leq r \leq \lfloor n/2 \rfloor, |\theta(r, l)| \leq \varepsilon\}, \quad W = (\{1, \dots, \lfloor n/2 \rfloor\} \times \mathbb{Z}) \setminus B.$$

Let $(a_i)_{i \geq 1}$ be iid with $\mathbb{P}(a_i = u) = 2^{-u}$ for $u \in \mathbb{N}_{>0}$, and put

$$S_\chi(m) = \mathbb{E}\chi\left(\sum_{v=1}^m 3^{v-1} 2^{-\sum_{w=1}^v a_w}\right) \quad (m \in \mathbb{N}_0),$$

with the empty-sum convention $S_\chi(0) = 1$. For independent copies a_1, a_2 of this geometric variable, define

$$\begin{aligned} f : \mathbb{Z}[1/2] \times \{2, 3, \dots\} &\longrightarrow \mathbb{C}, \\ f(x, b) &= \mathbb{E}[\chi(x(2^{a_2} + 3)) \mid a_1 + a_2 = b], \\ g : \mathbb{Z}[1/2] &\longrightarrow \mathbb{C}, \\ g(x) &= \mathbb{E}\chi(x2^{-a_1}). \end{aligned}$$

Let $q = \lfloor n/2 \rfloor$, $b_r = a_{2r-1} + a_{2r}$, and $B_r = \sum_{u=1}^r b_u$, with $B_0 = 0$. Then the b_r are iid with $\mathbb{P}(b_r = b) = (b-1)2^{-b}$ for $b \geq 2$. Expectation in the next two displays is over this joint Pascal law, and conditioning on the b_r gives the exact formulas

$$S_\chi(2q) = \mathbb{E} \prod_{r=1}^q f(3^{2r-2} 2^{-B_r}, b_r)$$

and

$$S_\chi(2q+1) = \mathbb{E} \left[\prod_{r=1}^q f(3^{2r-2} 2^{-B_r}, b_r) g(3^{2q} 2^{-B_q}) \right].$$

At every white point (r, l) ,

$$|f(3^{2r-2} 2^{-l}, 3)| \leq e^{-\varepsilon^3}.$$

The occurrences $b_r = 3$ form a renewal process with iid holding vector

$$\mathbf{H} = (1, 3) + \sum_{i=1}^R (1, Q_i), \quad \mathbb{P}(R = s) = \frac{1}{4} \left(\frac{3}{4} \right)^s \quad (s \in \mathbb{N}_0),$$

where R is independent of the iid Q_i , and the Q_i have the Pascal law conditioned on $Q_i \neq 3$. This vector has exponential tail, full lattice span, and mean $(4, 16)$.

Proof. The pairing identities follow by conditioning while retaining the random odd- n terminal factor inside the expectation. Conditional on $b = 3$, the fibres $(a_1, a_2) = (1, 2), (2, 1)$ have equal mass. Therefore

$$f(x, 3) = \frac{\chi(5x) + \chi(7x)}{2} = \frac{\chi(5x)}{2} (1 + \chi(2x)).$$

For $x = 3^{2r-2} 2^{-l}$, the corrected character identity is $\chi(2x) = e^{-2\pi i \theta(r, l)}$. Since $\theta(r, l) \in (-1/2, 1/2]$,

$$|f(x, 3)| = \cos(\pi \theta(r, l)).$$

The elementary bound $\cos(\pi u) \leq e^{-2u^2}$ for $|u| \leq 1/2$ gives, at a white point,

$$|f(x, 3)| \leq e^{-2\varepsilon^2} \leq e^{-\varepsilon^3},$$

where the last inequality uses $0 < \varepsilon \leq e^{-10} < 2$.

The success probability is $\mathbb{P}(b = 3) = 1/4$, giving the displayed failure-before-success representation. For $k = (k_1, k_2) \in \mathbb{R}^2$, put $M_Q(k) = \mathbb{E} e^{k_1 + k_2 Q}$. On the open neighbourhood of zero on which $M_Q(k) < \infty$ and $3M_Q(k) < 4$, the normalized moment generating function is

$$\mathbb{E} e^{\mathbf{H} \cdot k} = \sum_{s=0}^{\infty} \frac{1}{4} \left(\frac{3}{4} \right)^s e^{(1,3) \cdot k} M_Q(k)^s = \frac{e^{(1,3) \cdot k}}{4 - 3M_Q(k)}. \quad (252)$$

It equals one at $k = 0$ and is finite near zero. This corrects the off-by-one series at source line 1408 and proves the exponential tail. The support contains $(1, 3), (2, 5), (2, 7), (2, 8)$; its differences generate $(0, 1)$ and $(1, 0)$, so it is contained in no coset of a proper subgroup of $\mathbb{Z}^2$. Finally, $\mathbb{E} Q = 13/3$, $\mathbb{E} R = 3$, and direct substitution gives $\mathbb{E} \mathbf{H} = (4, 16)$. $\square$

Proposition 2.145 (corrected black-triangle decomposition). *Retain $n, \xi, \varepsilon, \theta, B, W$ from Proposition 2.144. After the forced corrections at v7 lines 1272, 1285–1287, and 1311–1313, B is a disjoint union of lattice triangles*

$$\Delta = \{(j, l) : j \geq j_\Delta, l \leq l_\Delta, (j - j_\Delta) \log 9 + (l_\Delta - l) \log 2 \leq s_\Delta\}.$$

Here $j_\Delta \in \mathbb{N}_{>0}$, $l_\Delta \in \mathbb{Z}$, $s_\Delta \in \mathbb{R}_{\geq 0}$, and the displayed set is intersected with the lattice $\mathbb{N}_{>0} \times \mathbb{Z}$. With $\delta = \frac{1}{10} \log(1/\varepsilon)$, every triangle satisfies

$$\Delta \subseteq \{(j, l) \in \mathbb{N}_{>0} \times \mathbb{Z} : j \leq n/2 - \delta\},$$

and distinct triangles have Euclidean distance at least δ .

Proof. The exact recurrences are

$$\theta(j+1, l) \equiv 9\theta(j, l), \quad \theta(j, l-1) \equiv 2\theta(j, l) \pmod{\mathbb{Z}}.$$

Call a point weakly black when $|\theta| \leq 1/100$. The recurrences give three exact propagation rules. First, if (j, l) is weakly black and $(j+1, l)$ is black, then $|9\theta(j, l)| \leq 9/100 < 1/2$, so there is no wrap and $|\theta(j, l)| = |\theta(j+1, l)|/9 \leq \varepsilon$. The same argument with $|2\theta(j, l)| \leq 2/100 < 1/2$ applies when $(j, l-1)$ is black. Second, if $(j+1, l)$ and $(j, l-1)$ are weakly black, then

$$|\theta(j, l)| \leq |\theta(j+1, l)| + 4|\theta(j, l-1)| \leq 5/100;$$

now $|2\theta(j, l)| \leq 1/10 < 1/2$, so the doubling recurrence has no wrap and $|\theta(j, l)| = |\theta(j, l-1)|/2 \leq 1/200$. Third, if $(j-1, l)$ and $(j, l-1)$ are weakly black, the factor-nine recurrence first gives $|\theta(j, l)| \leq 9/100$; hence $|2\theta(j, l)| \leq 18/100 < 1/2$, and doubling again gives $|\theta(j, l)| \leq 1/200$.

If (j, l) is black, multiplying its phase to the primitive residue at level 3^n gives

$$3^{n+1-2j}\varepsilon \geq \frac{1}{3}.$$

Taking logarithms gives

$$j \leq \frac{n}{2} + 1 - \frac{\log(1/\varepsilon)}{2 \log 3} \leq \frac{n}{2} - \frac{1}{10} \log(1/\varepsilon)$$

for $\varepsilon \leq e^{-10}$. There is no infinite upward black run: if (j, l') were black for every $l' \geq l$, repeated nonwrapping doubling would give $|\theta(j, l')| \leq 2^{l-l'}\varepsilon$, contradicting the same primitive residue lower bound as $l' \rightarrow \infty$.

Starting from a black (j, l) , let $l_* \geq l$ be the unique top of its consecutive black vertical run, and let $j_* \leq j$ be the unique left end of the consecutive black run on row l_* . Existence and uniqueness follow from the preceding no-infinite-run statement and the lower boundary $j = 1$. Because $2j_* - 2 < n$, while neither ξ nor 2 is divisible by three, the defining rational phase is not integral; hence $\theta(j_*, l_*) \neq 0$. Write $|\theta(j_*, l_*)| = \varepsilon e^{-s_*}$. On the corrected domain $j' \geq j_*$, $l' \leq l_*$, multiplication by $9^{j'-j_*} 2^{l_*-l'}$ gives

$$|\theta(j', l')| \leq \varepsilon e^{-s_* + (j'-j_*) \log 9 + (l_*-l') \log 2},$$

with equality before the right side reaches $1/2$. Thus all points of the anchored triangle are black. If an exterior Case 1 point is within δ , and $X = -s_* + (j' - j_*) \log 9 + (l_* - l') \log 2$, then

$$\varepsilon < \varepsilon e^X \leq \varepsilon^{1 - (\log 9 + \log 2)/10} < \frac{1}{2},$$

so $|\theta(j', l')| = \varepsilon e^X > \varepsilon$. The corrected Case 2 argument is as follows. If $l' > l_*$ and a nearby exterior point (j', l') were black, Euclidean distance at most δ gives

$$0 < (l' - l_*) \log 2 \leq \delta \log 2, \quad (j' - j_*) \log 9 \leq s_* + \delta \log 9.$$

Exact downward phase transport then gives

$$|\theta(j', l_* + 1)| \leq \varepsilon^{1 - \log 2/10} < 1/100.$$

When $j' > j$, the anchored phase bound also gives

$$|\theta(j' - 1, l_*)| \leq \varepsilon^{1 - \log 9/10} < 1/100.$$

The second propagation rule iterates left from $j' - 1$ to j . If $j' = j$, $(j, l_* + 1)$ is already the weakly black point supplied by the first display. Thus in either case $(j, l_* + 1)$ is weakly black, and blackness of (j, l_*) makes it black by the first rule, contradicting the definition of l_* . When $j' < j$, the black row from j_* through j and the third rule instead propagate weak blackness right from $(j', l_* + 1)$ to $(j, l_* + 1)$, giving the same contradiction.

In Case 3, $j' < j_*$. The same distance condition gives

$$|l' - l_*| \log 2 \leq s_* + \delta \log 2, \quad 0 < (j_* - j') \log 9 \leq \delta \log 9.$$

Assuming (j', l') black, exact horizontal phase transport gives

$$|\theta(j_* - 1, l')| \leq \varepsilon^{1 - \log 9/10} < 1/100.$$

If $l' \geq l_*$, vertical transport gives the explicit bound

$$|\theta(j_* - 1, l_*)| \leq \varepsilon^{1 - (\log 9 + \log 2)/10} < 1/100.$$

If $l' < l_*$, the anchored bound gives $|\theta(j_*, l' + 1)| \leq \varepsilon^{1 - \log 2/10} < 1/100$, and the second rule propagates upward until $(j_* - 1, l_*)$ is weakly black. In either subcase, the first propagation rule and blackness of (j_*, l_*) make $(j_* - 1, l_*)$ black, contradicting the defining minimality of j_* .

Hence every exterior point within δ is white. Since $\delta \geq 1$, the black horizontal and vertical paths defining the anchor cannot cross the boundary: the first black path point outside the triangle would be an exterior black point within distance one. Thus every original black point lies in its anchored triangle.

Conversely, every point of this triangle reconstructs the same anchor. For each column occurring in the triangle, the entire vertical segment up to row l_* is black and the point at height $l_* + 1$ is an exterior collar point, hence white. On row l_* , the entire horizontal segment back to j_* is black and, unless $j_* = 1$, the point at $j_* - 1$ is an exterior collar point, hence white. Therefore its reconstructed pair is exactly (j_*, l_*) . The anchored triangles partition B , and the white collar makes distinct members δ -separated.

Confinement of the entire abstract triangle follows directly from the anchor denominator, without presupposing confinement of its lattice points. The nonzero signed phase satisfies

$$|\theta(j_*, l_*)| \geq 3^{2j_* - 2 - n}.$$

Since its other expression is εe^{-s_*} ,

$$s_* \leq (n - 2j_* + 2) \log 3 - \log(1/\varepsilon).$$

Consequently its real right tip satisfies

$$j_* + \frac{s_*}{\log 9} \leq \frac{n}{2} + 1 - \frac{\log(1/\varepsilon)}{2 \log 3} \leq \frac{n}{2} - \delta.$$

Every point of the triangle lies weakly to the left of that tip, proving the asserted confinement and, in particular, the real-tip bound needed later. $\square$

Lemma 2.146 (first-passage Green kernel). *Let $(\mathbf{H}_i)_{i \geq 1}$ be iid copies of $\mathbf{H}$, let $\mathbf{V}_q = \sum_{i=1}^q \mathbf{H}_i$, and set $\mathbf{V}_0 = (0, 0)$. For $s \in \mathbb{N}_0$, put $\kappa_s = \inf\{q \in \mathbb{N}_{>0} : (\mathbf{V}_q)_2 > s\}$. For every $j, l \in \mathbb{Z}$ with $l > s$,*

$$\mathbb{P}(\mathbf{V}_{\kappa_s} = (j, l)) \ll \frac{e^{-c(l-s)}}{\sqrt{1+s}} G_{1+s}(c(j-s/4)).$$

Moreover, there is an absolute $C \geq 1$ such that the following typed post-passage estimate holds. Let $m \in \mathbb{N}_{>0}$ be sufficiently large, let $s, p \in \mathbb{N}_0$, let $r_0 \in [1, \infty)$, and let $h > 0$. Suppose

$$s > \frac{m}{\log^2 m}, \quad 0 \leq p \leq m^{0.1}, \quad 4r_0 \leq h \leq m^{0.4},$$

and let $\Sigma \subset \mathbb{Z}$ have pairwise spacing at least h . Then, for all sufficiently large m ,

$$\mathbb{P}(\text{dist}((\mathbf{V}_{\kappa_s+p})_1, \Sigma) \leq r_0) \leq C \frac{r_0}{h}.$$

Proof. Set $U(x, t) = \sum_{q \geq 0} \mathbb{P}(\mathbf{V}_q = (x, t))$ for $(x, t) \in \mathbb{Z} \times \mathbb{N}_0$. By Lemma 2.143,

$$\mathbb{P}(\mathbf{V}_q = (x, t)) \ll (q+1)^{-1} G_q(c((x, t) - q(4, 16))).$$

Writing $z = x - t/4$, the invertible coordinate identity

$$(x - 4q, t - 16q) = \left(z + \frac{t - 16q}{4}, t - 16q \right)$$

preserves norms up to two absolute constants. Put $\langle t \rangle = 1 + t$, and, after harmlessly weakening the denominator q in the Gaussian to $q + 1$, set

$$A_q(z, t) = \frac{1}{q+1} \exp \left(-c \frac{z^2 + (t - 16q)^2}{q+1} \right).$$

In the central range the complete Gaussian calculation is

$$\sum_{t/2 \leq 16q \leq 2t} A_q(z, t) \ll \langle t \rangle^{-1/2} e^{-c' z^2 / \langle t \rangle}.$$

Indeed $q + 1 \asymp \langle t \rangle$ there and the discrete Gaussian in $t - 16q$ has mass $O(\sqrt{\langle t \rangle})$. In the lower range, $|t - 16q| \gg t$ and $q + 1 \ll \langle t \rangle$; in the upper range use

$$(q+1) + \frac{z^2}{q+1} \gg \langle t \rangle + \min \left(\frac{z^2}{\langle t \rangle}, |z| \right).$$

Geometric summation therefore gives, respectively,

$$\sum_{16q < t/2} A_q(z, t) \ll \langle t \rangle^{-1/2} e^{-c' z^2 / \langle t \rangle}, \quad \sum_{16q > 2t} A_q(z, t) \ll \langle t \rangle^{-1/2} \left(e^{-c' z^2 / \langle t \rangle} + e^{-c' |z|} \right).$$

The exponential half of G_q is even smaller, since

$$\sum_{q \geq 0} \frac{e^{-c(|z|+|t-16q|)}}{q+1} \ll \langle t \rangle^{-1/2} e^{-c'|z|}.$$

The cases $t = 0, 1$ are absorbed by changing the absolute constants. Thus

$$U(x, t) \ll (1+t)^{-1/2} G_{1+t}(c'(x-t/4)).$$

At first passage the final vertical increment is at least $l-s$. Using its exponential tail and independence from the earlier increments gives

$$\mathbb{P}(\mathbf{V}_{\kappa_s} = (j, l)) \ll e^{-c(l-s)} \sum_{r=0}^s \sum_{u \geq 1} e^{-c(r+u)} U(j-u, s-r).$$

For

$$K_t(y) = (1+t)^{-1/2} \left(e^{-cy^2/(1+t)} + e^{-c|y|} \right),$$

the two needed convolution inequalities, with absolute $a, c_1, c_2 > 0$, are

$$\sum_{r=0}^s e^{-ar} K_{s-r}(x+r/4) \ll K_s(c_1 x), \quad (253)$$

$$\sum_{u \geq 1} e^{-au} K_s(x-u) \ll K_s(c_2 x). \quad (254)$$

For (253), split at $r = s/2$. In the first part the two variances are comparable and

$$ar + c \frac{(x+r/4)^2}{1+s-r} \geq c' \frac{x^2}{1+s} + c'r;$$

in the second part the factor e^{-ar} , together with the retained $x+r/4$ decay, obeys the explicit bound

$$ar + c \frac{(x+r/4)^2}{1+s-r} \geq c' \left(s + \min \left(\frac{x^2}{1+s}, |x| \right) \right) \quad (r > s/2).$$

Since $\sum_{r > s/2} (1+s-r)^{-1/2} e^{-c's} \ll (1+s)^{-1/2}$, this absorbs the lost square-root prefactor. For the exponential-kernel half, the corresponding inequalities are

$$ar + c|x+r/4| \geq c'(|x|+r) \quad (r \leq s/2), \quad ar + c|x+r/4| \geq c'(|x|+s) \quad (r > s/2).$$

For (254), split at $u = |x|/2$. In the first range $|x-u| \geq |x|/2$ whenever $x \geq 0$, while for $x < 0$ it holds for every $u \geq 1$. In the complementary range the geometric sum of e^{-au} is $O(e^{-a|x|/2})$. These statements apply separately to the Gaussian and exponential halves of K_s and prove the second displayed inequality. In the actual summand $U(j-u, s-r)$, the kernel coordinate is $j-s/4-u+r/4$. Thus first substitute $x = j-s/4-u$ in (253); after that summation, substitute $x = j-s/4$ in (254). This proves the first-passage display and fills the unexpanded calculation at source lines 1457–1463.

For the post-passage assertion, first sum the displayed first-passage bound over its vertical coordinate. The horizontal mass function is bounded by $CK_s(x-s/4)$, where

$$K_s(y) = (1+s)^{-1/2} \left(e^{-cy^2/(1+s)} + e^{-c|y|} \right).$$

If Σ' is h -separated and $r_0 \leq h/4$, the intervals of radius r_0 about its points are disjoint. Since $r_0 \geq 1$, direct summation of the Gaussian and exponential pieces gives

$$\sum_{\text{dist}(x, \Sigma') \leq r_0} K_s(x - s/4) \leq C \left(\frac{r_0}{\sqrt{1+s}} + \frac{r_0}{h} \right).$$

Here $h \leq m^{0.4} = o(\sqrt{s})$, because $s > m/\log^2 m$, so the right side is at most Cr_0/h . Conditional on the sum of the p later horizontal increments, the target set is merely a translate of Σ and has the same spacing. Averaging the uniform bound proves the assertion. The condition $h \leq m^{0.4}$ is essential; without it a singleton at the kernel peak would contradict a bare $O(r_0/h)$ claim. $\square$

Proposition 2.147 (repaired renewal monotonicity). *Retain n, ξ and the black-white partition from Proposition 2.145. There is an absolute $\varepsilon_0 > 0$, fixed in the exit-rectangle calculation below, and we assume $0 < \varepsilon \leq \varepsilon_0$. Extend 1_W by zero off $[n/2] \times \mathbb{Z}$. For $x \in \mathbb{Z}^2$, let*

$$Q(x) = \mathbb{E}_x \prod_{q \geq 0} e^{-\varepsilon^3 1_W(X_q)}, \quad X_q = x + \mathbf{V}_q,$$

and define

$$Q_m = \sup_{\substack{j \in \mathbb{N}_{>0}, \\ j \geq \lfloor n/2 \rfloor - m \\ l \in \mathbb{Z}}} \max(\lfloor n/2 \rfloor - j, 1)^A Q(j, l).$$

For every fixed sufficiently large A , there is an explicit finite threshold $C_{A, \varepsilon}$ such that

$$Q_m \leq Q_{m-1} \quad (m \in \mathbb{N}_{>0}, C_{A, \varepsilon} \leq m \leq \lfloor n/2 \rfloor).$$

The proof uses the weaker v7 Lemma 7.9 and does not assume (243).

Proof. If $m = \lfloor n/2 \rfloor$, the restrictions defining Q_m and Q_{m-1} both reduce to $j \in \mathbb{N}_{>0}$, so the asserted inequality is an equality. Hence suppose $m \leq \lfloor n/2 \rfloor - 1$. It then suffices to treat the genuine boundary row $j = \lfloor n/2 \rfloor - m \in \mathbb{N}_{>0}$. If (j, l) is white, the recursion

$$Q(j, l) = e^{-\varepsilon^3} \mathbb{E} Q((j, l) + \mathbf{H})$$

and

$$\max(m - r, 1)^{-1} \leq m^{-1} \exp \left(Cr \frac{\log m}{m} \right)$$

show that the immediate white factor absorbs the horizontal exponential moment for sufficiently large m . This is source Case 1.

Suppose next that (j, l) lies at depth $0 \leq s \leq m/\log^2 m$ in a black triangle. Stop at κ_s and apply the recursion once more than the source does:

$$Q(j, l) = \mathbb{E} \left[e^{-\varepsilon^3 \sum_{q=0}^{\kappa_s} 1_W(X_q)} Q(X_{\kappa_s+1}) \right].$$

The definition of Q_{m-1} now applies at the terminal point. With $\lambda = CA \log m/m$,

$$Q(j, l) \leq m^{-A} Q_{m-1} \mathbb{E} \left[e^{-\varepsilon^3 1_W(X_{\kappa_s})} e^{\lambda (\mathbf{V}_{\kappa_s+1})_1} \right].$$

Lemma 2.146 gives $\mathbb{E} e^{\lambda (\mathbf{V}_{\kappa_s})_1} = 1 + O_A(1/\log m)$. We now prove, rather than assume, the uniform exit probability. Summing that lemma first over $l - s > M_v$, and then over $|j - s/4| > M_h \sqrt{1+s}$, gives

$$\mathbb{P} \left(1 \leq (\mathbf{V}_{\kappa_s})_2 - s \leq M_v, |(\mathbf{V}_{\kappa_s})_1 - s/4| \leq M_h \sqrt{1+s} \right) \geq \frac{1}{2}$$

once the two absolute integers M_v, M_h are chosen sufficiently large. Write $J = (\mathbf{V}_{\kappa_s})_1$ and $L = (\mathbf{V}_{\kappa_s})_2$. In this rectangle $1 \leq L - s \leq M_v$, so the exit point is above the top row by at most M_v . Horizontally,

$$j + J - j_\Delta \geq \frac{s}{4} - M_h \sqrt{1+s} \geq -C_-(M_h),$$

while the triangle inequality and $J \leq s/4 + M_h \sqrt{1+s}$ give

$$\begin{aligned} (j + J - j_\Delta) \log 9 &\leq s_\Delta - s \log 2 + \frac{s \log 9}{4} + M_h \log 9 \sqrt{1+s} \\ &\leq s_\Delta + C_+(M_h). \end{aligned}$$

The last constants are finite because $\log 2 - \frac{1}{4} \log 9 > 0$, so a negative linear term absorbs the square-root term. Thus the horizontal coordinate lies within an absolute distance of the top-row segment $[j_\Delta, j_\Delta + s_\Delta / \log 9]$. Combining the horizontal and vertical bounds gives a Euclidean distance $C_{\text{exit}} = C(M_v, M_h)$ from the triangle. Choose the global $\varepsilon_0 \leq e^{-10}$ so that

$$\frac{1}{10} \log(1/\varepsilon_0) > C_{\text{exit}} + 1.$$

The white collar then gives $\mathbb{P}(X_{\kappa_s} \in W) \geq c_*$ with $c_* = 1/2$, uniformly in the triangle, its depth, n , and ξ .

Strong Markov factors the final holding increment. Its exponential moment is $1 + O_A(\log m/m)$. Since the exponential weight accumulated before that increment is at least one, the remaining expectation is at most

$$1 + O_A(1/\log m) - c_*(1 - e^{-\varepsilon^3}),$$

before multiplication by the final $1 + o(1)$ factor, and is at most one for $m \geq C_{A,\varepsilon}$. This repairs the unsupported source line 1556.

It remains to treat a starting point at depth $s > m/\log^2 m$. We first prove the two exact auxiliary bounds used in this case. Membership in the initial triangle and the real right-tip bound give the exact upper range

$$\frac{m}{\log^2 m} < s \leq \frac{\log 9}{\log 2} m. \quad (255)$$

Put $x = (j, l)$, retain the first-passage time κ_s , and reset the post-exit process by

$$\tilde{X}_p = X_{\kappa_s+p} = x + \mathbf{V}_{\kappa_s+p}, \quad \tilde{\mathbf{V}}_p = \mathbf{V}_{\kappa_s+p} - \mathbf{V}_{\kappa_s} \quad (p \in \mathbb{N}_0).$$

Conditionally on $\mathcal{F}_{\kappa_s}$, the increments of $\tilde{\mathbf{V}}$ are iid copies of $\mathbf{H}$ and are independent of $\tilde{X}_0$.

We first prove the v7 many-triangles estimate uniformly for a renewal process $Z_p = z + \mathbf{V}_p$ started at an arbitrary $z \in \mathbb{Z}^2$. Define t_1 as its first black time, allowing $t_1 = 0$. After entry into the i -th triangle Δ_i , let $\sigma_i > t_i$ be the first time whose vertical coordinate exceeds l_{Δ_i} , and let $t_{i+1} \geq \sigma_i$ be the first subsequent black time. Let $r \in \mathbb{N}_0 \cup \{\infty\}$ be the number of entries. The exit-rectangle argument gives, on $r \geq i$,

$$\mathbb{P}_z(Z_{\sigma_i} \in W \mid \mathcal{F}_{t_i}) \geq c_*.$$

Set $\rho = 1 - (1 - e^{-1})c_* \in (0, 1)$, and decrease the fixed ε_0 once more so that $e^{\varepsilon_0} \rho \leq 1$. For $q \in \mathbb{N}_0$, define the uniformly typed contraction

$$H_q(z) = \mathbb{E}_z \left[1_{\{r \geq q+1\}} \prod_{i=1}^q e^{-1_W(Z_{\sigma_i})} \right].$$

Since $H_0(z) \leq 1$, strong Markov at the first exit gives inductively

$$\begin{aligned} H_q(z) &= \mathbb{E}_z \left[1_{\{r \geq 1\}} e^{-1w(Z_{\sigma_1})} H_{q-1}(Z_{\sigma_1}) \right] \\ &\leq \rho^{q-1} \mathbb{E}_z \left[1_{\{r \geq 1\}} e^{-1w(Z_{\sigma_1})} \right] \leq \rho^q. \end{aligned}$$

On $r \geq R$, the distinct exits $\sigma_1, \dots, \sigma_{R-1}$ occur no later than t_R ; hence the complete white count dominates their indicators and

$$\mathbb{E}_z 1_{\{r \geq R\}} e^{-\sum_{p=1}^{t_R} 1w(Z_p) + \varepsilon R} \leq e^{\varepsilon R} \rho^{R-1} \leq e^\varepsilon. \quad (256)$$

This proves the weaker v7 Lemma 7.9 with a defined time-zero atom and the required uniform induction.

We next reconstruct the parameterised large-triangle estimate. For $p \in \mathbb{N}_0$ and $s' > 0$, define the event

$$E_{p,s'} = \{\tilde{X}_p \in \Delta' \text{ for some } \Delta' \in \mathcal{T} \text{ with } s_{\Delta'} \geq s'\}.$$

Let $D \geq D_0$, where D_0 is a sufficiently large absolute constant. The first-passage overshoot bound and the exponential tail of the next p increments give

$$\mathbb{P}((\tilde{X}_p)_2 \geq l_\Delta + 2D(1+p)) \leq C e^{-cD(1+p)}.$$

The horizontal first-passage kernel and the same increment tail, using $p \leq m^{0.1}$, $s > m/\log^2 m$, and $s' \leq m^{0.4}$, give

$$\mathbb{P}\left(\left|(\mathbf{V}_{\kappa_s+p})_1 - \frac{s}{4}\right| > 2s^{0.6}\right) \leq C \frac{D(1+p)}{s'}.$$

Outside these two exceptional events, membership of $\tilde{X}_p$ in an eligible Δ' forces its lower tip to satisfy

$$l_\Delta - 10 < l_{\Delta'} - \frac{s_{\Delta'}}{\log 2} \leq l_\Delta + 2D(1+p). \quad (257)$$

For if the left inequality failed, moving horizontally by $O(D(1+p))$ on the row l_Δ would produce a lattice point of Δ' ; the horizontal bound and $\frac{1}{4} \log 9 < \log 2$ put the same point in the initial Δ , contradicting disjointness. The right inequality follows directly from the triangle inequality for $\tilde{X}_p \in \Delta'$. Substituting (257) back into that triangle inequality gives

$$|(\tilde{X}_p)_1 - j_{\Delta'}| \leq CD(1+p).$$

If Δ', Δ'' are two eligible triangles, intersecting each with the row $l_\Delta + \lfloor s'/2 \rfloor$ produces disjoint horizontal integer intervals of length at least $cs' - CD(1+p)$. Thus, provided $s' \geq C_1 D(1+p)$, their top-left coordinates are at least $c_1 s'$ apart. Decrease the separation constant once, if necessary, so that $0 < c_1 \leq 1$. Apply the post-passage part of Lemma 2.146 with neighbourhood radius $r_0 = CD(1+p)$ and spacing $h = c_1 s'$. Enlarging C_1 enforces $4r_0 \leq h$, and $s' \leq m^{0.4}$ enforces the smoothing-scale bound. Consequently there are absolute $C_0, c_0, C_1 > 0$ such that

$$\mathbb{P}(E_{p,s'}) \leq C_0 \left(\frac{D(1+p)}{s'} + e^{-c_0 D(1+p)} \right) \quad (258)$$

whenever

$$0 \leq p \leq m^{0.1}, \quad C_1 D(1+p) \leq s' \leq m^{0.4}.$$

This exposes the lower-tip and separated-top steps preceding the post- p convolution at source lines 1805–1812.

Now put $\delta_A = 10^{-A-2}$, and choose

$$D_A \geq \max \left(D_0, \frac{1}{c_0} \log \frac{6C_0}{\delta_A(1-e^{-c_0})} \right), \quad K_A \geq \max \left(C_1 D_A, \frac{6C_0 D_A \zeta(2)}{\delta_A} \right).$$

Set $H = 10A/\varepsilon^3$ and

$$R = \left\lceil \frac{H + \log 3 + (A+2) \log 10 + \varepsilon + 1}{\varepsilon} \right\rceil.$$

Let $h = \lceil H \rceil$, put $u_1 = h$, and define, for $1 \leq i < R$,

$$u_{i+1} = u_i + \left\lceil 10K_A(1+u_i)^3 \right\rceil + h + 1, \quad P = u_R + 2.$$

All of $D_A, K_A, H, R, h, u_i, P$ depend only on A, ε . Enlarge $C_{A,\varepsilon}$ so that

$$P \leq m^{0.1}, \quad K_A(1+P)^3 \leq m^{0.4}. \quad (259)$$

Only now define the finite exceptional event

$$E_*^{(P)} = \bigcup_{p=0}^{P-1} E_{p, K_A(1+p)^3}.$$

Every term lies in the domain of (258), and the union bound gives

$$\begin{aligned} \mathbb{P}(E_*^{(P)}) &\leq \frac{C_0 D_A}{K_A} \sum_{p=0}^{P-1} \frac{1}{(1+p)^2} + C_0 \sum_{p=0}^{P-1} e^{-c_0 D_A(1+p)} \\ &\leq \delta_A/6 + \delta_A/6 = \delta_A/3. \end{aligned} \quad (260)$$

This replaces the source's insufficient $O(A^2 4^{-A})$: its ratio to 10^{-A-2} is $100A^2(10/4)^A$, which is unbounded.

Condition on $\tilde{X}_0$ and let r, t_i, σ_i be the preceding triangle statistics for the shifted process $(\tilde{X}_p)_{p \geq 0}$. Put

$$S_R = \sum_{p=1}^{t_R} 1_W(\tilde{X}_p), \quad F_* = \left\{ r \geq R, e^{-S_R + \varepsilon R} > 3\delta_A^{-1} e^\varepsilon \right\}.$$

The uniform estimate (256), integrated in $\tilde{X}_0$, and Markov's inequality give $\mathbb{P}(F_*) \leq \delta_A/3$. Outside F_* , the event $r \geq R$ forces

$$S_R \geq \varepsilon R - \log 3 - \log(\delta_A^{-1}) - \varepsilon > H.$$

Define the high-horizontal event

$$G = \{(\mathbf{V}_{\kappa_s+P})_1 \geq 0.9m\}.$$

By (255) and the first-passage kernel,

$$\mathbb{P}((\mathbf{V}_{\kappa_s})_1 \geq 0.8m) \leq C e^{-cm},$$

because $0.8 > \frac{1}{4} \frac{\log 9}{\log 2}$. Strong Markov and the exponential tail of the fixed sum $(\tilde{\mathbf{V}}_P)_1$ give

$$\mathbb{P}(G) \leq C_P e^{-cm}.$$

Enlarge $C_{A,\varepsilon}$ again so that

$$\mathbb{P}(G) \leq \delta_A/3, \quad m^A \mathbb{P}(G) \leq \frac{1}{2}. \quad (261)$$

Outside G , positivity of the horizontal increments and $j = \lfloor n/2 \rfloor - m$ keep every $\tilde{X}_p$, $0 \leq p < P$, inside the strip where $B \uplus W$ is the full partition.

Define the shifted white count

$$N_W = \sum_{p=0}^{P-1} 1_W(\tilde{X}_p).$$

Suppose $N_W \leq H$ and none of $E_*^{(P)}, F_*, G$ occurs. Among the $h+1$ indices $0, \dots, h$ at least one is black. Therefore the actual first entry time defined above satisfies $t_1 \leq h = u_1$. Induct on the recursively defined entry times, not on an arbitrary selection of black points. If $t_i \leq u_i$, then, outside $E_*^{(P)}$, its triangle Δ_i satisfies

$$s_{\Delta_i} < K_A(1+t_i)^3 \leq K_A(1+u_i)^3.$$

Every holding increment has vertical component at least three, so by time

$$b_i = t_i + \left\lceil 10K_A(1+u_i)^3 \right\rceil + 1$$

the process is strictly above l_{Δ_i} , and in particular $b_i \geq \sigma_i$. The block $b_i, \dots, b_i + h$ lies in $\{0, \dots, P-1\}$ and contains a black point; otherwise that block alone would contribute $h+1 > H$ white points. By the defining minimality of the next entry time,

$$t_{i+1} \leq b_i + h \leq u_i + \left\lceil 10K_A(1+u_i)^3 \right\rceil + h + 1 = u_{i+1}.$$

Thus t_{i+1} exists and is the entry into the next distinct triangle. Iteration gives R entries and $t_R \leq u_R \leq P-2$. Hence $S_R \leq N_W \leq H$, contradicting the preceding consequence outside F_* . Therefore

$$\mathbb{P}(N_W \leq H) \leq \mathbb{P}(E_*^{(P)}) + \mathbb{P}(F_*) + \mathbb{P}(G) \leq \delta_A. \quad (262)$$

It remains to connect this probability estimate to the recursion. Dropping the factors before κ_s and then iterating exactly P times gives

$$\begin{aligned} Q(j, l) &\leq \mathbb{E} \left[e^{-\varepsilon^3 N_W} Q(\tilde{X}_P) \right] \\ &\leq m^{-A} Q_{m-1} \mathbb{E} \left[e^{-\varepsilon^3 N_W} \max \left(1 - \frac{(\mathbf{V}_{\kappa_s+P})_1}{m}, \frac{1}{m} \right)^{-A} \right]. \end{aligned}$$

The second line uses the fact that the terminal horizontal coordinate has advanced by at least one, so the defining range of Q_{m-1} applies. On G the weight is at most m^A ; on G^c it is at most 10^A . Moreover,

$$\mathbb{E} e^{-\varepsilon^3 N_W} \leq \mathbb{P}(N_W \leq H) + e^{-\varepsilon^3 H} \leq \delta_A + e^{-10A} \leq 10^{-A-1}$$

for sufficiently large A . By (261), the complete weighted expectation is at most

$$m^A \mathbb{P}(G) + 10^A (\delta_A + e^{-10A}) \leq 1$$

after the stated enlargement of $C_{A,\varepsilon}$. This proves $Q_m \leq Q_{m-1}$ in the final case. $\square$

Corollary 2.148 (closure of the v7 fine-scale branch). *With the repairs proved above,*

$$Q(j, l) \ll_A \max(\lfloor n/2 \rfloor - j, 1)^{-A} \quad (j \in \mathbb{N}_{>0}, l \in \mathbb{Z}), \quad \mathbb{E}Q(\mathbf{H}) \ll_A n^{-A}.$$

Consequently the v7 implications

Proposition 7.8 $\Rightarrow$ Proposition 7.3 $\Rightarrow$ Proposition 7.1 $\Rightarrow$ Proposition 1.17 $\Rightarrow$ Proposition 1.14

hold at their printed uniform scope. The exact downstream dependency in the source is

Proposition 1.14 + Proposition 1.9 $\Rightarrow$ Proposition 1.11 $\Rightarrow$ Theorem 1.6 $\Rightarrow$ Theorem 1.3.

Proof. Monotonicity and the finite base range bound all Q_m independently of m . The first coordinate of $\mathbf{H}$ has exponential tail, so splitting at $\mathbf{H}_1 = n/4$ gives $\mathbb{E}Q(\mathbf{H}) = O_A(n^{-A})$. This is Proposition 7.3. The white-point pairing in Proposition 2.144 gives Proposition 7.1, whose reversed-variable form is Proposition 1.17. Finally Proposition 2.142 gives Proposition 1.14. The second display records every consequence edge leaving this branch. Proposition 2.152 now supplies the separately typed Proposition 1.9 input. The exact step combining it with Proposition 1.14 to prove Proposition 1.11, and the two later theorem edges, remain behind their named proof-audit gate. This is a dependency record, not a handwaving theorem claim. $\square$

Explicit nonclaim 2.149. Corollary 2.148 certifies this versioned proof branch; it does not replace the durable whole-paper release audit. By itself it does not certify Proposition 1.9, Proposition 1.11, Theorem 1.6, or Theorem 1.3; Proposition 2.152 supplies the first of those results by an independent argument. This branch does not prove the stronger v5/journal Lemma 7.9, use Baker’s theorem, establish exponential Fourier decay, prove natural-density mixing of the full affine map, give a fixed absolute orbit bound, or prove convergence to one.

2.21 The valuation-law gate: Tao’s Proposition 1.9

The proposition numbered 1.9 in the controlling v7 text is the distribution theorem for the finite Syracuse valuation tuple, not the later first-passage stabilisation theorem. Its statement is at v7 source lines 269–278 and its proof occupies lines 597–647 [32]. Proposition 1.11 is the first-passage result. This numbering matters in the dependency graph: Proposition 1.14 and Proposition 1.9 are the two inputs used to obtain Proposition 1.11.

The frozen routes identify the v5 and journal manifestations of the same work. Direct comparison uses v5 source lines 269–278 and 591–641, v7 source lines 269–278 and 597–647, and journal physical pages 6 and 17–18. Routing metadata is not used as mathematical evidence.

Source issue 2.150 (three-manifestation defects in Proposition 1.9). All three inspected manifestations print the same substantive residue-fibre argument and preserve three operative presentation defects. In the statement, the ambient ring is printed as $\mathbb{Z}/2^\ell\mathbb{Z}$, although ℓ is undefined; both the displayed law and every subsequent use require $\mathbb{Z}/2^{n'}\mathbb{Z}$. In the tail proof, one intermediate line changes the first-crossing event $A_k < n' \leq A_{k+1}$ into $A_k \leq n' < A_{k+1}$; the former is the event used before and after that line. Finally, the two central truncations are printed as $|\vec{a}| < m$, although no m is defined in that section; the proof requires $|\vec{a}| < n'$.

The source also invokes the empty prefix sum A_0 without extending its earlier nonempty-sum convention, and it does not dispatch $n = 0$. In the v5/v7 TeX proof of the prerequisite valuation-uniqueness lemma, a function is once written without its argument and $\text{Syr}^{N-1}(N)$ is printed where the induction requires $\text{Syr}^{n-1}(N)$. The reconstruction below supplies all of these data explicitly.

For $s \geq 1$, put

$$\mathcal{O}_s = (2\mathbb{Z} + 1)/2^s\mathbb{Z}.$$

Also set $\mathcal{O}_0 = \mathbb{Z}/\mathbb{Z}$, the singleton residue space, so that the $n = 0, n' = 0$ endpoint in the source statement is typed. For $s \geq 1$, $\mathcal{O}_s$ has 2^{s-1} elements. For $1 \leq t \leq s$, let

$$\pi_{s,t} : \mathcal{O}_s \longrightarrow \mathcal{O}_t, \quad [u]_{2^s} \longmapsto [u]_{2^t}$$

be reduction. Every fibre has exactly 2^{s-t} elements, so $(\pi_{s,t})_* \text{Unif}(\mathcal{O}_s)$ is $\text{Unif}(\mathcal{O}_t)$. For signed measures μ, ν on $\mathcal{O}_s$, the source's unhalved total-variation convention gives the exact contraction

$$\begin{aligned} \|(\pi_{s,t})_*\mu - (\pi_{s,t})_*\nu\|_1 &= \sum_{x \in \mathcal{O}_t} \left| \sum_{u \in \pi_{s,t}^{-1}(x)} (\mu(u) - \nu(u)) \right| \\ &\leq \sum_{u \in \mathcal{O}_s} |\mu(u) - \nu(u)|. \end{aligned} \tag{263}$$

Lemma 2.151 (valuation prefix and its exact residue fibre). *Let $n \geq 1$, let $N \in 2\mathbb{N} + 1$, and let $\vec{a} = (a_1, \dots, a_n) \in (\mathbb{N} + 1)^n$. Define*

$$A_0 = 0, \quad A_j = a_1 + \dots + a_j \quad (1 \leq j \leq n), \quad A = A_n,$$

and

$$C_n(\vec{a}) = \sum_{i=1}^n 3^{n-i} 2^{A_{i-1}}.$$

Then

$$\text{Aff}_{\vec{a}}(N) = \frac{3^n N + C_n(\vec{a})}{2^A}, \tag{264}$$

and

$$\vec{a}^{(n)}(N) = \vec{a} \iff \text{Aff}_{\vec{a}}(N) \in 2\mathbb{N} + 1. \tag{265}$$

Consequently the event in (265) is the single odd residue cylinder

$$N \equiv b_{\vec{a}} \pmod{2^{A+1}}, \quad b_{\vec{a}} \equiv 3^{-n}(2^A - C_n(\vec{a})) \pmod{2^{A+1}}. \tag{266}$$

If $A < n'$, the inverse image of this cylinder in $\mathcal{O}_{n'}$ has exactly $2^{n'-A-1}$ elements and hence uniform mass 2^{-A} .

Proof. Multiplying the source's affine-offset formula by 2^A gives (264). The forward implication in (265) is the defining Syracuse iteration. For the converse, induct on n . When $n = 1$, the equality $2^{a_1} \text{Aff}_{a_1}(N) = 3N + 1$ and the oddness of $\text{Aff}_{a_1}(N)$ give $a_1 = \nu_2(3N + 1)$ directly. Now let $n \geq 2$ and write

$$X = \text{Aff}_{a_1, \dots, a_{n-1}}(N), \quad Y = \text{Aff}_{\vec{a}}(N) \in 2\mathbb{N} + 1.$$

Both affine maps preserve positivity and $X \in \mathbb{Z}[1/2]$. The last step gives

$$2^{a_n} Y = 3X + 1.$$

Thus $3X$ is an odd integer. A rational number in $\mathbb{Z}[1/2]$ whose product by 3 is integral is itself integral; hence X is a positive odd integer. The induction hypothesis identifies the first

$n - 1$ valuation coordinates. Since $(3X + 1)/2^{a_n} = Y$ is odd, $a_n = \nu_2(3X + 1)$, proving the last coordinate.

An integer quotient in (264) is odd exactly when its numerator is congruent to 2^A modulo 2^{A+1} . Because 3^n is a unit modulo 2^{A+1} , this gives the unique residue in (266). It is odd: the first term of $C_n(\vec{a})$ is 3^{n-1} , every other term is even, and 2^A is even. Finally, reduction $\mathcal{O}_{n'} \rightarrow \mathcal{O}_{A+1}$ has fibres of size $2^{n'-(A+1)}$, proving the last assertion. $\square$

The exponential estimate used in both tails and in the accumulated cylinder error can be written without an unnamed appeal to Stirling's formula. Let

$$H(p) = -p \log p - (1-p) \log(1-p), \quad p_0 = \frac{1}{2+c_0/2} < \frac{1}{2}, \quad \eta(c_0) = \log 2 - H(p_0) > 0.$$

If $M = n' - 1$, $n' \geq (2+c_0)n$, and $n \geq 2/c_0$, then $n/M \leq p_0$. Set $z_0 = p_0/(1-p_0) \in (0, 1)$. Since $z_0^k \geq z_0^n$ for $0 \leq k \leq n$,

$$2^{-M} \sum_{k=0}^n \binom{M}{k} \leq z_0^{-n} \left(\frac{1+z_0}{2} \right)^M \leq \exp(-\eta(c_0)M). \quad (267)$$

The finitely many positive $n < 2/c_0$ are absorbed into a constant depending on c_0 . This proves a bound $C_{c_0} 2^{-c_1 n}$, for example with any fixed $0 < c_1 < \eta(c_0)/\log 2$.

Proposition 2.152 (Tao Proposition 1.9, residue-fibre reconstruction). *Let $n \in \mathbb{N}$, let $\mathbf{N}$ take values in $2\mathbb{N} + 1$, let $c_0 > 0$, and let $n' \in \mathbb{N}$ satisfy $n' \geq (2+c_0)n$. Suppose, with the unhalved total variation used by the source, that*

$$d_{\text{TV}}(\mathbf{N} \bmod 2^{n'}, \text{Unif}(\mathcal{O}_{n'})) \leq K 2^{-n'} \quad (268)$$

for a hypothesis constant $K = K(c_0) \geq 0$, independent of n, n' , and the law of $\mathbf{N}$. Then there are $c_1 = c_1(c_0) > 0$ and $C = C(c_0) > 0$ such that

$$d_{\text{TV}}(\vec{a}^{(n)}(\mathbf{N}), \text{Geom}(2)^n) \leq C(1+K)2^{-c_1 n}. \quad (269)$$

Thus the source statement follows with exactly the source-permitted c_0 -dependence of the implicit constants.

Proof. For $n = 0$, both laws are the point mass at the empty tuple, so the distance is zero. Assume $n \geq 1$, and write

$$\vec{a}^{(n)}(\mathbf{N}) = (\mathbf{a}_1, \dots, \mathbf{a}_n), \quad S_0 = 0, \quad S_j = \mathbf{a}_1 + \dots + \mathbf{a}_j.$$

Because every $\mathbf{a}_j$ is positive, the actual-valuation tail is the disjoint first-crossing union

$$\{S_n \geq n'\} = \bigsqcup_{k=0}^{n-1} \{S_k < n' \leq S_{k+1}\}. \quad (270)$$

Fix k and a positive prefix $(a_1, \dots, a_k)$, put $A_0 = 0$ and $A_j = \sum_{r=1}^j a_r$ for $1 \leq j \leq k$, and assume $A_k < n'$. On the corresponding crossing event, the exact affine numerator at time $k+1$ is divisible by $2^{S_{k+1}}$, hence

$$3^{k+1} \mathbf{N} + \sum_{i=1}^{k+1} 3^{k+1-i} 2^{A_{i-1}} \equiv 0 \pmod{2^{n'}}. \quad (271)$$

The displayed sum depends only on the fixed k -prefix, including $A_0 = 0$. Its first term is odd and all later terms are even, so (271) fixes one odd residue of $\mathbf{N}$ modulo $2^{n'}$. Its probability is at most $(2 + K)2^{-n'}$: the uniform mass of one element of $\mathcal{O}_{n'}$ is $2^{1-n'}$, and (268) controls the error.

Positive k -tuples with $A_k < n'$ correspond bijectively to the k -element subsets

$$\{A_1 < \dots < A_k\} \subseteq \{1, \dots, n' - 1\},$$

and therefore number $\binom{n'-1}{k}$, including the one empty prefix at $k = 0$. Summing (270) and applying (267) gives

$$\mathbb{P}(S_n \geq n') \ll_{c_0} (1 + K)2^{-c_1 n}. \quad (272)$$

For the model law, realize independent $\text{Geom}(2)$ coordinates as waiting times between successive successes in fair Bernoulli trials. Their sum is at least n' exactly when the first $n' - 1$ trials contain at most $n - 1$ successes. Hence

$$\mathbb{P}(|\text{Geom}(2)^n| \geq n') = 2^{-(n'-1)} \sum_{k=0}^{n-1} \binom{n'-1}{k} \ll_{c_0} 2^{-c_1 n}. \quad (273)$$

It remains to compare the two laws on positive tuples $\vec{a} = (a_1, \dots, a_n)$ with $A = |\vec{a}| < n'$. Lemma 2.151, the exact pushforward (263), and (268) give

$$\left| \mathbb{P}(\vec{a}^{(n)}(\mathbf{N}) = \vec{a}) - 2^{-A} \right| \leq K2^{-n'}. \quad (274)$$

The term 2^{-A} is exactly the mass of $\vec{a}$ under $\text{Geom}(2)^n$. Positive n -tuples with $A < n'$ are in bijection with n -element subsets of $\{1, \dots, n' - 1\}$, so their number is $\binom{n'-1}{n}$. Thus their accumulated error is bounded by

$$K2^{-n'} \binom{n'-1}{n} \ll_{c_0} K2^{-c_1 n}$$

using (267). Adding the two tail bounds (272) and (273) is justified by the exact decomposition

$$\begin{aligned} d_{\text{TV}}(\vec{a}^{(n)}(\mathbf{N}), \text{Geom}(2)^n) &\leq \sum_{\substack{\vec{a} \in (\mathbb{N}+1)^n \\ |\vec{a}| < n'}} \left| \mathbb{P}(\vec{a}^{(n)}(\mathbf{N}) = \vec{a}) - 2^{-|\vec{a}|} \right| \\ &\quad + \mathbb{P}(S_n \geq n') + \mathbb{P}(|\text{Geom}(2)^n| \geq n'). \end{aligned}$$

This proves (269). □

The proof exposes the exact morphism suppressed in the printed argument. Define

$$\mathcal{U}_{n,n'} = \left\{ u \in \mathcal{O}_{n'} : \begin{array}{l} \text{some positive odd representative } N \text{ of } u \text{ satisfies} \\ |\vec{a}^{(n)}(N)| < n' \end{array} \right\}.$$

Lemma 2.151 shows that the displayed condition and the resulting tuple are independent of the chosen positive representative: a tuple of total $A < n'$ is determined already modulo 2^{A+1} . Hence

$$V_{n,n'} : \mathcal{U}_{n,n'} \longrightarrow \mathcal{D}_{n,n'}, \quad V_{n,n'}([N]_{2^{n'}}) = \vec{a}^{(n)}(N),$$

is a well-defined map, where

$$\mathcal{D}_{n,n'} = \{\vec{a} \in (\mathbb{N} + 1)^n : |\vec{a}| < n'\}.$$

It is surjective, and its fibre over $\vec{a}$ is precisely the cylinder $b_{\vec{a}} \pmod{2^{A+1}}$, containing $2^{n'-A-1}$ odd residues modulo $2^{n'}$. Its set-theoretic kernel pair is

$$\mathcal{U}_{n,n'} \times_{\mathcal{D}_{n,n'}} \mathcal{U}_{n,n'} = \{(u, v) : V_{n,n'}(u) = V_{n,n'}(v)\},$$

namely equality of the first n valuation coordinates. There is an explicit section: for each $\vec{a}$, take the least positive odd representative $\tilde{b}_{\vec{a}}$ of the cylinder $b_{\vec{a}} \pmod{2^{A+1}}$ and put

$$s_{n,n'}(\vec{a}) = [\tilde{b}_{\vec{a}}]_{2^{n'}}, \quad V_{n,n'} \circ s_{n,n'} = \text{id}_{\mathcal{D}_{n,n'}}.$$

This section chooses all unrecovered higher binary digits to be zero. The map has no left inverse, and therefore no two-sided inverse: the tuple $(1, \dots, 1)$ has $A = n$ and fibre size $2^{n'-n-1} \geq 2$, because $n' \geq (2 + c_0)n$ with $n \geq 1$.

Explicit nonclaim 2.153. Proposition 2.152 certifies only the valuation-law theorem numbered Proposition 1.9. It neither assumes nor proves first-passage stabilisation. The separate edge combining this result with Proposition 1.14 to obtain Proposition 1.11 must still be checked before Theorem 1.6 or Theorem 1.3 is admitted as proved in this edition. No natural-density statement, convergence-to-one statement, or fixed universal orbit bound follows from this proposition.

2.22 The first-passage gate: Tao's Proposition 1.11

The controlling statement is Proposition 1.11 at v7 source lines 316–335, and the deduction from Propositions 1.9 and 1.14 occupies v7 lines 649–925 [32]. The corresponding comparison loci are v5 lines 643–919 and journal physical pages 18–27. The three manifestations have the same substantive modular, logarithmic-mass, and integer-window calculations. ArXiv v7 corrects the v5/journal interchange of passage time and passage location in the converse implication (v5 lines 789–791; journal physical page 23; v7 lines 795–797). No conclusion below is transferred silently between versions.

Source issue 2.154 (local defects at the Proposition 1.11 edge). The opening heuristic at v7 lines 651–652 writes Pass_x where the quantity being approximated is the time T_x , and its numerator requires $\log(N_y/x)$. Line 689 has $1.9n$ where the fixed descent time is $1.9n_0$. The periodic-orbit aside at line 699 requires a representative of $-\mathbf{m}$ modulo the least positive period, or a period multiple at least $\mathbf{m}$; that aside is not used here.

The additive endpoint deletion printed at lines 759–762 does not produce a time margin of order $(\log x)^{0.8}$. It has already been replaced by the multiplicative deletion in Proposition 2.135. The unsigned lower comparison, the event equality, the undefined m in the tuple length, the fixed- M affine equality, the two previous-iterate coefficients, the malformed progression estimate, the nonuniform weakened c_n -majorant, and the iid-versus-actual Chernoff citation have already been repaired in Proposition 2.136. This subsection supplies the remaining modular law, descent exponent, affine inverse, integer count, normalization cancellation, and arbitrary-event-to-total-variation arrows.

Throughout this subsection retain the source constants and floors

$$\alpha = \frac{1001}{1000}, \quad L = \log x, \quad \lambda = \log(4/3), \quad n_0 = \left\lfloor \frac{L}{10 \log 2} \right\rfloor, \quad m_0 = \left\lfloor \frac{\alpha - 1}{100} L \right\rfloor. \quad (275)$$

For $j \in \{1, 2\}$, put $y_j = x^{\alpha^j}$ and

$$\mathcal{R}_{y_j} = (2\mathbb{N} + 1) \cap [y_j, y_j^\alpha], \quad H_{y_j} = \sum_{N \in \mathcal{R}_{y_j}} \frac{1}{N}, \quad \mathbb{P}(\mathbf{N}_{y_j} = N) = \frac{1}{H_{y_j} N}. \quad (276)$$

Real endpoints cause no ambiguity: the intersection is with the integer set $2\mathbb{N} + 1$.

Lemma 2.155 (progression mass with real endpoints). *Let $1 \leq A < B$, let $q \geq 2$ be even, put $J = \log(B/A)$, and fix an odd residue $r \pmod{q}$. Then*

$$\left| \sum_{\substack{A \leq N \leq B \\ N \equiv r \pmod{q}}} \frac{1}{N} - \frac{J}{q} \right| \leq \frac{1}{A}. \quad (277)$$

If W_r denotes the sum in (277) and $H = \sum_{r \text{ odd } (q)} W_r$, then

$$\sum_{r \text{ odd } (q)} \left| W_r - \frac{J}{q} \right| \leq \frac{q}{2A}, \quad \left| H - \frac{J}{2} \right| \leq \frac{q}{2A}. \quad (278)$$

Proof. If the progression contributes points $t_0 < t_1 < \dots < t_s$, consecutive points differ by q . Monotonicity of $1/t$ gives

$$\sum_{i=0}^s \frac{1}{t_i} \leq \frac{1}{t_0} + \frac{1}{q} \int_{t_0}^{t_s} \frac{dt}{t} \leq \frac{1}{A} + \frac{J}{q}.$$

It also gives

$$\sum_{i=0}^s \frac{1}{t_i} \geq \frac{1}{q} \int_{t_0}^{t_s+q} \frac{dt}{t} \geq \frac{J}{q} - \frac{1}{q} \log \frac{t_0}{A} \geq \frac{J}{q} - \frac{1}{A},$$

because $0 \leq t_0 - A < q$ and $\log(t_0/A) \leq (t_0 - A)/A$. If the progression contributes no point, then $B - A < q$, whence $J/q < (B - A)/(qA) < 1/A$. This proves (277). There are exactly $q/2$ odd residue classes. Summing and using the triangle inequality proves (278). $\square$

Proposition 2.156 (exact modular input to Proposition 1.9). *Let $q = 2^{3n_0}$. Uniformly for $j = 1, 2$,*

$$d_{\text{TV}}(\mathbf{N}_{y_j} \bmod q, \text{Unif}((2\mathbb{Z} + 1)/q\mathbb{Z})) \ll q^{-1} = 2^{-3n_0}, \quad (279)$$

where total variation is the source's unhalved ℓ^1 distance. Hence Proposition 2.152, with

$$n = n_0, \quad n' = 3n_0, \quad c_0 = 1,$$

gives

$$d_{\text{TV}}(\tilde{a}^{(n_0)}(\mathbf{N}_{y_j}), \text{Geom}(2)^{n_0}) \ll x^{-c_2} \quad (280)$$

for one absolute $c_2 > 0$.

Proof. Set $A = y_j$, $B = y_j^\alpha$, and $J = (\alpha - 1) \log y_j$. For every odd $r \pmod{q}$, let W_r be the sum in Lemma 2.155. For sufficiently large x , $AJ \geq 2q$, and therefore $H_{y_j} \geq J/4$. Since uniform measure assigns mass $2/q$, not $1/q$, to each of the $q/2$ odd classes,

$$\begin{aligned} d_{\text{TV}} &= \sum_{r \text{ odd } (q)} \left| \frac{W_r}{H_{y_j}} - \frac{2}{q} \right| \\ &\leq \frac{1}{H_{y_j}} \left(\sum_{r \text{ odd } (q)} \left| W_r - \frac{J}{q} \right| + \left| H_{y_j} - \frac{J}{2} \right| \right) \leq \frac{2q}{AJ - q} \leq \frac{4q}{AJ}. \end{aligned} \quad (281)$$

The floor in n_0 is retained exactly:

$$\frac{x^{3/10}}{8} \leq q \leq x^{3/10}.$$

Consequently

$$\frac{4q}{AJ} \leq q^{-1} \frac{4x^{0.6-\alpha}}{(\alpha-1)\alpha L} = o(q^{-1}),$$

uniformly for $j = 1, 2$. This proves (279) with room to spare. The size hypothesis in Proposition 2.152 is the exact equality $3n_0 = (2 + c_0)n_0$. Finally,

$$2^{-c_1 n_0} \leq 2^{c_1} x^{-c_1/10},$$

so its conclusion has the form (280) despite the floor. $\square$

Proposition 2.157 (the finite-time descent estimate). *There is an absolute $c_3 > 0$ such that, uniformly for $j = 1, 2$,*

$$\mathbb{P}(T_x(\mathbf{N}_{y_j}) = +\infty) \ll x^{-c_3}. \quad (282)$$

More precisely, outside an event of probability $O(x^{-c_3})$, $T_x(\mathbf{N}_{y_j}) \leq n_0$.

Proof. For iid $G_1, \dots, G_{n_0} \sim \text{Geom}(2)$, the lower-tail Chernoff bound applied to

$$\mathbb{E}e^{-tG_1} = \frac{e^{-t}}{2 - e^{-t}}$$

gives $\mathbb{P}(G_1 + \dots + G_{n_0} \leq 1.9n_0) \leq e^{-\gamma n_0}$ for an absolute $\gamma > 0$. For example, take $e^{-t} = 18/19$; the base is $0.9(19/18)^{1.9} < 1$. Combining this with (280) gives

$$\mathbb{P}(|\vec{a}^{(n_0)}(\mathbf{N}_{y_j})| \leq 1.9n_0) \ll x^{-c_3} \quad (283)$$

after decreasing c_3 if necessary.

On the complementary event, the exact affine formula and $0 \leq F_{n_0} \leq 3^{n_0}$ give

$$\text{Syr}^{n_0}(\mathbf{N}_{y_j}) \leq 3^{n_0} 2^{-1.9n_0} x^{\alpha^3} + 3^{n_0}. \quad (284)$$

The numerical exponent check is strict, not heuristic. Put

$$\beta_1 = \alpha^3 + \frac{\log 3 - 1.9 \log 2}{10 \log 2}, \quad \beta_2 = \frac{\log 3}{10 \log 2}.$$

Rational interval arithmetic gives

$$\beta_1 < 0.972 < 1, \quad \beta_2 < 0.159 < 1. \quad (285)$$

One elementary certificate for these decimal inequalities uses

$$\log u = 2 \sum_{k=0}^M \frac{z^{2k+1}}{2k+1} + R_M, \quad z = \frac{u-1}{u+1}, \quad 0 < R_M < \frac{2z^{2M+3}}{(2M+3)(1-z^2)},$$

with $u = 2, 3$, $z = 1/3, 1/2$, and $M = 20$; every endpoint is rational. Since $n_0 \geq L/(10 \log 2) - 1$ and $\log 3 - 1.9 \log 2 < 0$, the first term on the right side of (284) is at most

$$e^{1.9 \log 2 - \log 3} x^{\beta_1}.$$

The positive coefficient in the second term instead uses $n_0 \leq L/(10 \log 2)$, and that term is at most x^{β_2} . Thus $e^{1.9 \log 2 - \log 3} x^{\beta_1} + x^{\beta_2} < x$ for all sufficiently large x . Thus passage occurs by time n_0 , and (283) proves the claim. $\square$

For the passage-location part, let $E \subseteq (2\mathbb{N} + 1) \cap [1, x]$. For either choice of $y = y_j$, retain the source interval

$$I_y = \left[\frac{\log(y/x)}{\lambda} + L^{4/5}, \frac{\log(y^\alpha/x)}{\lambda} - L^{4/5} \right] \quad (286)$$

and the source set, independent of y ,

$$E' = \{M \in 2\mathbb{N} + 1 : T_x(M) = m_0, \text{Pass}_x(M) \in E, \\ e^{-L^{7/10}}(4/3)^{m_0}x \leq M \leq e^{L^{7/10}}(4/3)^{m_0}x\}. \quad (287)$$

For an integer $n \in I_y$, put $r = n - m_0$. For sufficiently large x , every such n satisfies

$$m_0 \leq r \leq n_0 - m_0, \quad 1 \leq m_0 \leq r. \quad (288)$$

Proposition 2.158 (the singleton affine inverse). *Let $n \in I_y \cap \mathbb{Z}$, $r = n - m_0$, and $\vec{a} = (a_1, \dots, a_r) \in \mathcal{A}^{(r)}$. Put*

$$A_0 = 0, \quad A_k = a_1 + \dots + a_k, \quad A = A_r, \quad C_r(\vec{a}) = \sum_{i=1}^r 3^{r-i} 2^{A_{i-1}}, \quad F_r(\vec{a}) = 2^{-A} C_r(\vec{a}).$$

For $M \in E'$, the equation $\text{Aff}_{\vec{a}}(N) = M$ has at most one solution, namely

$$\Phi_r(\vec{a}, M) = \frac{2^A M - C_r(\vec{a})}{3^r} = 2^A \frac{M - F_r(\vec{a})}{3^r}. \quad (289)$$

It has a solution in $\mathcal{R}_y$ if and only if

$$M \equiv F_r(\vec{a}) \pmod{3^r} \quad (290)$$

in $\mathbb{Z}[1/2]/3^r\mathbb{Z}[1/2]$. When this congruence holds, $\Phi_r(\vec{a}, M)$ is a positive odd integer in $[y, y^\alpha]$,

$$\vec{a}^{(r)}(\Phi_r(\vec{a}, M)) = \vec{a}, \quad \text{Syr}^r(\Phi_r(\vec{a}, M)) = M,$$

and its logarithmic point mass is

$$\mathbb{P}(\text{Aff}_{\vec{a}}(\mathbf{N}_y) = M) = \frac{1 + O(x^{-c_4})}{\frac{\alpha-1}{2} \log y} \frac{2^{-A} 3^r}{M} \quad (291)$$

for one absolute $c_4 > 0$, uniformly in all displayed data.

Proof. The affine identity

$$\text{Aff}_{\vec{a}}(N) = \frac{3^r N + C_r(\vec{a})}{2^A}$$

proves uniqueness and forces (289). Congruence (290) is necessary. Conversely, it makes (289) an integer: $M - F_r(\vec{a})$ belongs to $3^r\mathbb{Z}[1/2]$, and its denominator divides 2^A . The integer is odd because $C_r(\vec{a})$ is odd, $2^A M$ is even, and 3^r is odd.

It remains to prove positivity and the interval bounds, rather than infer them from presentation. Since $r \leq n_0$,

$$0 \leq F_r(\vec{a}) \leq 3^r \leq x^{0.159}$$

by (285). From (287), $M \geq e^{-L^{7/10}}x$. Since $L^{7/10} \leq (141/1000)L$ for all sufficiently large L , the preceding two bounds give

$$0 \leq \frac{F_r(\vec{a})}{M} \leq x^{-7/10} < \frac{1}{2}. \quad (292)$$

Write $A = 2r + \delta$ and $M = x(4/3)^{m_0} e^\eta$. Membership in $\mathcal{A}^{(r)}$ and E' gives $|\delta| < L^{3/5}$ and $|\eta| \leq L^{7/10}$. Hence

$$\log \Phi_r(\vec{a}, M) = L + n\lambda + \delta \log 2 + \eta + \log \left(1 - \frac{F_r(\vec{a})}{M}\right). \quad (293)$$

The final three terms have absolute value at most $2L^{7/10}$ for all sufficiently large x . The endpoints of I_y give margins $\lambda L^{4/5}$, and $2L^{7/10} < \lambda L^{4/5}$ eventually. Thus $\log y \leq \log \Phi_r \leq \log y^\alpha$, proving both positivity and the claimed range. Substitution proves $\text{Aff}_{\vec{a}}(\Phi_r) = M$; the exact valuation-prefix lemma, Lemma 2.151, then proves the valuation and iterate identities.

Finally, Lemma 2.155 with modulus 2 gives

$$\left| H_y - \frac{\alpha - 1}{2} \log y \right| \leq \frac{1}{y}, \quad H_y = \frac{\alpha - 1}{2} \log y (1 + O(x^{-1})). \quad (294)$$

Using (289) exactly,

$$\frac{1}{\Phi_r(\vec{a}, M)} = \frac{2^{-A} 3^r}{M - F_r(\vec{a})}.$$

Equations (292) and (294) now give (291). $\square$

The last proposition is an exact morphism, not merely a pointwise formula. Define

$$\mathcal{D}_{r,y} = \{(\vec{a}, M) \in \mathcal{A}^{(r)} \times E' : M \equiv F_r(\vec{a}) \pmod{3^r}\}$$

and

$$\mathcal{X}_{r,y} = \{N \in \mathcal{R}_y : \vec{a}^{(r)}(N) \in \mathcal{A}^{(r)}, \text{Syr}^r(N) \in E'\}.$$

Then

$$\Theta_{r,y} : \mathcal{X}_{r,y} \longrightarrow \mathcal{D}_{r,y}, \quad N \longmapsto (\vec{a}^{(r)}(N), \text{Syr}^r(N)) \quad (295)$$

is a bijection with inverse Φ_r . Both compositions are identities; every fibre is a singleton; its set-theoretic kernel pair is the diagonal; and no coordinate or higher binary digit is discarded. This is the exact map underlying source lines 808–827.

Lemma 2.159 (the exact integer time-window count). *For $y_j = x^{\alpha^j}$, let the endpoints of I_{y_j} be*

$$a_j = \frac{\alpha^j - 1}{\lambda} L + L^{4/5}, \quad b_j = \frac{\alpha^{j+1} - 1}{\lambda} L - L^{4/5}.$$

For sufficiently large x , the interval is nonempty and

$$\begin{aligned} \#(I_{y_j} \cap \mathbb{Z}) &= \lfloor b_j \rfloor - \lceil a_j \rceil + 1 \\ &= \frac{\alpha - 1}{\lambda} \log y_j - 2L^{4/5} + \varepsilon_j, \quad -1 < \varepsilon_j \leq 1. \end{aligned} \quad (296)$$

Consequently

$$\frac{\#(I_{y_j} \cap \mathbb{Z})}{\frac{\alpha-1}{2} \log y_j} = \frac{2}{\lambda} - \frac{4}{(\alpha-1)\alpha^j} L^{-1/5} + O(L^{-1}) = \frac{2}{\lambda} + O(L^{-1/5}). \quad (297)$$

The leading term is independent of j ; the displayed finite- x buffer correction is not.

Proof. The first line is the exact count of integer points in a closed real interval. Subtracting the endpoints gives the second line with

$$\varepsilon_j = (\lfloor b_j \rfloor - b_j) + (a_j - \lceil a_j \rceil) + 1, \quad -1 < \varepsilon_j \leq 1.$$

Since $\log y_j = \alpha^j L$, division proves (297). It also shows why the source's variable exponent c must be decreased to at most $1/5$ at this step. The lower endpoint satisfies $a_j \geq 2m_0$, and

$$\frac{\alpha^3 - 1}{\lambda} < 0.010439 < 0.144269 < \frac{1}{10 \log 2};$$

therefore $I_y \subseteq [2m_0, n_0]$ for all sufficiently large x , which also proves (288) with every floor retained. $\square$

Theorem 2.160 (Tao Proposition 1.11, exact reconstructed edge). *Let $\alpha = 1.001$, and let $\mathbf{N}_y$ have the logarithmic law on the odd integers in $[y, y^\alpha]$. There is an absolute $c > 0$ such that, for all sufficiently large x ,*

$$\mathbb{P}(T_x(\mathbf{N}_y) = +\infty) \ll x^{-c} \quad (y = x^\alpha, x^{\alpha^2}) \quad (298)$$

and

$$d_{\text{TV}}(\text{Pass}_x(\mathbf{N}_{x^\alpha}), \text{Pass}_x(\mathbf{N}_{x^{\alpha^2}})) \ll L^{-c} \quad (299)$$

in the source's unhalved convention.

Proof. Equation (298) is Proposition 2.157. Fix an arbitrary $E \subseteq (2\mathbb{N} + 1) \cap [1, x]$, form E' by (287), and define

$$Z_E = \sum_{M \in E'} \frac{3^{m_0}}{M} \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^{m_0}\mathbb{Z}) \equiv M \pmod{3^{m_0}}). \quad (300)$$

This quantity depends on x and E , but not on whether $y = x^\alpha$ or $y = x^{\alpha^2}$.

Proposition 2.135 places the first-passage time in I_y outside an $O(L^{-1/5})$ event. The exact event equality and signed pre-passage estimate in Proposition 2.136, together with the bijection (295) and point mass (291), give

$$\mathbb{P}(\text{Pass}_x(\mathbf{N}_y) \in E) = \frac{1 + O(x^{-c_4})}{\frac{\alpha-1}{2} \log y} \sum_{n \in I_y \cap \mathbb{Z}} A_{n,E} + O(L^{-c_5}), \quad (301)$$

where, with $r = n - m_0$,

$$A_{n,E} = 3^r \sum_{\vec{a} \in \mathcal{A}^{(r)}} 2^{-|\vec{a}|} \sum_{\substack{M \in E' \\ M \equiv F_r(\vec{a}) \pmod{3^r}}} \frac{1}{M}.$$

All sums are over their displayed discrete sets; no conditional law or coupling is being inserted.

The uniform c_n -bound in Proposition 2.136 permits deletion of the bad iid path tube. Proposition 2.142, whose dependency branch is closed in Corollary 2.148, then replaces the law at level r by the exact uniform lift of its level- m_0 projection. The finite pairing is

$$\begin{aligned} \sum_{X \bmod 3^r} \left(3^r \sum_{\substack{M \in E' \\ M \equiv X \pmod{3^r}}} \frac{1}{M} \right) 3^{m_0-r} \mathbb{P}(\text{Syrac}(\mathbb{Z}/3^{m_0}\mathbb{Z}) \equiv X \pmod{3^{m_0}}) \\ = Z_E. \end{aligned} \quad (302)$$

Indeed, for each M there is exactly one residue $X = M \pmod{3^r}$, and $3^r 3^{m_0-r} = 3^{m_0}$. Thus, choosing the arbitrary mixing exponent large enough before summing over the $O(L)$ possible times,

$$A_{n,E} = Z_E + O(L^{-c_6}) \quad (303)$$

uniformly for both y 's and all $n \in I_y \cap \mathbb{Z}$. Moreover $A_{n,E} = O(1)$ by the uniform c_n -bound, so (303) gives $Z_E = O(1)$ without circular use of the probability conclusion.

Substitute (303) into (301). The $O(L)$ summation factor is cancelled by the denominator of size L ; after decreasing the common positive exponent,

$$\mathbb{P}(\text{Pass}_x(\mathbf{N}_y) \in E) = \frac{\#(I_y \cap \mathbb{Z})}{\frac{\alpha-1}{2} \log y} Z_E + O(L^{-c}).$$

Lemma 2.159 therefore yields the uniform, y -independent main term

$$\mathbb{P}(\text{Pass}_x(\mathbf{N}_y) \in E) = \frac{2}{\lambda} Z_E + O(L^{-c}). \quad (304)$$

Taking the difference of (304) for the two values of y , and then the supremum over all subsets E of the common state space $(2\mathbb{N} + 1) \cap [1, x]$, gives an $O(L^{-c})$ event distance. The convention $\text{Pass}_x(N) = 1$ when $T_x(N) = +\infty$ keeps both laws in this same state space. Finally the source inequality

$$d_{\text{TV}}(P, Q) \leq 2 \sup_E |P(E) - Q(E)|$$

proves (299) in the unhalved convention. $\square$

Explicit nonclaim 2.161. Theorem 2.160 proves exactly the first-passage stabilisation proposition and closes the edge

$$\text{Proposition 1.9} + \text{Proposition 1.14} \implies \text{Proposition 1.11}.$$

It does not assert exact residue uniformity, independence of actual valuation coordinates, arbitrary- y uniformity, natural-density transport, a coupling of the two passage locations, convergence to 1, exclusion of divergent or periodic orbits, or a fixed universal orbit bound. The finite- x time-window correction depends on y , even though its leading normalization does not. The exact later implications to Theorem 3.1, Theorem 1.6, and Theorem 1.3, including their source defects and repaired morphisms, are proved in the next subsection; they had not yet been proved at the checkpoint represented by this subsection.

2.23 The exact reduction from first passage to the main theorem

The source reduction from Proposition 1.11 to Theorems 3.1, 1.6, and 1.3 is v7 lines 535–593, with the definitions and statements at lines 159–206 [32]. The corresponding v5 loci are lines 529–587 and 159–206; after deletion of whitespace, the two reduction blocks are identical. The journal manifestation reproduces the same argument on physical pages 15–17, with its logarithmic-density and Syracuse setup on physical pages 3–4 and Proposition 1.11 on physical page 8. The comparison below is therefore version-pinned but not a claim that the complete manifestations are text-identical.

Source issue 2.162 (defects and suppressed arrows in the printed reduction). Eight points must be separated.

First, v7 lines 591–593 define $\tilde{f}(x) = \inf_{N \geq x} f(N)$ and then use that lower envelope on the range $N \leq x$, where it supplies no comparison. The displayed bound is false: on the odd positive integers, take $f(1) = 0$ and $f(N) = e^N$ for $N \geq 3$. The term $N = 1$ contributes one to the printed left side, while its asserted right side is $O(\log x/x^c)$. Second, the complement of $\text{Syr}_{\min}(N) < f(N)$ is $\text{Syr}_{\min}(N) \geq f(N)$, not the printed strict inequality in the other direction; equality cannot be discarded. Third, the statement permits real-valued f , while line 591 silently restricts its codomain to $[0, +\infty)$.

Fourth, line 206 assigns logarithmic density 2^{-a} to the stratum $\nu_2(N) = a$ and $1 - 2^{-a_0}$ to the union $0 \leq a \leq a_0$. The exact values are respectively $2^{-(a+1)}$ and $1 - 2^{-(a_0+1)}$; the function on the odd coordinate must moreover be $M \mapsto f(2^a M)$, not $f(M)$.

Fifth, the iteration at lines 569–580 omits the possible value $J = 0$. Sixth, its probabilistic blocks need not be defined at small scales because the narrow interval can contain no odd integer. Seventh, the parenthetical at line 552 says that finite passage is redundant when $x \geq N_0$; this is false under the source convention $\text{Pass}_x = 1$ when $T_x = +\infty$, since $1 \in E_{N_0}$. The displayed proof does retain the finite-passage condition. Eighth, the substitution at lines 569–574 produces $O((\alpha^{j-2} \log y)^{-c})$, not the printed $O((\alpha^j \log y)^{-c})$. Their ratio is the fixed factor α^{2c} , so the iteration survives only after that factor is explicitly absorbed.

The same passage also suppresses the indexed first-passage morphism, the actual multiplicative block cover, its harmonic normalisation, and the dyadic weight transport. These are proved below. None of the repairs is silently attributed to the source.

For a real cutoff $X \geq 1$, set

$$H_{\text{odd}}(X) = \sum_{\substack{1 \leq M \leq X \\ M \text{ odd}}} \frac{1}{M}, \quad H(X) = \sum_{1 \leq N \leq X} \frac{1}{N}.$$

The integral test, with the integer cutoffs retained, gives

$$H_{\text{odd}}(X) = \frac{1}{2} \log X + O(1), \quad H(X) = \log X + O(1). \quad (305)$$

For a real threshold $N_0 \geq 2$, write

$$E_{N_0} = \{M \in 2\mathbb{N} + 1 : \text{Syr}_{\min}(M) \leq N_0\}. \quad (306)$$

Lemma 2.163 (the indexed nested-threshold morphism). *Let $1 \leq u \leq v$, and put*

$$\mathcal{D}_u = \{N \in 2\mathbb{N} + 1 : T_u(N) < +\infty\}.$$

For $N \in \mathcal{D}_u$, define

$$d_{u,v}(N) = T_u(N) - T_v(N) \in \mathbb{N}, \quad \sigma_d(k) = k + d.$$

Then

$$\text{Pass}_u(N) = \text{Syr}^{d_{u,v}(N)}(\text{Pass}_v(N)). \quad (307)$$

More precisely, for the indexed tails

$$\mathcal{O}_w^N(k) = \text{Syr}^k(\text{Pass}_w(N)), \quad k \in \mathbb{N},$$

one has the exact identity

$$\mathcal{O}_u^N = \mathcal{O}_v^N \circ \sigma_{du,v(N)}. \quad (308)$$

The shift σ_d is injective, has image $d + \mathbb{N}$, has singleton fibres over that image, and has diagonal equality kernel. Hence the value-set of the u -tail is contained in the value-set of the v -tail. In particular,

$$T_u(N) < +\infty, \quad \text{Pass}_u(N) \in E_{N_0} \implies \text{Pass}_v(N) \in E_{N_0}. \quad (309)$$

Proof. Every iterate at most u is also at most v , so $T_v(N) \leq T_u(N) < +\infty$. The semigroup law for the iterates gives

$$\text{Syr}^{T_u(N)}(N) = \text{Syr}^{T_u(N)-T_v(N)}(\text{Syr}^{T_v(N)}(N)),$$

which is (307). Applying a further k iterates gives (308). If the later tail from $\text{Pass}_u(N)$ reaches N_0 , the containing tail from $\text{Pass}_v(N)$ reaches the same value, proving (309). No injectivity of $\mathcal{O}_v^N$ is used or asserted: a periodic orbit can repeat values at different indices. $\square$

For every scale s for which the odd logarithmic block is nonempty, retain

$$\mathcal{R}_s = (2\mathbb{N} + 1) \cap [s, s^\alpha], \quad H_s = \sum_{N \in \mathcal{R}_s} \frac{1}{N}, \quad \mathbb{P}(\mathbf{N}_s = N) = \frac{1}{H_s N}.$$

Define the numerical event probability

$$b_{N_0}(s) = \mathbb{P}(T_s(\mathbf{N}_{s^\alpha}) < +\infty, \text{Pass}_s(\mathbf{N}_{s^\alpha}) \in E_{N_0}). \quad (310)$$

Proposition 2.164 (the exact block recurrence). *There are absolute constants $X_*, C, c > 0$ such that, whenever $s \geq X_*$,*

$$b_{N_0}(s^\alpha) \geq b_{N_0}(s) - C(\log s)^{-c}. \quad (311)$$

The cutoff X_ is chosen so that Proposition 1.11 applies and every block occurring here contains an odd integer.*

Proof. On the probability space supporting $\mathbf{N}_{s^{\alpha^2}}$, let

$$A_s = \{T_s(\mathbf{N}_{s^{\alpha^2}}) < +\infty, \text{Pass}_s(\mathbf{N}_{s^{\alpha^2}}) \in E_{N_0}\}.$$

Lemma 2.163, with $u = s$ and $v = s^\alpha$, gives

$$A_s \subseteq \{T_{s^\alpha}(\mathbf{N}_{s^{\alpha^2}}) < +\infty, \text{Pass}_{s^\alpha}(\mathbf{N}_{s^{\alpha^2}}) \in E_{N_0}\}.$$

The probability of the event on the right is $b_{N_0}(s^\alpha)$. Proposition 1.11 and the event inequality for its unhalved total-variation convention therefore give

$$\begin{aligned} b_{N_0}(s^\alpha) &\geq \mathbb{P}(A_s) \\ &\geq \mathbb{P}(\text{Pass}_s(\mathbf{N}_{s^{\alpha^2}}) \in E_{N_0}) - Cs^{-c} \\ &\geq \mathbb{P}(\text{Pass}_s(\mathbf{N}_{s^\alpha}) \in E_{N_0}) - C(\log s)^{-c} - Cs^{-c} \\ &\geq b_{N_0}(s) - C'(\log s)^{-c}. \end{aligned}$$

Increasing C proves (311). This uses the two marginal passage-location laws and failure control; it does not construct a coupling. $\square$

Theorem 2.165 (repaired exact form of Tao's Theorem 3.1). *There are absolute constants $C, c > 0$ such that, for all real $N_0 \geq 2$ and all real $X \geq 2$,*

$$\sum_{\substack{M \in 2\mathbb{N}+1, \ M \leq X \\ \text{Syr}_{\min}(M) > N_0}} \frac{1}{M} \leq C \frac{\log X}{(\log N_0)^c}, \quad (312)$$

and

$$\sum_{\substack{N \in \mathbb{N}+1, \ N \leq X \\ \text{Col}_{\min}(N) > N_0}} \frac{1}{N} \leq C \frac{\log X}{(\log N_0)^c}. \quad (313)$$

Consequently the corresponding good sets have, uniformly in X , odd relative and full logarithmic probabilities at least $1 - O((\log N_0)^{-c})$.

Proof. Choose X_* as in Proposition 2.164. We first prove the block bound

$$\mathbb{P}(\text{Syr}_{\min}(\mathbf{N}_s) > N_0) \ll (\log N_0)^{-c}. \quad (314)$$

For bounded N_0 , this follows after increasing the implied constant. Thus assume N_0 is large enough that $N_0^{1/\alpha^3} \geq X_*$.

If $s \leq N_0^{1/\alpha}$, every integer in $[s, s^\alpha]$ is at most N_0 , so the probability in (314) is exactly zero. This is the omitted $J = 0$ case.

Now suppose $s > N_0^{1/\alpha}$. Let $J \geq 1$ be least such that

$$y = s^{\alpha^{-J}} < N_0^{1/\alpha}.$$

Minimality gives

$$y^\alpha \geq N_0^{1/\alpha}, \quad y \geq N_0^{1/\alpha^2}, \quad y^\alpha < N_0. \quad (315)$$

Put $r_0 = y^{1/\alpha}$. Then $r_0 \geq N_0^{1/\alpha^3} \geq X_*$, and the input in $b_{N_0}(r_0)$ is $\mathbf{N}_y$. On finite passage below r_0 , the passage location is less than N_0 , so Proposition 1.11 gives

$$b_{N_0}(r_0) \geq 1 - Cr_0^{-c}. \quad (316)$$

For $j = 1, \dots, J$, apply (311) at the exact scale $y^{\alpha^{j-2}}$. The accumulated logarithmic error is

$$\begin{aligned} \sum_{j=1}^J (\alpha^{j-2} \log y)^{-c} &\leq (\log y)^{-c} \sum_{j=1}^{\infty} \alpha^{-c(j-2)} \\ &= \frac{\alpha^c}{1 - \alpha^{-c}} (\log y)^{-c} \ll (\log N_0)^{-c}. \end{aligned} \quad (317)$$

Here (315) was used only after the exact factor in the first line was retained. Equations (316) and (317) yield

$$b_{N_0}(s^{1/\alpha}) \geq 1 - C(\log N_0)^{-c}.$$

Its input variable is exactly $\mathbf{N}_s$. Whenever its event holds, the orbit of $\mathbf{N}_s$ reaches E_{N_0} , proving (314).

For $s \geq 2$, the integral test gives $H_s \ll \log s$, with an absolute constant because $\alpha = 1001/1000$ is fixed. An applied cover scale with $1 < s < 2$ has no odd integer in $[s, s^\alpha]$, so its local sum is zero; every nonempty cover block to which the estimate is applied satisfies $s \geq 2$. Hence

$$\sum_{\substack{M \in (2\mathbb{N}+1) \cap [s, s^\alpha] \\ \text{Syr}_{\min}(M) > N_0}} \frac{1}{M} \ll \frac{\log s}{(\log N_0)^c}. \quad (318)$$

For a target $X > N_0$, set $z_k = X^{\alpha^{-k}}$, and let $L \geq 1$ be least such that $z_L \leq N_0$. Then

$$z_{L-1} > N_0, \quad z_L > N_0^{1/\alpha}, \quad [z_L, X] = \bigcup_{k=1}^L [z_k, z_k^\alpha].$$

Every bad starting value is greater than N_0 , since its orbit contains the starting value. Thus the leftover interval $[1, z_L]$ is bad-free, and endpoint duplication in the displayed cover can only enlarge an upper bound. Finally,

$$\sum_{k=1}^L \log z_k = \log X \sum_{k=1}^L \alpha^{-k} \leq \frac{\log X}{\alpha - 1}. \quad (319)$$

Summing (318) proves (312). If $X \leq N_0$, its left side is zero.

For the all-integer assertion, write every N uniquely as $N = 2^a M$, with $a \in \mathbb{N}$ and M odd. The exact minimum identity proved again below gives

$$\sum_{\substack{N \leq X \\ \text{Col}_{\min}(N) > N_0}} \frac{1}{N} = \sum_{a \geq 0} 2^{-a} \sum_{\substack{M \leq X/2^a, M \text{ odd} \\ \text{Syr}_{\min}(M) > N_0}} \frac{1}{M}. \quad (320)$$

When $X/2^a < 2$, the inner bad sum vanishes because $N_0 \geq 2$. For the remaining terms, (312) and the truncated, termwise nonnegative estimate

$$\sum_{\substack{a \geq 0 \\ X/2^a \geq 2}} 2^{-a} \log(X/2^a) \leq \log X \sum_{a \geq 0} 2^{-a} \leq 2 \log X$$

prove (313). The probability statements follow from (305); no identity of unnormalised and normalised harmonic mass is being asserted. $\square$

Theorem 2.166 (repaired exact form of Tao's Theorem 1.6). *Let $f : 2\mathbb{N} + 1 \rightarrow \mathbb{R}$ satisfy $f(M) \rightarrow +\infty$. Then*

$$\text{Syr}_{\min}(M) < f(M) \quad (321)$$

for a set of odd positive integers of relative odd logarithmic density one. Equivalently, the set of odd M on which $\text{Syr}_{\min}(M) \geq f(M)$ has relative odd logarithmic density zero.

Proof. Fix an integer $K \geq 2$. Since $f(M) \rightarrow +\infty$, there is a finite cutoff Y_K such that

$$f(M) > K \quad (M \geq Y_K, M \text{ odd}).$$

Let $D_f = \{M \text{ odd} : \text{Syr}_{\min}(M) \geq f(M)\}$. For every X , the exact tail inclusion gives

$$\begin{aligned} \sum_{\substack{M \leq X \\ M \in D_f}} \frac{1}{M} &\leq H_{\text{odd}}(Y_K) + \sum_{\substack{M \leq X, M \text{ odd} \\ \text{Syr}_{\min}(M) > K}} \frac{1}{M} \\ &\leq H_{\text{odd}}(Y_K) + C \frac{\log X}{(\log K)^c}. \end{aligned} \quad (322)$$

Divide first by $H_{\text{odd}}(X)$, let $X \rightarrow +\infty$ with K fixed, and only then let $K \rightarrow +\infty$. Equation (305) gives

$$\limsup_{X \rightarrow \infty} \frac{\sum_{M \leq X, M \in D_f} 1/M}{H_{\text{odd}}(X)} \leq \frac{2C}{(\log K)^c} \rightarrow 0.$$

This proves the strict inequality in (321), includes the equality part of its complement, permits arbitrary nonmonotone real f , and confines every negative or otherwise exceptional value of f to the finite prefix. No tail lower envelope is used. $\square$

Proposition 2.167 (the exact dyadic stratum morphism). *Let*

$$\mathcal{O} = 2\mathbb{N} + 1, \quad \mathcal{S}_a = \{N \in \mathbb{N} + 1 : \nu_2(N) = a\}.$$

For each $a \in \mathbb{N}$, the maps

$$\iota_a : \mathcal{O} \longrightarrow \mathcal{S}_a, \quad \iota_a(M) = 2^a M, \quad \pi_a : \mathcal{S}_a \longrightarrow \mathcal{O}, \quad \pi_a(N) = N/2^a$$

are two-sided inverses. Their fibres are singletons, their equality kernels are diagonal, and they lose no starting-number coordinate information. Globally,

$$\mathbb{N} + 1 \longleftrightarrow \coprod_{a \in \mathbb{N}} \{a\} \times \mathcal{O}, \quad N \longmapsto (\nu_2(N), N/2^{\nu_2(N)}) \quad (323)$$

is a bijection with inverse $(a, M) \mapsto 2^a M$.

For every $A \subseteq \mathcal{O}$ and real $X \geq 1$, the cutoff and harmonic weight transport exactly as

$$\sum_{\substack{N \in \iota_a(A) \\ N \leq X}} \frac{1}{N} = 2^{-a} \sum_{\substack{M \in A \\ M \leq X/2^a}} \frac{1}{M}. \quad (324)$$

In particular,

$$\sum_{\substack{N \leq X \\ \nu_2(N) = a}} \frac{1}{N} = 2^{-a} H_{\text{odd}}(X/2^a) = 2^{-(a+1)} \log X + O_a(1). \quad (325)$$

Finally,

$$\text{Col}_{\min}(2^a M) = \text{Syr}_{\min}(M). \quad (326)$$

Proof. The defining condition $\nu_2(N) = a$ says uniquely that $N = 2^a M$ with M odd, which proves every bijection and fibre assertion. Replacing N by $2^a M$ in the finite sum proves (324); then (305) proves (325).

It remains to prove the minimum identity rather than infer it from notation. The first a Collatz steps starting at $2^a M$ are divisions by two and terminate at M ; every value on that

initial segment is at least M . If M_j is an odd Syracuse state and $q_j = \nu_2(3M_j + 1)$, then the corresponding Collatz segment is

$$M_j, 2^{q_j} M_{j+1}, 2^{q_j-1} M_{j+1}, \dots, 2M_{j+1}, M_{j+1}, \quad M_{j+1} = \text{Syr}(M_j).$$

Every suppressed intermediate value is at least its terminal odd value. Thus the minimum over the complete Collatz orbit is exactly the minimum over the odd Syracuse states, proving (326). The map is not time-preserving. The full Collatz time coordinate is reconstructed by expanding each Syracuse step using $q_j = \nu_2(3M_j + 1)$, which is determined by its odd source state M_j . $\square$

Theorem 2.168 (repaired exact form of Tao's Theorem 1.3). *Let $f : \mathbb{N} + 1 \rightarrow \mathbb{R}$ satisfy $f(N) \rightarrow +\infty$. Then*

$$\text{Col}_{\min}(N) < f(N) \tag{327}$$

for a set of positive integers of logarithmic density one.

Proof. For each fixed $a \in \mathbb{N}$, define

$$g_a(M) = f(2^a M) \quad (M \in \mathcal{O}).$$

Then $g_a(M) \rightarrow +\infty$, so Theorem 2.166 says that

$$B_a = \{M \in \mathcal{O} : \text{Syr}_{\min}(M) \geq g_a(M)\}$$

has harmonic mass $o(\log T)$ below T . By Proposition 2.167, the bad set for (327) in $\mathcal{S}_a$ is exactly $\iota_a(B_a)$, and its mass below X is

$$2^{-a} \sum_{\substack{M \in B_a \\ M \leq X/2^a}} \frac{1}{M} = o(\log X)$$

for this fixed a .

Fix $A \in \mathbb{N}$. Splitting the global bad mass into the first $A + 1$ strata and the remaining tail gives

$$\begin{aligned} \sum_{\substack{N \leq X \\ \text{Col}_{\min}(N) \geq f(N)}} \frac{1}{N} &\leq \sum_{a=0}^A 2^{-a} \sum_{\substack{M \in B_a \\ M \leq X/2^a}} \frac{1}{M} + \sum_{\substack{N \leq X \\ \nu_2(N) > A}} \frac{1}{N}, \\ \sum_{\substack{N \leq X \\ \nu_2(N) > A}} \frac{1}{N} &= 2^{-(A+1)} H(X/2^{A+1}). \end{aligned} \tag{328}$$

The second identity follows from the unique formula $N = 2^{A+1}Q$, with Q now an arbitrary positive integer. Divide by $H(X)$ and let $X \rightarrow +\infty$ at fixed A . The finite first sum vanishes in the limit and (305) gives

$$\limsup_{X \rightarrow \infty} \frac{\sum_{N \leq X, \text{Col}_{\min}(N) \geq f(N)} 1/N}{H(X)} \leq 2^{-(A+1)}.$$

Only now send $A \rightarrow \infty$. The bad logarithmic density is zero, which proves the strict inequality (327). No uniformity in a , and no interchange of the two limits, has been used. $\square$

As an independent cross-check, one may apply the fixed- K argument in (322) directly to the Collatz estimate (313). This gives Theorem 2.168 without passing through the dyadic strata, and yields the same strict bad-set complement.

Explicit nonclaim 2.169. Theorems 2.165, 2.166, and 2.168 prove the exact repaired chain

$$\text{Proposition 1.11} \implies \text{Theorem 3.1} \implies \text{Theorem 1.6} \implies \text{Theorem 1.3}.$$

They prove logarithmic-density statements, not natural-density statements; they do not give an absolute bound valid for almost all starting values, force an individual orbit through a threshold, exclude divergent or periodic exceptional orbits, prove convergence to one, construct an invariant logarithmic measure, or produce a coupling. The dyadic bijection preserves all starting-number information—it is a bijection on the displayed starting coordinates—and preserves the orbit minimum at the displayed scope. It is not itself a time-preserving conjugacy; Collatz time is recovered by the explicit valuation expansion above. This closes the main-theorem reduction, not the durable whole-paper, whole-corpus, formalisation, or release gates.

2.24 Computational verification: exact coordinates, certificates, and chronology

The computational literature does not use one unnamed Collatz map. In this subsection

$$U(n) = \begin{cases} 3n + 1, & n \text{ odd}, \\ n/2, & n \text{ even}, \end{cases} \quad T(n) = \begin{cases} (3n + 1)/2, & n \text{ odd}, \\ n/2, & n \text{ even} \end{cases}$$

denote respectively the unshortened and shortened maps. One odd T -step is two U -steps and one even T -step is one U -step. Barina also uses the auxiliary chart

$$T_1(n) = \begin{cases} (n + 1)/2, & n \text{ odd}, \\ 3n/2, & n \text{ even}. \end{cases}$$

Its exact branchwise bridges are

$$T(n) = \begin{cases} T_1(n + 1) - 1, & n \text{ odd}, \\ n/2, & n \text{ even}, \end{cases} \quad T_1(n) = \begin{cases} T(n - 1) + 1, & n \text{ odd}, \\ 3n/2, & n \text{ even}. \end{cases}$$

They move the additive term between coordinate charts; they do not identify the maps or their clocks. In particular, the 2025 paper's page-4 *trajectory on $n + 1$* , $(7, 4, 6, 9, 5, 3, 2)$, is the shifted-coordinate presentation associated with the trajectory from 12, not the T -orbit of 13 [35, pp. 2, 4].

Proposition 2.170 (Barina's block map, with fibres). *For $x \in \mathbb{N}_{>0}$, put*

$$\alpha = \nu_2(x + 1), \quad u = \frac{x + 1}{2^\alpha}, \quad y = 3^\alpha u - 1, \quad \beta = \nu_2(y),$$

and $B(x) = y/2^\beta$. Then $B(x)$ is a positive odd integer and

$$B(x) = T^{\alpha+\beta}(x). \tag{CV.1}$$

For every odd $z > 0$, the fibre $B^{-1}(z)$ is exactly the set of

$$x = 2^\alpha \frac{2^\beta z + 1}{3^\alpha} - 1 \quad (\text{CV.2})$$

over integers $\alpha \geq 0, \beta \geq 1$ satisfying $3^\alpha \mid 2^\beta z + 1$. Hence B is surjective, has infinite fibres, and has no left or two-sided inverse after the block boundary and elapsed clock have been discarded.

Proof. The integer u is odd. For $0 \leq j \leq \alpha$, direct induction gives

$$T^j(x) = 3^j \frac{x+1}{2^j} - 1.$$

The state at $j = \alpha$ is y ; the following β steps divide it by two, proving (CV.1). Conversely, (CV.2) gives $x+1 = 2^\alpha u$, with $u = (2^\beta z + 1)/3^\alpha$ odd, and $3^\alpha u - 1 = 2^\beta z$. Thus the two valuations are exactly α, β and $B(x) = z$. The section $z \mapsto 2z$ proves surjectivity. Taking $\alpha = 0$ in (CV.2) gives the distinct points $2^\beta z, \beta \geq 1$, in the same fibre. $\square$

On the odd 2-adic integers with normalized Haar measure, $\Pr(\alpha = a) = 2^{-a}$ and $\Pr(\beta = b \mid \alpha = a) = 2^{-b}$ for $a, b \geq 1$. Multiplication by the odd unit 3^a proves the conditional statement, so the two valuations are independent and

$$\mathbb{E}(\alpha + \beta) = 4.$$

This is the exact measure-theoretic content of the articles' phrase *average four iterations*; it is not an assertion that every finite interval has raw arithmetic mean four [34, pp. 3–5].

Source issue 2.171 (the exceptional start 1 in the printed algorithms). Equation (CV.1) has $B(1) = 1$, representing the shortened cycle $1 \mapsto 2 \mapsto 1$. The printed endpoints differ by version. The 2021 Algorithm 1 uses strict **until** $\mathbf{n} < \mathbf{n}_0$ and loops at $n_0 = 1$. Its Algorithm 2 uses **until** $\mathbf{n} = 1$: it terminates after one complete $1 \mapsto 1$ macro, but a literal delay accumulator gives 2 rather than the conventional stopping time 0. Both 2025 Algorithms 1 and 2 use the strict endpoint and loop at 1 [34, pp. 4–5] [35, pp. 4–5]. The exact repairs are a separate base case, or the domain $n_0 > 1$ where only descent is required. The production inner loop tests starts 3 mod 4, so this exceptional input is not encountered there.

Proposition 2.172 (the dyadic affine-prefix coordinate). *Fix $k \geq 0, h \geq 0$, and $0 \leq r < 2^k$, and let*

$$a_j(r) = \#\{0 \leq i < j : T^i(r) \text{ is odd}\}.$$

For every $0 \leq j \leq k$,

$$T^j(2^k h + r) = 3^{a_j(r)} 2^{k-j} h + T^j(r). \quad (\text{CV.3})$$

In particular,

$$T^k(2^k h + r) = 3^{a_k(r)} h + T^k(r). \quad (\text{CV.4})$$

Proof. For $j < k$, the high term in (CV.3) is even. The full state and the low state $T^j(r)$ therefore have the same parity and take the same next branch. Applying that branch proves the induction. This is simultaneously Oliveira e Silva's Proposition 1 and Barina's equation (6), after their maps and variables have been typed rather than identified [36, p. 191][34, p. 5]. $\square$

Write $a = a_k(r)$, $d = T^k(r)$, $L = 3^a - 2^k$, and $R = r - d$. The exact necessary-and-sufficient condition for

$$T^k(2^k h + r) < 2^k h + r \quad (h = 1, 2, \dots)$$

is

$$(L < 0 \text{ and } L < R) \quad \text{or} \quad (L = 0 \text{ and } R > 0). \quad (\text{CV.5})$$

Indeed the inequality is $Lh < R$, whose most restrictive positive h is $h = 1$ when $L < 0$; it is constant when $L = 0$, and eventually fails when $L > 0$. The generator's `is_convergent` implements only the first disjunct. It is conservative in the equality case; it does not falsely certify it.

If $r' < r$ and

$$a_k(r') = a_k(r), \quad T^k(r') = T^k(r),$$

then (CV.4) proves, for every h ,

$$T^k(2^k h + r') = T^k(2^k h + r), \quad 2^k h + r' < 2^k h + r. \quad (\text{CV.6})$$

Thus a dead sieve bit has a finite descent, coalescence, inherited-prefix, or restricted-predecessor certificate. A live bit is merely the complement of the implemented certificates. It is not a residue class proved not to converge; the 2025 page-6 wording *do not converge* must be read at this finite algorithmic scope [35].

Source issue 2.173 (cardinality and generator-history defects). Within one half-open work unit of width 2^{40} , a fixed class modulo 2^k contains 2^{40-k} starts, while there are 2^k classes. The 2025 Section 3.4 reverses these quantities twice. The 2021 page-6 example and `src/worker/worker.c:336--369` use the correct 2^{40-34} .

The separate `collatz-sieve` history has a dated implementation defect. Commit `49c5524d8d42867c866e863030a8e5d86237bf17` (10 April 2020) passed the odd-step count `c_` as both the third and fourth arguments of `is_convergent`; the fourth argument had to be the endpoint `d_`. Commit `b3d2f61d89eec55008d61c32577a9e8c8b3874ab` (6 April 2023) repaired it. The baseline committed maps predate the faulty interval, while the `hesieve` and `h2esieve` maps postdate the repair. The defect governs regeneration from intermediate source history; it is not silently attributed to blobs outside that interval.

The first exact lower-start reductions are

$$T(2q + 1) = 3q + 2,$$

and

$$T^3(8q + 3) = 9q + 4, \quad T^6(64q + 7) = 81q + 10, \quad T^7(128q + 95) = 243q + 182. \quad (335)$$

Their odd-step counts are respectively 1, 2, 4, 5. Every source on the left is strictly smaller than its target. The first three families omit 37 of 81 initial residues: the first two give $\{2, 4, 5, 8\} \bmod 9$, their lifts modulo 27 are

$$\{2, 4, 5, 8, 11, 13, 14, 17, 20, 22, 23, 26\},$$

and the $81q + 10$ class is disjoint from them. The fourth target is already $2 \bmod 3$ as an initial residue, but is not redundant when it occurs as an intermediate low-coordinate state.

Proposition 2.174 (the Hercher coalescence lift). *Let x be odd and put*

$$\alpha = \nu_2(x+1), \quad u = (x+1)/2^\alpha, \quad y = 3^\alpha u - 1, \quad \beta = \nu_2(y).$$

If $\beta \geq 2$ and $m = (x-1)/2$, then

$$T^{\alpha+\beta-1}(m) = T^{\alpha+\beta}(x) = y/2^\beta. \quad (\text{CV.7})$$

If $x = T^s(r)$, the prefix contains c odd steps, and $m < r$, then for every $N = 2^k h + r$ the lower lifted start

$$Q = 3^c 2^{k-s-1} h + m$$

satisfies $Q < N$ and its trajectory coalesces with that of N .

Proof. Since $y \equiv 0 \pmod{4}$, direct iteration of m gives (CV.7). The affine prefix has nonnegative intercept, positive when $c > 0$. From $m < r$ one has $x < 2r$, hence $3^c/2^s < 2$. This is exactly the strict high-coefficient comparison for $Q < N$; (CV.3) and (CV.7) then prove coalescence. $\square$

The additional Oliveira–Roosendaal example is likewise an identity:

$$T^6(64q + 15) = T^6(64q + 14) = T^5(32q + 7) = 81q + 20. \quad (\text{CV.8})$$

Theorem 2.175 (finite closure of the recursive high-coefficient check). *Consider **collatz-sieve's** restricted recursive closure generated by the four inverse transports above. For every invocation arising in its packed implementation domain $k \leq 34$, a returned lower representative $P < r$ has a uniformly lower full-width lift. This assertion is finite in k ; it does not prove an unbounded recursive invariant.*

Proof. A recursive word with d odd steps and t total shortened steps gives

$$T^t(P) = L.$$

For the four primitive words, (d, t, H) is

$$(1, 1, 1), (2, 3, 5), (4, 6, 73), (5, 7, 211),$$

and the inverse is the exact floor map

$$\phi(L) = \frac{2^t L - H}{3^d} = \left\lfloor \frac{2^t L}{3^d} \right\rfloor.$$

A nonempty composite word has

$$P = \frac{2^t L - H_w}{3^d} = \frac{2^t}{3^d} L - h_w, \quad 0 < h_w < \ell(w) \leq d. \quad (\text{CV.9})$$

Suppose $L = T^s(r)$ after c odd steps. Substitution in (CV.3) shows that the unique coefficient-correct full lift is

$$Q = 3^{c-d} 2^{k-s+t} h + P,$$

and $Q < 2^k h + r$ for every $h > 0$ precisely when $P < r$ and

$$2^t 3^{c-d} \leq 2^s. \quad (\text{CV.10})$$

For the recursive branch applied after Proposition 2.174, the corresponding condition, at the preceding prefix (s_0, c_0) , is

$$2^t 3^{c_0-d} \leq 2^{s_0+1}. \quad (\text{CV.11})$$

The C function returns only P , retains neither d nor t , and tests neither inequality.

An exact finite reduction supplies the missing check. In the relevant exponent boxes the smallest coefficient strictly above one is

$$C_{\min} = \frac{3^{12}}{2^{19}} = \frac{531441}{524288}, \quad C_{\min} - 1 = \frac{7153}{524288}. \quad (\text{CV.12})$$

If (CV.10) failed while $P < r$, equation (CV.9) and the affine prefix imply $P(C-1) < h_w < 34$ and $r - P \leq 33$; hence $P \leq 2492$. For (CV.11), the extra half-step contributes less than $1/2$, so an unsafe witness has $P \leq 2528$ and $r - P \leq 34$.

The deterministic certificate exhausts these proved supersets. In the current-state branch it checks 7105 restricted word states, 1557 identity endpoint pairs, 2983 nonidentity common-endpoint pairs, and all 2585 budget-admissible pairs. In the coalescence branch it checks 7216 word states, 1003 common-endpoint pairs, including 480 identity and 523 nonidentity pairs, and all 487 budget-admissible nonidentity pairs. Every required comparison satisfies (CV.10) or (CV.11). The finite minimum (CV.12), rather than an asymptotic claim, is what closes the implementation domain. $\square$

Source issue 2.176 (remaining code and witness boundaries). The name `smallest_predecessor` means the minimum only in the four-map recursive closure, not among all Collatz predecessors. At residue zero, the unsigned expression $(L_{\text{last}} - 1)/2$ wraps; it does not clear zero but has no positive-integer predecessor meaning. The effective $k \leq 34$ domain is the packed-field assertion $b < 2^{34}$, not a function precondition; disabling assertions outside it permits the 34-bit residue field to alias. The `ctzu64` helper assumes that `unsigned long` has 64 bits and traps on LLP64. Finally, a dead-map bit stores no predecessor word, step counts, or elimination-reason witness, so the bit alone cannot reconstruct its v3 certificate.

Proposition 2.177 (finite convergence from certified descent). *If, for every $2 \leq n \leq X$, a computation exhibits a finite iterate $T^{j(n)}(n) < n$, then every positive start through X reaches 1.*

Proof. Strong induction on n . The displayed iterate is a smaller positive start, so its orbit reaches 1 by the induction hypothesis; concatenate the two finite orbit segments. The base $n = 1$ is immediate. $\square$

This proposition explains why sieve elimination and stopping-time descent can establish a finite range without individually iterating every start to one. It does not establish that a reported machine run occurred.

Source	exact reported range	status in this edition
Oliveira e Silva (2010)	$n \leq 20 \cdot 2^{58}$	sourced finite result
Barina (2021)	$n < 2^{68}$	sourced finite result
Barina (2025)	$n < 2^{71}$, 15 Jan. 2025	sourced finite result
Project API, 28 Aug. 2026	$2075 \cdot 2^{60}$	later dynamic assertion

Oliveira e Silva’s printed inclusive endpoint is

$$20 \cdot 2^{58} = 5764607523034234880.$$

The twenty directly processed intervals are naturally half-open, but the boundary itself is even and maps to the already covered $10 \cdot 2^{58}$; hence the inclusive theorem follows. The chapter reports a depth-46 pruned parity tree with 269949796982 surviving leaves, 2^{12} -start blocks, variable-step tables with $\Delta = 14$ and tolerance $\tau = 10$, two independent workers per item, and exact agreement of candidate data and a cyclic-redundancy checksum. One mismatch in about 81 CPU-years was invalidated and recomputed [36, pp. 189–198]. The underlying worker source, completed-work ledger, and duplicate outputs were not acquired, so the range is not relabelled an F031 reproduction.

The 2021 article reports the 2^{68} run from September 2019 to May 2020 [34, p. 7]. The 2025 Table 10 records the strict boundary $n < 2^{71}$ and completion date; its surrounding phrase *up to* does not change the half-open work units

$$[j2^{40}, (j+1)2^{40}).$$

The boundary 2^{71} itself is a power of two and independently descends, but was not in the first 2^{31} units. Reported CPU/GPU years, throughput, and speedups are measurements on named configurations, not hardware-independent complexity theorems. The page-2 throughput convention also counts starts eliminated by sieve certificates whose trajectories were not individually run to one [35].

The dated API snapshot reports

$$2075 \cdot 2^{60} = 2392312122059207475200 \approx 2^{71.0188956211},$$

with the first incomplete unit beginning at

$$2175975546 \cdot 2^{40} = 2392510414583230365696.$$

It is later than the paper, is not a completed-work ledger, and is not backdated into the 2025 theorem [39].

Source issue 2.178 (the public-code validation boundary). The 2025 paper pins the repository `xbarin02/collatz` at commit `53c2a0608075d6fe3f10cc6eeef50e400c86338`. It does not pin the separate `collatz-sieve` dependency. The bootstrap performs unpinned clone-or-pull operations. The edition’s sieve commit and map hashes are therefore separately attributed reconstruction pins, not historical facts invented for the run [38].

The default GPU kernel is the Git symlink `kernel.c1` to `kernel132.c1`. A GPU overflow zeroes a checksum lane; the host aborts the unit and `mclient` attempts a CPU rerun. GMP continuation exists only when that CPU worker was compiled with `_USE_GMP`. The generic bootstrap does so, whereas the Lumi and Meta GPU-only scripts build no CPU worker. Thus GPU–CPU–GMP is a conditional deployment chain, not GPU multiprecision or a universal local fallback.

The optional/test `kernel132-precalc.c1` has a sieve-dependent `continue` immediately before a work-group barrier. A multi-lane group with differing sieve bits violates barrier convergence. This defect is scoped to that variant and is not transferred to the default kernel.

The server first rejects stale returns with the wrong stored client ID and rejects zero or non-whitelisted checksum prefixes. For an accepted current return, however, it marks the unit

complete before comparing a previous checksum, then logs and overwrites a mismatch; maximum offsets are handled in the same order. McIent transmits the α -sum and discards the β -sum. The verification utility recomputes only the trajectory maximum at the stored offset, not the work unit. These are consistency signals, not Oliveira e Silva's independent duplicate-run protocol or a cryptographic certificate.

No acquired immutable ledger binds every unit to its code tree, map hash, binary, compiler, OpenCL environment, and return value. Selected regression tests and public source permit code-level audit; they do not independently authenticate the full historical 2^{71} computation.

Proposition 2.179 (shortened and unshortened maximum excursions). *For an odd $n > 1$, if the orbit maxima are finite, then*

$$M_U(n) = 2M_T(n). \quad (\text{CV.13})$$

The nontrivial record-start sequence is the same under both conventions, although its maximum values differ by the factor two.

Proof. An odd shortened state $x > 1$ cannot be the shortened maximum because $(3x + 1)/2 > x$. A maximal shortened state cannot have an even predecessor, which would be twice as large. It is therefore the output $y = (3x + 1)/2$ of an odd predecessor, and the unshortened orbit contains the intermediate state $2y$. Every state omitted by shortening is twice a shortened state, proving (CV.13). No even $n > 2$ begins a new unshortened record, because the lower odd start $n - 1$ has first image $3n - 2 > n$. Thus record starts are odd and multiplication of every maximum by two preserves their order. $\square$

The 2025 abstract says four new records, while Section 6 lists five and calls them new. The first listed start, 274133054632352106267, already appears in the 2021 paper. The exact reconciliation is four records new after 2021 and five cumulative project records; Section 6's phrase *five new* is imprecise rather than an unexplained numerical contradiction [34, p. 7][35, pp. 12–13]. Finite record data are consistent with the Lagarias–Weiss prediction but do not prove its asymptotic form.

Proposition 2.180 (Roosendaal's modular maximum theorem). *Let $N > 2$ be odd. If its unshortened maximum $q = M_U(N)$ is finite and $q > 3N + 1$, then*

$$q \equiv 16 \pmod{36}.$$

Proof. The maximum q is even and cannot be $2 \pmod{4}$, since then the next two states $q/2$ and $3q/2 + 1$ exceed it. Because $q > 3N + 1$, it is neither the start nor its first odd output; halving cannot create a maximum, so $q = 3p + 1$ for an odd predecessor and $q \equiv 1 \pmod{3}$. The remaining classes modulo 36 are 4, 16, 28. If $q = 36a + 4$, its forced preceding states include $48a + 4 > q$. If $q = 36a + 28$, they include $48a + 36 > q$. Only $16 \pmod{36}$ remains [37]. $\square$

The page's following sentence applies this congruence to P_i , but the theorem concerns $M_U(P_i)$, not the starting values. Its Conjecture 3 also prints $N < 1$ against the surrounding $N > 1$ domain. Neither defect is silently normalized. Theorems 2–3 each begin with a residue-bound hypothesis that is not proved on the page; this edition does not establish that hypothesis and therefore does not import either conclusion.

Oliveira e Silva additionally studies the one-branch-per-iterate map

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ n/3, & n \not\equiv 0 \pmod{2}, n \equiv 0 \pmod{3}, \\ 5n + 1, & \gcd(n, 6) = 1. \end{cases} \quad (\text{CV.14})$$

Its clock is not the T - or U -clock.

Source issue 2.181 (Oliveira e Silva's Proposition 2 is false as printed). The proposition asks the first i iterates of m_i to determine counts k_2, k_3, k_5 , puts

$$r_i = 2^{k_2} 3^{k_3}, \quad m_i = n_0 \bmod r_i, \quad n_i = \lfloor n_0 / r_i \rfloor,$$

and asserts

$$C^i(n_0) = 5^{k_5} n_i + C^i(m_i).$$

At $i = 1$, $n_0 = 1$, $r_i = 1$ forces $m_i = 0$, whose division-by-two branch demands $r_i = 2$; $r_i = 2$ or 3 forces $m_i = 1$, whose $5n + 1$ branch demands $r_i = 1$. Thus the circular definition has no fixed point. If one instead takes the branch of n_0 first, then $r_i = 1$, $m_i = 0$, $n_i = 1$ and the asserted equation reads $6 = 5$. No domain qualification repairs it [36, p. 200].

The chapter defines $\lceil x \rceil$ as the largest integer not smaller than x , which is ceiling language. Proposition 1's unique dyadic decomposition requires $\lfloor n_0 / 2^i \rfloor$. Replacing the bracket by a floor repairs that use; it does not repair Proposition 2.

Proposition 2.182 (the corrected 6^i branch-cylinder coordinate). *For $i \geq 0$, put*

$$a_i = n_0 \bmod 6^i, \quad Q_i = \left\lfloor \frac{n_0}{6^i} \right\rfloor.$$

Let k_2, k_3, k_5 be the branch counts in the first i iterates of a_i , set $R_i = 2^{k_2} 3^{k_3}$, and

$$N_i = \frac{n_0 - a_i}{R_i} = \frac{6^i}{R_i} Q_i.$$

Then N_i is an integer, n_0 and a_i have the same first i branches, and

$$C^i(n_0) = 5^{k_5} N_i + C^i(a_i). \tag{CV.15}$$

The representative a_i is a branch-cylinder representative, generally not the canonical remainder modulo R_i .

Proof. If $x \equiv y \pmod{6^j}$, the two points take the same branch. After division by two their difference is divisible by $2^{j-1} 3^j$, after division by three by $2^j 3^{j-1}$, and after $5x + 1$ by 6^j ; every image pair is congruent modulo 6^{j-1} . Induction gives the common length- i branch word.

For a fixed word w , induction on its branches gives an integer B_w with

$$C^i(x) = \frac{5^{k_5} x + B_w}{R_i}$$

throughout its cylinder. A division branch multiplies R_i by two or three, while a $5x + 1$ branch replaces B_w by $5B_w + R_i$. Since $R_i \mid 6^i$, subtracting the formula at n_0 and a_i proves (CV.15). Reducing a_i modulo R_i is valid only after a separate proof that the reduced point retains the same branch word. $\square$

The source reports exhaustive computations through 10^{16} for (CV.14) and 10^{14} for the analogous $C_{7,\{2,3,5\}}$, taking about five and two CPU-months respectively on one 450 MHz Pentium III [36, pp. 200–204]. It does not say that the replicated-worker protocol of the $3x + 1$ run was used here. Its finite pruning counts, affine tables, transition matrix, and stationary vector

are reproducible finite calculations. Section 4 expressly calls the passage from a Chernoff upper bound to an asymptotic equality non-rigorous; the printed values

$$\rho_5 \approx 1.442762198279, \quad \rho_7 \approx 2.215508001593$$

remain conjectured record-growth exponents, not proved rates.

Explicit nonclaim 2.183. The deterministic F031 certificate pins the exact manifestations, code trees, bundles, selected maps, and route artifacts; reproduces the basic 2^{16} sieve byte for byte; proves Theorem 2.175; and checks the map, fibre, affine, transport, maximum, modular, and corrected branch-cylinder identities above. It does not reproduce a historical distributed run, authenticate a completed-work database, independently regenerate the full 2^{24} or 2^{34} maps, certify hardware or compilers, justify the Markov asymptotics, or prove universal convergence. The eight legacy unresolved-dependency records are audit-schema records rather than mathematical claims; this section does not use them to preserve or conditionalize an unproved conclusion.

Explicit nonclaim 2.184. Crandall’s Theorem 6.1 proves only that $\pi(X) > X^c$ for some $c > 0$ and all sufficiently large X , where π counts positive odd $m > 1$ known to reach 1 in accelerated time; the repaired source argument permits the edition witness $c = 1/100$ (Theorem 2.46). It is not a positive-density theorem. Tao’s theorem gives logarithmic-density-one descent below every prescribed divergent threshold, but does not say those orbits reach 1. Neither result, nor their conjunction, yields logarithmic-density-one convergence to 1.

3 Modern Chatnotes: the four-label coefficient algebra

This section is an editorial reconstruction of raw Chatnotes, not an attribution to the authors of the external literature. The controlling local witness is the raw export *Z_n Symmetries of Collatz*, physical lines 5712–6220. The external source actually cited in that interval is Aizawa–Isaac–Segar’s construction of $\mathbb{Z}_2 \times \mathbb{Z}_2$ color superalgebras inside universal enveloping algebras [40]. The elementary coefficient results below are proved directly. The cited paper controls only the color definition and its own enveloping-algebra example.

3.1 Four affine labels and the state-dependent evaluator

Put

$$V = \mathbb{F}_2^2, \quad F_{a,b}(X) = \frac{(1+2a)X+b}{2} \quad ((a,b) \in V).$$

Thus the four labels give

$$F_{00}(X) = \frac{X}{2}, \quad F_{01}(X) = \frac{X+1}{2}, \quad F_{10}(X) = \frac{3X}{2}, \quad F_{11}(X) = \frac{3X+1}{2}.$$

Let $\text{Aff}_{\mathbb{Q}}^{\times}$ denote the group of affine maps $X \mapsto mX + c$ with $m \in \mathbb{Q}^{\times}$, represented by coordinates (m, c) and composition

$$(m, c)(m', c') = (mm', mc' + c).$$

Proposition 3.1 (exact type of the four-label map). *The map of sets*

$$\lambda : V \longrightarrow \text{Aff}_{\mathbb{Q}}^{\times}, \quad \lambda(a, b) = \left(\frac{1+2a}{2}, \frac{b}{2} \right),$$

is injective. It is not a homomorphism from the additive group V to the affine group under composition. For every $n \in \mathbb{Z}$,

$$F_{a,b}(n) \in \mathbb{Z} \iff n \equiv b \pmod{2}.$$

Proof. Equality of two affine maps gives equality of their intercepts, hence $b = b'$, and equality of their slopes, hence $a = a'$. This proves injectivity. The identity of the affine group is $(1, 0)$, whereas the zero of V is sent to $(1/2, 0)$. A group homomorphism must preserve the identity, so λ is not one. Finally, $1 + 2a$ is odd, and therefore

$$(1 + 2a)n + b \equiv n + b \pmod{2}.$$

Division by two is integral exactly when this numerator is even, equivalently $n \equiv b \pmod{2}$. $\square$

Define

$$\partial : V \longrightarrow \mathbb{F}_2, \quad \partial(a, b) = a + b, \quad \Delta = \ker \partial = \{00, 11\}.$$

For $n \in \mathbb{Z}$, let $\bar{n}$ be its class in $\mathbb{F}_2$ and set

$$\iota(n) = (\bar{n}, \bar{n}) \in \Delta.$$

The map $\iota : \mathbb{Z} \rightarrow \Delta$ is a surjective group homomorphism with kernel $2\mathbb{Z}$ and fibres

$$\iota^{-1}(00) = 2\mathbb{Z}, \quad \iota^{-1}(11) = 2\mathbb{Z} + 1.$$

Proposition 3.2 (typed diagonal evaluation). *Let*

$$D = \{(n, v) \in \mathbb{Z} \times \Delta : v = \iota(n)\}, \quad \eta : \mathbb{Z} \longrightarrow D, \quad \eta(n) = (n, \iota(n)).$$

Then η is a bijection with inverse the first projection. The formula

$$E : D \longrightarrow \mathbb{Z}, \quad E(n, v) = F_v(n),$$

is well-defined, and the shortened Collatz map is exactly

$$H = E \circ \eta, \quad H(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n + 1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

Proof. The definition of D makes the two inverse identities for η and the first projection immediate. If $(n, v) \in D$, then $v = (\bar{n}, \bar{n})$, so Proposition 3.1 proves $F_v(n) \in \mathbb{Z}$. Substitution of $v = 00$ on the even fibre and $v = 11$ on the odd fibre gives the displayed formula for H . $\square$

The last proposition is a coefficient-label encoding with an exact evaluator. It does not make λ an action, and it does not turn iteration of H into addition in V .

3.2 Function algebra and group algebra without conflation

There are two four-dimensional algebras in the raw passage. Write

$$A = \text{Map}(V, \mathbb{C})$$

with pointwise multiplication, and let δ_u be the point mass at $u \in V$. Then

$$\delta_u \delta_w = \mathbf{1}_{u=w} \delta_u, \quad 1_A = \sum_{u \in V} \delta_u.$$

Write

$$B = \mathbb{C}[V]$$

for the group algebra, with group-element basis g_v . Its multiplication is

$$g_v g_w = g_{v+w}, \quad 1_B = g_{00}.$$

The point masses are therefore not the group-element basis.

Use the symmetric bilinear form

$$\langle u, v \rangle = u_1 v_1 + u_2 v_2 \in \mathbb{F}_2$$

and define the Fourier idempotents

$$p_u = \frac{1}{4} \sum_{v \in V} (-1)^{\langle u, v \rangle} g_v.$$

Theorem 3.3 (exact Fourier algebra isomorphism). *The linear map*

$$\mathcal{F} : A \longrightarrow B, \quad \mathcal{F}(\delta_u) = p_u,$$

is a unital algebra isomorphism. Its inverse is fixed on the group basis by

$$\mathcal{F}^{-1}(g_v) = \sum_{u \in V} (-1)^{\langle u, v \rangle} \delta_u.$$

Equivalently,

$$p_u p_w = \mathbf{1}_{u=w} p_u, \quad \sum_{u \in V} p_u = g_{00}, \quad g_v = \sum_{u \in V} (-1)^{\langle u, v \rangle} p_u.$$

Proof. For $z \in V$, the coefficient of g_z in $p_u p_w$ is

$$\frac{1}{16} \sum_{v \in V} (-1)^{\langle u, v \rangle + \langle w, z-v \rangle} = \frac{(-1)^{\langle w, z \rangle}}{16} \sum_{v \in V} (-1)^{\langle u-w, v \rangle}.$$

Character orthogonality makes the last sum equal to 4 when $u = w$ and 0 otherwise. This proves $p_u p_w = \mathbf{1}_{u=w} p_u$. Applying the same orthogonality relation to $\sum_u p_u$ leaves only g_{00} . It also gives the displayed expansion of every g_v in the p_u basis. Thus the four p_u form a basis, the proposed inverse identities hold, and $\mathcal{F}$ preserves multiplication and the unit. $\square$

This isomorphism loses no algebraic information, but it does not preserve the natural V -grading

$$B = \bigoplus_{v \in V} \mathbb{C} g_v, \quad \deg g_v = v.$$

Every p_u has a nonzero component in all four homogeneous summands. There is also a basis-independent obstruction to declaring the point masses homogeneous in their labels. If a nonzero idempotent e in a direct V -graded algebra had degree $u \neq 00$, then e^2 would have degree $u + u = 00$, while $e^2 = e$ would have degree u . Directness of the grading would force $e = 0$, a contradiction.

3.3 What the diagonal projector actually returns

In $A \otimes_{\mathbb{C}} \mathbb{C}[X]$, define

$$\mathfrak{F}(X) = \sum_{v \in V} \delta_v \otimes F_v(X), \quad P_{\Delta} = \delta_{00} + \delta_{11}.$$

Pointwise multiplication gives

$$P_{\Delta} \mathfrak{F}(X) P_{\Delta} = \delta_{00} \otimes \frac{X}{2} + \delta_{11} \otimes \frac{3X+1}{2}. \quad (345)$$

The right-hand side is a two-component labelled polynomial; it is not a scalar map $\mathbb{Z} \rightarrow \mathbb{Z}$. If $\text{ev}_v : A \rightarrow \mathbb{C}$ denotes evaluation at v and $\text{ev}_{X=n} : \mathbb{C}[X] \rightarrow \mathbb{C}$ denotes substitution, then the exact scalar recovery is

$$H(n) = (\text{ev}_{\iota(n)} \otimes \text{ev}_{X=n})(P_{\Delta} \mathfrak{F}(X) P_{\Delta}). \quad (346)$$

The state-dependent evaluator in (346) cannot be omitted: projection removes the two off-diagonal labels but does not choose between 00 and 11.

Under Theorem 3.3, the same projector is

$$\mathcal{F}(P_{\Delta}) = p_{00} + p_{11} = \frac{1}{2}(g_{00} + g_{11}). \quad (347)$$

This is a proved change of basis, not an identification of the two basis multiplications or gradings.

3.4 The exact color-algebra consequence of the cited paper

Aizawa–Isaac–Segar define a color superalgebra $\mathfrak{g} = \bigoplus_{v \in V} \mathfrak{g}_v$ with bracket

$$[[X_v, Y_w]] = X_v Y_w - (-1)^{\langle v, w \rangle} Y_w X_v$$

inside an enveloping algebra [40]. Starting from an $\mathcal{N} = 1$ superconformal Galilei algebra, they introduce

$$P_{nm} = \{P_n, P_m\}, \quad \Lambda_{nm} = \{P_n, X_m\}, \quad X_{nm} = [X_n, X_m]$$

and assign degrees

$$\begin{array}{l|l} 00 & H, D, K, P_{nm}, X_{nm}, \\ 01 & P_n, \\ 10 & Q, S, \Lambda_{nm}, \\ 11 & X_n. \end{array}$$

Their relation tables and Jacobi check establish closure for that specific enveloping-algebra construction.

Lemma 3.4 (totally isotropic diagonal restriction). *Let $C = \bigoplus_{v \in V} C_v$ be an associative V -graded algebra. Then*

$$C_{\Delta} = C_{00} \oplus C_{11}$$

is closed under multiplication. On C_{Δ} , the color commutator defined by the displayed pairing is the ordinary commutator.

Proof. The set Δ is a subgroup, so $C_u C_v \subseteq C_{u+v} \subseteq C_{\Delta}$ for $u, v \in \Delta$. Moreover $\langle 00, 00 \rangle = \langle 00, 11 \rangle = \langle 11, 00 \rangle = \langle 11, 11 \rangle = 0$ in $\mathbb{F}_2$. Hence the color sign is +1 on every pair of diagonal homogeneous elements, and the bracket is $XY - YX$. $\square$

The lemma explains a valid formal feature of the repeated degree group. It does not supply a Collatz color superalgebra. The raw passage defines no graded algebra of Collatz states or paths, no bracket on such an algebra, no degree-preserving bracket morphism to either algebra in the cited paper, and no action of those algebras on Collatz states or observables. The coefficient algebra A also cannot be given the asserted grading merely by declaring its nonzero point idempotents to have their index degrees.

3.5 The two involutions and the chosen orthogonal block

Coordinate swap

$$s : V \longrightarrow V, \quad s(a, b) = (b, a),$$

fixes the diagonal pointwise. In contrast, translation by 11,

$$\tau(v) = v + 11,$$

exchanges 00 and 11. Translation is an involution of the underlying two-point diagonal set, but it is not a group automorphism because $\tau(00) = 11$.

On the real vector space $W_\Delta = \mathbb{R}\delta_{00} \oplus \mathbb{R}\delta_{11}$, one may choose

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$S^2 = I, \quad J^2 = -I, \quad SJS^{-1} = -J, \quad \det S = -1, \quad \det J = 1.$$

Consequently $\{e^{\theta J} : \theta \in \mathbb{R}\} \cong SO(2)$ and adjoining S produces the standard semidirect product $SO(2) \rtimes \langle S \rangle \cong O(2)$.

This matrix calculation is exact, but J is additional structure. Neither s nor the Collatz branch evaluator selects it; indeed s fixes the two diagonal labels rather than exchanging them. No universal property, dynamical action, lifting problem, or determinant anomaly is constructed by the raw passage. More precisely, the raw passage supplies neither a homomorphism between the two ambient group systems nor a commutative diagram relating the quotient map

$$q : \text{Spin}(4) \times SU(2) \longrightarrow (\text{Spin}(4) \times SU(2)) / \langle (-1, -1) \rangle$$

to the coefficient homomorphism

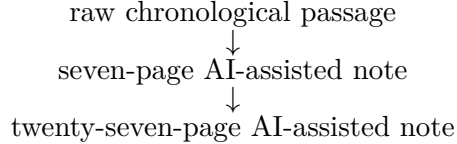
$$\partial : V \longrightarrow \mathbb{F}_2, \quad \partial(a, b) = a + b, \quad \Delta = \ker \partial.$$

The occurrence of order-two data in both constructions is therefore not, by itself, a defined comparison arrow. No identification or theorem transfer is admitted without those arrows and their compatibility identities.

Explicit nonclaim 3.5. The results of this section do not construct a $\mathbb{Z}_2 \times \mathbb{Z}_2$ action on Collatz dynamics, a Collatz color superalgebra, an intrinsic $O(2)$ action or anomaly, a $\text{Spin}-SU(2)$ quotient for Collatz, or a reformulation of the Collatz conjecture. They establish the exact coefficient encoding, the Fourier algebra isomorphism, the required state-dependent evaluator, the isotropic-diagonal lemma, and the stated obstructions at those scopes only. The raw continuation turns to weighted paths; its composition algebra must be read before the natural local programme boundary is fixed.

4 Weighted odd-step paths, exact completions, and finite residue actions

This section reconstructs the next coherent programme in the raw Z_n *Symmetries* export, physical lines 6221–9596. The controlling local witness is the chronological export itself. Two later PDFs entitled *The Odd-Step Collatz Path Algebra* are recorded as AI-assisted restatements of parts of that passage, not as scholarly sources and not as theorem authority. Their order in the local lineage is



The passage contains a substantial finite mathematical core, but it also uses mutually incompatible composition orders, calls nonunital finite semigroups monoids, puts infinite sums in a finite-support algebra, and identifies a prefix 3-adic cocycle with a forward affine intercept without the required word reversal. We retain the objects and calculations at their proved scope and repair each of those defects explicitly.

The general algebraic vocabulary is crosswalked against the standard definitions of free monoids, semirings, formal power series, weighted paths, and transformation monoids in [42, 41, 43]. Those sources do not contain the Collatz-specific theorems below; every such theorem is proved here.

4.1 The odd first-return category and its local block category

Let

$$\mathcal{O} = \{1, 3, 5, \dots\}, \quad a(x) = \nu_2(3x + 1), \quad C(x) = \frac{3x + 1}{2^{a(x)}} \quad (x \in \mathcal{O}).$$

Thus C is the positive-odd first-return map. Write

$$o(n) = \frac{3n + 1}{2} \quad (n \text{ odd}), \quad e(n) = \frac{n}{2} \quad (n \text{ even})$$

for the two legal branches of the shortened map, with their domains retained.

Definition 4.1 (first-return path category). The category $\mathcal{E}_C$ has object set $\mathcal{O}$. For every $x \in \mathcal{O}$ it has a generating arrow

$$\epsilon_x : x \longrightarrow C(x), \quad \lambda(\epsilon_x) = a(x),$$

and its morphisms are finite composable strings of these arrows, including the empty path 1_x at each object. A path

$$p : x_0 \xrightarrow{a_1} x_1 \xrightarrow{a_2} \dots \xrightarrow{a_n} x_n$$

has length and total exponent

$$\ell(p) = n, \quad A(p) = a_1 + \dots + a_n,$$

and weight

$$w(p) = u^{\ell(p)} v^{A(p) - \ell(p)}.$$

The category $\mathcal{D}_C^{\text{blk}}$ has the same objects. Its morphisms are precisely the legal shortened-map paths between odd endpoints whose local word has the form

$$(oe^{m_1})(oe^{m_2}) \dots (oe^{m_n}), \quad m_i \geq 0.$$

For a block path q , put

$$\#_o(q) = \text{number of } o\text{-steps}, \quad \#_e(q) = \text{number of } e\text{-steps}, \quad w_{\text{blk}}(q) = u^{\#_o(q)} v^{\#_e(q)}.$$

Composition in both categories is path concatenation.

The set of strict first-return edges by itself is a directed graph, not a category: a composite of two strict first-return edges is usually not a single strict first-return edge. The free path category $\mathcal{E}_C$ is the typed categorical object used below.

Theorem 4.2 (exact odd-block category isomorphism). *For a positive exponent put*

$$\iota(a) = oe^{a-1}.$$

There is a functor

$$\iota : \mathcal{E}_C \longrightarrow \mathcal{D}_C^{\text{blk}}$$

which is the identity on objects and replaces each exponent edge by its local block. It is an isomorphism. Its inverse is

$$U((oe^{m_1}) \cdots (oe^{m_n})) = (m_1 + 1, \dots, m_n + 1).$$

Moreover

$$\#_o(\iota(p)) = \ell(p), \quad \#_e(\iota(p)) = A(p) - \ell(p), \quad w_{\text{blk}}(\iota(p)) = w(p).$$

In particular, this block passage loses no local trajectory information.

Proof. Fix $x \in \mathcal{O}$ and put $a = a(x)$. The first local step gives $(3x + 1)/2$. If $a > 1$, this value and the following $a - 2$ intermediate values are even, so exactly $a - 1$ legal e -steps follow. Their endpoint is $(3x + 1)/2^a = C(x)$, which is odd. Conversely, if a legal local path from an odd x has word oe^m and first returns to an odd state at its endpoint, then $3x + 1$ is divisible by 2^{m+1} and not by 2^{m+2} . Hence $a(x) = m + 1$.

Every displayed block word has a unique parsing: each occurrence of o starts a block and every following run of e 's ends immediately before the next o or at the word endpoint. Therefore $U\iota$ and ιU are the identity on every morphism. Both maps preserve endpoints and concatenation, so they are mutually inverse functors. Counting the one o and the $a_i - 1$ copies of e in each block proves the three weight identities. $\square$

The theorem corrects a recurrent description of U as a proper forgetful quotient. It would be information-losing on an unrestricted local word category, but on the explicitly defined legal odd-to-odd block category it has the displayed two-sided inverse.

4.2 Two different $\mathbb{F}_2^2$ degrees

The weighted block counts give a degree

$$d_{\text{count}}(p) = (\ell(p) \bmod 2, (A(p) - \ell(p)) \bmod 2).$$

The coefficient-label convention in Section 3, which assigns 00 to the even branch and 11 to the odd branch, instead gives

$$d_{\text{coeff}}(p) = (\ell(p) \bmod 2, \ell(p) \bmod 2).$$

Proposition 4.3 (exact comparison of the two degrees). *The homomorphism*

$$\kappa : \mathbb{F}_2^2 \longrightarrow \mathbb{F}_2^2, \quad \kappa(r, s) = (r, r),$$

satisfies $d_{\text{coeff}} = \kappa \circ d_{\text{count}}$. Its kernel, image, and fibres are

$$\begin{aligned} \ker \kappa &= \{00, 01\}, & \text{im } \kappa &= \{00, 11\}, \\ \kappa^{-1}(00) &= \{00, 01\}, & \kappa^{-1}(11) &= \{10, 11\}, \end{aligned}$$

with empty fibres over 01 and 10. Thus κ is an idempotent projection onto the diagonal $\Delta = \{00, 11\}$. Equivalently, its codomain restriction induces the quotient isomorphism $\mathbb{F}_2^2 / \ker \kappa \simeq \Delta$. It is not a quotient onto the whole displayed codomain, and the two degrees are not interchangeable.

Proof. Substitution gives $\kappa(\ell, A - \ell) = (\ell, \ell)$ modulo two. The kernel, image, and four fibres follow by evaluating the four elements of $\mathbb{F}_2^2$. $\square$

4.3 Composition order and the ordinary category algebra

We use the standard target-left convention throughout this section. If $p : x \rightarrow y$ and $q : y \rightarrow z$, then $q \circ p$ means first p , then q , and the category-algebra product is

$$[q][p] = [q \circ p], \quad [q][p] = 0 \quad \text{when } t(p) \neq s(q). \quad (348)$$

Consequently

$$[p] = e_{t(p)}[p]e_{s(p)}, \quad e_x = [1_x],$$

and for the path in Definition 4.1,

$$[p] = [\epsilon_{x_{n-1}}] \cdots [\epsilon_{x_0}].$$

Chronological word concatenation is the opposite written order from this target-left product. Several raw formulas use one order in the path algebra and the other order in a transformation or pullback algebra without an opposite construction; Subsection 4.9 gives the forced repair.

Let R be a commutative ring. The ordinary weighted category algebra is

$$\mathfrak{A}_C = \bigoplus_{p \in \text{Mor}(\mathcal{E}_C)} R[u, v][p]$$

with multiplication (348). Every element has finite path support. Therefore neither the sum of all object idempotents nor the sum of all finite paths is generally an element of $\mathfrak{A}_C$.

4.4 Endpoint series and the exact path completion

For $x, y \in \mathcal{O}$, define

$$Z_{x \rightarrow y}(u, v) = \sum_{\substack{n \geq 0 \\ C^n(x) = y}} u^n v^{A_n(x) - n} \in R[v][[u]], \quad (349)$$

where $A_0(x) = 0$ and $A_n(x) = \sum_{j=0}^{n-1} a(C^j(x))$. The coefficient of each u^n is either zero or one monomial in v ; no convergence assertion is needed for this formal series.

Proposition 4.4 (exact hit-time alternatives). *For fixed $x, y \in \mathcal{O}$, the set*

$$H_{x,y} = \{n \geq 0 : C^n(x) = y\}$$

is empty, a singleton, or

$$r + d\mathbb{N}_0$$

for a unique first hit r and the least positive period d of y . In the last case, if

$$B = \sum_{j=0}^{d-1} a(C^j(y)),$$

then

$$Z_{x \rightarrow y}(u, v) = \frac{u^r v^{A_r(x)-r}}{1 - u^d v^{B-d}} \quad (350)$$

as a formal geometric series. Every positive d -cycle satisfies

$$2^B > 3^d. \quad (351)$$

Hence the numerical specialization $(u, v) = (3/2, 1/2)$ converges on every cyclic endpoint series. In contrast,

$$Z_{1 \rightarrow 1}(1, 1) = \sum_{n \geq 0} 1$$

diverges.

Proof. If $m < n$ are two hit times, then $C^{n-m}(y) = y$, so y is periodic. Once the orbit first reaches a periodic y , its later returns occur exactly at multiples of the least period. This proves the alternatives and the uniqueness of r, d . Each traversal of the cycle adds d to the length and B to the exponent sum, which gives (350).

Write the cycle as $y_i \xrightarrow{a_i} y_{i+1}$ with indices modulo d . For every i ,

$$2^{a_i} y_{i+1} = 3y_i + 1 > 3y_i.$$

Multiplication and cancellation of the positive product $\prod_i y_i$ give $2^B > 3^d$. At $(3/2, 1/2)$ the geometric ratio is $3^d/2^B < 1$. Finally $C(1) = 1$ and $a(1) = 2$, so every term of $Z_{1 \rightarrow 1}$ has weight $(uv)^n$. $\square$

The completion needed for global path sums is not the preceding direct sum.

Definition 4.5 (coefficientwise path completion). Put

$$\widehat{\mathfrak{A}}_C = \prod_{p \in \text{Mor}(\mathcal{E}_C)} R[u, v][p].$$

For $F = (F_p)$ and $G = (G_p)$, define the coefficient at a path r by

$$(FG)_r = \sum_{r=q \circ p} F_q G_p.$$

This is a finite sum: a path of length n has exactly $n + 1$ cuts.

The finite-cut statement proves that multiplication is well defined for arbitrary path support. Associativity follows by identifying both parenthesizations with the finite set of two-cut decompositions of a fixed path. The completion contains

$$E = \sum_{x \in \mathcal{O}} e_x, \quad W = \sum_{x \in \mathcal{O}} uv^{a(x)-1}[\epsilon_x], \quad G = \sum_p w(p)[p].$$

Every nonempty path has a unique first edge and a unique last edge, so

$$G = E + WG = E + GW. \quad (352)$$

Because R is a ring, this says $G = (E - W)^{-1}$ inside the displayed completion. If S is instead a commutative semiring, repeat Definition 4.5 with coefficients $S[u, v]$. The finite-cut convolution and the identities $G = E + WG = E + GW$ remain valid there and define the Kleene star W^* ; the subtraction and inverse notation are not used. None of E, W, G is silently placed in an ordinary finite-support algebra.

4.5 The predecessor transfer operator

For odd y define

$$\mathcal{P}(y) = \left\{ a \geq 1 : x_a(y) = \frac{2^a y - 1}{3} \text{ is an integer} \right\}.$$

Lemma 4.6 (exact predecessor fibres). *For odd y ,*

$$\mathcal{P}(y) = \begin{cases} \emptyset, & y \equiv 0 \pmod{3}, \\ \{2, 4, 6, \dots\}, & y \equiv 1 \pmod{3}, \\ \{1, 3, 5, \dots\}, & y \equiv 2 \pmod{3}. \end{cases}$$

For every $a \in \mathcal{P}(y)$, $x_a(y)$ is positive odd and

$$C(x_a(y)) = y, \quad a(x_a(y)) = a.$$

Every positive odd predecessor of y occurs exactly once in this list.

Proof. Modulo three, $2^a y = 1$ is impossible when $3 \mid y$. Since 2^a is 1 for even a and 2 for odd a modulo three, the other two cases have the stated parities. The numerator is positive and odd; division by three preserves oddness. The identity $3x_a(y) + 1 = 2^a y$ with y odd proves both displayed first-return identities. Solving that same identity for x proves uniqueness. $\square$

Define

$$(Lf)(y) = u \sum_{a \in \mathcal{P}(y)} v^{a-1} f(x_a(y)). \quad (353)$$

This formula has several distinct exact domains.

Theorem 4.7 (algebraic, formal, and analytic transfer domains). *Let R be a commutative ring.*

- (i) *Formula (353) defines an $R[u, v]$ -linear endomorphism of the finite-support module $R[u, v]^{(\mathcal{O})}$.*

(ii) It defines a continuous endomorphism of

$$\mathcal{M} = (R[[v]][[u]])^{\mathcal{O}}.$$

All v -series in this product are interpreted coefficientwise. The module $\mathcal{M}$ carries the complete linear, uniform u -adic topology with

$$F^k \mathcal{M} = (u^k R[[v]][[u]])^{\mathcal{O}} \quad (k \geq 0).$$

For a fixed coefficient of v^j , only $a \leq j+1$ can contribute, and L obeys $L(F^k \mathcal{M}) \subseteq F^{k+1} \mathcal{M}$. Consequently

$$(I - L)^{-1} = \sum_{n \geq 0} L^n$$

is well defined u -adically and

$$[(I - L)^{-1} \delta_x](y) = Z_{x \rightarrow y}(u, v). \quad (354)$$

(iii) Let $u, v \in \mathbb{C}$ and $u \neq 0$. On counting-measure sequence spaces, L is bounded on $\ell^1(\mathcal{O})$ exactly when $|v| \leq 1$, and then

$$\|L\|_{\ell^1 \rightarrow \ell^1} = |u|.$$

For $1 < p \leq \infty$, let q be conjugate to p , with $q = 1$ when $p = \infty$. Then L is bounded on $\ell^p(\mathcal{O})$ exactly when $|v| < 1$, and

$$\|L\|_{\ell^p \rightarrow \ell^p} = |u|(1 - |v|^{2q})^{-1/q}. \quad (355)$$

The same strict condition and norm hold on $c_0(\mathcal{O})$.

If $u = 0$, the zero operator is defined for every finite v .

Proof. For (i), a finitely supported f has only finitely many states x with $f(x) \neq 0$, and each such state has one image $C(x)$. Thus the apparently infinite predecessor formula is a finite sum on this domain.

For (ii), the coefficient of v^j can receive a term only from $a-1 \leq j$. This proves coefficientwise finiteness. The factor u proves the filtration statement and hence the Neumann identity. Starting from δ_x , each application of L follows the unique forward edge from the current odd state and multiplies by uv^{a-1} . Induction gives

$$(L^n \delta_x)(y) = \begin{cases} u^n v^{A_n(x)-n}, & C^n(x) = y, \\ 0, & C^n(x) \neq y, \end{cases}$$

which proves (354).

For (iii), the input coordinates split into disjoint fibres $C^{-1}(y)$. The absolute row weights over a fibre are either

$$|u|(1, |v|^2, |v|^4, \dots) \quad \text{or} \quad |u|(|v|, |v|^3, |v|^5, \dots),$$

or the empty sequence. For $1 < p \leq \infty$, Hölder's inequality gives the row-functional norm as the ℓ^q norm of this weight sequence. The first row is the larger one, and its norm is exactly the right side of (355). Finite truncations with phases aligned to the weights approach equality, so the

upper bound is sharp. If $|v| \geq 1$, those same finite truncations have unbounded row-functional norm.

For $p = 1$, each input state occupies exactly one column and has column sum $|u||v|^{a(x)-1}$. All positive exponents occur by Lemma 4.6. The supremum is $|u|$ when $|v| \leq 1$ and is infinite when $|v| > 1$. Finally, for $|v| < 1$, the same row estimate maps c_0 to itself: split the series into finitely many exponents, whose predecessors tend to infinity with y , and a uniformly small geometric tail. Finite-support phase-aligned vectors prove sharpness. When $|v| \geq 1$, choose values tending slowly to zero along one predecessor fibre and align their phases; the defining row sum fails to be bounded. This proves the c_0 assertion. $\square$

At $u = v = 1$, the operator is therefore norm one on ℓ^1 , but it is unbounded on ℓ^∞ and is not defined on the constant function one by the raw infinite-sum formula. An arbitrary product of polynomial-valued functions is likewise not an algebraic domain for L : the missing v -adic completion is essential.

4.6 Prefix cocycles and forward affine intercepts

For a finite exponent word $w = a_1 \cdots a_n$, put

$$A_k(w) = a_1 + \cdots + a_k, \quad A(w) = A_n(w),$$

and define the finite prefix coordinate

$$\rho(w) = \sum_{k=1}^n 3^{k-1} 2^{-A_k(w)} \in \mathbb{Z}_3. \quad (356)$$

For the empty word, $A(\emptyset) = 0$ and $\rho(\emptyset) = 0$.

Proposition 4.8 (typed prefix cocycle and its exact fibres). *For exponent words w, z in chronological concatenation order,*

$$\rho(wz) = \rho(w) + 3^{|w|} 2^{-A(w)} \rho(z). \quad (357)$$

Let

$$P = \{(m, A) \in \mathbb{N}_0^2 : A \geq m\}, \quad \theta_{(m,A)}(r) = 3^m 2^{-A} r \quad (r \in \mathbb{Z}_3).$$

With addition on P and multiplication

$$(r, p) \star (s, q) = (r + \theta_p(s), p + q),$$

the semidirect product $\mathbb{Z}_3 \rtimes_{\theta} P$ is a monoid and

$$\chi_{\text{pre}}(w) = (\rho(w), (|w|, A(w)))$$

is an injective monoid homomorphism from positive-exponent words.

More explicitly, put $R(w) = 2^{A(w)} \rho(w) \in \mathbb{Z}$. If $n > 1$, then

$$a_n = \nu_2(R(w) - 3^{n-1}), \quad R(a_1 \cdots a_{n-1}) = \frac{R(w) - 3^{n-1}}{2^{a_n}}. \quad (358)$$

Together with $a_1 = A(w)$ when $n = 1$, this is a two-sided recursive inverse on the image. Every fibre of χ_{pre} is therefore a singleton.

Proof. Splitting the sum (356) after $|w|$ terms and factoring $3^{|w|}2^{-A(w)}$ from the remainder proves (357). The endomorphisms satisfy $\theta_p\theta_q = \theta_{p+q}$, so the displayed semidirect multiplication is associative with identity $(0, (0, 0))$; the cocycle identity proves multiplicativity of χ_{pre} .

Multiplying (356) by 2^A gives

$$R(w) = \sum_{k=1}^n 3^{k-1} 2^{A-A_k}.$$

The last term is 3^{n-1} . After subtracting it, the preceding term has exact 2-adic valuation a_n , while every earlier term is divisible by $2^{a_{n-1}+a_n}$. Hence no cancellation changes that valuation, proving the first formula in (358). Division gives the same integer for the prefix word. Recursion recovers all exponents, which proves injectivity and the inverse identities. $\square$

The acting object here is a monoid, not a group: $\theta_{(m,A)}$ is not surjective on $\mathbb{Z}_3$ when $m > 0$. In the coordinate pair (m, A) its multiplier is $3^m 2^{-A}$, not $3^m 2^{-(m+A)}$.

The preceding injectivity is a statement about exponent words. Erasing the source and target of a path is a different operation and has exact infinite fibres.

Proposition 4.9 (exact source fibres of an exponent word). *For a word $w = (a_1, \dots, a_n)$ with $n \geq 0$, put $A = A(w)$ and*

$$B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1+\dots+a_{i-1}} \in \mathbb{Z}.$$

The word w is the exact successive valuation word of a positive odd source x if and only if

$$x \equiv 3^{-n}(2^A - B_w) \pmod{2^{A+1}}. \quad (359)$$

Thus every exponent word has one odd residue class of positive sources modulo 2^{A+1} , hence infinitely many path-category realizations. Retaining the source object selects one of them.

Proof. If $n = 0$, the claim is the tautology $x \equiv 1 \pmod{2}$ for a positive odd source, and the word is empty. Now let $n \geq 1$. We prove directly, by induction on n , that w is the exact valuation word of x if and only if

$$3^n x + B_w \equiv 2^A \pmod{2^{A+1}}. \quad (360)$$

For $n = 1$, this is $3x + 1 \equiv 2^{a_1} \pmod{2^{a_1+1}}$, exactly the assertion $\nu_2(3x + 1) = a_1$.

For $n > 1$, write $w' = (a_2, \dots, a_n)$, $A' = A - a_1$, and let $B_{w'}$ be its displayed integer. Then

$$B_w = 3^{n-1} + 2^{a_1} B_{w'}.$$

Reducing (360) modulo 2^{a_1} first shows that 2^{a_1} divides $3x + 1$. Put $x_1 = (3x + 1)/2^{a_1}$. Dividing the full congruence by 2^{a_1} gives

$$3^{n-1} x_1 + B_{w'} \equiv 2^{A'} \pmod{2^{A'+1}}.$$

Here $A' \geq 1$ and $B_{w'}$ is odd, so reduction modulo two shows that x_1 is odd; positivity is immediate from $x > 0$. By induction the displayed congruence is equivalent to x_1 being positive odd with exact valuation word w' . In particular x_1 is odd, so the first valuation is exactly a_1 . Reversing the argument proves the converse.

Solving (360) for x is legitimate because 3^n is a unit modulo 2^{A+1} , and gives the unique class (359). For $n \geq 1$, the integer B_w is odd: its first summand is odd and every later summand is

even. Hence the class is odd; the $n = 0$ class was handled separately. Adding positive multiples of 2^{A+1} gives infinitely many positive sources. This direct induction also supplies, without using it as a proof dependency, the finite characterization appearing in Tao's source [32, source lines 455–470]. $\square$

Now put

$$g_a(X) = \frac{3X + 1}{2^a}$$

and apply a_1 first. The forward chronological composite is

$$G_w = g_{a_n} \circ \cdots \circ g_{a_1} = 3^n 2^{-A(w)} X + \beta(w), \quad (361)$$

where

$$\beta(w) = \sum_{i=1}^n 3^{n-i} 2^{-(a_i + \cdots + a_n)}. \quad (362)$$

Thus $B_w = 2^{A(w)} \beta(w)$, now as a consequence of the two displayed formulas rather than as a definition used before β is introduced.

Proposition 4.10 (exact reversal to Tao's offset). *For every finite positive-exponent word,*

$$\beta(w) = \rho(w^{\text{rev}}).$$

Moreover $\beta(w)$ is exactly Tao's finite offset $F_n(a_1, \dots, a_n)$ in the forward convention of [32, source lines 208–229 and 384–393]. Thus the prefix coordinate and forward intercept are connected by word reversal; they are not equal on the same chronological word in general.

Proof. Reverse the index in (356). The k th prefix sum of w^{rev} is the k -term suffix sum of w , and its coefficient 3^{k-1} becomes 3^{n-i} . This is exactly (362). Induction in (361) gives the same sum, and direct comparison with Tao's displayed definition gives the final identity. This also agrees with the pointwise reversal already proved in Proposition 2.132. $\square$

For example,

$$\rho(1, 4) = \rho(3, 1, 1) = \frac{19}{32}, \quad \beta(1, 4) = \frac{5}{32}, \quad G_{(1,4)}(X) = \frac{9X + 5}{32}.$$

The distinction is not cosmetic: $G_{(1,4)}(3) = 1$, whereas the unreversed prefix constant would give the wrong value. The first equality also shows that ρ alone is not injective. The decoration in χ_{pre} is essential: the two words have respective $(|w|, A(w))$ coordinates $(2, 5)$ and $(3, 5)$.

Because target-left category multiplication writes later arrows on the left, χ_{pre} becomes covariant on the path algebra only after its coefficient monoid is replaced by the opposite monoid. This typing is made explicit in Proposition 4.17 below.

4.7 Finite residue transformation semigroups

Fix $N \geq 1$ and write

$$R_N = \mathbb{Z}/3^N \mathbb{Z}, \quad U_N = R_N^\times, \quad \varphi_N = |U_N| = 2 \cdot 3^{N-1}.$$

Since

$$\text{ord}_{3^N}(2) = 2 \cdot 3^{N-1},$$

the elements 2^{-a} , $a \geq 1$, run through all of U_N . For $\eta \in U_N$ put

$$G_\eta(r) = \eta(3r + 1) = 3\eta r + \eta.$$

Let S_N be the image of all nonempty words in the transformations G_η , and let

$$T_N = S_N \sqcup \{\text{id}_{R_N}\}.$$

Every nonempty word sends 0 to a unit, so the identity is not in S_N . Thus S_N is a transformation semigroup and T_N is its identity-adjoined transformation monoid.

For chronological $\boldsymbol{\eta} = (\eta_1, \dots, \eta_k)$, with η_1 applied first, direct expansion gives

$$G_{\boldsymbol{\eta}}(r) = 3^k \left(\prod_{i=1}^k \eta_i \right) r + \sum_{j=1}^k 3^{k-j} \prod_{i=j}^k \eta_i. \quad (363)$$

The constant term is congruent to η_k modulo three and is therefore a unit.

Lemma 4.11 (order of two modulo powers of three). *For every $N \geq 1$, $\text{ord}_{3^N}(2) = 2 \cdot 3^{N-1}$.*

Proof. By Euler's theorem the order divides $\varphi(3^N) = 2 \cdot 3^{N-1}$, and reduction modulo three shows that it is even. Hence it has the form $2 \cdot 3^j$ with $0 \leq j \leq N-1$. The lifting-the-exponent identity gives

$$\nu_3(2^{2 \cdot 3^j} - 1) = \nu_3(4^{3^j} - 1) = \nu_3(4 - 1) + \nu_3(3^j) = j + 1.$$

The congruence $2^{2 \cdot 3^j} \equiv 1 \pmod{3^N}$ therefore holds exactly when $j \geq N-1$. The only possible order is $2 \cdot 3^{N-1}$; the case $N = 1$ is included with $j = 0$. $\square$

For $N \geq 2$ define

$$L_{N,1} = \{r \mapsto 3br + b : b \in U_N\},$$

and for $2 \leq k < N$ define

$$L_{N,k} = \{r \mapsto 3^k sr + b : s \in (\mathbb{Z}/3^{N-k}\mathbb{Z})^\times, b \in U_N\}.$$

Finally let

$$\mathcal{C}_N = \{c_b : b \in U_N\}, \quad c_b(r) = b.$$

Theorem 4.12 (exact finite residue normal form). *For $N = 1$,*

$$S_1 = \mathcal{C}_1 = \{c_1, c_2\}.$$

For every $N \geq 2$, there is a disjoint union

$$S_N = L_{N,1} \sqcup \bigsqcup_{k=2}^{N-1} L_{N,k} \sqcup \mathcal{C}_N. \quad (364)$$

More precisely, $L_{N,k}$ is exactly the set represented by words of length $k < N$, every word of length at least N is constant, and every unit-valued constant already has a representative of length N .

Proof. Formula (363) shows that a word of length $k < N$ has slope of exact 3-adic valuation k and unit intercept; at length at least N it is constant. At length one, slope and intercept satisfy the exceptional relation $\alpha = 3b$, giving $L_{N,1}$.

We prove surjectivity onto every later stratum constructively. First suppose $N \geq 3$. At length two, prescribe $s \in (\mathbb{Z}/3^{N-2}\mathbb{Z})^\times$ and $b \in U_N$. Put

$$q = sb^{-1} \pmod{3^{N-2}}, \quad \eta_1 = \frac{q}{1-3q} \pmod{3^{N-2}},$$

choose any unit lift of η_1 to U_N , and set

$$\eta_2 = \frac{b}{1+3\eta_1} \pmod{3^N}.$$

Then

$$\eta_2(3\eta_1 + 1) = b, \quad \eta_2\eta_1 \equiv s \pmod{3^{N-2}},$$

so $G_{\eta_2} \circ G_{\eta_1}(r) = 9sr + b$. When $N = 2$, choose an arbitrary $\eta_1 \in U_2$ and put $\eta_2 = b/(1+3\eta_1)$ in U_2 ; the resulting length-two word is the constant c_b .

Suppose all length- $(k-1)$ forms occur, where $3 \leq k < N$. Given a desired $3^k s'r + b'$, choose any prior unit intercept b_0 , put

$$\eta_k = \frac{b'}{3b_0 + 1},$$

and choose the prior slope parameter s_0 so that $s_0 \equiv s'\eta_k^{-1} \pmod{3^{N-k}}$. Applying G_{η_k} on the left gives the prescribed slope and intercept. To obtain a prescribed constant $c_{b'}$ at length $N \geq 3$, choose any already constructed length- $(N-1)$ form with intercept b_0 and apply G_{η_N} with $\eta_N = b'/(3b_0 + 1)$; its slope vanishes modulo 3^N . Equality of two affine maps is equivalent to equality of their values at 0 and 1, hence of intercept and slope; distinct strata have different slope valuations. This proves both completeness and disjointness. $\square$

Corollary 4.13 (cardinalities and ranks). *For $N \geq 2$,*

$$|S_N| = \varphi_N(3^{N-2} + 1), \quad |T_N| = 1 + \varphi_N(3^{N-2} + 1). \quad (365)$$

The first six semigroup cardinalities are

$$2, 12, 72, 540, 4536, 39852,$$

and the corresponding monoid cardinalities are

$$3, 13, 73, 541, 4537, 39853.$$

An element of $L_{N,k}$ has image $b + 3^k R_N$ and rank 3^{N-k} ; the constants have rank one, while the adjoined identity has rank 3^N .

Proof. The stratum sizes are

$$|L_{N,1}| = \varphi_N, \quad |L_{N,k}| = \varphi_N \varphi_{N-k} \quad (2 \leq k < N), \quad |\mathcal{C}_N| = \varphi_N.$$

Since $\sum_{j=1}^{N-2} \varphi_j = 3^{N-2} - 1$, their sum is (365). The image and rank statements follow by varying $3^k sr$ over the ideal $3^k R_N$. $\square$

Proposition 4.14 (the minimum ideal). *The rank-one stratum $\mathcal{C}_N$ is the minimum two-sided ideal of both S_N and T_N . Its multiplication is left-zero:*

$$c_b c_d = c_b.$$

Proof. For every $f \in T_N$ and unit b ,

$$c_b f = c_b, \quad f c_b = c_{f(b)} \in \mathcal{C}_N,$$

because every transformation in T_N sends units to units. Hence $\mathcal{C}_N$ is a two-sided ideal. If a nonempty ideal contains f , then it contains $c_b f = c_b$ for every $b \in U_N$, so it contains $\mathcal{C}_N$. This direct proof agrees with the general minimum-rank description for faithful finite transformation monoids [43, Proposition 13.1 and Lemma 13.2]. $\square$

4.8 Reduction maps and the exact inverse limit

For $M > N$, coefficient reduction gives

$$\pi_{M,N} : S_M \longrightarrow S_N, \quad (\alpha, b) \longmapsto (\alpha \bmod 3^N, b \bmod 3^N).$$

Theorem 4.15 (surjectivity and exact reduction fibres). *The maps $\pi_{M,N}$ are surjective semigroup homomorphisms. Put $d = M - N$. If $N \geq 2$, then*

$$|\pi_{M,N}^{-1}(s)| = \begin{cases} 3^d, & s \in L_{N,1}, \\ 3^{2d}, & s \in L_{N,k}, \quad 2 \leq k < N, \\ 3^{2d}, & s \in \mathcal{C}_N. \end{cases} \quad (366)$$

For the bottom reduction $S_M \rightarrow S_1$, each of the two fibres has size

$$3^{M-1}(3^{M-2} + 1).$$

The maps extend to $T_M \rightarrow T_N$, and the identity has the singleton fibre $\{\text{id}\}$.

Proof. Reduction intertwines every generator and therefore every word. The normal form theorem supplies a lift, proving surjectivity. A target in $L_{N,1}$ has exactly the 3^d unit lifts of its intercept; its slope remains forced to be three times that intercept. In an intermediate stratum, the intercept and unit slope parameter each have 3^d lifts, giving 3^{2d} .

A constant target receives 3^d constant lifts and lifts from every source stratum of slope valuation $k \geq N$. Their number is

$$3^d \left(1 + \sum_{k=N}^{M-1} \varphi_{M-k} \right) = 3^d \left(1 + \sum_{j=1}^d \varphi_j \right) = 3^{2d}.$$

At level one both target elements are constants. Symmetry of the unit intercepts modulo three splits $|S_M|$ equally, giving the stated bottom fibre. A nonempty word has unit intercept, so it cannot reduce to the identity, which proves the last assertion. $\square$

Theorem 4.16 (exact 3-adic inverse limit). *Under coefficientwise reduction,*

$$\varprojlim_N S_N \cong \{x \mapsto 3bx + b : b \in \mathbb{Z}_3^\times\} \sqcup \{x \mapsto \alpha x + b : \alpha \in 9\mathbb{Z}_3, b \in \mathbb{Z}_3^\times\}. \quad (367)$$

The second family includes $\alpha = 0$. Moreover

$$\varprojlim_N T_N = \{\text{id}_{\mathbb{Z}_3}\} \sqcup \varprojlim_N S_N.$$

For $\eta \in \mathbb{Z}_3^\times$ define $G_\eta(x) = \eta(3x + 1)$, and distinguish the two semigroups

$$\mathcal{S}_{\text{unit}} = \langle G_\eta : \eta \in \mathbb{Z}_3^\times \rangle, \quad \mathcal{S}_{\text{exp}} = \langle G_{2^{-a}} : a \geq 1 \rangle.$$

In the coefficientwise 3-adic topology,

$$\overline{\mathcal{S}_{\text{exp}}} = \overline{\mathcal{S}_{\text{unit}}} = \varprojlim_N S_N.$$

Proof. A compatible transformation family determines unique compatible coefficients

$$b_N = f_N(0), \quad \alpha_N = f_N(1) - f_N(0),$$

hence unique $b \in \mathbb{Z}_3^\times$ and $\alpha \in \mathbb{Z}_3$. At every level $N \geq 2$, the normal form says either $\alpha_N = 3b_N$ or $\alpha_N \in 9R_N$. A higher-level element of the second kind cannot reduce to a level- N element of the first kind, whose slope has valuation exactly one. Thus a compatible family lies entirely in the first branch from level two onward, giving $\alpha = 3b$, or entirely in the second, giving $\alpha \in 9\mathbb{Z}_3$. Conversely every displayed coefficient pair reduces to the finite normal form at every level, so it defines a compatible family. Composition of affine coefficient pairs is

$$(\alpha, b) \circ (\gamma, d) = (\alpha\gamma, \alpha d + b).$$

In the displayed family b, d are units and $3 \mid \alpha$, so $\alpha d + b$ is a unit, while $\alpha\gamma \in 9\mathbb{Z}_3$. Hence the family is closed under composition, and coefficient extraction respects composition. This proves the asserted semigroup isomorphism. The identity-fibre statement in the preceding theorem prevents a mixed identity sequence.

For arbitrary unit generators, the constructive normal-form proof works unchanged over $\mathbb{Z}_3$: a word of length one gives $\alpha = 3b$, and words of each fixed length $k \geq 2$ give every pair with $\nu_3(\alpha) = k$ and unit intercept. Closing these strata adds exactly the zero slopes. Thus $\overline{\mathcal{S}_{\text{unit}}}$ is the displayed inverse limit.

For the actual exponent generators, Lemma 4.11 shows that their reductions 2^{-a} fill U_N at every level N . Given any cylinder determined by an element of S_N , choose an exponent separately for every unit letter in one representing word. The resulting element of $\mathcal{S}_{\text{exp}}$ has the prescribed reduction. Therefore $\mathcal{S}_{\text{exp}}$ meets every finite-level cylinder and is dense in the same inverse limit. This proves both closure equalities without identifying the countable exponent semigroup with the all-unit semigroup. $\square$

The slope condition cannot be weakened to $\alpha \in 3\mathbb{Z}_3$ with arbitrary unit intercept. At valuation one the exact relation is $\alpha = 3b$; for example $x \mapsto 3x + 2$ is already excluded modulo nine.

4.9 Typed finite actions and the two raw obstructions

The specific object-erasing assignment that sends each path to its residue transformation cannot extend linearly to an algebra representation in one transformation algebra. For distinct objects $x \neq y$,

$$e_x e_y = 0,$$

whereas erasing the object coordinate sends both empty paths to the same identity transformation. The product of their images is then nonzero.

There is a separate variance defect. For a map $h : R_N \rightarrow R_N$, let $P_h f = f \circ h$. Then

$$P_h P_k = P_{k \circ h},$$

so pullback reverses composition. This is visible on the actual composable path

$$3 \xrightarrow{1} 5 \xrightarrow{4} 1.$$

Modulo nine,

$$G_1(r) = 6r + 5, \quad G_4(r) = 3r + 4,$$

but

$$G_4 \circ G_1 = c_1, \quad G_1 \circ G_4 = c_2.$$

Thus the two orderings are genuinely different transformations, not merely different notation.

Let $R_0 = R[u, v]$ and use the monoid algebra $R_0[T_N]$ with multiplication induced by ordinary target-left transformation composition. Put

$$\mathcal{B}_N = \text{Mat}_{\mathcal{O}}^{\text{fin}}(R_0[T_N]),$$

the algebra of matrices having finite support in the infinite object index. For the decorated prefix coordinate, put

$$\mathcal{B}_{\text{pre}} = \text{Mat}_{\mathcal{O}}^{\text{fin}}(R_0[(\mathbb{Z}_3 \rtimes_{\theta} P)^{\text{op}}]),$$

where the bracket denotes the finite-support monoid algebra.

Proposition 4.17 (typed forward and pullback representations). *For $p : x \rightarrow y$, let G_p be its forward residue transformation. The map*

$$\Pi_N([p]) = w(p)E_{y,x} \otimes [G_p] \tag{368}$$

is an R_0 -algebra homomorphism $\mathfrak{A}_C \rightarrow \mathcal{B}_N$.

On the explicitly based free module

$$\mathcal{V}_N = \bigoplus_{x \in \mathcal{O}} \bigoplus_{r \in R_N} R_0 e_{x,r}$$

it has the concrete forward action

$$\Pi_N([p])e_{x,r} = w(p)e_{y, G_p(r)},$$

and is zero on every other object fibre.

For pullback, put

$$\mathcal{F}_N = \text{Fun}(R_N, R_0), \quad \mathcal{K}_N = \text{Mat}_{\mathcal{O}}^{\text{fin}}(\text{End}_{R_0}(\mathcal{F}_N)).$$

The typed pullback map

$$K_N([p]) = w(p)E_{x,y} \otimes P_{G_p}$$

from $\mathfrak{A}_C$ to $\mathcal{K}_N$ is an anti-homomorphism:

$$K_N(ab) = K_N(b)K_N(a).$$

Finally, the decorated prefix map has the covariant typed realization

$$\Psi([p]) = w(p)E_{t(p), s(p)} \otimes [\chi_{\text{pre}}(\text{word}(p))]_{\text{op}}$$

as an R_0 -algebra homomorphism $\Psi : \mathfrak{A}_C \rightarrow \mathcal{B}_{\text{pre}}$.

Proof. If $p : x \rightarrow y$ and $q : z \rightarrow x$, then matrix-unit multiplication and forward composition give

$$\Pi_N([p])\Pi_N([q]) = w(p)w(q)E_{y,z} \otimes [G_p \circ G_q] = \Pi_N([p \circ q]).$$

For noncomposable paths the matrix units multiply to zero. Also $\Pi_N(e_x) = E_{x,x} \otimes [\text{id}]$, so the object idempotents remain orthogonal. This proves the algebra and concrete-module statements.

The identity $P_h P_k = P_{k \circ h}$ proves the pullback anti-homomorphism, with the transposed matrix unit sending target functions back to source functions. For the last statement, the chronological word of $p \circ q$ is the word of q followed by the word of p . Proposition 4.8 therefore gives

$$\chi_{\text{pre}}(\text{word}(p \circ q)) = \chi_{\text{pre}}(\text{word}(q)) \star \chi_{\text{pre}}(\text{word}(p)),$$

which is exactly the product in the opposite semigroup algebra after the matrix units have enforced composability. $\square$

Theorem 4.18 (exact kernels of the typed representations). *Let R be a nonzero commutative ring. The decorated prefix representation Ψ is injective. For $F = \sum_p f_p[p] \in \mathfrak{A}_C$, the finite-residue representation has the exact kernel criterion*

$$F \in \ker \Pi_N \iff \sum_{\substack{p: x \rightarrow y \\ G_p = h}} f_p w(p) = 0 \quad \text{for every } x, y \in \mathcal{O} \text{ and } h \in T_N. \quad (369)$$

In particular, every Π_N is noninjective. If $\epsilon_1 : 1 \rightarrow 1$ is the exponent-two loop, then

$$uv[\epsilon_1^N] - [\epsilon_1^{N+1}] \in \ker \Pi_N \cap \ker K_N \quad (370)$$

is nonzero.

Proof. Distinct paths have either different source-target matrix units or different exponent words. In the latter case the decorated map χ_{pre} separates them by Proposition 4.8. Hence distinct path basis elements have distinct matrix-monoid basis coordinates under Ψ . Multiplication by the monomial $w(p)$ only shifts polynomial exponents and is injective in $R[u, v]$. Finite linear independence proves that Ψ is injective.

For Π_N , collect the image of a finite sum by its matrix coordinate (y, x) and transformation basis element $[h]$. Its coefficient is exactly the sum on the right of (369); independence of those basis coordinates proves both directions of the criterion.

Finally $a(1) = 2$, so $w(\epsilon_1) = uv$. In R_N , where 4 is invertible, the residue transformation $H = G_{\epsilon_1}$ satisfies

$$H(r) - 1 = \frac{3}{4}(r - 1).$$

Therefore $H^N(r) - 1 = (3/4)^N(r - 1) = 0$ in R_N , so $H^N = H^{N+1} = c_1$. The two terms in (370) consequently have the same image $(uv)^{N+1}E_{1,1} \otimes [c_1]$. Their pullback images likewise agree, because both use P_{c_1} and the same matrix unit. The original paths are distinct basis elements and R is nonzero, so their displayed difference is nonzero. No equality between the full kernels of K_N and Π_N is asserted. $\square$

The target $\mathcal{B}_N$ is finite in each residue fibre but not finite-dimensional: it retains the infinite object set $\mathcal{O}$. A genuinely finite quotient would require a separately proved categorical congruence. Merely reducing object labels modulo 3^N identifies orthogonal idempotents and does not provide one. Theorem 4.12 classifies the image transformations and their strata, not the word fibres. The exact linear collision criterion is instead (369).

4.10 What the raw programme does and does not construct

Several later slogans in the passage fail before any Collatz-specific issue arises. The exact failures are useful because they determine which algebraic structures have actually been built.

Source issue 4.19 (four algebraic type failures). First, the natural reduction maps

$$\mathbb{Z}/3^{N+1}\mathbb{Z} \longrightarrow \mathbb{Z}/3^N\mathbb{Z}$$

form an inverse system, not the direct system printed in an early draft. Its limit is $\mathbb{Z}_3 = \varprojlim_N \mathbb{Z}/3^N\mathbb{Z}$; a residue class modulo 3^m has no canonical lift modulo 3^{m+n} .

Second, $\mathbb{R}_{>0}$ under its usual addition and multiplication is not a semiring because it has no additive zero. The path weights may instead take values in the multiplicative monoid $\mathbb{R}_{>0}$ or in the semiring $\mathbb{R}_{\geq 0}$.

Third, ordinary deconcatenation is a coalgebra coproduct on words but is not multiplicative for ordinary concatenation. For two letters a, b ,

$$\Delta(a)\Delta(b) = 1 \otimes ab + a \otimes b + b \otimes a + ab \otimes 1,$$

whereas

$$\Delta(ab) = 1 \otimes ab + a \otimes b + ab \otimes 1.$$

The uncanceled term $b \otimes a$ proves that the printed data do not define a bialgebra. An unshuffle coproduct, a braided tensor product, or another explicit compatibility law would be a different construction.

Fourth, the two-element set of local affine formulas $\{X/2, (3X+1)/2\}$ is not closed under affine composition: already $(X/2) \circ (X/2) = X/4$. It is therefore not an algebra acting on the path algebra, and no crossed product with it is defined by the passage.

At the proved scope, the programme supplies:

- (1) an exact equivalence between exponent paths and legal local odd-to-odd blocks;
- (2) a genuine two-variable path weight and two explicitly compared $\mathbb{F}_2^2$ degrees;
- (3) endpoint formal series, a coefficientwise path completion, and exact resolvent identities;
- (4) a predecessor transfer operator with precise algebraic, formal, and analytic domains;
- (5) an injective decorated prefix map χ_{pre} and its exact reversal relation to the forward affine/Tao offset;
- (6) finite residue transformation semigroups with complete normal forms, cardinalities, ranks, minimum ideals, reduction fibres, and inverse limit;
- (7) typed forward representations and pullback anti-representations that retain object idempotents and composition order.

Explicit nonclaim 4.20. The directly read passage does not construct a Collatz path groupoid of local homeomorphisms, a groupoid C^* -algebra, a Leavitt or Cuntz–Krieger algebra, a quotient category QGr , a Hopf or bialgebra structure, a crossed coefficient product, an operator-algebra completion, a BCM system, an adelic system, a KMS state, or a uniqueness or canonicity theorem for any such object. Names and analogies appearing after the finite constructions do not supply the missing objects or maps. The rejected BCM/adelic material in the later local AI-assisted PDF is not imported here. None of the theorems in this section proves the Collatz conjecture, and no priority claim is made for the independently reconstructed results.

The finite certificate `certificates/chatnotes_weighted_path_finite_actions.py` pins the raw passage, all three human references, both local AI-assisted lineage witnesses, Tao's v7 source, and the existing literature bridge. It verifies bounded instances of the block inverse, grading fibres, cocycle and reversal formulas, the noninjective undecorated ρ example, source congruences, predecessor fibres, endpoint coefficients, finite path-cut counts, the exact generated residue semigroups through $N = 6$, their ranks and adjacent reduction fibres, finite projections of the inverse-limit classification, typed path products, and the explicit fixed-point-loop kernel relation. It does not independently prove the infinite, analytic, faithfulness, or exact-kernel theorems printed above.

5 The Chatnotes arctangent programme: Gaussian torsion and its boundary

This section reconstructs the arctangent programme in the directly read raw chronology at physical lines 9597–11087 of the same 17,437-line Chatnotes export used in Sections 3 and 4. The controlling file has 538,005 bytes and SHA-256 `9a80f53764a9da42c0f8e0b6d19ce61f978980a448a683b9c45f04038716160d`. The direct-locator audit is `intake/chatnotes/ZN-SYMMETRIES-ARCTANGENT-DIRECT-LOCATOR-AUDIT-05.md`. The passage is read chronologically: its initial exploratory rhetoric is not used to infer a theorem, and its later correction controls its earlier terminology.

At physical lines 9597–9602 the user proposed testing the rational Collatz prefix pieces against Heegner-field and Machin/arctangent constructions. The response then fixed a finite positive exponent word

$$w = (a_1, \dots, a_n), \quad A_k(w) = a_1 + \dots + a_k,$$

and, at lines 9611–9621, isolated the rational quantities

$$r_k(w) = 3^{k-1} 2^{-A_k(w)}. \quad (371)$$

These are the summands of the prefix coordinate $\rho(w)$ in (356); they are not the terms of an Euler product.

5.1 The exact multiplicative torsion test

At lines 9649–9685 and again at lines 10181–10246, the passage introduced principal arctangent angles and their Cayley transforms. With notation chosen to avoid the existing action $\theta_{(m,A)}$ and integer $R(w)$, put

$$\vartheta_k(w) = \arctan r_k(w), \quad z_k(w) = 2^{A_k(w)} + i3^{k-1}, \quad u_k(w) = \frac{z_k(w)}{\overline{z_k(w)}} = \frac{1 + ir_k(w)}{1 - ir_k(w)}. \quad (372)$$

The arctangent is the principal value, so $0 < \vartheta_k(w) < \pi/2$ and

$$u_k(w) = e^{2i\vartheta_k(w)} \in \mathbb{Q}(i)^\times. \quad (373)$$

Proposition 5.1 (repaired torsion formulation). *For every $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$,*

$$\sum_{k=1}^n m_k \vartheta_k(w) \in \pi\mathbb{Q} \iff \prod_{k=1}^n u_k(w)^{m_k} \in \mu_4, \quad \mu_4 = \{1, -1, i, -i\}. \quad (374)$$

Consequently the raw question defines the exact lattice

$$\text{Rel}_\pi(w) = \left\{ m \in \mathbb{Z}^n : \sum_{k=1}^n m_k \vartheta_k(w) \in \pi\mathbb{Q} \right\}.$$

It is the kernel of the homomorphism

$$\mathbb{Z}^n \longrightarrow \{x \in \mathbb{Q}(i)^\times : x\bar{x} = 1\} / \mu_4, \quad m \longmapsto \left[\prod_{k=1}^n u_k(w)^{m_k} \right].$$

Proof. Equation (373) gives

$$\prod_k u_k(w)^{m_k} = \exp\left(2i \sum_k m_k \vartheta_k(w)\right).$$

If the sum is in $\pi\mathbb{Q}$, the product is a root of unity. It also lies in $\mathbb{Q}(i)^\times$, whose complete root-of-unity group is μ_4 . Conversely, if the product is i^j , then $2 \sum_k m_k \vartheta_k(w) \equiv j\pi/2 \pmod{2\pi}$; hence the sum lies in $(\pi/4)\mathbb{Z}$. Multiplication of the u_k 's proves the homomorphism and kernel statements. Every $u_k \bar{u}_k = 1$, so the displayed codomain is exact. $\square$

The raw versions wrote μ_∞ . The replacement by μ_4 is not a change of mathematical object: it records the exact torsion subgroup of the already printed field $\mathbb{Q}(i)$. Calcut's Main Lemma and Corollary 3 give the same $(\pi/4)\mathbb{Z}$ restriction for rational arctangent sums [51].

5.2 The correction internal to the chronology

The passage initially called the product condition a “finite logarithmic unit equation.” At physical lines 10866–10936 it explicitly corrected that phrase: a classical unit equation is additive, whereas (374) is a multiplicative torsion test in $\mathbb{Q}(i)^\times$. The same correction says that Baker's theorem may be applied after a particular logarithmic linear form has been proved nonzero, but supplies no universal nonvanishing theorem for the arctangent family. The critical edition therefore retains the exact criterion and does not turn its kernel into a theorem of triviality.

This distinction matters already for the legal word $w = (1, 1)$. Its first two angles satisfy

$$2 \arctan(1/2) + \arctan(3/4) = \frac{\pi}{2}. \quad (375)$$

The identity follows from $(2 + i)^2(4 + 3i) = 25i$, and it appears, after division by two, in the two-term classification of Gasull, Luca, and Varona [52, Theorem 1]. Thus the strongest possible blanket independence assertion is false; the exact question is classification of the relation lattice.

5.3 Published fixed-instance machinery and the remaining uniform question

Conway, Radin, and Sadun reconstruct Størmer's $d = 1$ theory by unique factorization in $\mathbb{Z}[i]$: the arguments of one oriented Gaussian prime above each rational prime $p \equiv 1 \pmod{4}$, together with $\pi/4$, give independent coordinates for rational-tangent angles [50, Theorems 1–2 and Section 2]. Hence every fixed finite instance of (374) is algorithmically decidable by factoring the finitely many Gaussian integers $z_k(w)$. Calcut proves the exact root-of-unity restriction and the single-angle non-torsion criterion [51, Main Lemma and Corollaries 1–3]. Gasull, Luca, and Varona classify

their stated two-term, nonzero-right-hand-side formulas with 2-integer arguments and derive the relevant zero double-angle family [52, Section 2].

These sources do not state a uniform rank theorem for the parametric Gaussian integers

$$2^{A_k(w)} + i3^{k-1} \quad (k = 1, \dots, n).$$

Exact index and web searches did not locate that formulation. This negative search result is not a priority certificate. It identifies the precise point at which the source-faithful reconstruction ends and separately authored research begins.

Explicit nonclaim 5.2. The raw chronology proves neither universal non-torsion nor a Collatz endpoint statement. Its initial references to zeta, Heegner fields, modular forms, monster symmetry, CM arithmetic, transfer operators, and period theory do not establish a map from any of those structures to the lattice $\text{Rel}_\pi(w)$. The fixed-word Gaussian valuation theorem, the modular non-torsion proof, the exact bounded lattice calculation, and the all-length rigidity conjecture belong to the separately authored research companion and are not silently attributed to the raw response.

6 The exponent survivor coding, its arithmetic locus, and two geometric boundaries

This section audits the final coherent passage of the raw Z_n *Symmetries* export, physical lines 14012–17437. The passage contains three different layers which must not be merged:

- (i) an exact finite exponent-word calculus and its inverse-limit survivor coding;
- (ii) proposed tropical, operator-algebraic, and prismatic interfaces; and
- (iii) a later correction of a proposed residue transducer.

The first layer has a rigorous core, but its stated map on all odd 2-adic integers misses an exceptional backward orbit. The second layer contains useful source directions but no constructed Collatz morphism. The third layer finds a genuine counterexample to its own earlier transition rule, but then makes broader impossibility assertions which the calculation does not prove. We give the exact survivor-coding theorem, retain the explicit counterexample, construct the correct refinement tree, and prove the two geometric typing boundaries.

The finite residue-class theorem used below is Proposition 4.9. The tropical square is crosswalked against Connes–Consani [44]; the Cartier–Witt functor against Bhatt–Lurie [45]; the one-variable cyclotomic calculus against Gros–Le Stum–Quirós [46]; and the countable-shift hypotheses against Sarig [47, 49, 48]. None of those sources contains the Collatz-specific deductions proved here.

The terminology matters. Positive odd integers meet every odd residue class modulo every power of two, so their metric completion is the whole space $\mathcal{O}_2 = 1 + 2\mathbb{Z}_2$. The domain constructed below is instead the survivor subspace $D = \mathcal{O}_2 \setminus E$. Its coding by nested residue balls is an inverse-limit survivor extension, not the metric completion of the positive dynamics.

6.1 The partial odd 2-adic first-return map

Put

$$\mathcal{O}_2 = 1 + 2\mathbb{Z}_2, \quad x_* = -\frac{1}{3} \in \mathcal{O}_2.$$

For $x \in \mathcal{O}_2 \setminus \{x_*\}$, the nonzero number $3x + 1$ has a finite positive 2-adic valuation, and we define

$$a_2(x) = \nu_2(3x + 1), \quad C_2(x) = \frac{3x + 1}{2^{a_2(x)}} \in \mathcal{O}_2. \quad (376)$$

Thus C_2 is initially a partial map. For $n \geq 0$, let

$$E_n = \left\{ x \in \mathcal{O}_2 : \begin{array}{l} C_2^j(x) \text{ is defined for } 0 \leq j < n, \\ C_2^n(x) = x_* \end{array} \right\}, \quad E = \bigcup_{n \geq 0} E_n, \quad D = \mathcal{O}_2 \setminus E.$$

Here $E_0 = \{x_*\}$. The set D is the *survivor domain*: every finite forward iterate is defined there.

Lemma 6.1 (the exceptional set cannot be omitted). *The map C_2 restricts to a self-map $C_2 : D \rightarrow D$. The set E is countable. For every $y \in D$ its complete fibre in D is*

$$C_2^{-1}(\{y\}) \cap D = \left\{ \frac{2^a y - 1}{3} : a \geq 1 \right\}.$$

For every $a \geq 1$ the odd 2-adic integer

$$x_a = -\frac{2^a + 3}{9}$$

satisfies

$$\nu_2(3x_a + 1) = a, \quad C_2(x_a) = -\frac{1}{3}.$$

Consequently, deleting only $-1/3$ does not suffice to define every infinite itinerary.

Proof. If $x \in D$, then $x \neq x_*$, so $C_2(x)$ is defined. If $C_2(x) \in E_n$, then $x \in E_{n+1}$, a contradiction; hence $C_2(x) \in D$. Every $y \in \mathcal{O}_2$ has precisely the inverse branches

$$x = \frac{2^a y - 1}{3} \quad (a \geq 1)$$

in $\mathcal{O}_2$: the denominator is a 2-adic unit, the numerator is odd, and $3x + 1 = 2^a y$ gives exact valuation a . Each level E_n is therefore countable, and so is their union. If $y \in D$ and one of these inverse branches lay in E , applying C_2 would put y in E ; hence all the displayed branches lie in D , proving the fibre formula. Substituting $y = -1/3$ gives the displayed x_a and

$$3x_a + 1 = -\frac{2^a}{3},$$

which proves both identities. □

6.2 Exact finite cylinders and the survivor-space conjugacy

For a finite word $w = (a_1, \dots, a_n) \in \mathbb{N}_{>0}^n$, retain the notation

$$A(w) = \sum_{i=1}^n a_i, \quad B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1 + \dots + a_{i-1}}.$$

Let r_w be the unique representative in $\{1, 3, \dots, 2^{A(w)+1} - 1\}$ of

$$r_w \equiv 3^{-n} (2^{A(w)} - B_w) \pmod{2^{A(w)+1}}. \quad (377)$$

For the empty word put $A(\emptyset) = B_\emptyset = 0$ and $r_\emptyset = 1$, with its class interpreted modulo 2.

Lemma 6.2 (2-adic extension of the finite-prefix criterion). *For $w \in \mathbb{N}_{>0}^n$, the set of $x \in \mathcal{O}_2$ whose first n successive valuations are defined and equal to w is exactly the ball*

$$U_w = r_w + 2^{A(w)+1}\mathbb{Z}_2. \quad (378)$$

If $a \geq 1$, then $U_{wa} \subset U_w$. Distinct words of the same length have disjoint balls.

Proof. The induction proving Proposition 4.9 uses only congruences modulo powers of two and division by the odd unit 3; positivity is used there only to select positive representatives. The same induction in $\mathbb{Z}_2$ proves that the first n valuations are exactly w if and only if

$$3^n x + B_w \equiv 2^{A(w)} \pmod{2^{A(w)+1}}.$$

Since 3^n is a unit modulo $2^{A(w)+1}$, this is precisely (378). Appending one valuation preserves the earlier valuations, which proves nesting. Two words which first differ at position j prescribe different exact valuations at the same j th iterate, so their balls are disjoint. $\square$

Lemma 6.3 (exact inverse branches and terminated partition). *For a finite word $w = (a_1, \dots, a_n)$, write*

$$h_a(y) = \frac{2^a y - 1}{3}, \quad h_w = h_{a_1} \circ \dots \circ h_{a_n},$$

with $h_\emptyset$ the identity. Then

$$h_w(y) = \frac{2^{A(w)} y - B_w}{3^n}, \quad h_w : \mathcal{O}_2 \xrightarrow{\sim} U_w. \quad (379)$$

The displayed map is an affine homeomorphism, and $C_2^n|_{U_w}$ is its inverse while the n displayed branches are read. There is an exact disjoint decomposition

$$U_w = \{h_w(x_*)\} \sqcup \bigsqcup_{a \geq 1} U_{wa}. \quad (380)$$

Moreover, E is dense in $\mathcal{O}_2$.

Proof. For $y \in \mathcal{O}_2$, the numerator $2^a y - 1$ is odd and division by the 2-adic unit 3 keeps it in $\mathcal{O}_2$. Directly,

$$3h_a(y) + 1 = 2^a y, \quad \nu_2(3h_a(y) + 1) = a, \quad C_2(h_a(y)) = y.$$

Induction on n , using $B_w = 3^{n-1} + 2^{a_1} B_{(a_2, \dots, a_n)}$, gives the affine formula in (379). It also shows that every $h_w(y)$ reads the exact word w and hence lies in U_w . Conversely, if $x \in U_w$, then $y = C_2^n(x) \in \mathcal{O}_2$, and reversing the n exact branch equations gives $x = h_w(y)$. Thus h_w is a continuous bijection from the compact space $\mathcal{O}_2$ to the Hausdorff space U_w and is a homeomorphism; the displayed restriction of C_2^n is its inverse.

For $x \in U_w$, put $y = C_2^n(x)$. Either $y = x_*$, giving $x = h_w(x_*)$, or $3y + 1 \neq 0$ has one finite positive valuation a . In the second case $y \in U_a$ and hence $x \in h_w(U_a) = U_{wa}$. Exact valuations make the alternatives disjoint, proving (380).

Finally, let $V \subset \mathcal{O}_2$ be nonempty and open. It contains an odd residue ball $m + 2^M \mathbb{Z}_2$ with a positive odd representative m . The positive orbit of m has an infinite exponent itinerary. Choose a prefix w with $A(w) + 1 \geq M$. Then $m \in U_w \subset m + 2^M \mathbb{Z}_2 \subset V$, while $h_w(x_*) \in E \cap U_w$. Thus every nonempty open set meets E . $\square$

Define the itinerary on D by

$$\varepsilon_2(x) = (a_2(x), a_2(C_2x), a_2(C_2^2x), \dots).$$

Write σ for the left shift on $\mathbb{N}_{>0}^{\mathbb{N}_0}$.

Theorem 6.4 (exact odd 2-adic survivor coding). *The map*

$$\varepsilon_2 : D \longrightarrow \mathbb{N}_{>0}^{\mathbb{N}_0}$$

is a bijection and satisfies

$$\varepsilon_2 \circ C_2 = \sigma \circ \varepsilon_2.$$

With the subspace topology on $D \subset \mathbb{Z}_2$ and the product topology on the full countable shift, it is a homeomorphism.

For $\mathbf{a} = (a_1, a_2, \dots)$, its inverse is the unique point $x(\mathbf{a}) \in \mathbb{Z}_2$ in the nested intersection

$$\{x(\mathbf{a})\} = \bigcap_{n \geq 1} \left(r_{(a_1, \dots, a_n)} + 2^{A_n+1} \mathbb{Z}_2 \right), \quad A_n = a_1 + \dots + a_n. \quad (381)$$

That point automatically lies in D .

Proof. The prefix balls in (381) are nested by Lemma 6.2. Their radii tend to zero because $A_n \geq n$. Completeness and separatedness of $\mathbb{Z}_2$ therefore give a unique intersection point $x(\mathbf{a})$. The same lemma says that this point realizes every finite prefix. It cannot lie in E : if $C_2^k(x) = x_*$, then its valuation at time $k+1$ is undefined, whereas membership in the ball for $(a_1, \dots, a_{k+1})$ makes that valuation the finite integer a_{k+1} . Hence $x(\mathbf{a}) \in D$ and $\varepsilon_2(x(\mathbf{a})) = \mathbf{a}$, proving surjectivity.

If $x, y \in D$ have the same itinerary, then for every n they lie in the same ball of modulus 2^{A_n+1} . Since $A_n \rightarrow \infty$, one has $x - y \in \bigcap_n 2^{A_n+1} \mathbb{Z}_2 = \{0\}$, proving injectivity. The conjugacy identity follows by deleting the first valuation.

For a finite word w , the inverse image of its shift cylinder is $D \cap U_w$, which is clopen in D ; hence ε_2 is continuous. Conversely, if a 2-adic neighborhood prescribes a residue modulo 2^M , choose n with $A_n + 1 \geq M$. The cylinder determined by the first n symbols maps into a single residue class modulo 2^{A_n+1} and hence into that neighborhood. Thus the inverse is continuous. $\square$

Source issue 6.5 (the raw all-odd-domain statement). The raw theorem at physical lines 16227–16255 and its repetitions replace D by all of $\mathcal{O}_2$. The point $x_* = -1/3$ and the explicit preimages in Lemma 6.1 disprove that domain. The corrected theorem is not a weakening by convention: it identifies the maximal forward-invariant subset on which every displayed finite valuation exists.

6.3 Positive integers inside the survivor coding

For the positive odd system of Section 4, put

$$\Sigma_C^+ = \varepsilon_2(\mathcal{O}) \subset \mathbb{N}_{>0}^{\mathbb{N}_0}.$$

Positive odd orbits never meet E , since every iterate is positive.

Theorem 6.6 (full finite language and exact shift image). *Every finite positive-exponent word occurs as the initial word of infinitely many positive odd orbits. Consequently Σ_C^+ is countable, dense, proper, and nonclosed in $\mathbb{N}_{>0}^{\mathbb{N}_0}$.*

It is forward-shift stable, but not shift-surjective. More precisely,

$$\sigma(\Sigma_C^+) = \varepsilon_2(\{y \in \mathcal{O} : 3 \nmid y\}) \subsetneq \Sigma_C^+. \quad (382)$$

Proof. Proposition 4.9 gives one odd residue class modulo $2^{A(w)+1}$ for every finite word w , and every sufficiently large positive representative of that class realizes w . Thus every basic shift cylinder meets Σ_C^+ , proving density. The image is countable because $\mathcal{O}$ is countable, and it is proper because the full shift is uncountable. A dense proper subset of this Hausdorff space is not closed.

The conjugacy identity gives forward-shift stability. By Lemma 4.6, a positive odd integer y has a positive odd predecessor exactly when $3 \nmid y$. Therefore the first equality in (382) holds. It is strict: $3 \in \mathcal{O}$ has no predecessor, and injectivity of ε_2 on D prevents its itinerary from being the itinerary of a different starting point. $\square$

For $\mathbf{a} \in \mathbb{N}_{>0}^{\mathbb{N}_0}$, let $r_n(\mathbf{a})$ be the canonical positive odd representative (377) of its length- n prefix. The nesting theorem gives

$$r_{n+1}(\mathbf{a}) \equiv r_n(\mathbf{a}) \pmod{2^{A_n+1}}. \quad (383)$$

Because both representatives lie in their canonical intervals, there is a unique integer k_n such that

$$r_{n+1} = r_n + k_n 2^{A_n+1}, \quad 0 \leq k_n < 2^{a_{n+1}}. \quad (384)$$

Thus (r_n) is nondecreasing as a sequence of ordinary integers.

Theorem 6.7 (the exact positivity selector). *For $\mathbf{a} \in \mathbb{N}_{>0}^{\mathbb{N}_0}$ the following are equivalent:*

- (i) $\mathbf{a} \in \Sigma_C^+$;
- (ii) *the sequence of canonical representatives $r_n(\mathbf{a})$ is eventually constant;*
- (iii) *the sequence $r_n(\mathbf{a})$ is bounded in the ordinary order.*

When these conditions hold, the eventual constant is the unique positive odd integer x with $\varepsilon_2(x) = \mathbf{a}$.

Proof. Let $x = x(\mathbf{a})$ be the 2-adic inverse from Theorem 6.4. If x is a positive odd integer, then for all n with $2^{A_n+1} > x$, the unique representative of x modulo 2^{A_n+1} in the prescribed interval is x itself. Hence $r_n = x$ thereafter. Conversely, if $r_n = r$ for all $n \geq n_0$, then (381) has intersection point r . This is positive and odd, and the survivor-coding theorem gives $\varepsilon_2(r) = \mathbf{a}$. Finally, eventual constancy implies boundedness, while boundedness implies eventual constancy for the nondecreasing integer sequence (384). $\square$

This is an asymptotic prefix-stabilization criterion for the entire rooted prefix chain, not a tail-invariant property. Indeed, $(2, 2, 2, \dots)$ is the itinerary of the positive fixed point 1, but the tail-equivalent sequence $(1, 2, 2, \dots)$ is the itinerary of $h_1(1) = (2 \cdot 1 - 1)/3 = 1/3$, which is not an integer.

The properness and nonclosedness in Theorem 6.6 also have an explicit witness. The constant sequence $(1, 1, 1, \dots)$ corresponds to $-1 \in D$, since $C_2(-1) = -1$. Its canonical prefix representatives are $2^{n+1} - 1$, so it is not in Σ_C^+ ; nevertheless every cylinder $[1^n]$ contains positive odd realizations.

Construction 6.8 (the exact prefix-refinement tree). Let $\mathcal{T}$ have one vertex

$$v_w = (w, A(w) + 1, r_w)$$

for every finite positive-exponent word w , and one edge $v_w \rightarrow v_{wa}$ for every $a \geq 1$. Retaining w is part of the state. The ball attached to v_{wa} is contained in the ball attached to v_w by Lemma 6.2. Infinite rooted paths are therefore identified, coordinate for coordinate, with $\mathbb{N}_{>0}^{\mathbb{N}_0}$, and their nested-ball endpoint is the inverse map (381). The positive ends are exactly those for which the decorations r_w stabilize, by Theorem 6.7.

This construction is an exact rooted-tree model; no appeal to a general inverse-sequence analogy is needed to prove it.

6.4 Pressure on the full survivor shift

On $\Sigma = \mathbb{N}_{>0}^{\mathbb{N}_0}$ define the one-coordinate potential

$$\phi(\mathbf{a}) = \log 3 - a_1 \log 2.$$

Theorem 6.9 (complete pressure domain and Bernoulli law). *For real β , the unrestricted length- n partition sum is*

$$Z_n(\beta) = \sum_{(a_1, \dots, a_n) \in \mathbb{N}_{>0}^n} \exp \left(\beta \sum_{j=1}^n (\log 3 - a_j \log 2) \right).$$

It is finite exactly when $\beta > 0$. On that domain,

$$Z_n(\beta) = \left(\frac{(3/2)^\beta}{1 - 2^{-\beta}} \right)^n, \quad (385)$$

$$P(\beta) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n(\beta) = \beta \log \frac{3}{2} - \log(1 - 2^{-\beta}). \quad (386)$$

The probability vector

$$p_\beta(a) = (1 - 2^{-\beta})2^{-\beta(a-1)} \quad (a \geq 1) \quad (387)$$

defines a Bernoulli measure μ_β satisfying the exact cylinder identity

$$\mu_\beta[a_1, \dots, a_n] = \exp \left(\beta \sum_{j=1}^n \phi(\sigma^{j-1} \mathbf{a}) - nP(\beta) \right).$$

In particular it is an invariant Gibbs measure with Gibbs constant one. At $\beta = 1$, $p_1(a) = 2^{-a}$.

Proof. All summands are nonnegative, so Tonelli factorization gives

$$Z_n(\beta) = \left(\sum_{a \geq 1} 3^\beta 2^{-a\beta} \right)^n.$$

The inner geometric series converges exactly for $2^{-\beta} < 1$, equivalently $\beta > 0$, and then equals the base in (385). Taking logarithms proves (386). Dividing the one-symbol weight $3^\beta 2^{-a\beta}$ by its sum gives (387); multiplying these normalized weights over a cylinder gives the displayed Gibbs identity. $\square$

This $P(\beta)$ is also the Gurevich pressure used by Sarig. Indeed, fix a symbol b and put

$$S_\beta = \sum_{a \geq 1} e^{\beta \phi(a)}.$$

The period- n partition sum based at b is

$$Z_n^G(\beta\phi, b) = \sum_{\substack{\sigma^n \mathbf{a} = \mathbf{a} \\ a_1 = b}} e^{\beta \sum_{j=0}^{n-1} \phi(\sigma^j \mathbf{a})} = e^{\beta \phi(b)} S_\beta^{n-1}. \quad (388)$$

For $\beta > 0$, its exponential growth rate is $\log S_\beta = P(\beta)$. For $\beta \leq 0$, $S_\beta = \infty$, so the based periodic sum is already infinite for every $n \geq 2$ and the Gurevich pressure is $+\infty$.

More explicitly, this is the one-sided topological Markov shift of the complete directed graph on $\mathbb{N}_{>0}$, hence it is topologically mixing and has BIP: any one fixed symbol supplies both the finite big-image and big-preimage set; in particular, it satisfies the one-sided big-images condition (13) of Sarig's 1999 article. The potential $\beta\phi$ is one-cylinder locally constant, so every variation from level one onward is zero and the variations are summable; for $\beta > 0$ one also has $\sup \beta\phi = \beta \log(3/2) < \infty$ and finite pressure by (386). Thus the hypotheses of Sarig's 2003 Gibbs/BIP theorem [49] are individually verified on the full shift when $\beta > 0$. Sarig's distinct 1999 Theorem 8 [47] additionally uses positive recurrence and an RPF eigenfunction uniformly bounded above and below, together with its one-sided big-images condition; it is not the BIP theorem. None of these hypotheses is transferred to Σ_C^+ or the prefix tree. The direct calculation above fixes the measure and Gibbs constant without using either theorem as a black box.

Corollary 6.10 (the full-shift Gibbs law does not select positive integers). *For every $\beta > 0$, the countable dense set Σ_C^+ has μ_β -measure zero. Under the conjugacy of Theorem 6.4, the corresponding invariant probability measure on D assigns measure zero to $\mathcal{O}$.*

Proof. The largest one-symbol mass is $1 - 2^{-\beta} < 1$. Hence the mass of any single infinite sequence is bounded by $(1 - 2^{-\beta})^n$ for every n and is zero. Countable additivity and countability of Σ_C^+ finish the proof; the second assertion is its pushforward form. $\square$

6.5 The failed compressed transition and its exact replacement

The raw passage proposes states (N, r) , with r an odd representative modulo 2^N , and the update

$$(N, r) \xrightarrow{a} (N + a, r'), \quad 3r' + 1 \equiv 2^a r \pmod{2^{N+a}}. \quad (389)$$

Proposition 6.11 (explicit obstruction to the proposed update). *Rule (389) does not propagate the canonical residue of the initial value. Starting at $(1, 1)$ and reading the word $(1, 1, 2)$ gives*

$$(1, 1) \longrightarrow (2, 3) \longrightarrow (3, 7) \longrightarrow (5, 9).$$

But the exact initial-source class for $(1, 1, 2)$ is

$$x \equiv 7 \pmod{32}.$$

The congruence in (389) is a predecessor relation for r' over r , not the refinement relation $r_{wa} \equiv r_w \pmod{2^{A(w)+1}}$.

Proof. Successive solution of the three raw congruences gives 3, 7, and 9 as displayed. For $w = (1, 1, 2)$ one has $A(w) = 4$ and

$$B_w = 3^2 + 3 \cdot 2 + 2^2 = 19.$$

Equation (377) becomes $27x + 19 \equiv 16 \pmod{32}$, whose unique odd solution is $x \equiv 7$. Finally, the raw equation has the literal form $3r' + 1 \equiv 2^a r$; it solves for a branch predecessor r' of r . It contains no condition involving the previous prefix composite applied to an initial-source lift. $\square$

The exact local update is the edge $v_w \rightarrow v_{wa}$ in Construction 6.8, with r_{wa} recomputed by (377). This retains the prefix and has no ambiguity. The counterexample proves failure of the displayed raw update; it does *not* prove that every possible residue encoding, Markov extension, quotient, or recoding is impossible.

Proposition 6.12 (canonical residue pairs retain the prefix). *On the canonical state set the pair itself retains the word. If*

$$(A(w) + 1, r_w) = (A(z) + 1, r_z),$$

then $w = z$. Consequently a successor operation on canonical pairs is well-defined by decoding w and passing to wa .

Proof. Any positive representative of the common residue class realizes both w and z . Determinism makes the shorter word a prefix of the longer. Equality of the total exponents then forces equal lengths, since every additional exponent is positive. Hence the words are equal. $\square$

What Proposition 6.11 rules out is the printed predecessor congruence as that successor, not the existence of every pair-based presentation.

6.6 The tropical degree map and the object-idempotent obstruction

Let $\mathbf{BN}_0^2$ denote the one-object category whose endomorphism monoid is $(\mathbb{N}_0^2, +)$. Define

$$\deg(p) = (\ell(p), A(p) - \ell(p)).$$

Put $Q = \mathbb{R}_{\geq 0}^2$. Connes–Consani define $\text{Conv}(\mathbb{Z}^2)$ to consist of the empty set and the closed convex sets $C \subset \mathbb{R}^2$ satisfying $C + Q = C$ and $C \subset z + Q$ for some $z \in \mathbb{R}^2$, with integral extreme points. Its addition is convex hull of union, its multiplication is Minkowski sum, its additive zero is $\emptyset$, and its multiplicative unit is Q [44].

Proposition 6.13 (the exact tropical shadow). *The object-erasing assignment*

$$\deg : \mathcal{E}_C \longrightarrow \mathbf{BN}_0^2$$

is a functor. The translated-quadrant map

$$j : \mathbb{N}_0^2 \longrightarrow \text{Conv}(\mathbb{Z}^2), \quad j(r, s) = (r, s) + Q,$$

is an injective monoid homomorphism for addition on $\mathbb{N}_0^2$ and Minkowski multiplication in $\text{Conv}(\mathbb{Z}^2)$. After viewing that multiplicative monoid as a one-object category, $j \circ \deg$ is a typed functorial degree shadow.

For $r \geq 1$, the exponent-word part of the fibre over (r, s) consists of the $\binom{r+s-1}{r-1}$ compositions of $r + s$ into r positive parts; each word has the infinite source fibre proved in Proposition 4.9. Over $(0, 0)$ the fibre consists exactly of all object identities, while every fibre over $(0, s)$ with $s > 0$ is empty. Thus the degree shadow deliberately loses objects, exponent order, and arithmetic source data.

Proof. Length and total exponent are additive under composable concatenation, so both coordinates of $\deg$ are additive and identities map to $(0, 0)$. For every $(r, s) \in \mathbb{N}_0^2$, the set $(r, s) + Q$ is closed and convex, obeys $((r, s) + Q) + Q = (r, s) + Q$, is contained in the translate $(r, s) + Q$, and has the unique integral extreme point (r, s) . It therefore belongs to $\text{Conv}(\mathbb{Z}^2)$ under the cited definition. Translated upper quadrants satisfy

$$((r, s) + Q) +_{\text{Mink}} ((r', s') + Q) = (r + r', s + s') + Q,$$

because $Q + Q = Q$. The unique extreme point of $j(r, s)$ is (r, s) , proving injectivity, and $j(0, 0) = Q$ is the multiplicative unit. Finally, words of length r and total exponent $r + s$ are exactly the positive compositions of that integer into r parts, giving the stated count. $\square$

Proposition 6.14 (no object-erasing category-algebra homomorphism). *Let K be a nonzero commutative semiring and let $K[\mathcal{E}_C]$ denote the finite-support category semialgebra. The K -linear map*

$$\pi_{\deg} : K[\mathcal{E}_C] \longrightarrow K[\mathbb{N}_0^2], \quad [p] \longmapsto X^{\ell(p)} Y^{A(p) - \ell(p)}$$

is not multiplicative.

Separately, let $\text{Mor}(\mathcal{E}_C)^0$ be the path semigroup with an absorbing zero, so noncomposable paths multiply to 0. The typed set map

$$\theta : \text{Mor}(\mathcal{E}_C)^0 \longrightarrow \text{Conv}(\mathbb{Z}^2), \quad \theta(0) = \emptyset, \quad \theta(p) = j(\deg p),$$

is multiplicative on composable path pairs but is not a semigroup homomorphism. No coefficient-semiring map into $\text{Conv}(\mathbb{Z}^2)$ is asserted.

Proof. Choose distinct objects $x \neq y$. Their identities are orthogonal in the category semialgebra: $[1_x][1_y] = 0$. Both monomial images are the target unit, so their product is $1 \neq 0$. For the separately typed map θ , one has

$$\theta(1_x 1_y) = \theta(0) = \emptyset, \quad \theta(1_x) \theta(1_y) = Q +_{\text{Mink}} Q = Q \neq \emptyset.$$

Thus θ also fails multiplicativity, now without any unprovided coefficient map. $\square$

Connes–Consani use $\text{Conv}(\mathbb{Z} \times \mathbb{Z})$, not a silently substituted $\text{Conv}(\mathbb{N}_0^2)$, and explicitly retain a noncancellativity caveat before passing to the reduced square. The Collatz image above occupies only translated upper quadrants anchored at nonnegative lattice points. The typed matrix representations of Proposition 4.17, not object erasure, are the algebraic repair.

6.7 Cartier–Witt functoriality and the missing formal-scheme map

Bhatt–Lurie attach a groupoid-valued Cartier–Witt stack WCart_X to a bounded p -adic formal scheme X and make the construction functorial in formal-scheme maps [45]. For $X = \text{Spf}(\mathbb{Z}_p)$, finality identifies WCart_X with the base Cartier–Witt stack. Their Witt-vector Frobenius is a separate endomorphism in that construction.

Proposition 6.15 (the raw branch is not a self-map of $\text{Spf}(\mathbb{Z}_3)$). *Every formal-scheme endomorphism of $\text{Spf}(\mathbb{Z}_3)$ induced by a continuous unital ring endomorphism of $\mathbb{Z}_3$ is the identity. For every $a \geq 1$, the affine function*

$$g_a(x) = \frac{3x + 1}{2^a}$$

is not such a ring endomorphism: $g_a(0) = 2^{-a} \neq 0$. Therefore Bhatt–Lurie functoriality does not produce a morphism called $\text{WCart}(g_a)$ from a branch self-map of $\text{Spf}(\mathbb{Z}_3)$.

Proof. A unital ring endomorphism $f : \mathbb{Z}_3 \rightarrow \mathbb{Z}_3$ fixes every integer. The integers are dense in $\mathbb{Z}_3$, so continuity forces f to be the identity. Independently, every ring homomorphism sends zero to zero, whereas the displayed affine function does not. $\square$

Proposition 6.16 (the exact formal-affine-line repair). *Put*

$$\hat{\mathbb{A}}_{\mathbb{Z}_3}^1 = \mathrm{Spf}(\mathbb{Z}_3\langle T \rangle).$$

The continuous coordinate-ring endomorphism fixing $\mathbb{Z}_3$ and sending $T \mapsto (3T + 1)/2^a$ defines an endomorphism of this formal affine line, so Cartier–Witt functoriality may be applied to that map. On $\mathbb{Z}_3$ -valued points it is injective with image $2^{-a} + 3\mathbb{Z}_3$; the inverse on that image is $y \mapsto (2^a y - 1)/3$.

This endomorphism does not by itself encode the Collatz branch partition. In $\mathbb{Z}_3$, the element 2 is a unit, so $(2^m) = \mathbb{Z}_3$ for every m ; hence the 2-power ideals supply no selector filtration on the formal point.

Proof. The coefficient 2^{-a} lies in $\mathbb{Z}_3$, so substitution by the displayed restricted polynomial gives a continuous $\mathbb{Z}_3$ -algebra endomorphism of $\mathbb{Z}_3\langle T \rangle$ and hence, contravariantly, the formal-scheme map. The pointwise fibre statement follows by solving the affine equation. The ideal identity $(2^m) = \mathbb{Z}_3$ follows from invertibility of 2. $\square$

The ideal calculation alone does not prove that the integer valuation strata fail to be 3-adic congruence strata. The exact obstruction is constructive.

Proposition 6.17 (the integer valuation selector is 3-adically dense and codense). *For $a \geq 1$, let*

$$\mathcal{D}_a^+ = \{n \in \mathcal{O} : \nu_2(3n + 1) = a\}.$$

For every $k \geq 1$ and every residue $b \pmod{3^k}$, the residue ball $b + 3^k\mathbb{Z}_3$ contains a positive odd integer in $\mathcal{D}_a^+$ and a positive odd integer outside $\mathcal{D}_a^+$; the latter can be chosen in $\mathcal{D}_{a+1}^+$. Thus both $\mathcal{D}_a^+$ and its complement inside $\mathcal{O}$ meet every finite 3-adic residue ball. In particular there is no clopen $U \subset \mathbb{Z}_3$ for which $U \cap \mathcal{O} = \mathcal{D}_a^+$.

Proof. Let

$$c_a \equiv 3^{-1}(2^a - 1) \pmod{2^{a+1}}.$$

It is an odd residue and $3c_a + 1 \equiv 2^a \pmod{2^{a+1}}$. Since 3^k and 2^{a+1} are coprime, the Chinese remainder theorem gives an integer n with

$$n \equiv b \pmod{3^k}, \quad n \equiv c_a \pmod{2^{a+1}}.$$

Adding a sufficiently large positive multiple of $3^k 2^{a+1}$ makes n positive without changing either congruence. It remains odd, and the second congruence says exactly that $\nu_2(3n + 1) = a$. Repeating the construction with c_{a+1} modulo 2^{a+2} gives a positive odd m in the same 3-adic residue ball with $\nu_2(3m + 1) = a + 1$, so $m \notin \mathcal{D}_a^+$.

If a clopen U cut out $\mathcal{D}_a^+$ on $\mathcal{O}$, choose $n \in \mathcal{D}_a^+ \subset U$. Openness supplies a finite residue ball $n + 3^k\mathbb{Z}_3 \subset U$. The preceding construction places a positive odd $m \notin \mathcal{D}_a^+$ in the same ball, contradicting $U \cap \mathcal{O} = \mathcal{D}_a^+$. $\square$

Moving from the base formal point to the affine line repairs the type of the polynomial map. The CRT theorem proves that its valuation selector cannot be recovered from a finite clopen 3-adic congruence condition on the embedded positive odd integers; the required piecewise arithmetic dynamics therefore needs additional 2-adic selector data.

For $W = W(k)$ with k a perfect field of characteristic $p > 0$ and a primitive p th root ζ , Gros–Le Stum–Quirós distinguish the relevant one-variable categories precisely: Theorem 7.5 uses complete modules with Δ -hyperstratification, while Theorem 7.12 identifies finite-free weakly nilpotent Δ -modules with the stated prismatic vector bundles; with Frobenius, their later comparison reaches crystalline representation lattices [46]. This source fixes explicit hypotheses and the one-variable coordinate ring $W[[q-1]]$. It contains no two-variable ring $W[[q_2-1, q_3-1]]$, no two independent Collatz coordinate directions or commuting connection operators, no pair of Collatz branch operators, and no theorem identifying their difference with a drift observable.

Explicit nonclaim 6.18 (interfaces not admitted at this stage). The raw groupoid C^* -algebra, KMS, BCM, motivic, two-coordinate toric, Cartier–Witt, prismatic, crystalline, and Langlands statements are not theorems of this edition. The exact degree functor, translated-quadrant Newton-polygon map, affine-line branch maps, and survivor-space symbolic conjugacy are the proved interfaces. No further arrow is obtained by similarity of words or by the fact that several constructions are spectral.

6.8 What the audited passage actually establishes

The rigorous content retained from this passage is now exact:

- (1) finite exponent words are exact residue cylinders, and their nested limits give a homeomorphic conjugacy between the survivor domain D and the full one-sided countable shift;
- (2) the missing set is the full eventual backward orbit of $-1/3$, not merely one exceptional point;
- (3) positive integer itineraries form a countable dense nonclosed subset, characterized exactly by stabilization of canonical prefix representatives and stable only in the forward-shift sense;
- (4) the full-shift pressure and Bernoulli law are explicit exactly for $\beta > 0$, while the positive arithmetic locus has measure zero for those full-shift laws;
- (5) the proposed (N, r) update fails on $(1, 1, 2)$, and the prefix-refinement tree supplies a correct replacement without claiming that every other recoding is impossible;
- (6) the tropical degree is a genuine functor and embeds as translated upper Newton quadrants, but object erasure obstructs the proposed category-algebra homomorphism; and
- (7) the raw $\mathrm{Spf}(\mathbb{Z}_3)$ branch map does not exist. A formal affine-line map exists, but it does not encode the 2-adic valuation strata required by Collatz dynamics.

None of these statements proves termination, excludes divergent positive orbits, or identifies the Collatz system with the cited geometric theories.

7 The Chatnotes groupoid and motivic interval: exact maps and source boundaries

This section audits physical lines 11088–14011 of the directly read 17,437-line Z_n *Symmetries* export. It closes the only interval of that file which lacked a direct-locator audit. The chronology consists of four successive expansions:

- (i) lines 11088–11720 attach the 1-motive and Nori sources and propose a symbolic/groupoid ladder and a period ladder;
- (ii) lines 11721–12206 repeat the grading and period formulas and replace the unnamed groupoid by an inverse-edge shift groupoid;
- (iii) lines 12207–12660 attach a much broader operator and motivic vocabulary without constructing the proposed comparison maps; and
- (iv) lines 12677–14011 discover genuine partial homeomorphisms of $\mathbb{Z}_3$, then incorrectly identify their compact operator model with the positive-integer path algebra and promote that identification to a larger arithmetic and equilibrium package.

The chronology contains real mathematics. Its two torus coordinate systems, its 3-adic branch maps, and its logarithmic and arctangent formulas can all be made exact. The exact results require maps which the raw response did not supply, and those independently proved repairs are explicitly separated from the source assertions below.

The controlling raw file has 538,005 bytes and SHA-256 9a80f53764a9da42c0f8e0b6d19ce61f978980a448a683b9c45f04038716160d. The locator report is `intake/chatnotes/ZN-SYMMETRIES-GROUPOID-MOTIVIC-DIRECT-LOCATOR-AUDIT-06.md`.

7.1 What is already proved before this interval

The raw grading

$$\deg(p) = (\ell(p), A(p) - \ell(p))$$

and its torus action at lines 11130–11265 are correct in the target-left category algebra already defined in Section 4. The finite maps at lines 11269–11298 repeat the maps audited in Theorem 4.12 and 4.17; the raw text again erases object idempotents and again calls the nonempty-word image a monoid before adjoining an identity. No second finite representation is admitted from that repetition.

The full countable-word pressure calculation at lines 11302–11354 and 13637–13684 is the calculation already proved in Section 6. It applies to the full label shift. The raw response does not verify the hypotheses needed to transfer a Gibbs or equilibrium theorem to the positive-integer image.

The groupoid arrow at lines 11358–11391,

$$R[\mathcal{E}_C] \rightsquigarrow \mathcal{G}_C \rightsquigarrow C^*(\mathcal{G}_C),$$

does not declare a topology, Haar system, full or reduced completion, or an algebra map. The inverse-edge version at lines 11813–11885 declares a path space and shift but still gives no topology and no proof that the shift is a local homeomorphism. The later 3-adic formulas supply a different exact route; it must not be identified with either of those arrows without the maps proved below.

7.2 The literature actually used

Deaconu begins with the set of triples attached to a map and proves his topological theorem when the map is a covering of a compact Hausdorff space [53, printed pp. 1779–1781]. The positive-odd set with the discrete topology is not compact and C is not a self-covering, so that theorem is not invoked. Its set-theoretic groupoid formula is used as a reference point; the required discrete topology and Haar assertions are proved directly here.

Huber–Wüstholz define a 1-motive $[L \rightarrow G]$ and its morphisms as maps of complexes, compute $\log \alpha$ from the Kummer motive $[\mathbb{Z} \xrightarrow{1 \mapsto \alpha} \mathbf{G}_m]$, and identify all 1-periods with periods of 1-motives [54, Definition 8.1, Section 10.2, and Proposition 12.10]. Their Theorem 10.4 explicitly uses characters $(x, y) \mapsto x^n y^m$ of $\mathbf{G}_m^2$. Huber–Müller-Stach define a Nori vertex as a triple (X, Y, i) and permit exactly two primitive edge types: a contravariant edge from an actual map of pairs and a coboundary edge from an actual nested triple [55, Definition 9.1.1]. Concatenation of integration paths is not a third edge type.

Barbieri-Viale–Kahn define a Deligne 1-motive and its morphisms by a commutative square in source TeX lines 728–750, prove the fully faithful curve-generated embedding in source lines 1736–1751, and define $LAlb$ in source lines 3796–3800 [56]. Their source provides the general machinery, not the particular point $(2, 3)$, the character $x^A y^{-\ell}$, or a curve realizing that object. The concrete squares below are therefore independent constructions.

7.3 The exact positive-integer groupoid

For $n \geq 0$ write

$$S_n a(x) = \sum_{j=0}^{n-1} a(C^j x), \quad S_0 a(x) = 0.$$

Define

$$\mathcal{G}_{\text{arith}} = \{(x, k, y) : x, y \in \mathcal{O}, \exists m, n \geq 0, k = m - n, C^m(x) = C^n(y)\}. \quad (390)$$

Put $r(x, k, y) = x$, $s(x, k, y) = y$,

$$(x, k, y)(y, l, z) = (x, k + l, z), \quad (x, k, y)^{-1} = (y, -k, x),$$

and give the set the discrete topology.

Theorem 7.1 (independent arithmetic groupoid repair). *The object (390) is a countable Hausdorff, locally compact, étale groupoid with counting Haar system. The formula*

$$c_a(x, k, y) = S_m a(x) - S_n a(y)$$

is independent of the representative (m, n) . The maps

$$D(x, k, y) = (-k, -c_a(x, k, y) + k), \quad \delta_{\mathcal{G}}(x, k, y) = -c_a(x, k, y) \log 2 + k \log 3 \quad (391)$$

are additive groupoid cocycles.

For a path $p : x \rightarrow y$,

$$J(p) = (y, -\ell(p), x)$$

defines an injective functor and

$$D(J(p)) = (\ell(p), A(p) - \ell(p)), \quad \delta_{\mathcal{G}}(J(p)) = A(p) \log 2 - \ell(p) \log 3.$$

Therefore $[p] \mapsto 1_{\{J(p)\}}$ is an injective algebra map $\mathbb{C}[\mathcal{E}_C] \rightarrow C_c(\mathcal{G}_{\text{arith}})$. The two cocycles give respectively a $\mathbb{T}^2$ gauge action and an $\mathbb{R}$ time action on both the full and reduced groupoid C^* -algebras.

Proof. If (m, n) is a representative and $u = C^m(x) = C^n(y)$, extending both times by t adds the same sum $S_t a(u)$ to the two terms defining c_a . Any two representatives have the same difference and admit a common extension, so c_a is well-defined. For composable representatives (m, n) and (p, q) , extend them until the two middle times equal $\max\{n, p\}$. The middle Birkhoff sums cancel, proving additivity of both k and c_a , hence of the two displayed linear combinations.

Every singleton is a compact open bisection in the discrete topology, which proves all topological and Haar assertions. The pair $(0, \ell(p))$ represents $J(p)$ and has $c_a = -A(p)$, proving the cocycle formulas. Source and length determine a path of the deterministic first-return category, so J is injective. Point masses respect the target-left product and are linearly independent. Multiplication by the characters of D and by $e^{it\delta g}$ gives convolution $*$ -automorphisms. Universal norms give the full extensions and diagonal unitaries in regular representations give the reduced extensions. $\square$

7.4 The exact 3-adic object and the missing arithmetic map

Lines 12801–12928 introduce

$$X = \mathbb{Z}_3, \quad Y_a = \{y : 2^a y \equiv 1 \pmod{3}\}, \quad g_a(x) = \frac{3x+1}{2^a}, \quad \beta_a(y) = \frac{2^a y - 1}{3}. \quad (392)$$

This is a genuine construction.

Proposition 7.2 (exact branch homeomorphisms). *For every $a \geq 1$, $Y_a = 2^{-a} + 3\mathbb{Z}_3$ is clopen and $g_a : X \rightarrow Y_a$ is a homeomorphism with inverse β_a . For a word $w = (a_1, \dots, a_n)$,*

$$h_w(x) = g_{a_n} \circ \dots \circ g_{a_1}(x) = \frac{3^n x + B_w}{2^{A(w)}}, \quad B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1 + \dots + a_{i-1}},$$

and h_w maps X homeomorphically onto a clopen set.

Proof. The residue class is clopen because 2^a is a 3-adic unit. Both inverse identities follow by direct substitution, and the affine word formula follows by induction. $\square$

Let $\mathcal{H}_3$ be the groupoid of germs of the pseudogroup generated by these homeomorphisms and their inverses. Each pseudogroup element is locally affine. Two affine maps agreeing on a nonempty 3-adic open set have equal slope and intercept; this proves that the germ groupoid is Hausdorff. Its germ bisections make it second countable, locally compact, and étale.

Theorem 7.3 (independent arithmetic-to-3-adic germ functor). *For $(x, m - n, y) \in \mathcal{G}_{\text{arith}}$, let $p_m(x)$ and $p_n(y)$ denote the actual exponent words along the indicated positive orbits. The formula*

$$\Xi(x, m - n, y) = [h_{p_m(x)}^{-1} \circ h_{p_n(y)}, y] \quad (393)$$

defines an injective groupoid functor $\Xi : \mathcal{G}_{\text{arith}} \rightarrow \mathcal{H}_3$. If u is the rational slope of the affine germ, then

$$u = 3^{n-m} 2^{S_m a(x) - S_n a(y)}, \quad D = (\nu_3 u, -\nu_2 u - \nu_3 u), \quad \delta g = -\log |u|_\infty.$$

Indeed $\nu_3 u = n - m = -k$ and $\nu_2 u = c_a$, so these are exactly the coordinates $(-k, -c_a + k)$.

Proof. Adding a common future word on both sides of (393) cancels that affine map against its inverse, so the germ is independent of (m, n) . Common-middle extension proves functoriality. Its slope has $\nu_3(u) = n - m$ and $\nu_2(u) = S_m a(x) - S_n a(y)$. Source, range, and $-\nu_3(u)$ recover the triple, proving injectivity and the cocycle formulas. $\square$

The raw recovery on $\ell^2(\mathcal{O})$ at lines 12995–13042 is false. Indeed $g_1(1) = 2 \notin \mathcal{O}$. Under the printed formula for V_1 , the equation $\beta_1(y) = 1$ has the unique solution $y = 2$, so $V_1 \delta_1 = 0$ and $V_1^* V_1 \neq 1$. The subsequent claims that the old path algebra is exactly recovered, that an unspecified semigroup crossed product equals $C^*(\mathcal{H}_3)$, and that prefixing defines the printed endomorphism therefore do not follow. The exact replacement is the functor Ξ ; because $\mathbb{Z}_3$ is nondiscrete, it is not a point-mass embedding into $C_c(\mathcal{H}_3)$.

The prefix defect can be proved before any completion. In the free-monoid algebra let $L_q[p] = [qp]$. For nonempty q ,

$$L_q(1 \cdot 1) = [q], \quad L_q(1)L_q(1) = [qq],$$

and these are distinct basis vectors. Thus left prefixing is not an algebra endomorphism. A genuine endomorphism would require a different declared map and a proof that it respects every covariance relation.

7.5 The two torus coordinates and the actual 1-motive squares

The raw text uses the degree character $(\ell, A - \ell)$ and the arithmetic character $(A, -\ell)$ as if they were the same coordinates. They are related by a specific unimodular torus isomorphism.

Theorem 7.4 (independent torus and Kummer repair). *Let*

$$T_{\text{ar}} = (\mathbf{G}_m)_{x,y}^2, \quad T_{\text{deg}} = (\mathbf{G}_m)_{\lambda,\mu}^2,$$

and

$$\Phi(x, y) = (x/y, x), \quad \Phi^{-1}(\lambda, \mu) = (\mu, \mu/\lambda).$$

For a path p put

$$\chi_p(x, y) = x^{A(p)} y^{-\ell(p)}, \quad \psi_p(\lambda, \mu) = \lambda^{\ell(p)} \mu^{A(p) - \ell(p)}.$$

Then $\psi_p \Phi = \chi_p$, with lattice matrices

$$\begin{pmatrix} A \\ -\ell \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \ell \\ A - \ell \end{pmatrix}, \quad \begin{pmatrix} \ell \\ A - \ell \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} A \\ -\ell \end{pmatrix}.$$

Over $\mathbb{Q}$, define

$$M_{\text{ar}} = [\mathbb{Z} \rightarrow T_{\text{ar}}], \quad n \mapsto (2^n, 3^n), \quad K(\alpha_p) = [\mathbb{Z} \rightarrow \mathbf{G}_m], \quad n \mapsto \alpha_p^n,$$

where $\alpha_p = 2^{A(p)} 3^{-\ell(p)}$. Then $(\text{id}, \chi_p) : M_{\text{ar}} \rightarrow K(\alpha_p)$ is a morphism of 1-motives and, for $\gamma(s) = (2^s, 3^s)$,

$$\int_{\gamma} \chi_p^* \frac{dz}{z} = A(p) \log 2 - \ell(p) \log 3.$$

The path fixes the logarithm branch. The same square with $\chi_{p,k}(x, y) = x^{-A_k} y^{k-1}$ gives the period $\log(3^{k-1} 2^{-A_k})$.

Proof. Substitution proves the torus inverse and character identity; multiplication of the integral matrices gives the identity. The square of group schemes commutes because

$$\chi_p(2^n, 3^n) = (2^{A(p)} 3^{-\ell(p)})^n.$$

Finally $\chi_p^*(dz/z) = A(p) dx/x - \ell(p) dy/y$, and integration on the declared path proves the period formula. The prefix calculation is identical. $\square$

The Cartier dual specialization is

$$M_{\text{ar}}^\vee = [\mathbb{Z}^2 \rightarrow \mathbf{G}_m], \quad (r, s) \mapsto 2^r 3^s.$$

Thus the alternate raw object $[\mathbb{Z}^2 \rightarrow \mathbf{G}_m], (m, n) \mapsto 2^m 3^{-n}$, is the same dual after the declared basis involution $(m, n) \mapsto (m, -n)$. It is not a replacement for the commutative square just proved.

7.6 The arctangent pair and the Nori edge direction

Over $K = \mathbb{Q}(i)$, put $U_K = \mathbf{P}_K^1 \setminus \{i, -i\}$. The Cayley map and inverse are

$$\kappa(z) = \frac{1+iz}{1-iz}, \quad \kappa^{-1}(w) = \frac{w-1}{i(w+1)}.$$

Proposition 7.5 (independent Cayley pair repair). *The map κ is an isomorphism $U_K \simeq \mathbf{G}_{m,K}$. For $r \in \mathbb{Q}_{>0}$ and $\alpha(r) = (1+ir)/(1-ir)$ it is an isomorphism of pairs*

$$(U_K, \{0, r\}) \xrightarrow{\sim} (\mathbf{G}_{m,K}, \{1, \alpha(r)\}), \quad \kappa^* \frac{dw}{w} = \frac{2i dz}{1+z^2}.$$

The resulting Nori functoriality edge is contravariant:

$$(\mathbf{G}_{m,K}, \{1, \alpha(r)\}, 1) \longrightarrow (U_K, \{0, r\}, 1).$$

Over $\mathbb{Q}$, the same integral is a 1-period of $(\mathbf{P}^1 \setminus V(z^2+1), \{0, r\}, 1)$.

Proof. Solving the fractional-linear equation proves the inverse. The deleted points map to 0 and ∞ , while $0 \mapsto 1$ and $r \mapsto \alpha(r)$. Logarithmic differentiation proves the differential identity. The Nori edge direction is exactly the contravariant functoriality direction in Definition 9.1.1 of [55]. The rational form $dz/(1+z^2)$ is regular away from $V(z^2+1)$. At infinity the coordinate $t = 1/z$ gives

$$\frac{dz}{1+z^2} = -\frac{dt}{1+t^2},$$

so it is regular there as well. Hence the rational relative curve pair supplies the period. $\square$

The raw curve at lines 11515–11518 deletes infinity unnecessarily. Under κ , deleting infinity also deletes -1 from $\mathbf{G}_m$, so it is not the full pair isomorphism printed above.

The raw assertion at lines 11574–11588 of a path-indexed Nori diagram is not constructed. It omits the degree i , switches the attached Nori volume's convention $(k, \mathbb{Q})\text{-Vect}$ to $(\mathbb{Q}, k)\text{-Vect}$, and gives neither maps of pairs nor nested triples. Concatenating singular paths gives a relation among homology classes, not a map of ambient algebraic pairs.

There is nevertheless an exact multiplication map. For composable p, q ,

$$\chi_{q \circ p} = \chi_q \chi_p, \quad \alpha_{q \circ p} = \alpha_q \alpha_p.$$

Put

$$N_{p,q} = [\mathbb{Z} \rightarrow \mathbf{G}_m^2], \quad n \mapsto (\alpha_p^n, \alpha_q^n).$$

Then

$$M_{\text{ar}} \xrightarrow{(\text{id}, (\chi_p, \chi_q))} N_{p,q} \xrightarrow{(\text{id}, m)} K(\alpha_p \alpha_q), \quad m(z_1, z_2) = z_1 z_2,$$

is a pair of commutative 1-motive squares. The map of pairs

$$m : (\mathbf{G}_m^2, \{(1, 1), (\alpha_p, \alpha_q)\}) \rightarrow (\mathbf{G}_m, \{1, \alpha_p \alpha_q\})$$

gives the genuine contravariant Nori edge

$$(\mathbf{G}_m, \{1, \alpha_p \alpha_q\}, 1) \xrightarrow{m^*} (\mathbf{G}_m^2, \{(1, 1), (\alpha_p, \alpha_q)\}, 1),$$

and

$$m^* \frac{dz}{z} = \text{pr}_1^* \frac{dz}{z} + \text{pr}_2^* \frac{dz}{z}$$

proves period additivity. This is the exact replacement for the missing “concatenation arrow.”

Finally, for $r > 0$,

$$\text{Im} \frac{1 + ir}{1 - ir} = \frac{2r}{1 + r^2} > 0,$$

whereas $2^{A(p)} 3^{-\ell(p)}$ is positive rational. The two marked Kummer points are therefore unequal; no direct edge between those literal pairs is obtained by identifying them.

7.7 Status of the remaining assertions

The status of the raw interval is now exact:

- the bidegree, torus action, finite residue maps, full-shift pressure, logarithmic formulas, and arctangent integrals survive at the proved scopes stated here and in the preceding sections;
- the discrete arithmetic groupoid, the 3-adic germ groupoid, the faithful functor Ξ , the torus coordinate isomorphism, the Kummer squares, and the Cayley and multiplication maps are separately authored repairs proved in this edition;
- the printed $\ell^2(\mathcal{O})$ isometry and exact path-algebra recovery are disproved by $g_1(1) = 2$;
- the unnamed crossed-product equality, prefix endomorphism, equilibrium classification, cyclic-homology chain, motivic coaction, absolute-arithmetic interpretation, and local-factor package are not constructed by the raw formulas and are not admitted here.

Explicit nonclaim 7.6. The exact maps above do not prove amenability, equality of full and reduced groupoid completions, existence or classification of equilibrium states, a universal path-indexed Nori category, a motivic Galois action on the path algebra, or a Collatz endpoint theorem. These are precise nonclaims about this audited interval, not assertions that no such mathematics can be constructed elsewhere.

8 The affine-packet programme: source reconstruction and exact repairs

This chapter reconstructs the affine-packet mathematics in the 3,021-line modern Chatnotes export *Implications of Affine Formula* and of its immediate 61-page local antecedent, the OpenAI technical note *Affine Packets, Cyclic Orbit Packets, Dyadic Dilation, Lambert–Mahler Series, Hydra/Numen Structure, and Exact Rational Periodic Orbit Enumeration in the Odd-Step Collatz Algebra*. Neither local item is used as a substitute for published literature. The purpose of the reconstruction is to identify exactly which coordinates are classical, which maps have now been proved, which finite calculations reproduce, and which statements in the antecedent require repair.

The raw export has SHA-256

abb1abbb948dc140394de3d35f0efbde9b3facdf718c24b065b8d0adb951366e

and 3,021 physical lines. The local technical note has SHA-256

13f49961a60b870fa7a6f0f646758b4d72c1a4e4267f347606cc8a3238d552f1.

All page and line locators below refer to those exact manifestations.

8.1 The local sources

The raw export has the following chronological organization. Physical line numbers refer to the export itself, not its generated attachment labels.

Physical lines	source content
1–613	First synthesis of the attached local technical note; source names and theorem numbers here are routing prompts, not independent evidence.
614–1373	Recheck of the fixed-return algebra, cyclic-shift bridge, primitive necklaces, signed rational counts, packet envelopes, and the length-four argument.
1375–2001	Length-five argument; cycle product identity; exact minimum-point strip; reduction of period eight to $(m, A) = (8, 13)$.
2003–2679	Further derivation of the strip and candidate packet list.
2680–3020	First-letter residue recursion, the claimed $R_{8,13}(233)$ calculation, the 99 cyclic classes, and exhaustive candidate packet scans through $m = 16$.
3021	Exporter footer; no mathematical content.

The export refers to attached CSV, Markdown, and summary tables, but those objects are not embedded in the 3,021-line file. Accordingly no calculation was admitted from a link label. The 99-class table was independently recreated as `research_companion/data/period_8_A_13_classes.csv`, and every row is checked by the portable certificate.

The antecedent develops affine blocks, translations, cyclic transport, necklace counting, and several analytic constructions. Its Theorems 10.1 and 10.3 supply the cyclic-transport and primitive-necklace arguments examined below. The indexed source TeX and a second, 62-page PDF agree with the controlling PDF in the twelve statement/proof blocks collated for this chapter; the observed differences there are populated cross-references and repagination. This is a passage-level correspondence, not an identification of every unexamined statement or of the exact producing builds.

8.2 Published antecedents and the exact coordinate bridge

The core return formula predates the local corpus.

- Böhm–Sontacchi, printed pp. 263–264, give the formal affine composition and fixed-return candidate for a prescribed parity pattern [13].
- Crandall, printed pp. 1287–1290, gives the accelerated affine cycle formula, the minimum-orbit estimate, and a cycle-length exclusion based on the then-current verification range [1]. Thus the product/minimum strip in raw lines 1581–1620 and 1829–1908 is a direct form of a classical cycle inequality, not a new structural principle.
- Lagarias, printed pp. 34–39, proves that every binary parity word gives a unique rational periodic point, displays its numerator and denominator, defines irreducibility by cyclic rotation, and counts irreducible binary necklaces [17]. In particular, a new vocabulary for the same numerator cannot enlarge the theorem.
- Urata, printed pp. 8–11, writes the odd-step exponent numerator, proves the periodic-exponent/periodic-orbit correspondence, and asks exactly when

$$\frac{C(p)}{2^{a_1+\dots+a_m} - 3^m}$$

is an integer [20].

- Laarhoven–de Weger, source lines 303–321, relate rational cycles to binary necklaces/Lyndon words and to primitive $3n + b$ cycles [19]. Their statements retain the source defects and scope boundaries recorded in Chapter 2; they are not used to erase the primitive-denominator conditions in Lagarias.

For a positive exponent packet $p = (a_1, \dots, a_m)$, $m \geq 1$, of total $A \geq m$, define

$$E(p) = 1, 0^{a_1-1} \dots 1, 0^{a_m-1}.$$

This word uses Lagarias’s shortened map T , whose odd branch is $(3x + 1)/2$ and whose even branch is $x/2$, on the rationals with odd denominator. It is a bijection from compositions of A into m positive parts to binary words of length A , weight m , and initial bit one. If $S_j = a_1 + \dots + a_j$, the ones occur at positions $S_0, \dots, S_{m-1}$, so Lagarias’s numerator is exactly

$$N(E(p)) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{S_j} = C(p).$$

Packet rotation by j letters is not one-bit rotation: it is binary rotation by the prefix length S_j . The rotation comparison is the groupoid isomorphism

$$(p, j) \longmapsto (E(p), S_j(p)), \quad E(\sigma^j p) = \rho^{S_j(p)} E(p),$$

with inverse given by cyclic gaps between successive ones. The composition identity

$$S_{j+k \bmod m}(p) \equiv S_j(p) + S_k(\sigma^j p) \pmod{A}$$

proves functoriality. It also proves that primitive exponent packets and primitive binary parity words are the same periodic objects in different coordinates. The complete domain, codomain, inverse, fibres, and stabilizers are proved in the companion's theorems *Packet-parity coordinate isomorphism* and *Exact cyclic groupoid morphism*, in its section *Affine packets, parity necklaces, and exact residue automata*. Here the acting groups are $\mathbb{Z}/m\mathbb{Z}$ and $\mathbb{Z}/A\mathbb{Z}$; rotation arrows with equal targets remain distinct. For $p = u^r$, with u primitive of length d and sum B , the stabilizer map is $kd \bmod m \mapsto kB \bmod A$ for $0 \leq k < r$.

For the full map F , with odd branch $3x + 1$ and even branch $x/2$, the corresponding word is instead

$$E_F(p) = 1, 0^{a_1} \dots 1, 0^{a_m}.$$

It has length $A + m$. Inserting a zero after each one of $E(p)$, or deleting that zero from $E_F(p)$, gives mutually inverse pointed-word maps. The exact odd-return times are

$$T^{S_j}(x_p) = \Phi^j(x_p) = F^{S_j+j}(x_p).$$

For the primitive root u , the exact periods are d odd returns, B shortened steps, and $B + d$ full steps. The companion's theorem *Exact comparison of three clocks* proves the intermediate values and minimal periods, rather than identifying the clocks by terminology.

8.3 Cyclic affine transport and exact valuations

The antecedent already proves the affine arrow explicitly in Theorem 10.1, physical p. 36: $g_{a_1}(x_p)$ is fixed by $g_{\sigma p}$, whose fixed point is unique. Raw lines 959–1003 give the corresponding numerator calculation. The additional check needed to identify the actual odd-return branch is its exact valuation. With

$$D = 2^A - 3^m, \quad x_p = \frac{C(p)}{D}, \quad \sigma p = (a_2, \dots, a_m, a_1),$$

the first-letter recurrences, with $C(\emptyset) = 0$ for the $m = 1$ tail, give

$$3C(p) + D = 2^{a_1}C(\sigma p), \quad 3x_p + 1 = 2^{a_1}x_{\sigma p}.$$

Because $C(\sigma p)$ and D are odd,

$$\nu_2(3x_p + 1) = a_1, \quad \Phi(x_p) = x_{\sigma p}.$$

This proves an actual odd-step morphism; it is not merely equality of two formal affine expressions. Since $\gcd(D, 6) = 1$, both 2 and 3 are units modulo D . The first identity therefore gives $D \mid C(p)$ if and only if $D \mid C(\sigma p)$, proving invariance under cyclic rotation.

The exact positive-cycle criterion is therefore:

$$\left. \begin{array}{l} p \text{ primitive,} \\ 2^A > 3^m, \\ 2^A - 3^m \mid C(p) \end{array} \right\} \iff \text{a positive odd integral orbit of exact odd-step period } m$$

with exponent necklace $[p]$. If p is imprimitive, the same rational point belongs to the orbit of its primitive root; a length- m return is not thereby an exact period- m orbit.

8.4 Exact repairs to the 61-page local antecedent

The following are mathematical repairs, not stylistic preferences.

Source issue 8.1 (Periodic-tail calculation, physical p. 30). Theorem 9.2 asserts that an integral orbit with an eventually repeated exponent block p is already fixed under the block map from the beginning of that exponent tail. This assertion is correct. Write $D = 2^A - 3^m$, $C = C(p)$, and $z_j = Dy_j - C$ for the integral block samples $y_{j+1} = (3^m y_j + C)/2^A$. The required calculation is

$$z_{j+1} = \frac{3^m Dy_j + DC - 2^A C}{2^A} = \frac{3^m (Dy_j - C)}{2^A} = \frac{3^m}{2^A} z_j.$$

The printed intermediate numerator replaces DC by $3^m C$; the following line already gives the correct result. For $p = (2)$ and $y_j = 1$, the incorrect line equals $1/2$, whereas $z_{j+1} = 0$.

After the displayed repair, $2^{jA} z_j = 3^{jm} z_0$ and $\gcd(2^{jA}, 3^{jm}) = 1$ give $2^{jA} \mid z_0$ for every j . Since $A \geq 1$, a nonzero integer z_0 cannot have this divisibility: choose j with $2^{jA} > |z_0|$. Hence $z_0 = 0$ and $y_j = C/D$ for all j . The intermediate odd states repeat because the map is deterministic. This proves exact repetition, not only convergence. The hypothesis that the exponent tail repeats is retained; no eventual periodicity of an arbitrary orbit is inferred.

The same intermediate error occurs in the indexed 62-page PDF, p. 31, and in the indexed TeX at line 1977. These particular passages were collated directly; no whole-version identity is needed.

Source issue 8.2 (Complex Abel coordinate, physical pp. 11–12). On a simply connected invariant domain, arbitrary choices of a holomorphic logarithm need not satisfy $\text{Log}(\chi w) = \log \chi + \text{Log}(w)$ literally; the two sides can differ by $2\pi i k$. The transport statement requires a branch-compatible choice, or an explicit domain argument forcing $k = 0$. Later portions of the note impose compatibility, but Theorem 4.2 as printed does not.

Source issue 8.3 (Concentration calculation, physical p. 18). For $f(t) = t^2 \log^2 2$ and $x \geq y \geq 2$,

$$f(x+1) + f(y-1) - f(x) - f(y) = 2 \log^2 2 (x - y + 1) > 0.$$

The stated concentration conclusion is correct, but the displayed application in Corollary 6.5 reverses the sign in its rearrangement. The identity above is the repaired proof.

Source issue 8.4 (Maximizer multiplicity, physical p. 19). At the boundary $A = m$ there is only the all-ones packet, so the maximizing word has multiplicity one. More generally, the multiplicity is one when $m = 1$ or $A = m$, and is m when $m \geq 2$ and $A > m$, when the unique large entry can occupy any of the m positions. Corollary 6.7 omits this boundary case.

Source issue 8.5 (The $m = 1$ proof boundary, physical pp. 21–23). The displayed formula in Theorem 6.10 remains correct for $m = 1$, but the printed proof sets $r = m - 2$, whereas its auxiliary lemma on p. 20 assumes $r \geq 0$. For $m = 1$ there is exactly one packet, (A) , and hence

$$\mathbb{E}[a_1^2] = A^2 = \frac{A(2A - 1 + 1)}{1(1 + 1)}, \quad \mathbb{E}[Q] = (A \log 2 - \log 3)^2.$$

These are precisely the displayed formulas at $m = 1$, completing that case.

Source issue 8.6 (Pointed periodic words, physical p. 37). The last sentence of the proof of Theorem 10.3 incorrectly makes equality of infinite pointed periodic words equivalent to cyclic conjugacy of their primitive roots. For example, $(1, 2)^\infty \neq (2, 1)^\infty$. The correct statement is

equality of their primitive roots as pointed words. Indeed, extend a common periodic sequence to all integer indices. Its periods form a subgroup of $\mathbb{Z}$, so their least positive element divides every period; each word is a power of that common initial block. Primitive words therefore equal that block. In the source proof the base points have already been aligned, so this stronger conclusion supplies the needed implication. Passing back to unpointed orbits gives cyclic conjugacy, as stated in the theorem.

Source issue 8.7 (One-letter Hamiltonian domain, physical p. 55). The source Hilbert space is $\ell^2(\mathbb{N}_{\geq 1})$, and the multiplier is $u(a) = a \log 2 - \log 3$. The self-adjoint operator is

$$\mathcal{D}(H_1) = \left\{ \xi \in \ell^2(\mathbb{N}_{\geq 1}) : \sum_{a \geq 1} |u(a)|^2 |\xi_a|^2 < \infty \right\},$$

$$(H_1 \xi)_a = u(a) \xi_a.$$

To verify self-adjointness, test the adjoint identity on each basis vector: an adjoint-domain vector η must have adjoint image $(u(a)\eta_a)_a \in \ell^2$, and hence belongs to the displayed domain. Conversely this condition suffices by the Cauchy–Schwarz inequality. Thus the adjoint has exactly the same domain and action. Truncating at $a \leq N$ converges both in ℓ^2 and after application of H_1 ; finite-support vectors form a core, giving an essentially self-adjoint restriction, not a second unspecified self-adjoint operator.

The resolvent at i is the diagonal operator with entries $(u(a) - i)^{-1}$. Since $|u(a)/(u(a) - i)| \leq 1$, this inverse maps ℓ^2 into the stated operator domain. Its entries tend to zero, and its truncations converge in operator norm, so the resolvent is compact. Finally, for every $\beta > 0$,

$$\mathrm{Tr}(e^{-\beta H_1}) = \sum_{a \geq 1} 3^\beta 2^{-\beta a} = \frac{3^\beta}{2^\beta - 1},$$

as in Theorem 19.1. Specifying the domain completes this operator statement without changing its spectrum or partition function.

The note also contains unresolved bibliography/cross-reference markers and named analogies that are not accompanied by typed maps. Those analogies are not mathematical identifications and are not propagated. No theorem is discarded merely because it uses local vocabulary; each formula is instead crosswalked through its explicit coordinates.

8.5 What reproduces from the raw Chatnotes

8.5.1 Signed fixed-content necklace counts

Raw lines 1041–1083 give a valid fixed-label refinement of the classical primitive-necklace count. For $m \geq 1$, $A \geq m$, let $W(m, A)$ count the primitive words in $\Pi_{m,A}$. Unique primitive-root decomposition and Möbius inversion give

$$W(m, A) = \sum_{d \mid \gcd(m, A)} \mu(d) \binom{A/d - 1}{m/d - 1}.$$

The cyclic action is free on primitive words, so the number of rational Φ -orbits of exact odd-return period m at this packet label is

$$N_{\mathrm{orb}}(m, A) = \frac{W(m, A)}{m} = \frac{1}{A} \sum_{d \mid \gcd(m, A)} \mu(d) \binom{A/d}{m/d}.$$

The second equality is the fixed-Hamming-weight binary-necklace formula. Stanley’s multivariate Lyndon series enumerates primitive necklaces by their colour multiplicities; coefficient extraction at binary content $(m, A - m)$ gives exactly this expression [18, Exercise 7.89(a), equations (7.149)–(7.150), and the solution on printed p. 555]. Lagarias gives the unrestricted primitive binary-necklace and rational-cycle coordinates, and Laarhoven–de Weger restate the unrestricted count in Lyndon-word language [17, pp. 38–39][19]. The new step in the present reconstruction is not the enumeration formula: it is the exact transport through the packet–parity groupoid followed by the Collatz-denominator sign partition.

Every point belonging to the label (m, A) is $C(\sigma^j p)/(2^A - 3^m)$ with positive numerator. Hence the whole orbit is positive when $2^A > 3^m$ and negative when $2^A < 3^m$; equality is impossible. Multiplication by the relevant sign indicator gives the two signed orbit counts, and multiplication by m gives the two point counts. The companion states and proves all three typed counts separately. Raw line 1083 instead calls every word a periodic point. The exact statement is that a primitive word determines a distinguished point, while its m rotations determine the m distinct points of one exact-period orbit. When $\gcd(m, A) = 1$, every word is primitive and the orbit and point counts are respectively $m^{-1} \binom{A-1}{m-1}$ and $\binom{A-1}{m-1}$.

8.5.2 The analytic packet family

The raw series at lines 432–475 are substantive content, not merely motivation. In the source coordinates $c(p) = C(p)/2^A$, put

$$S_m(A) = \sum_{p \in \Pi_{m,A}} c(p), \quad X_m(A) = \sum_{p \in \Pi_{m,A}} \frac{C(p)}{2^A - 3^m}, \quad F_m(q) = \sum_{A \geq m} S_m(A) q^A.$$

The source’s Theorems 16.2–16.5 give, with $u = q/(1 - q)$ and $v = q/(2 - q)$,

$$F_{m+1} = v(3F_m + u^m), \quad F_1 = v, \quad F_m = \sum_{k=0}^{m-1} 3^{m-1-k} u^k v^{m-k}.$$

These identities follow by appending a letter, using $c(p, a) = 2^{-a}(3c(p) + 1)$, and summing a finite convolution at every coefficient. They supply the stated two-state linear representation. The companion section *Packet fixed-point series and their dyadic poles* reconstructs this family with its full proofs.

Source issue 8.8 (Generalized dilation equation, physical p. 51). Theorem 17.1 proves the coefficientwise identity

$$X_m(2q) - 3^m X_m(q) = F_m(2q), \quad X_m(q) = \sum_{A \geq m} X_m(A) q^A,$$

and its uniqueness, because $2^A - 3^m \neq 0$. The additional claim that the defining series therefore give this identity throughout $|q| < 1$ fails when $m \geq 2$. Already for $m = 2$,

$$X_2(A) = \frac{2^A + 3A - 5}{2^A - 9} \longrightarrow 1,$$

so $X_2(2q)$ diverges at $q = 3/4$. The phrase also occurs in the indexed PDF, p. 52, and in the indexed TeX at line 3376; it is not present in the raw Chatnotes statement.

The exact common Taylor-series domain is $|q| < 1/2$ for $m \geq 2$; for $m = 1$ it is $|q| < 1$. Indeed, for large A the total $X_m(A)$ is positive and bounded above by a constant times $\binom{A-1}{m-1}$. If $m \geq 2$, a word ending in 1 contributes at least $1/2$, so the radius of X_m is exactly one. For $m = 1$, $X_1(A) = (2^A - 3)^{-1}$ has radius two. The companion gives the coefficient bounds and the inverse of the formal dilation map explicitly.

The source already continues X_1 meromorphically with simple poles at $2, 4, 8, \dots$. The companion extends this calculation to the whole packet family: for $m \geq 2$, X_m has a pole of order $m - 1$ at 1 and a pole of order m at every 2^j , $j \geq 1$, with no others. The construction subtracts a finite coefficient prefix and sums scaled copies of the remaining rational function with a uniform geometric tail bound. It proves a meromorphic identity, not convergence of the original series outside its radius. No claim of priority over the broader literature accompanies this extension of the local source calculation.

8.5.3 The finite cycle calculations

The length-four and length-five arguments at raw lines 1209–1365 and 1567–1825 are arithmetically correct after the classical lineage above is made explicit. For length five the strip forces $A = 8$; the denominator is 13; and the seven cyclic representatives have numerator residues

$$3, 6, 12, 4, 10, 6, 7 \pmod{13}.$$

None is integral.

The packet residue recurrence at raw lines 2683–2790 is also exact. For a positive odd modulus q ,

$$R_{m,A}(q) = \{C(p) \bmod q : p \in \Pi_{m,A}\}$$

satisfies

$$R_{m,A}(q) = \bigcup_{a=1}^{A-m+1} \left(3^{m-1} + 2^a R_{m-1,A-a}(q) \right), \quad R_{1,A}(q) = \{\bar{1}\} \quad (A \geq 1).$$

The recurrence is for $m \geq 2$, $A \geq m$; $R_{m,A}(q) = \emptyset$ when $A < m$. For column count vectors the transition matrix is explicitly $M_{m,a} e_r = e_{3^{m-1} + 2^a r}$, with inverse $M_{m,a}^{-1} e_s = e_{2^{-a}(s - 3^{m-1})}$. Its dependence on a is periodic modulo $\lambda = \text{ord}_q(2)$, taking $\lambda = 1$ for $q = 1$. Summing the periodic matrix blocks proves the rational generating function in the companion, with every coefficient a finite sum.

For $(m, A) = (8, 13)$, direct enumeration and the independent recursion agree:

$$R_{8,13}(233) = \mathbb{Z}/233\mathbb{Z} \setminus \{0, 138\}.$$

The 792 packets form 99 free cyclic classes. Exactly 15 minimal representatives have numerator divisible by 7, none has numerator divisible by 233, and none is integral. The reproduced exhaustive table through $m = 16$ contains 4,734,620 packets and zero integral packets in the eight strip-selected labels

$$(5, 8), (8, 13), (10, 16), (11, 18), (13, 21), (14, 23), (15, 24), (16, 26).$$

The resulting exclusion of nontrivial exact odd-step periods $2 \leq m \leq 16$ is valid, but it is a finite certificate rather than a novelty claim; the published literature already contains much stronger cycle-length exclusions.

8.6 Reproducibility and admission boundary

The companion additionally retains the common exponent word when comparing residues modulo coprime factors. Its *Synchronized CRT transport* theorem gives the explicit isomorphism and transition identity, and *Information discarded by the marginals* computes the kernel of the marginal map. At $(m, A) = (10, 16)$ the complete joint table has 3473 occupied cells modulo $(13, 499)$ but no occupied $(0, 0)$ cell, although both marginal supports are full. All 35 words with numerator divisible by 499 are squares of packets in $\Pi_{5,8}$. Thus 499 already rules out an integral periodic point for every primitive word at this label. These are independently reproduced finite calculations, not a failure of the Chinese remainder theorem.

The root certificate `certificates/affine_packet_source_and_cycle_checks.py` pins the raw export, local antecedent, and exact primary manifestations of Böhm–Sontacchi, Crandall, Lagarias, Urata, and Laarhoven–de Weger. It then executes the portable companion certificate `research_companion/certificates/affine_packet_residue_checks.py`. The latter uses integer and exact rational arithmetic only, checks the packet–parity morphism on 12,089 packets, including 155,047 shortened steps, 233,525 full steps, and 78,478 odd returns. It checks exact periods for primitive and repeated packets, groupoid inverses and composition, 702 residue multiplicity vectors and 546 rational-series coefficient vectors, verifies the 99-row CSV, and runs two independent full packet scans. Within the portable `research_companion/` tree, the separate certificate `certificates/affine_signed_necklace_checks.py` checks the two signed fixed-content formulas on 100 parameter pairs containing 39,202 words and 39,022 primitive words, together with the cyclic freeness, packet–binary necklace bijection, point/orbit distinction, coprime case, and recovery of the unrestricted Moreau sum. All three scripts reject optimized Python.

The admitted mathematical content is therefore separated as follows.

- The rational fixed-return and parity-necklace theorems are attributed to their published sources.
- The packet–parity groupoid isomorphism, its fixed-content signed count, and the displayed residue recursion are direct deductions proved in the companion, with classical necklace lineage retained explicitly.
- The period-five, period-eight, and through-period-sixteen statements are finite exact calculations with pinned reproducible certificates.
- The local technical note remains a local antecedent, not a publication and not an authority for enlarging any published theorem.
- No statement here proves global termination, excludes all nontrivial cycles, or assigns a result to an author who did not state it.

9 Source register for admitted statements

This working register lists only sources already used in mathematical text. Machine-readable identities, hashes, manifestations, and reading status are maintained in `state/source_registry.jsonl`.

References

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quantifier, ineffective threshold, two order coordinates, Pillai dependency repair, separately attributed all-integer extension, source label defect, unpublished-computation boundary, and distinct Siegel-1929 dependency are recorded in `qa/HERSCHELD-1936-F018-eventual-uniqueness-audit.md`; deterministic finite checks are in `certificates/herschfeld_1936_checks.py`. Crandall’s reference 10 spelling “Herschfeld” is not the spelling on the primary byline. The conditional primitive-root paragraph printed on p. 1288 is reconstructed from the full monotone return equation, with its exact domain and two residue branches, in `qa/CRANDALL-1978-F020-primitive-root-audit.md`; the finite selector certificate is `certificates/primitive_root_return_checks.py`.

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exact convergent-error formula, law of best approximations, unimodular equivalence, Serret theorem, Markoff formula, and starred Exercise 5.22 were read from the local published-book PDF. The Exercise 5.22 solution, printed matrix defects, exact clock crosswalk, and nonclaims are recorded in `qa/STEUDING-2005-F011-exercise-522-audit.md`; exact finite quadratic-field checks are in `certificates/steuding_522_checks.py`. The exact transfer from Theorems 3.1, 3.2, 3.6, and 3.8 to Crandall’s Section 7 inequalities, including the strict $0 < y < q_n$ range and the $y = q_n$ equality boundary, is recorded in `qa/STEUDING-2005-F021-crandall-continued-fraction-audit.md`; its finite arithmetic and pinned-source checks are in `certificates/continued_fraction_dependency_checks.py`. This independent support is not an identification of Crandall’s cited Roberts or Khinchin manifestations.

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displayed definition of the analogue map $F : \mathbb{Z} \rightarrow \mathbb{Z}$ in Theorem 2 omits some divisible residue classes (for example 6 mod 8 when $d = 2, u = 3$); the live reader states the unique bounded residue correction as a separately attributed repair before using the same-slopes/opposite-dynamics obstruction.

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replaced by the proved multiplicative interval in Proposition 2.135; its dependency use has been checked through line 925, with the signed lower bound at lines 790–793 and the malformed line 889 estimate also repaired. The event chain, fixed- M affine equivalence, positive valuation domain, the distinct line 868 and line 899 previous-iterate coefficients, line 917 probability transfer, the line 689 $1.9n_0$ index, and the non-uniform line 913 summation are repaired in Proposition 2.136. This certifies source Proposition 5.2 and the uniform c_n lemma, and checks their Section 5 dependency use through line 925. The exact audit is in `qa/TA0-V7-F001-proof-text-audit.md`; the pinned-source, locator, and arithmetic checks are in `certificates/tao_v7_endpoint_buffer_checks.py` and `certificates/tao_v7_prop52_cn_checks.py`. Fine-scale mixing, Fourier decay, and renewal are reconstructed in Propositions 2.142–2.147 and Corollary 2.148; the line-level and version-comparison record is `qa/TA0-V5-V7-JOURNAL-F024-fourier-renewal-audit.md`; the exact finite lift, offset, quotient-frequency, holding-law, and entry-budget kernels are in `certificates/tao_v7_fourier_renewal_checks.py`. This closes the v7 fine-scale branch through Proposition 1.14 at its printed uniform scope. The independent residue-fibre proof of Proposition 1.9 is Proposition 2.152; its three-manifestation comparison is `qa/TA0-V5-V7-JOURNAL-F025-prop19-valuation-audit.md`, and its exact finite kernels are checked by `certificates/tao_prop19_valuation_checks.py`. The exact source edge from Propositions 1.14 and 1.9 to Proposition 1.11 is reconstructed in Theorem 2.160. Its three-manifestation comparison, source-defect boundary, affine path–endpoint bijection, integer window, and exact mixing collapse are recorded in `qa/TA0-V5-V7-JOURNAL-F026-prop111-first-passage-audit.md`; its finite integration kernels are checked by `certificates/tao_prop111_transport_checks.py`. Proposition 1.11 is therefore closed at its printed v7 scope. Its exact deductions through Theorem 3.1 and Theorem 1.6 to Theorem 1.3, including the indexed nested-threshold morphism, fixed-tail repair, dyadic stratum bijection and harmonic transport, are reconstructed in Subsection 2.23. The three-manifestation defect record is `qa/TA0-V5-V7-JOURNAL-F027-main-theorem-reduction-audit.md`, and the finite exact kernels are checked by `certificates/tao_main_reduction_checks.py`. The remaining whole-version comparison is recorded in `qa/TA0-V5-V7-JOURNAL-F028-whole-version-citation-audit.md`. It classifies all 28 non-whitespace v5–v7 hunks, the journal’s 15-v7/12-v5/one-hybrid manifestation matrix, the exact terminal boundary, and all active citations. Its Section 6 calculation proves that the v5 and journal half-window is too small for the printed Archimedean argument, whereas v7’s 0.99-window closes; see Source Issue 2.138. The deterministic source, bibliography, citation, and finite-margin checks are in `certificates/tao_whole_version_checks.py`. The substantive cited literature not yet present locally and the durable whole-paper release gate remain open.

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also drops the factor $\log 2$ in the printed conversion. An explicit concentration-admissible family makes one claimed sub- 3^n summand exceed 3^n . This invalidates that printed size argument without establishing a residue collision or disproving the standalone conclusion. V5 also prints $1_{\overline{E}_k \cap B_k \cap C_{k,l}}$ where the factorization requires $1_{E_k \cap B_k \cap C_{k,l}}$. Both defects are repaired in v7 and reconstructed in Source Issue 2.138. V5 and the journal also state the stronger unconditional form of Lemma 7.9, while v7 states an indicator-restricted form. The stronger displayed induction has an unresolved $r = 1 < R$ gap and is not used in this edition; see Source Issue 2.137. After whitespace deletion, its main reduction at lines 529–587 is exactly equal to v7 lines 535–593; this local equality does not identify the complete versions. The shared reduction defects and the independently authored exact repair are recorded in `qa/TA0-V5-V7-JOURNAL-F027-main-theorem-reduction-audit.md` and Subsection 2.23. The complete source comparison, including the journal’s mixed-version boundary and the exact bibliography, is F028; local equality of the bibliography and main-reduction blocks still does not identify the complete manifestations.

- [34] David Barina, *Convergence verification of the Collatz problem*, The Journal of Supercomputing **77** (2021), no. 3, 2681–2688, doi:10.1007/s11227-020-03368-x; published online 1 July 2020. All eight publisher pages and the distinct eight-page author postprint were read. Their SHA-256 hashes are respectively 737a225fbc291cc182e940cc7ca31da4d2eb99b64e51a33e380213d8ed6b33a3 and 6df53d8f2a41d032f215209f06a021d35e447f6bdf222f1d557be730b64409dc. The map charts, block algorithm, affine sieve coordinate, reported strict 2^{68} bound, record datum, exceptional-start defects, and nonclaims are reconstructed in Subsection 2.24 and `qa/BARINA-OLIVEIRA-ROOSENDAAL-F031-computational-verification-audit.md`.
- [35] David Barina, *Improved verification limit for the convergence of the Collatz conjecture*, The Journal of Supercomputing **81** (2025), article 810, doi:10.1007/s11227-025-07337-0; accepted 21 April 2025 and published online 2 May 2025. All fourteen publisher pages were read and rendered; the controlling PDF has SHA-256 764fd732ad79545e71440f74bf738bf315b479eb404f68bae2c3b109785a5d06. The FIT record’s volume 81, no. 1, pp. 1–14 coordinate is retained as a metadata discrepancy rather than substituted for the publisher’s article number. Table 10’s strict $n < 2^{71}$ boundary and 15 January 2025 completion date control over the surrounding phrase *up to*. The two reversed class cardinalities, three exceptional-start cases, optional-kernel and validation boundaries, and exact four-new/five-cumulative record reconciliation are stated separately from the reported finite result.
- [36] Tomás Oliveira e Silva, *Empirical Verification of the $3x+1$ and Related Conjectures*, in Jeffrey C. Lagarias, editor, *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, 2010, 189–207. Every printed page was read from a third-party scan whose page images remain temporary inspection witnesses and are not copied into the topical shelf or reader. Identity and pagination were crosschecked against the author’s University of Aveiro bibliography and the AMS volume record. The chapter’s inclusive $20 \cdot 2^{58}$, 10^{16} , and 10^{14} reports, duplicate-worker protocol, generalized maps, finite tables, and expressly non-rigorous Markov argument are retained at their stated scopes. Its floor sentence is repaired locally. Proposition 2 is not normalized: the exact $i = 1, n_0 = 1$ counterexample and independently proved 6^i branch-cylinder replacement are Source Issue 2.181 and Proposition 2.182.
- [37] Eric Roosendaal, *On the $3x+1$ problem*, continuously revised author web page, snapshot identifying itself as last modified 30 June 2026. The retained snapshot has SHA-256 e3ff21

- c62b1fe108ecaa9086a2c8679207339b3e086289a159b56099515f157d. It was captured over plain HTTP after the HTTPS endpoint presented a certificate-name failure, so transport authentication is explicitly absent. The modular-maximum argument is reconstructed in Proposition 2.180; the page’s maximum-variable and $N < 1$ defects are not silently corrected in attribution.
- [38] xbarin02, *collatz* and *collatz-sieve*, public source-code lineages. The 2025 article pins *collatz* commit 53c2a0608075d6fe3f10cc6eeef50e400c86338, tree 82b9731dcd5108298a2fb2085ba1d76c837de6f5; the earlier reconstruction boundary is commit 0fabf721434b20f00af820564e3b3781dcabe768, tree f8597ec3867e5734cd6a9fb57379aef2c670f857. Complete history is retained in a verified Git bundle with SHA-256 e7b655972c628f060ecbc7aaffaa381eeeb6e6868567d14c4f4524c18559cc3dd. The separate, article-unpinned sieve dependency is frozen for this audit at commit e5f0c264ba1f5f9c0a7d32a10c3aa61b225c70d1, tree 4b5a3b5afdf35596409c91da7f226e338c7c763a; its complete bundle has SHA-256 687f00faba696d2a3dc8b0a6f556c9cc6da56756bdb36535ac8f3ed49575983f. These pins support code-level reconstruction and the finite Theorem 2.175; without a completed-work ledger they do not authenticate a historical distributed run.
- [39] David Barina’s Collatz project, *verification status API*, dated snapshot generated 28 August 2026 at 19:24:59 +0200. The 198-byte JSON manifestation has SHA-256 daf454223878d578dc7a87dc0677aa1925eba602ae295f8547ce25927ce5c5ab and reports `verified_up_to_times_2_60=2075`, `work_unit_exp=40`, and first incomplete/unassigned unit 2175975546. It is a dynamic project assertion, not a paper result or an independently acquired server ledger.
- [40] N. Aizawa, P. S. Isaac, and J. Segar, $\mathbb{Z}_2 \times \mathbb{Z}_2$ *generalizations of $\mathcal{N} = 1$ superconformal Galilei algebras and their representations*, Journal of Mathematical Physics **60** (2019), 023507, doi:10.1063/1.5054699; arXiv:1808.09112v1 (submitted 28 August 2018). The complete 1,127-line source $\text{\TeX}$ was read. The paper supplies the color bracket, its enveloping-algebra construction, complete relation tables, central extension, decompositions, adjoints, and representations. It contains no Collatz map or action; Section 3 uses it only at the exact color-algebra scope and proves the coefficient crosswalk independently.
- [41] Manfred Droste, Werner Kuich, and Heiko Vogler, editors, *Handbook of Weighted Automata*, Springer, 2009, ISBN 978-3-642-01491-8, eISBN 978-3-642-01492-5. The content read for Section 4 comprises physical PDF pages 3–7, 20–22, 27, 84, 89, 134–135, and 191–192: free monoids and morphisms, semirings, formal power series and polynomials, path-label products, cycle-free and complete-semiring summation conditions, finite weighted automata, and matrix/formal-series behavior. The book supplies this standard framework, not the Collatz-specific theorems of that section.
- [42] Werner Kuich and Arto Salomaa, *Semirings, Automata, Languages*, EATCS Monographs on Theoretical Computer Science, volume 5, Springer, 1986. Physical PDF pages 1–4 of the local chapter-I witness were read for the definitions of free monoids, monoid morphisms, semirings, formal series, finite-support polynomials, and Cauchy multiplication. No Collatz result is attributed to this source.
- [43] Benjamin Steinberg, *Representation Theory of Finite Monoids*, Universitext, Springer, 2016, ISBN 978-3-319-43930-3, eBook ISBN 978-3-319-43932-7. Physical PDF pages 4–5,

- 22, and 203–204 were read for transformation monoids, transformation modules, rank, Proposition 13.1, and Lemma 13.2. The minimum-ideal argument in Proposition 4.14 is proved directly and then crosswalked against Steinberg’s general minimum-rank theorem.
- [44] Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, *Advances in Mathematics* **291** (2016), 274–329, doi:10.1016/j.aim.2015.11.045; arXiv:1502.05580. The complete local source $\text{T}_{\text{E}}\text{X}$ was used for the present crosswalk. The relevant source coordinates are the tensor square and Frobenius maps, source lines 1151–1214; the semiring $\text{Conv}(\mathbb{Z} \times \mathbb{Z})$ and its quotient, lines 1364–1400; and the reduced-square factorization, lines 1470–1519. The Frobenius correspondence construction and its composition law occur at lines 1522–1902, with the three cases stated at lines 1747–1755: $\Psi(\lambda) \circ \Psi(\lambda') = \Psi(\lambda\lambda')$ when $\lambda\lambda'$ is irrational; the same equality when both parameters are positive rational; and $\Psi(\lambda) \circ \Psi(\lambda') = \text{Id}_{\varepsilon} \circ \Psi(\lambda\lambda')$ when both parameters are irrational but their product is positive rational. In particular, the source does not define a Collatz path algebra or identify a nonnegative bidegree cone with the arithmetic-site square.
- [45] Bhargav Bhatt and Jacob Lurie, *The Prismaticization of p -adic Formal Schemes*, preliminary postscript. The local source $\text{T}_{\text{E}}\text{X}$ was read at its definition of WCart_X , lines 480–483; functoriality in formal-scheme maps, lines 523–530; the separate Witt Frobenius and Hodge–Tate material, lines 532–550; and the derived/classical warnings, lines 1450–1511. Section 6 uses only that stated functorial scope and independently proves that the raw affine branch is not an endomorphism of $\text{Spf}(\mathbb{Z}_3)$.
- [46] Michel Gros, Bernard Le Stum, and Adolfo Quirós, *Absolute Calculus and Prismatic Crystals on Cyclotomic Rings*, arXiv:2310.13790v3, 30 June 2025. The controlling 72-page PDF was read at Theorems 7.5 and 7.12, the de Rham and Hodge–Tate comparisons in Sections 8–9, and the prismatic F -crystal/crystalline-representation results in Section 10. Its base is the one-variable cyclotomic ring $W[[q - 1]]$ with k a perfect field of characteristic $p > 0$ and under its explicit root-of-unity hypotheses. No two-coordinate Collatz calculus is attributed to it.
- [47] Omri M. Sarig, *Thermodynamic Formalism for Countable Markov Shifts*, *Ergodic Theory and Dynamical Systems* **19** (1999), no. 6, 1565–1593, doi:10.1017/S0143385799146820. The official 33-page author preprint was read at physical pages 3, 5–7, 18, and 24–25. These pages contain the countable topological Markov shift and mixing definitions, fixed-base periodic definition of Gurevich pressure, condition (13), and Theorem 8. Condition (13) is the one-sided big-images condition: finitely many states $b_1, \dots, b_n$ receive an edge from every state. It is not BIP. At the theorem’s stated topological-mixing, locally Hölder, and finite-pressure scope, Theorem 8 says that a Gibbs measure exists exactly when the potential is positive recurrent, its RPF eigenfunction is uniformly bounded above and below, and condition (13) holds; then the Gibbs measure is $h d\nu$ and its pressure parameter is the Gurevich pressure. This theorem is not silently replaced by the later BIP criterion.
- [48] Omri M. Sarig, *Thermodynamic Formalism for Countable Markov Shifts*, in *Hyperbolic Dynamics, Fluctuations and Large Deviations*, Proceedings of Symposia in Pure Mathematics **89**, American Mathematical Society, 2015, 81–117, doi:10.1090/pspum/089/01485. The controlling local file is the 36-page 2015 overview despite its misleading filename. Physical pages 14, 16, 18, and 19 were read for Theorems 4.3, 4.9, 5.3, Definition 5.8, and Theorem 5.9;

physical pages 35–36 explicitly list the 1999 article as a distinct source. The overview is therefore neither a manifestation nor a theorem locator for the 1999 article.

- [49] Omri Sarig, *Existence of Gibbs Measures for Countable Markov Shifts*, Proceedings of the American Mathematical Society **131** (2003), 1751–1758, doi:10.1090/S0002-9939-03-06927-2. All eight pages were read. Theorem 1 gives the Gibbs/BIP equivalence under topological mixing, summable variations including finite first variation, and finite Gurevich pressure. In Theorem 6.9 these hypotheses are checked only for the completed full shift; the positive arithmetic locus is handled separately.
- [50] John H. Conway, Charles Radin, and Lorenzo Sadun, *On Angles Whose Squared Trigonometric Functions are Rational, Discrete & Computational Geometry* **22** (1999), no. 3, 321–332, doi:10.1007/PL00009463; arXiv:math-ph/9812019v1. The versioned arXiv source was acquired to the topical shelf. Its controlling Plain-TeX file has 42,239 bytes and SHA-256 0b37130b7e958c58c1e9f70f855f9181699b0f42d78ac5d04c29329aae0db21d. Theorems 1–2 are at source lines 151–168. Lines 251–306 reconstruct the $d = 1$ Størmer theory from unique factorization in $\mathbb{Z}[i]$, orient one Gaussian prime over each split rational prime, and prove the corresponding angle coordinates rationally independent modulo π . This supplies the fixed-instance relation algorithm used in Section 5, not a uniform Collatz-prefix rank theorem.
- [51] Jack S. Calcut, *Gaussian Integers and Arctangent Identities for π* , *The American Mathematical Monthly* **116** (2009), no. 6, 515–530, doi:10.1080/00029890.2009.11920967. The controlling 16-page PDF has 169,888 bytes and SHA-256 d650da61527c78c67caca06758363280bac7303cb2b553fc786914ddf3a4886f. The Main Lemma is on physical page 5 / printed page 519; Corollaries 1–3 are on physical page 6 / printed page 520. They prove that the relevant rational angle is in $(\pi/4)\mathbb{Z}$ and that a single rational tangent gives a rational multiple of π only for tangent 0 or ± 1 .
- [52] Armengol Gasull, Florian Luca, and Juan L. Varona, *Three Essays on Machin’s Type Formulas, Indagationes Mathematicae* **34** (2023), no. 6, 1373–1396, doi:10.1016/j.indag.2023.07.002; arXiv:2302.00154v2. The versioned arXiv source was acquired to the topical shelf. Its controlling source-TeX file has 82,898 bytes and SHA-256 53f19ec4bf58c9e64ec51a2397401cb51fc450c7d9eb5ec61d2cf0372c162baf. Section 2 begins at source line 230. Theorem 1 is at lines 246–267 and contains $(1, 1/2, 1/2, 3/4)$; lines 278–312 give the Gaussian root-of-unity reformulation; and lines 341–352 derive the relevant zero double-angle family and its exceptional $a_1 = 1$ specialization. Its literal theorem scope is two terms with nonzero right-hand side, not all zero relations or arbitrary Collatz-prefix length.
- [53] Valentin Deaconu, *Groupoids Associated with Endomorphisms, Transactions of the American Mathematical Society* **347** (1995), no. 5, 1779–1786. The controlling nine-page PDF has 742,789 bytes and SHA-256 63a34e76b5404328d6eb5e7516660af357511d824a6de4b4a6286112e77cfdae. Physical pages 2–4 / printed pages 1779–1781 were read, and physical pages 2–3 were rendered. They supply the set of triples, the groupoid operations, and the compact-Hausdorff self-covering hypotheses of Theorem 1. The positive-odd discrete theorem in Section 7 is proved independently because those hypotheses do not hold there.
- [54] Annette Huber and Gisbert Wüstholz, *Transcendence and Linear Relations of 1-Periods*, Cambridge Tracts in Mathematics, volume 227, Cambridge University Press, 2022. The

controlling 265-page PDF has 5,175,455 bytes and SHA-256 1743c141fd0078e7ac19805a5174078a26551d978bf6d1cb623bf3b237da3d17. Physical pages 90, 116–120, and 133–140 / printed pages 69, 95–99, and 112–119 were read; physical pages 119 and 138 were rendered. Definition 8.1 fixes 1-motives and morphisms as complexes and commutative squares, Section 10.2 supplies the Kummer motive and path-selected logarithm, and Proposition 12.10 supplies the 1-period comparison. The concrete point $(2, 3)$ and all path characters in the live chapter are independent specializations.

- [55] Annette Huber and Stefan Müller-Stach, with contributions by Benjamin Friedrich and Jonas von Wangenheim, *Periods and Nori Motives*, Springer, annotated version dated 1 July 2025. The controlling 402-page PDF has 1,855,231 bytes and SHA-256 6d08014f21a616abb4ba7e177effad4d15e4a215b0f234ce3a4605c20a02b731. Physical pages 161–164, 231–232, and 323–324 / printed pages 139–142, 209–210, and 301–302 were read. Definition 9.1.1 states exactly two primitive edge types for effective pairs: contravariant functoriality from a map of pairs and the coboundary edge from a nested triple. It also confirms the volume’s $(k, \mathbb{Q})$ -Vect convention. No path-concatenation edge is attributed to the source.
- [56] Luca Barbieri-Viale and Bruno Kahn, *On the Derived Category of 1-Motives*, *Astérisque* **381** (2016); arXiv:1009.1900v2. The versioned source archive was acquired to the topical shelf. Its controlling source file `der1motast-pb1.tex` has 564,248 bytes and SHA-256 f884b90f3760ac666c1047a975e6788dfc02a1eb7f047e8eae8b57df6f013c18. Source lines 728–750 define a Deligne 1-motive and its commutative-square morphisms; lines 1529–1618 give the Cartier-dual character-lattice machinery; lines 1736–1751 state the fully faithful curve-generated embedding; and lines 3796–3820 define the Albanese functor and record the integral adjunction boundary. The source contains no Collatz-specific $(2, 3)$ specialization or Kummer character used in the live chapter.